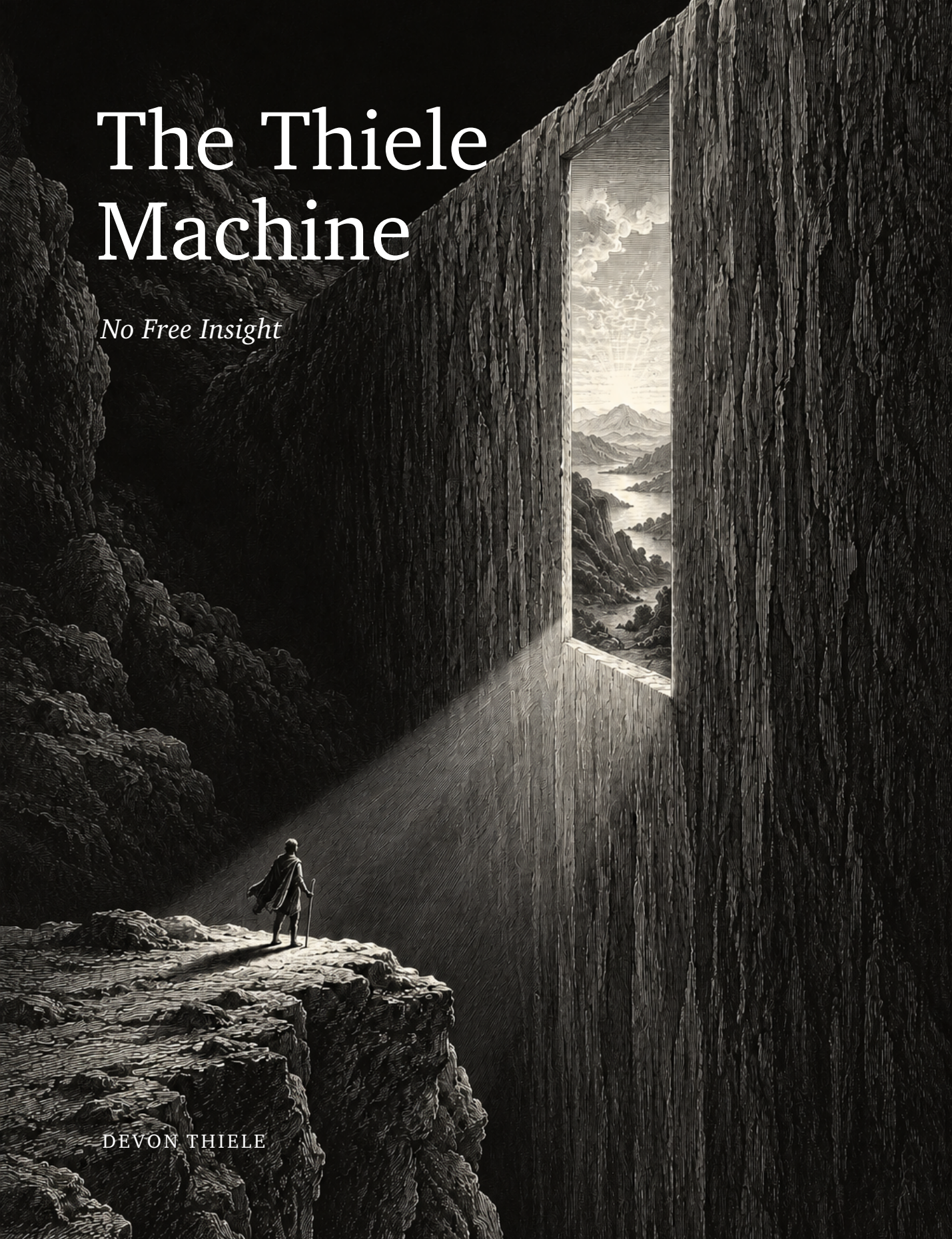
The Thiele Machine

No Free Insight



DEVON THIELE

DEVON THIELE

The Thiele Machine

No Free Insight

The Thiele Machine: No Free Insight

Devon Thiele

Copyright 2025–2026 Devon Thiele. Licensed under Creative Commons Attribution-ShareAlike 4.0 (CC BY-SA 4.0).

The source of this book, its proofs and its build are public: github.com/sethirus/The-Thiele-Machine. Archived at doi.org/10.5281/zenodo.17316437.

Set in Charter. Built from its source with pdfTeX.

The cover and part illustrations were generated with AI in the engraving tradition of Gustave Doré. They are new illustrations, not reproductions of his work.

For Julie, and for my dad.

Julie,

Three years ago, a few months into the rendering engine, I came out of my office and told you I thought I was onto something. You said okay. You didn't ask what it was, or what it was for, or when it would be done, and you didn't ask me for anything I could have shown you, because there was nothing to show. There was a rendering program doing things I hadn't built it to do, and a man who didn't yet know what a Turing machine was.

You gave me the three years anyway. The nights after the lot closed. The weekends. The hours I was in the house and not in the house, because my head was in a wall I couldn't get past. Every dead end, and there were more of those than there are pages here. You never once asked me to stop, and you never once said it was fine when it wasn't. You just kept giving me the next evening.

This book is what I did with them. It's the one thing I've built from nothing that I'm sure of, because I threw out every piece of it I couldn't prove, and what's left holds. I know that isn't what you gave the years for. You gave them for me. But it's what I have to show, and I wanted it to be worth showing.

Dad,

The name on the cover was yours first. I asked you before I used it and you said go ahead. You and Mom taught me that nothing is true just because somebody says it is. This book is that lesson, hundreds of pages long. I put our name on it because I wanted one thing in the world with that name on it that nobody handed me, that I couldn't have faked, and that I can stand behind in front of anyone. Here it is.

Julie carries the name now too. So the machine has it third. This is for the two of you who had it first.

Contents

What this is 13

1 Any machine that steps 16
- 1.1 Is the pencil thing a computer? 16
- 1.2 How far does that go? 17
- 1.3 Try to break it 18
- 1.4 Now ask it about keeping 19
- 1.5 The machine that stops 19
- 1.6 What Turing actually proved 20

2 Running out of room 22
- 2.1 Four people, three chairs 22
- 2.2 This is very old 23
- 2.3 The hotel that never runs out 25
- 2.4 Cut the strip 27
- 2.5 How long can a finite machine hold off? 28
- 2.6 What's being counted 28

3 Why charge for it 29
- 3.1 A rule you can refuse 29
- 3.2 The first door: endless paper 30
- 3.3 The second door: let it come back down 30
- 3.4 The third door: don't charge for it 31
- 3.5 How much 32
- 3.6 Found, assumed, chosen 33

4 The smallest one 35
- 4.1 Clay, seals, and seven witnesses 35
- 4.2 Two columns and a stamp 36
- 4.3 Closer to now 37
- 4.4 Five jobs 38
- 4.5 Two boxes and a coin 39
- 4.6 What the tally is and isn't 40
- 4.7 Any machine can wear one 41
- 4.8 Which question gets charged? 41
- 4.9 The cheapest price list 42

5 More settled 43
- 5.1 A row of cups 43

5.2 Naming what's on the table 45
5.3 The toll on the whole direction 46
5.4 One cup isn't enough 47
5.5 Somebody got here first, from the side 48
5.6 Why crossing a line has to cost 48

6 Looking isn't keeping 50
6.1 A small being at a small door 50
6.2 Bennett moves where the cost is 51
6.3 The demon on a sheet of paper 52
6.4 Two rows of dots 53
6.5 Who holds the pen 54
6.6 Learning after the first look 55
6.7 Finishing together 55
6.8 The certainty count 56
6.9 Who gets charged is partly bookkeeping 56
6.10 Somebody got there first, goddamn it 57
6.11 What the toll is for, said at last 58

7 Looking through a window 60
7.1 Two boxes under a camera 60
7.2 A cave, and the people who got good at it 60
7.3 The shadow of a cube 61
7.4 The whole fact 62
7.5 When does one exist? 62
7.6 People who found this on their own 63
7.7 Notes passed across a room 64
7.8 Slips handed out at desks 66
7.9 Moves that keep to the tabletop 66
7.10 The clerk with two photographs 67
7.11 Look before the concrete covers it 68
7.12 Three ways past the window 68
7.13 What gets through 71

8 Established 72
8.1 Two computers that end up the same 72
8.2 Look harder 73
8.3 So keep the trace 73
8.4 More boxes to tick 75
8.5 Computed, stored, established 76
8.6 The clerk who forgets everything 76
8.7 Information doesn't have it either 77

8.8 Statues that walk off 78
8.9 What “could have failed” means 79
8.10 Computing already had a word for it 80
8.11 The two computers, put back together 82
8.12 What putting it in the step buys 83

9 Settled 85
9.1 Settled, whether it’s true or not 85
9.2 A proof is an act 87
9.3 Not a switch 88
9.4 How a statement gets established 90
9.5 What settling looks like 91

10 What structure is 93
10.1 A string, a square, and a drowning nobody can place 94
10.2 Shapes poured into nothing 96
10.3 Prove it from what you and I both accept 97
10.4 The statue and the bronze 98
10.5 Where things are, without numbers 98
10.6 Naming a thing by saying when two are the same 99
10.7 Choose the changes, keep what survives 100
10.8 An invariance comes with a conserved quantity 101
10.9 All of mathematics as structures 101
10.10 A set with some named relations 102
10.11 The structure is in the arrows 103
10.12 Structure is the part that compresses 105
10.13 No place for a grouping 108
10.14 Laying them all down 109
10.15 The smallest piece 113
10.16 Buckets 114

11 The counter 118
11.1 A stick, split down the middle 118
11.2 Why it has to climb 119
11.3 The total is the charges and nothing else 121
11.4 Only one honest count 121
11.5 The toll is an old trick with a new name 122
11.6 What the counter is, said once 124

12 What a ticked box lets you lean on 126
12.1 The coin that moved 126
12.2 Plato nearly had it 127
12.3 Three things on hand 128

12.4 Changing your mind about the cups 130
12.5 Count the rivals 132
12.6 Narrower means a stronger test 133
12.7 What you've got 134
12.8 What standing behind a grouping gets you 135

13 What insight is 136
13.1 Sultan and the sticks 136
13.2 It isn't a feeling 137
13.3 What happens when a computation learns 138
13.4 Test, then commit 139
13.5 What the word means 140

14 A box over any machine 141
14.1 A cup and a die 141
14.2 What "answers only to the machine" has to mean 142
14.3 A base that never merges 143
14.4 The clean start 144
14.5 Both rules do work 146
14.6 Arguing with cups 147
14.7 A box that can change its mind 148
14.8 One shape, two choices nobody forces 149
14.9 What leaving the base open buys 150

15 The machine with the record dropped 151
15.1 What's left when you cover the cups 151
15.2 Pricing through a window, with a whole row 152
15.3 What a paid move destroys 152
15.4 What's left over is the ordinary computer 153
15.5 Shortcuts 154

16 Reading the box 155
16.1 The shortcut 155
16.2 Four doors 156
16.3 Running all four 157
16.4 Why the other doors exist 158
16.5 What the doors say 159

17 What a machine can't decide about itself 161
17.1 Five names and a table 161
17.2 The same move, again and again 162
17.3 Where the diagonal gets misused 165
17.4 Now point it at the box 166
17.5 What this does to the book 167

17.6 Two walls 168

18 Five real systems 169
18.1 Proof-of-stake finality: the price is on going back 170
18.2 Gas: the one that charges a fee at the write 172
18.3 Attestation: a quote about the past 174
18.4 Certificate transparency: visible, unpriced 177
18.5 Proof-carrying code: no toll required 179
18.6 The same question, five ways 179

19 Which box is worth keeping 181
19.1 A dial and the light off it 181
19.2 The claim 182
19.3 How much of the borrowed shape survived 184
19.4 What the thesis asks of the examples 186
19.5 Where the argument has got to 186
19.6 Does the list of requirements pick out the toll? 187

20 Heat 189
20.1 The stamp at zero 189
20.2 What Landauer said 190
20.3 Who doubts it 190
20.4 Where the difference goes 192
20.5 Two meanings of reversible 193
20.6 Counting in bits 194
20.7 How big is that 196
20.8 Three roads, two strengths 197
20.9 The coin under the cup 198
20.10 Going faster gives off more heat 200
20.11 A simulation in code 201
20.12 What's been measured 203
20.13 Why the rule sits on the raise 204

21 Keeping without the heat 205
21.1 A cell that's written once 205
21.2 Noisy steps 206
21.3 Quantum steps 207
21.4 How much room keeping takes 209
21.5 Two levers 210
21.6 Keeping every history 212
21.7 The counter and the heat 213
21.8 What you're entitled to 214
21.9 The shape of the second law 216

21.10 At the bottom: what I suspect and can't prove 217

22 What's shown, what's chosen, what's argued 219
22.1 What it is, and what it's for 219
22.2 The objection a theorist makes in four seconds 220
22.3 Bolting two together 221
22.4 Steps that branch 221
22.5 What doesn't follow 222
22.6 The ledger 223
22.7 Whose is what 229
22.8 The Thiele thesis 230
22.9 What isn't settled 231
22.10 Back to the warehouse 231
22.11 The blindfold 232

Afterword: how I got here 233

A Things I had to learn first 238
A.1 Counting 238
A.2 Machines and programs 239
A.3 Heat 240
A.4 Structure 241
A.5 Words this book builds 242

B Every assumption in one place 245
B.1 The definition 245
B.2 What a machine is 245
B.3 Prices 246
B.4 Conditions put on a machine 246
B.5 Assumptions about the world 247
B.6 Theses 248

C What's mine, and what isn't 249
C.1 The prior-work map 250
C.2 What I'm claiming 253

Sources 255

What this is

I think in shapes, so I'll start with one.

Take a thing. Anything. A coin on a table, a light switch, a page in a notebook. Right now it's one way. It could be another.

Now do something to it. Flip the coin, hit the switch, write on the page. That's a move: a direction that takes the thing from how it is to how it is next.

Flip here means turn it over by hand. When something else picks what happens, chance with a die, your hand in a bucket of numbers, a person at a button, a message coming in, call each way it can go its own move, with exactly one next thing each. Whatever does the picking sits outside the machine, and chapter twenty-two takes on what that changes.

Here's a computer you can run with a pencil. Write 3 on a scrap of paper. The move is: erase the number and write the next one up. Do it twice. 3, 4, 5. You just added 2 and 3. Nothing happened there but a thing, a move, and the next thing. Hold on to that picture; whether it covers every computer is something I'll have to earn, and I will, later, with you watching.

Here's the part I couldn't let go of. Some moves just shuffle. Flip the coin and you can flip it back, and nothing is different for good. But some moves make a difference that stays. Write a word in pen. Blow a fuse. Check an answer and mark it checked. After that move something is the case that wasn't before, and it stays the case.

I figured there had to be a cost in that step. Not in the shuffling. In the step where something goes from not settled to settled and stays there. That was a hunch, so test it on the pencil computer.

Tell it when to stop: count up, and when you reach 5, stop. Ask one question of each number: is it done? 3, no. 4, no. 5, yes. And once it says yes it keeps saying yes, because the machine has stopped.

A machine that has stopped is still somewhere on the next tick. It stays where it is: 5 goes to 5. Draw the arrows. 4 goes to 5. 5 goes to 5. Two different starting places, one landing place. Standing on 5, you can't tell whether you just got there or have been there all along.

Could you dodge that? Try letting the stopped machine rock back and forth instead of sitting still: 5-tick, 5-tock, 5-tick, both of them saying done. Draw the arrows. 4 goes to 5-tick. 5-tock goes to 5-tick. Two arrows land on 5-tick, the spot where you came in. Rocking only moved the place where they meet. The one way out is to never land anywhere twice: make every tick write something new, a tally

mark, the time, anything, so the paper never repeats. That works as long as there's paper. On a finite sheet you run out, and two arrows land on one spot again.

So whenever this machine settles its answer and keeps it, something gives. Either two states become one, or it uses up fresh paper on every tick for as long as it runs. That part isn't a hunch. You drew it. You drew it for one small counter; the book proves it for any machine that goes from one state to a next, and shows you where.

What I do with it is a choice, and I'll say so. I charge for the move where the answer goes from not done to done. Charging for it is up to me. That something has to give once it makes that move isn't. The two arrows don't always meet on that very move. Give done two spots, 5a and 5b: 4 goes to 5a, 5a goes to 5b, and 5b stays put. Now the arrows from 5a and 5b meet on 5b, a tick after the answer settled. But after the move from not done to done, there's no avoiding it. Two arrows meet somewhere, or the paper never repeats.

Two more things have to be pinned down before the rule means anything, and you can see both with what's already on the table.

First, take the coin again. Flip it, flip it back, forever, free. Ask whether it's settled. It never is. A rule that says "a move that settles something has to cost" never asks the coin for anything, because the coin never settles anything. So the rule alone would let a coin that does nothing count. The machine has to actually settle something, at least once.

Second, suppose a later move could pay you back. The move that settles charges you one, the next move hands you one back, and the whole run comes out free. So prices can be nothing or something, never a refund.

One last thing before the whole definition. Settled doesn't have to be just yes or no. Take a checklist with three boxes. None ticked, one, two, all three: each tick settles it a little more, and ticks don't come off. "Done or not" is the smallest case, a checklist with one box. So when I say "more settled" from here on, picture the checklist.

So here's the Thiele Machine, all of it. A thing. Moves that take it to the next thing. A way to tell how settled the thing is, where settled never goes back to unsettled. Some run, from a start you pick, that actually settles it more than it was. A price on every move, nothing or more, never a refund. And one rule: a move that settles something more than it was costs something.

That's the whole definition. The rest of the book works out what follows from it, one step at a time, and tries to break it.

Now look back at the pencil computer with a stop in it. That's Turing's machine, in miniature. Turing's machine, from his 1936 paper, is a tape and a table of rules: from where it is, the table says where it goes next. A thing and moves. It already has something that settles and stays settled: whether it has stopped, the same question you asked of 3, 4 and 5. What Turing never put in is a price on the move that stops

it. Put one in, and a Turing machine that ever stops fits every line above. So this isn't a rival to Turing's machine, and it computes nothing his can't. With that price in, his machine that stops is one kind of Thiele machine.

Now try this. Give two people the same messy list of numbers. The first sorts it, then goes back down it checking that every number is no bigger than the next. The second copies a sorted list off somebody else's sheet. Now look at the two sheets. Same numbers, same order. Look as hard as you like: nothing on either sheet tells you which one was checked. The difference was real, and neither sheet kept it. It's the same thing you saw standing on 5: you can see where you are, and not how you got there.

So put a box on each sheet that says checked. A plain Thiele machine charges for ticking the box, but nothing in the rules above stops the copier from paying and ticking it anyway. Paying for the tick doesn't make it true. So add one more condition: the box can only be ticked right after a check of that very list passes. Now the copier's tick is refused, and a ticked box tells you a check of that list passed. I call a box like that established. It's a condition you put on the machine. The bare definition doesn't hand it to you.

One more picture to carry through the book. Write "checked" on Turing's tape, and nothing in his model stops the next rule from rubbing it out, or from printing it without the check; a particular table might leave it alone, but the model doesn't make it hold. Cover a Thiele machine's record with your hand and set the price of settling to zero, and what's left is a bare machine that steps: every one of them, Turing's included, is a Thiele machine with a blindfold on.

It's called the Thiele Machine. The afterword says why.

CHAPTER 1

Any machine that steps

When I started all this I didn't know what a Turing machine was. I knew it had something to do with computers, and that's it. I had a program that had done something I didn't expect, and a hunch, and no words for either. Most of the words in this book I found the same way: I'd hit a wall, go looking for whoever had been there before me, and not move on until I could watch the idea actually work. This chapter is the first of those trips. I'll take you on it the way I took it.

The question I had was the one I left you with on page one. I'd been calling that pencil-and-scrap-of-paper thing a computer. Is it? Or did I just make it small enough to look like one?

1.1 Is the pencil thing a computer?

Start with what the word used to mean. Before about 1940 a computer wasn't a machine. It was a job. In the late 1800s the Harvard Observatory hired rooms full of women to do the arithmetic of the sky, cataloging stars by hand for a fraction of what a man was paid. Some of them found things nobody else had; Henrietta Leavitt worked out a way to tell how far away a star is. (I got that from Dava Sobel's history of those women and from the historians at the American Institute of Physics.) Rooms like that worked out artillery tables in both world wars; that one's from David Alan Grier's history of human computers.

So the pencil thing wasn't a toy version of a computer. For a long time it was the only kind there was. Good start, but that's history, not proof. A person with a pencil can do a lot of things that aren't following rules. I wanted to know what was left if you took the person away and kept only the rules.

That's exactly the question Alan Turing asked, in a paper in 1936. Going in, all I knew was that it had something to do with computers. What's in there is a clerk. He wrote down, as plainly as he could, everything a person sitting at a desk with a pencil does when they follow a procedure, and threw out everything else. What was left had four parts, and you can build it on the table in front of you right now.

Draw a strip of three squares. That's the tape. Write 1 in the first square, 1 in the second, leave the third blank.

Put a coin on the first square. That's where the clerk is looking. Turing called it the head.

On a scrap of paper write the letter A. That's which step of the procedure the clerk is on. Turing called it the state.

Now the rules. Write these two on a card:

> If you're on A and looking at a 1: leave it, move the coin one square right, stay on A.
>
> If you're on A and looking at a blank: write a 1 there, don't move, and change the letter to STOP.

That's the whole machine: a tape, a head, a state, and a card of rules. Now run it. Each tick, read the card and do what it says. After each tick, write down everything about the table: what's on each square, where the coin is, and the letter. I'll put brackets around the square the coin is on.

[1]	1	␣	A
1	**[1]**	␣	A
1	1	**[␣]**	A
1	1	**[1]**	STOP

It started with two 1s and ended with three. It added one. That's the same thing the pencil computer did when it went from 4 to 5.

Now look at what you wrote down on each line. The whole tape, where the coin is, and the letter. Is there anything about this machine at that moment that isn't on the line? Try to think of something. The rules don't change. The tape, the coin and the letter are all there is. So one line is the whole thing, and the card takes one line to the next line. Thing, move, next thing.

So the pencil computer wasn't a cheat. Turing's machine is the pencil computer with a bigger piece of paper.

1.2 How far does that go?

It would be nice if the answer were "every computer, done." It isn't quite. Push on it and it came apart into three different claims that people tend to run together. Here they are, and what each one stands on.

The first one is the one we just did. Every machine you can write down as a whole state and a rule for the next state has the shape from page one. You checked that yourself for Turing's machine: one line, one card, next line.

The second is that the other ways people have written computation down, with registers instead of a tape, or with functions handed to functions (that's Alonzo Church's lambda calculus, from the same year as Turing's paper), compute exactly the same things Turing's machine does. I didn't prove that, and I won't pretend I did. Other people proved it, one pairing at a time; Turing himself sketched the

match with Church's version in an appendix to his 1936 paper. What you can check yourself is that each of those is also a whole state and a rule for the next one, the same way you just did with the coin.

The third is the big one, and here's where I had to be careful. Does this cover everything a person could ever do by following rules? Does it cover every machine anybody could ever build out of real stuff? Those turned out to be claims nobody can prove, and for a good reason. "Everything a person could do by following rules" isn't a precise enough thing to prove anything about. That claim has a name, the Church-Turing thesis. "Every machine you could build" is a claim about physics, and the only way to knock it down would be to build one that beats Turing's, which nobody has done. Neither is a theorem, and in this book neither will get treated like one.

So the debt from page one comes out like this. For every machine you can write down as a whole state and a rule for the next state, paid: you just checked it. For every machine anybody could ever build, that's a claim about the world, it's labeled as a claim, and I'm not going to lean on it.

1.3 Try to break it

Now the question that actually mattered to me. Page one showed that when the counter settled on 5, two arrows had to meet, unless it used fresh paper forever. That was one little counter. Is it true of every machine with this shape?

Don't take my word for it. Get a sheet of paper and do this.

Draw some dots. Ten, twenty, as many as you like. Those are states. From every dot, draw exactly one arrow to some dot. It can point anywhere, even back at the dot it starts from. That's the one rule: every state has exactly one next state.

Now pick a dot to start on and follow the arrows. Keep a list of every dot you've been on.

Do it a few times, with different drawings and different starts. Watch what happens at each step. You'll find that each step does one of three things.

It lands on a dot that isn't on your list yet. A new one.

It lands on the dot you started from. Now you're going around in a circle, and since every dot has only one arrow out, you'll go around that same circle forever.

Or it lands on some other dot that's already on your list. Look at the first time that happens. The first time you got to that dot, you came in on an arrow from the dot right before it on your list. Now you've come in again, on an arrow from the dot you were just on. Those have to be two different dots, because until this step you hadn't been on any dot twice. Two arrows, from two different dots, landing on one. That's a merge, the same thing as 4 and 5 both going to 5. And from there you're on a path you've walked before, so you walk it again, forever.

Now try to make it do a fourth thing. Really try. Draw whatever you want.

Here's why you can't. Wherever the next arrow points, that dot is either on your list or it isn't. If it isn't, it's new. If it is, it's either the start or it isn't the start. There's no other place for the arrow to point. Three cases, and they cover everything.

And notice what you never had to know: how many dots there were. Ten dots or a billion or an endless sheet, it's the same three cases. What you did need was that every dot has exactly one arrow out. Hold on to that, because it's about to matter.

1.4 Now ask it about keeping

Go back to your dots. Color the dot you start on white. Pick some other dots and color them black, and make one rule about the coloring: no arrow is allowed to go from a black dot to a white one. Once you're on black, you stay on black. That's "settled stays settled." White is not done, black is done.

Follow a run from your white start until it lands on black. Now which of the three things can happen after that?

It can't circle back to the start. The start is white, and you can never get back to white once you're on black.

So it's one of the other two. Either it keeps landing on new dots forever, or two arrows meet somewhere.

Try to draw a counterexample: a run that goes from white to black, stays black, never lands on a new dot forever, and never has two arrows meet. You'll be drawing for a while. There isn't one, and the three cases above are the reason.

That's what page one promised: "the book proves it for any machine that goes from one state to a next." You drew it for one counter there. Now you've drawn it for any machine at all, any number of dots.

What it doesn't tell you is anything about heat. Whether two arrows meeting gives off heat in a real device is a question about the device. People argue about it, and they get their own chapter. This is dots and arrows and nothing else.

1.5 The machine that stops

Here's where the "exactly one arrow out" part bites.

Go back to the tape and the coin. The last line was 1 1 [1], STOP. What does the card say to do next? Nothing. There's no rule for STOP. The machine just stops. In dots and arrows: the STOP dot has no arrow out.

That breaks the one thing the three cases needed. So try the keeping question on a machine that stops. Draw two dots. "Working" is white. "Stopped" is black. One arrow, from working to stopped, and no arrow out of stopped.

Follow it. White, then black, then nowhere. Did two arrows meet? No, there's only one arrow. Did it use up new dots forever? No, it stopped. The answer settled and nothing gave.

Now ask it the obvious question: what does a stopped machine do on the next tick? Not "nothing happens," because something is still there. The usual answer is that it stays where it is. So give stopped an arrow back to itself. Now look again. Working points to stopped. Stopped points to stopped. Two arrows, one dot. The merge is back.

What if you don't want it to stay put? Point stopped at a new dot, "parked," and point parked at itself. Now stopped and parked both point at parked. Two arrows, one dot again. The merge moved one dot over. It didn't go away. That's the rocking machine from page one.

So a machine that stops only dodges this by not answering what happens next. Answer it any way you like, and you're back in the three cases: either two arrows meet, or you draw new dots forever.

1.6 What Turing actually proved

Turing is usually credited with the famous halting problem: that no machine can look at every other machine and say whether it stops. Joel David Hamkins, a set theorist, and Theodor Nenu open their paper on it by questioning "the attribution commonly given to Turing's 1936 paper." Since the stopping machine is right in the middle of this chapter, I went and looked at what he actually wrote, with an AI walking me through the paper and the links it turned up, but it isn't quite what happened.

Turing's own machines in 1936 weren't built to stop at all. The ones he cared about printed out the digits of a number, forever. What he proved no machine could always decide was a question about those: whether a machine keeps printing digits, and whether it ever prints a given symbol. The second one is a few lines away from the halting problem, so in a real sense he had it. But he never wrote it down that way. Stephen Kleene stated it in 1952. Martin Davis seems to be the first to have called it "the halting problem," in 1958. That's from the same paper by Hamkins and Nenu, who went back to Turing's 1936 paper and checked, line by line.

So the machine that stops, the one sitting right in the hole, is Turing's machine the way people later learned to read it. It's the one page one meant when it said Turing's machine already has something that settles and stays settled: whether it has stopped.

Put it together. Read a Turing machine through "has it stopped?" Give the stopped line a next line by letting it stay put. Then the step that stops it is a step

where two arrows meet, which you just drew. That's the step I charge for. Charging for it is still my choice. That something gives there isn't.

The dots in this chapter could go on forever. The next question is what happens when they can't, when the paper runs out.

CHAPTER 2

Running out of room

The last chapter left one door open. A run that settles its answer and keeps it either has two arrows meet somewhere, or it lands on a new dot every step, forever. Endless fresh paper. On a sheet that never ends, that's a real way out.

But nobody has a sheet that never ends. So what happens when the paper is finite?

2.1 Four people, three chairs

Do this one with real chairs, or with coins on a table. It takes about a minute.

Draw a box on the floor and call it the done room. Put three chairs inside it and one chair outside, by the door. Sit four people down, one per chair. The people are the states the machine could be in. The three inside are already done. The one by the door isn't.

One move of the machine moves everybody at once: each person gets up and goes to whatever chair the rule sends them to. There are two rules about where they can go.

Nobody inside may end up outside. Done stays done.

The person by the door has to end up inside. This is the move that settles their answer.

Now try to finish the move with one person in each chair.

You can't. The three inside have to stay inside, so they fill the three inside chairs between them. The person from the door has to come in too, and there's no fourth chair. Somebody has to share. Two people, one chair: two arrows, one landing place.

Try it with more chairs. Ten inside, one by the door. Eleven people have to fit into ten chairs. A thousand inside, one by the door: a thousand and one people, a thousand chairs. It doesn't matter how big the room is, as long as it has a size. The insiders take every inside chair, and the newcomer has nowhere to go but somebody's lap.

Notice who ends up sharing. It doesn't have to be the newcomer. The rule could send the newcomer to an empty-looking chair and send two insiders to the same

chair somewhere else. Page one already showed you that: 4 went to 5a, and the arrows met a tick later on 5b. What's forced is that two arrows meet somewhere in that move. Where they meet depends on the rule.

Two words I'll need from here on. A move is one-to-one when two different starting places always end up in two different places: nobody shares a chair. A move that isn't one-to-one is a merge: two arrows, one landing place.

Now say what the chairs showed in those words. Take any move that keeps done states done and takes one state from not done to done, on a machine with K done states. If the move were one-to-one, the K people already inside would fill all K inside chairs with nobody sharing, and the newcomer would have nowhere to land. So it can't be one-to-one. It merges.

That's the whole argument, and it's the chairs. The only thing finiteness did was make "it fills all K done states" true: K people into K chairs with no sharing leaves no empty chair. That's the one place the size of the room mattered.

2.2 This is very old

The piece of mathematics doing the work there is old, but it isn't mine. Put more things into boxes than there are boxes, and some box gets two. People call it the pigeonhole principle. The earliest use of it I could find is Jean Leurechon in 1622, stating in one sentence, with no argument, that somewhere two people must have exactly the same number of hairs on their heads. The first name for it I could find came from Dirichlet, who used it in 1842 and later called it the drawer principle. It's the kind of fact a child accepts and a mathematician spends a whole career finding new uses for.

And the use I just made of it turned out to be old too. I found that out twice, and the second time stung more.

The first time was physics. In 1876 Josef Loschmidt pointed out, against Ludwig Boltzmann's account of how a gas settles down, that the laws of motion run just as well backwards: reverse every molecule and the gas retraces its steps. Twenty years later, in 1896, a young assistant of Max Planck's in Berlin pushed harder. He used Henri Poincaré's recurrence theorem from 1890, which says a system like that comes back as close as you like to where it started, again and again, to argue that nothing in it can climb forever. Give it time and it comes back down. The assistant was Ernst Zermelo, the same Zermelo who would put set theory on firm axioms twelve years later. Boltzmann answered the same year, not very politely, that you'd have to wait so long for a return that nobody would ever see one. (Jos Uffink, who wrote the Stanford Encyclopedia's entry on Boltzmann's physics, calls that reply drenched in sarcasm and bitterness, and calls Zermelo's paper a careful logical argument.)

You can see Zermelo's point on the dots from the last chapter. Draw a finite number of dots where no two arrows ever land on the same dot. Every dot has one arrow in and one arrow out. Follow any run. Can it ever reach a dot it's been on before, other than the start? No: that would take two arrows into one dot. So since the dots are finite and it can't keep finding new ones, it has to come back to its start. Every run goes around in a loop. Now color the start white and some dots black, done stays done. Can a run go from white to black? If it did, it would have to come back around to its white start, and black can't go back to white. So it can't. A machine where no two arrows ever meet, on a finite sheet, can never settle anything new for good.

That's Zermelo's objection, with the infinities taken out. On a finite set you don't need Poincaré's theorem. And the wait isn't enormous; it's at most as many steps as there are dots. Zermelo was aiming it at the second law of thermodynamics, and settling an answer never came into it. A quantum version came in 1957, when P. Bocchieri and A. Loinger proved that a closed quantum system comes back close to where it started. That's a neighbor, not the same proof: even a small quantum system has endlessly many states, and you'd have to say how a record is read off it.

The second time was computer science, and that's the one that stung. Picture a small machine that reads letters, one move per letter. If every letter just reshuffles its states, no two arrows meeting, people call it a permutation automaton, and the sets of inputs such machines can pick out are called group languages. Jean-Éric Pin's 1992 paper on reversible automata takes a wider class, where a letter may also leave some states with no move at all, and sorts out what those machines can recognize. In that world the chairs argument is folklore: a machine like that can't recognize "has seen a b at some point," because recognizing it would mean a done that stays done. Going quantum doesn't get around it. Alex Brodsky and Nicholas Pippenger showed in 2002 that a quantum version that only measures once, at the end, recognizes exactly the same group languages. One that's allowed to measure after every letter, and stop on what it sees, does better, and "has seen a b" is one of the things it picks up. A measurement that can stop the machine is a record that stays. That's the extra thing that lets it in.

Then the closest one, and it's recent. In 2023 Thomas Ouldridge and David Wolpert worked out the physics of a small machine built the way real computers are: the same physical step run over and over. They sorted what such machines recognize into two kinds. If the smallest machine for a job has a letter where two arrows meet, every machine for that job does, and on their derivation running it gives off entropy that no cleverer design can get back. If the smallest machine only reshuffles, they find it can run with no unavoidable entropy at all. One of their worked examples is exactly "has not seen a b," which is "has seen a b" turned inside out. So the counting, and a physical price on it, were in their paper three years

before this book. Their price comes out of the physics of the device. Mine, in the next chapter, is a choice I make and say so.

There's a cousin that speaks a different language. Constructor theory is David Deutsch and Chiara Marletto's way of stating physics as which transformations a device that runs in a cycle can and can't bring about, and in it a task is irreversible when it's possible and its reverse isn't. Marletto, Vlatko Vedral and their coauthors in 2022 built an irreversible task out of a machine of quantum bits that swap a little with each other, and ran it with photons. Maria Violaris and Marletto argued the same year that this machine, run in a cycle, can turn a pure quantum bit into a mixed one and never back, even though every piece of it can be reversed, and that this suggests erasing may cost more than the usual bound. The chairs are a small, exact, classical relative of that: a set of states the step can't leave, a task that's possible, and no way back. Theirs is quantum and approximate and aimed at physics. Mine is counted. Deutsch and Marletto's paper on information, from 2015, has another neighbor in its section 8.6: measuring one quantity disturbs another for good, and nothing can undo the disturbance while the result is still being kept. That's a second way of saying that a record that stays makes the step impossible to run backwards.

And Morgan Rogers, in his papers of 2019 and 2021 on what mathematicians call toposes of monoid actions, gives the general home for one more way to see it: if every move can be undone by some other sequence of moves, you can never enter a room you're not allowed to leave, because the undoing sequence would have to take you back out. On a finite set, a move where no two arrows meet can always be undone that way: repeat it enough times and everybody comes home.

So the counting isn't mine. It had homes in physics and in automata long before me, and a physical price on it a few years before me. What I do with it starts in the next chapter.

2.3 The hotel that never runs out

Now the way out. If the chairs run out, give the room more chairs. Endlessly many.

Take a strip of paper that never ends. Each move writes a 1 in the next empty box. Done means "there's at least one 1 on the strip." Check it against everything so far. Done stays done: once there's a 1, there's always a 1. The first move takes the empty strip to done. And no two arrows ever meet: two different strips always become two different strips, because you can always erase the last 1 and get back where you were. So here's a machine that settles something, keeps it, and never merges anything.

It got out of the chairs. It didn't get out of paying. Look at what every move does: it writes in a box nobody has used before. The first move does it, and so does every

move after that, forever. That's the other column from the last chapter: fresh paper on every tick.

If the endless strip feels like a cheat, it's a famous one. David Hilbert told a story in a lecture course in the winter of 1924 to 1925 about a hotel with endlessly many rooms, every one of them taken, where a new guest still gets a room. Everybody moves from room n to room n + 1, and room 1 is left empty. Nobody shares. The one-to-one move leaves a gap. That's the strip.

The part worth checking is that Hilbert never published it. The hotel got famous because George Gamow put it in a popular book in 1947, and for a long time it wasn't clear Hilbert had told it at all. The historian of science Helge Kragh first guessed that Gamow made it up. Then Tilman Sauer, who had helped edit the volume of Hilbert's lectures on physics, pointed him to Hilbert's lecture notes on the infinite, which had just been printed in another volume in 2013, and in 2014 Kragh took his guess back in a paper called "The True (?) Story of Hilbert's Infinite Hotel." Keep the question mark. Even Kragh's paper says January 1924 twice, in its summary and its text, and January 1925 twice, in a heading and its conclusion. It can't be 1924: the course ran in the winter semester of 1924 to 1925, which hadn't started yet in January 1924. January 1925 fits, though Kragh is the only one I've found who gives the month. Still, this time a story everybody repeated had a real lecture behind it. I've learned not to count on that.

Bring it back to the chairs. In the closed room the person from the door had nowhere to sit because the chairs ran out. In Hilbert's hotel they never do. So the whole argument was about running out of room, all along.

You can run the chairs backwards to see exactly how much room. Take a move that keeps done states done, settles one new state, and never lands two arrows on one place. Everybody already in the done room moves to a different done chair. The newcomer gets a done chair too, and nobody already inside is moving into it, because that would be two arrows meeting. So everybody already inside got seated in the done room with at least one chair to spare. With n chairs, that's n people into n - 1 chairs, which can't be done. So a move like that needs a done room with no end to it. Give the room any fixed size, and the move merges.

That's Hilbert's front desk run backwards. It's also exactly how Richard Dedekind defined "infinite" in 1888: a collection is infinite when you can match it one-to-one with a part of itself that leaves something out. So taking fresh paper isn't a way out of paying. It's the other way to pay, and only a machine with endlessly many states can keep paying it.

2.4 Cut the strip

Now cut the strip to eight boxes. Same rule: fill the next empty box from the left, never touch the earlier marks. There are nine possible strips, from empty to full. Write them down:

0 marks	`00000000`	next: `10000000`
1 mark	`10000000`	next: `11000000`
⋮	⋮	⋮
7 marks	`11111110`	next: `11111111`
8 marks	`11111111`	next: ?

After eight moves the strip is full. What should the ninth move do? Try each answer.

It stays full: then the strip with seven marks and the full strip both go to the full strip. Two arrows, one landing. The chairs.

It stops: then it has no next state, and that's the stopping hole from the last chapter.

It clears the strip: then done went back to not done.

Every answer gives up one of the things the chairs needed. None of them gets past the chairs.

Keeping the whole history like this is the trick Charles Bennett used in 1973, when he showed that any computation can be done without two arrows ever meeting, if you let it keep a copy of everything it did. Rolf Landauer, back in 1961, had already written that for a machine that runs a program that doesn't stop, the space to hold every step "cannot be there." Keeping the history is a real way to pay. It costs a memory that never fills, and every machine you can buy has a memory that fills.

Bennett wasn't the first there, or the last. Yves Lecerf showed in a short note to the French Academy of Sciences in 1963 that machines with no merging can imitate any Turing machine, and hardly anybody noticed. Tommaso Toffoli, and then Edward Fredkin and Toffoli together, both at MIT, built logic gates and circuits that compute anything and lose nothing, by carrying extra inputs in and extra outputs out. Their basic gate wasn't new even then: Landauer's own 1961 paper writes one down, calls it reversible and able to do an AND, and doubts anything like it could be useful. What Toffoli added was the proof that it computes everything. And Kenichi Morita proved that even a tiny machine with two counters can compute everything with no two arrows ever meeting. Every one of them keeps what it doesn't forget somewhere: a history, extra outputs, a counter that codes the past. They all pay in fresh paper.

2.5 How long can a finite machine hold off?

One more question, because the 5a and 5b example on page one hinted at it. If the machine has a few spare done states, can it put off the merge for a while? Yes, and you can count exactly how long.

At three o'clock a screen says 42. The program is finished. Leave the camera running. At 3:01 it says 42. At 3:02 it says 42. The screen hasn't changed, but the box behind it still has a next state every tick. It can stay exactly where it is, or something inside it can move while the screen stays put. Picture a wheel turning from "first holding position" to "second holding position."

Say the box has K different ways of holding the answer 42. Start one tick before it finishes. Suppose the run goes through its holding ticks without two arrows ever meeting along the way. Could two of those holding ticks be the exact same state? If they were, follow the arrows backwards from both, one tick at a time. Since no two arrows meet along the way, being on the same state now means they were on the same state a tick ago. Keep walking both back. Eventually the earlier one reaches the tick before it finished, the state that wasn't done yet, while the later one is still on a done state. A state that isn't done would have to be the same state as one that is: impossible. So every holding tick needs its own state, and with K of them, it can last at most K ticks. Within K + 1 ticks of finishing, two arrows meet.

You can build one that lasts exactly that long: the unfinished state, then K holding states in a row, the last one stepping to itself. The answer holds for K ticks with nothing merging, and on the next tick the last two arrows meet. A bigger memory puts off the merge. It doesn't make it go away.

2.6 What's being counted

One thing to keep straight before going on. Whether a move merges depends on what you count as the state. The endless strip keeps its whole history in the state, so nothing ever merges; cut it to eight boxes and the same rule merges. Keep a list of every earlier state alongside the current one, and any machine's moves become one-to-one, because two different histories never look the same; a later chapter proves that. Keep only the state one step back, but it isn't enough: two states that differ only in what came before them still land together.

So "this move merges" is always a fact about a move on a stated set of states. Every merge in this book is counted on the whole state the machine says it has, and nothing outside that state is counted.

What I've got now is counting and nothing else: on a finite machine, settling something for good means two arrows meet. The next chapter is about what I do with that, and which parts of it are my choice.

CHAPTER 3

Why charge for it

Everything so far has been counting, and the counting isn't mine. On a finite machine, settling something for good means two arrows meet somewhere. On an endless one, it can mean fresh paper forever instead. Nobody gets to choose that. You drew it.

Page one also said something I do get to choose: I charge for the move where the answer goes from not done to done. This chapter is about how much of a choice that is. If it's nothing but my preference, then the whole machine is a long way of telling you my favorite color, and you should treat it that way. So I want to find out what it rests on.

3.1 A rule you can refuse

I'm not going to ask you to believe a price. I'm going to write one down as a rule, in the smallest form I can, and you can accept it or refuse it.

Any move where two arrows meet costs at least one.

That's all it says. It doesn't mention done or not done. It doesn't mention checking or settling. It puts a price on one thing: a move that sends two different states to the same place.

It isn't my rule. Charles Bennett, who spent decades working out what computing has to cost, put it this way in 2003: what has to cost is "the erasure of a bit or the merging of two computation paths." Two arrows meeting is two paths merging. I'm taking his rule as it stands and writing it on the move.

Now put it next to the chairs. The chairs said: on a finite machine, a move that settles something for good is a move where two arrows meet. The rule says: a move where two arrows meet costs at least one. Put those two sentences side by side and read them together. A move that settles something for good costs at least one.

That's the rule from page one, the one I said was my choice. I call it the toll. That was my word for it before I could prove any of this: there had to be a toll, a cost in the transition, in the step itself. And look at where it came from just now. Nobody handed it to the machine. It fell out of the chairs and Bennett's rule together.

So it isn't only my preference. But it isn't free of choices either. Count what went into it. There are three, and you can refuse each one.

The done states have to be finitely many, so the chairs run out.

Done has to stay done.

And two arrows meeting has to cost something.

Accept all three and you don't get to refuse the toll. It comes with them. Refuse any one, and the toll goes back to being my choice. So the question "did I have to pick it?" has turned into three smaller questions, and those are better things to argue about. For each one there's a machine that walks out through that door, and you should see each one, because a rule you can't picture breaking is a rule you're taking on faith.

3.2 The first door: endless paper

You've already seen this machine. It's the strip from the last chapter that never runs out. Each move writes a 1 in the next empty box. Done stays done, the first move settles it, and no two arrows ever meet.

So Bennett's rule never charges it anything. Two arrows never meet, so there's nothing to charge. The strip settles its answer at no cost on the price list.

But it didn't get out of paying. It pays in fresh paper, on every move, forever. A price list that only charges for arrows meeting just isn't looking at that column. And on any real machine, the paper runs out, and the chairs come back.

3.3 The second door: let it come back down

Take one coin. The only move is: turn it over. Heads is done, tails is not done.

Draw the arrows. Tails goes to heads. Heads goes to tails. Two dots, two arrows, but they never meet. So Bennett's rule never charges it.

And it does settle something: tails goes to heads, not done to done. At no cost.

What it broke is the second condition. The same move that took tails to heads takes heads back to tails. Done doesn't stay done. So it's not keeping anything. It's flipping a switch back and forth, and the page-one definition already throws it out: a Thiele machine's done never goes back.

Here's the part of this I found sharpest. Forget about done staying done entirely, and keep only the finite chairs. Take any move that takes some state from not done to done. Now go through the done states, one at a time, and ask: does this move send any of them back to not done?

If it does, you've found the coin: the move takes something back somewhere.

If it doesn't, then this move keeps every done state done. That's all the chairs needed, for this one move. So two arrows meet.

There's no third answer. On a finite machine, a move that settles something either has two arrows meet, or it takes something back somewhere else. A move that settles for free and never takes anything back doesn't exist.

That's narrower than "every record costs." It's still true, and I'll take it.

One more picture, so you don't think it's tidier than it is. Three cards, T0, T1 and T2, with a star on T1. The move sends T0 to T1, T1 to T2, and T2 to T1. Look at what it does. T0 gains the star, so something settled. T1 loses the star, so something was taken back. And T0 and T2 both land on T1, so two arrows meet. So Bennett's rule charges this move, and yet it doesn't keep anything for good. Charged moves and permanent moves aren't the same list. What's true is one direction. On a finite machine, every move that settles something for good is a move Bennett's rule charges.

And there's an exact way to say which moves Bennett's rule charges no matter what the rest of the price list looks like. A move where two arrows meet gets charged by every price list that follows the rule; that's what the rule says. A move where they never meet doesn't have to be: price that one move at zero and every other move at one, and the rule is still followed. So the moves the rule forces you to charge are exactly the moves where two arrows meet. That one holds on any machine, finite or not.

3.4 The third door: don't charge for it

Take a sheet of paper. It's either blank or stamped. One move: stamp it. Blank goes to stamped, and stamped stays stamped.

Check the conditions. Two states, so the chairs run out. Stamped stays stamped. Blank goes to stamped, so something settles. And draw the arrows: blank goes to stamped, stamped goes to stamped. Two arrows meet. Everything the chairs needed is here.

Now set the price of the stamp to zero.

Nothing in counting stops you. The stamp merges two states, and you just decided merges are free. The answer settled, and the bill says zero.

Price the stamp at one instead, and every condition holds at once, so the three can live together. That stamp at one is the smallest Thiele machine there is, and it gets its own chapter. But the third door asks whether they have to live together, and counting can't say so. The only thing that could close this door is a claim about the world: that two arrows meeting really does cost something, out there, in a real device.

The nearest such claim is Rolf Landauer's, from 1961: that forgetting gives off heat. It's measured for a few devices and argued over for everything else, and it gets its own chapter, with the people who doubt it named. I'll say one thing about it now,

because it matters here. Bennett's rule as I wrote it charges every move where two arrows meet, whether or not you already knew which state you were in. Landauer's heat, as that chapter will show, is only owed for what you didn't know. The rule I'm using is the worst case on purpose: a move that has to work whatever it's handed pays for the worst it could do.

So that's where the toll stands. Accept finite chairs, done stays done, and a price on two arrows meeting, and the toll isn't a choice; it's what the three say together. Each of the three is a choice, and I've shown you a machine walking out through each door. The physics that would close the third door is a premise, and I'll always call it one.

3.5 How much

The toll says at least one. Counting can say more, if you let it.

Take four cards and write A, B, C and DONE on them. Pick one without showing me. Now cover its face with a label that says DONE. Whatever you picked, the face now says DONE. Before the label, how many yes-or-no questions would it take me to find out which card you had? Two: "Is it A or B?" and then one more. After the label, I don't need to ask anything, because there's only one thing the face can say. Two questions' worth of difference got squeezed out.

That's four states landing on one. Each question you no longer need halves the number of places you could have been: four to two, two to one. Two halvings.

If you're willing to go one step further than Bennett's rule, and say each unit of cost pays for at most one halving, then that move costs at least two. And two is enough. In general: if m states were already done and the move settles k more, all m + k of them land on the m done ones, and the move has to pay at least as many halvings as it takes to get from m + k down to m. One newcomer next to one done state is one halving: the toll again.

(The label can be peeled off, so the cards only show the squeeze on what the face says. Whether anything physical got erased is a different question, for the heat chapter.)

There's a plainer way to say what the halving rule asks for. Look at one move and find the place where the most arrows land. Call that its worst pile-up. A cost of one allows piles of two. A cost of two allows piles of four. A pile of three already needs two. That's why the m + k states in the done room had to pay at least that much: spread m + k arrows across m chairs as evenly as you like, and the fullest chair still gets at least (m + k) / m of them. So the rule asks every move to pay for the worst it could ever do, whichever state it's actually handed.

That worst case has a home in physics. Philippe Faist, Frédéric Dupuis, Jonathan Oppenheim and Renato Renner worked out in 2015 the least work any logical step

needs when it has to succeed every time, on every input, with no averaging. It comes to exactly that logarithm of the worst pile-up. An AND gate sends three of its four inputs to 0, so it needs about 1.6 halvings' worth, two in whole units. My rule counts the same thing in units of cost. Joules don't come into it.

You can also count the whole run this way. Write N different starting states on cards. Run each card through the rule once, and stack the cards that land on the same state into one pile. Do it again with the piles. Each tick, count how many different states you're down to. The drops add up exactly: what you started with equals everything lost along the way plus what's left at the end. If you end up with fewer than you started with, some tick had two arrows meet. Turn the cards over first, though. Two runs can show the same answer on the front and still be different underneath, and those are different states.

A word about the k in all this. Physicists write kT for heat at a temperature, and when the counting meets the physics in the heat chapter you'll see k with a little B hung under it, for Boltzmann. That letter has a story I didn't expect. Boltzmann's grave in Vienna has S = k log W carved on it: entropy is a constant times the logarithm of the number of ways the parts could be arranged. The idea is his. But the constant, and the equation written that way, came from Max Planck, around 1900. In his Nobel lecture in 1920 Planck said the constant "is frequently termed Boltzmann's constant, although to the best of my knowledge Boltzmann himself never introduced it." So the most famous constant in this corner of physics wears the name of a man who never wrote it down. Take the constant away and what's left is the logarithm of a count, which is a number of halvings, which is what the cards just gave you. The constant is the exchange rate between halvings and heat. Whether the world honors that rate is the premise.

3.6 Found, assumed, chosen

That's the place to sort everything so far into three piles, because the rest of the book depends on where each thing falls.

I see it like this. I didn't invent a machine. I found one. That walks straight into one of the oldest fights in mathematics, so here's a short trip into it. G. H. Hardy, the Cambridge number theorist who wrote A Mathematician's Apology in 1940, came down on the found side about as hard as anyone ever has. He wrote that mathematical reality lies outside the people who study it, and the theorems they like to call their creations are really notes on what they saw. The German mathematician Leopold Kronecker supposedly split the difference: God made the whole numbers, and everything else is the work of people. Supposedly, because lean on that line and it gets thin fast. It comes down through Heinrich Weber, in a memorial piece

written after Kronecker died, pointing back to a lecture from 1886, and that's the only source I could find anybody citing. It's a great line with one witness.

None of that settles whether I found the Thiele Machine or made it up, and a proof wouldn't settle it either. So it's a conviction, and I'll call it one. What it changes is how I hold the thing: like something that was already there, which means I can be wrong about its shape and the shape won't care. And "found" only covers what counting forced. Here are the three piles.

Found: what counting forces on any machine.

Every step of a machine with one next state for each state lands somewhere new, circles back to its start, or has two arrows meet, and after it comes back it repeats forever. You drew that in chapter one.

A run that settles something and keeps it can't circle back to its start, so it has two arrows meet or takes fresh states forever. When only finitely many states are done, the move that settles it has two arrows meet. You sat people in chairs for that.

A machine that stops, with no next state, slips past all of it, and gets caught the moment you say what it does on the next tick.

Assumed: what the found pile is taken to stand on in the world.

That two arrows meeting in a real device, when the device really could have been in either state, gives off heat. That's Landauer's principle, and it's an assumption here, never proved, with its critics named in its chapter.

That a real device can only tell apart finitely many states in any bounded space, so its fresh paper runs out. Nobody has measured where that ceiling is.

That the machine starts clean: if done means "this happened," that's only because it started at not done. That start is put in by hand.

Chosen: what I add, each of which could have gone another way.

To put the price inside the step, as part of what a step is.

That done stays done. That's in the definition because I put it there.

The toll: at least one on the move that settles something. With finite chairs and done staying done, it follows from a price on two arrows meeting, and that price is a choice too.

Which thing gets to count as settling, and what the price list charges. Nothing I've shown you forces either one.

Everything after this chapter is built on those three piles, and I'll keep saying which pile each piece comes from.

CHAPTER 4

The smallest one

The last chapter put a price on the move that settles something. Before I build the smallest machine that carries that price, there's a question I had to ask first. People have been making marks that other people are supposed to rely on for a very long time. Did any of them put a price on the moment the mark goes on and stays? I wanted to know what they thought they were doing, and what it cost them. So here's what I could reach, and where the search ran out.

4.1 Clay, seals, and seven witnesses

Start a little over five thousand years ago, in the Near East, with people who counted sheep with little clay tokens. To send a count across a distance without the courier changing it, they put the tokens inside a hollow clay ball, pressed the shapes of the tokens into the outside while the clay was soft, and sealed it. The archaeologist Denise Schmandt-Besserat has spent a career on these, and she puts the practice at around 3300 BC; the ball is called a bulla.

You can do a version of it with a cookie tin. Glue a paper seal across the lid and the rim, let it dry, and then try to open the lid without tearing the seal. The bare lid would come straight off. The seal is what turns opening it into something you can see afterwards. A substitute seal could fool you, so the tin shows that somebody disturbed it, not that nobody could ever fake it. The clay ball does the same: the outside shows what's inside, and opening it breaks it. Keep that idea, because it comes back: a record is worth something only if it can't be quietly undone.

Now law, which is where people get serious about who's allowed to say what happened. The book of Deuteronomy, chapter nineteen: nobody is convicted on the word of one witness; it takes "two or three." The Roman jurist Gaius, writing in the second century, says a will the court will honor is one sealed by seven witnesses. Why seven? Gaius doesn't say. The legal historian Henry Maine, in 1861, had an answer: the older Roman will was a mock sale, done in front of five witnesses, a man holding the scales, and the buyer of the estate, seven people in all, and when the ceremony fell away the seven seals kept the count. Maine's is the only account of that I found, so treat it as his.

Try it. Write “paid five dollars, signed A” on a napkin and ask a friend to forge it. It takes them ten seconds. Now have two other people initial it at the moment of payment and each keep a copy. If whoever reads it later compares those copies and trusts the people holding them, changing your napkin alone gets caught. Stolen copies, forged initials or a reader who never compares them can still beat it. The claim didn’t change. What stands behind it did.

Which brings me to seals, and the medieval historian Michael Clanchy, whose book is called From Memory to Written Record. England between the Norman Conquest and the early 1300s was learning to trust writing. Until then it had trusted people who remembered. The lesson was painful: seals were forged, charters were forged, and historians today have a whole craft of telling real charters from fakes because there are so many fakes. A seal is only as good as whoever holds the stamp. In 2011 a company called DigiNotar, which issued the certificates web browsers trust, found out the same thing at the cost of its existence: a mathematically perfect signature is worth exactly what the custody of the key is worth, and intruders had taken over the machines that signed. It’s the same failure nine hundred years apart, and I find that oddly comforting.

And then the obvious next question: if seals could be forged, what caught the forgeries? It wasn’t the seals. The most famous forged record in European history is the Donation of Constantine, a document in which the emperor Constantine supposedly handed rule over Rome and the western empire to the Pope. It was most likely written in the eighth century, some four hundred years after Constantine died, and it was leaned on for centuries. Around 1440 a scholar named Lorenzo Valla took it apart by its Latin: the language belonged to a much later age, and it used words, “satraps” among them, that no Roman of Constantine’s day would have used for his officials. (Valla wasn’t even first. Nicholas of Cusa, a German churchman who later became a cardinal, had already called it a forgery in 1433. Valla’s is the version people remember, which is a pattern that keeps turning up.)

Look at what kind of check that was. The question wasn’t “is the stamp real.” It was “could these words have been written when they say they were.” A record can pass the first and fail the second. That was my first hint that what I kept calling one thing, a record, was doing more than one job.

4.2 Two columns and a stamp

The one I’d never have guessed mattered is bookkeeping. Around 1300, the ledger of a Florentine company called Farolfi, kept at its branch in Provence, shows the earliest complete example anyone has found of a trick merchants were working out: write every transaction twice, once where the money came from and once where it went, and the two columns have to balance. The treasurers of the city of Genoa

kept their ledger for 1340 that way. And in 1494 a friar named Luca Pacioli printed the method in a mathematics book, and it spread.

Do it for one day. Keep two columns of every dollar that moves, and total them at night. Have a friend secretly change one entry in one column. The totals disagree; you catch it. Now have them change the matching entry in both columns. The totals agree, and you've caught nothing. So that's what double entry actually is: a check for slips, which a careful liar walks straight through.

Did Pacioli see that? A scholar named Forrester, on the University of Strathclyde's accounting-history pages, reads him as reporting that in many places the books were taken to the local consul to be stamped before a single entry went in. The stamp is a witness again, from outside. The two columns are a method. The stamp is somebody else agreeing to stand behind the fact that these are the books. Keep those two apart, because one is a habit and the other is a commitment.

Then I found where this got turned into a motto. The Royal Society, founded in London in 1660 and still running, picked as its motto Nullius in verba: take nobody's word for it. Where did they get it? From somebody's words, naturally. It's half a line of the Roman poet Horace, from a letter to his patron, saying he isn't bound to swear by the words of any master. So the society that takes nobody's word for it took its motto from a poet and cut the line to fit. That's not a contradiction. It's a nice picture of the method: borrow what's useful, say whose it was, and don't swear by it.

The historian Steven Shapin wrote about what that motto meant in practice, and the answer is more interesting than the slogan. Robert Boyle—pumping the air out of a glass globe to see what happens to a candle or a bird—had three problems: the machine, the report, and the witnesses. He solved the third by having gentlemen in the room who'd sign their names, and the second by writing the report so richly that a reader who wasn't there could picture the scene and count as a witness anyway. Shapin calls that virtual witnessing, and he notes, dryly, that the air pump was very expensive, the big science of the 1600s. So there was a cost. It just wasn't a cost of declaring the fact. It was the cost of the instrument and of getting enough people into the room. Keep that difference in front of you. The rest of the book hangs on it.

4.3 Closer to now

The rest of the lineage points the same way, so I'll go through it quickly.

Regulators who made stockbrokers keep their electronic records on media that can't be rewritten (from 1997 the SEC allowed records kept electronically only on media written once and read many times, and in 2022 it allowed an audit trail as another way) were paying for the hardware. The rule put no charge on the act of writing.

Stuart Haber and Scott Stornetta, in 1991, chained time-stamped records together so that changing an old entry breaks every entry after it. Bitcoin's footnotes lead back to them. Try the idea with a running total. Take five entries, 1, 2, 3, 4, 5, and add each to the total so far: 1, 3, 6, 10, 15. Now change the second entry to 3. The totals become 1, 4, 7, 11, 16. The change travels all the way down. Now also lower the third entry to 2, and the last total is 15 again. Simple addition can be fooled by changes that cancel out. A real chain uses a much harder function than addition, and somebody has to keep a trusted copy of the last value to compare against. (Ralph Merkle's hash tree, from his 1979 Stanford thesis, lets one short value stand for a great many records the same way.) What the running total shows you is how each entry depends on the ones before it. It doesn't put a price on writing one.

Proof of work, which Cynthia Dwork and Moni Naor proposed in 1992 to make junk mail expensive, and which Bitcoin later chained, is the one place where a record really does cost something. But what it charges for is a search: how many tries it took to find a lucky number. An empty block and a full one face the same target. The work pays for the search. Nothing extra gets charged for the moment anything becomes settled.

Robert Floyd and Tony Hoare, asking in 1967 and 1969 how you'd prove a program does what it's supposed to, attached claims to points in a program: "whenever you get here, this is true." The claim is backed by a proof, and the proof is paid for somewhere else, off the machine.

The certificates of complexity theory, going back to Stephen Cook in 1971, are answers designed to be quick to check: hand me the certificate and I can confirm the claim fast. Whether they're also hard to find is the famous open question of the field, P versus NP, and nobody knows. Either way, what's bounded is the checking. Nothing is charged for raising a flag.

And from statistics, a ledger that runs the other way. Dean Foster and Robert Stine's alpha-investing, from 2008, runs a long string of tests against a budget of allowed false alarms: every test spends some of the budget, and every test that finds something earns some back. That's a budget for how often you're allowed to be wrong. A reward there doesn't mean the discovery was true.

4.4 Five jobs

Lay them all down together, and what shows through isn't one thing. It's five different jobs the word "record" has been doing, and I'd been letting one word carry all five. Most of them come with a corpse.

Durability: the record can't be quietly undone. The clay ball, the write-once disk and the hash chain do this one. The corpse is the editable log, and the company that

kept its debts off the books: Enron hid debt in side companies set up to keep it off the balance sheet, which is how the Justice Department described it in 2002, and its auditor, Arthur Andersen, shredded paper. Andersen was convicted that year, and the Supreme Court threw the conviction out in 2005 over how the jury had been instructed. Notice that this job doesn't need a witness at all. A copy will do, as long as somebody else holds it.

Authenticity: the record came from who it says. Seals, signatures and the consul's stamp do this one. The corpse is the forged charter, and DigiNotar. A perfectly authentic entry can still be erased, and a perfectly durable one can still be a forgery. They're different jobs.

Evidence: enough independent people stand behind the claim that lying gets expensive. Two witnesses, seven seals and Boyle's gentlemen. This one is about how many people put their own risk next to the claim. Paper has nothing to do with it.

Accounting: the record checks itself for slips. The two columns. And the second column isn't an outside witness. It's you, checking you. It catches a dropped digit, and a careful liar gets past it.

Quick checking: a certificate, signature or fingerprint that makes one particular check fast. Checking doesn't have to be cheaper than making, and I once thought it had to. Plenty of records are checked by doing the whole thing over, and that's fine when the thing is small.

Now the thing none of them says. Not one of those jobs puts a price on the moment a record goes on and stays. Proof of work prices a search. Bookkeeping prices a habit. Witnesses and seals price showing up and guarding the stamp. I looked, with everything I had, and that's where it ran out. If somebody priced that moment itself, they're on a shelf I never found.

So I had to try it myself, on the smallest thing I could build.

4.5 Two boxes and a coin

Take a sheet of paper. Draw two boxes and label them NO and YES. NO is not done, YES is done. Put a coin in NO. Draw a line underneath for tally marks; that's where you keep score.

You get one move. Call it the stamp. From NO, the coin slides to YES. From YES, it stays in YES. Every time you make the move, whichever box you're in, you put down one tally mark.

Play. Start in NO, make the move as many times as you like, and look at the tally. Now try to get to YES with no marks at all. You can't, and you can see why: the only way from NO to YES is the stamp, and the stamp makes a mark.

Now cross out the mark in the stamp's rule and play again. You get to YES with no marks, every time. That's how little the mark is. The board doesn't force it.

Chapter three is the reason it's there: finite chairs, done stays done, and Bennett's rule. Here it's written on a card.

Draw the arrows for the stamp. NO goes to YES. YES goes to YES. Two arrows, one landing place. It's the same picture as 4 and 5 both going to 5 on page one.

That's a Thiele machine. A thing (where the coin is). A move (the stamp). A way to tell how settled it is (NO or YES), where YES never goes back to NO. A run that actually settles it (start in NO, stamp once). A price on the move (one mark, never a refund). And the toll holds: the move that takes NO to YES costs one.

Is there a smaller one? Try.

One box? Then there's nothing to settle. The coin is always in the same place, so no run ever takes it from not done to done.

Two boxes, then. A move either sends the two boxes to two different boxes, or it sends both to the same one. If it keeps them apart, there are only two ways to do that: leave each where it is, or swap them. Leaving them where they are never settles anything. Swapping them is the coin from the second door: it takes NO to YES, but it also takes YES back to NO, and done has to stay done. So a two-box machine that settles anything has to have a move that sends both boxes to the same box, and since YES has to stay YES, that box is YES. That's the stamp.

So nothing smaller works, and look at what that means. Chapters one and two found that keeping something takes two arrows meeting or endless fresh paper. Here there are only two boxes, so there's no fresh paper to be had, and the arrows meet. The smallest machine there is already has the merge inside it, in plain view. Nobody added it. A fuse that blows, a bead that drops into a dip and stays, a stamp pressed onto a box: each one is two states and one move that sends both to the done side, and each one, with that move priced at one, is this machine.

I'll call it the latch, because that's what it is: something that catches and holds.

4.6 What the tally is and isn't

Why tally marks? Because they're about the oldest receipt there is. The real ones, notched sticks split down the middle, get their own chapter later, fire and all. The tally on your paper is nothing like those. Nobody holds the other half. You keep it, beside the board, and you could rub a mark out and nobody would ever know.

Remember that one. Of the five jobs, the latch does a small piece of the first one: none of its moves takes YES back to NO. It has no witnesses, no signature, no second column, and nothing in it checks anything. YES means the coin is in YES. That's all it means. Whether a YES can ever be made to mean "this was checked and can be relied on" is a real question, and it's the one the second half of the book is about. For now the tally is a scorekeeper's receipt: you know what you paid, and nobody else has any reason to believe you.

4.7 Any machine can wear one

Here's something the latch lets you do to any machine at all.

Take any machine, any states, any moves. Pick one state, call it x, that some move lands on. Now build a new machine right beside it: the same states and the same moves, with one extra coin riding along in its own two boxes, NO and YES. The coin starts in NO. The first time a move lands on x, the coin goes to YES, and it never comes back. Price every move at one.

Check it against the definition. The coin never goes back to NO. Start anywhere the move that lands on x can be made, with the coin in NO, and make it: the coin goes to YES. Every move costs one, so the toll holds. It's a Thiele machine.

Now cover the extra coin with your hand and watch the rest. What you see is the machine you started with, move for move. Nothing about it changed. The coin rode along and the old machine never knew.

And with your hand over the coin, you can't tell which box it's in. Let any spot with the coin in NO count as a place to start. Now find a moment where the old machine is sitting on x. Did it just arrive there, which put the coin up? Or did it start there, with the coin still down? Both can happen, and the part you can see looks exactly the same. The coin is the only place the difference lives.

So every machine that has a move at all is what's left of a Thiele machine once you cover the coin. Even a machine with one state and one move that goes nowhere: the move lands on that one state, so the coin goes up, and what you've built is the latch. Turing's machine read through "has it stopped?" doesn't even need the extra coin: chapter one showed it already has a done that stays done, and stopping is already a step where two arrows meet. Both ways, it points the same direction page one did. The machine with the price in it is the general thing. The computer is what's left of it when you cover the coin and set the price of settling to zero.

4.8 Which question gets charged?

One more puzzle, because it shows why the definition picks one question and charges only that one.

Put a tile on a table and slide it one place to the right. Let that move be free. Now ask a new question: is the tile in the place it just reached? Before the slide, no. After it, yes. If every question had to follow the toll, that slide would have to cost at least one.

Try it with any free move that changes anything. Ask "is the machine where this move just put it?" The answer goes from no to yes. So if the toll applied to every question you could ask, every move that changes anything would cost something,

and nothing could ever be free. That's why the toll belongs to one question, the one the machine says is "how settled," and not to every question you could think of.

And for that one question, a free move can't touch it. It can't take NO to YES, because the toll charges that. It can't take YES to NO, because done stays done. So a free move leaves the settled question exactly where it found it. That's the whole reason the free moves and the settling moves come apart so cleanly.

4.9 The cheapest price list

Last thing about the latch. Put three price cards next to a button: zero, one and two. Somewhere in the machine, pressing that button takes NO to YES. Cross out zero; the toll won't allow it. One works. Two works too. The toll picked the smallest card it could, without forbidding the bigger ones.

Now take a button that never takes NO to YES, from anywhere. Zero works. So go through the buttons: one on each button that can settle something somewhere, zero on each button that never does. That's the cheapest price list that follows the toll, and there's only one cheapest. Notice the price stays on the button: the stamp costs one even when you press it on YES and nothing changes. The toll is a floor, not an exact bill.

And it counts. Every time a run goes from NO to YES, that step costs at least one, so a run costs at least as much as the number of times it settled something. On the latch that's at most once, because YES stays YES. On a switch that can be turned off for free, you could settle it and unsettle it a hundred times, and you'd pay at least a hundred. A latch pays once, and keeps it.

CHAPTER 5

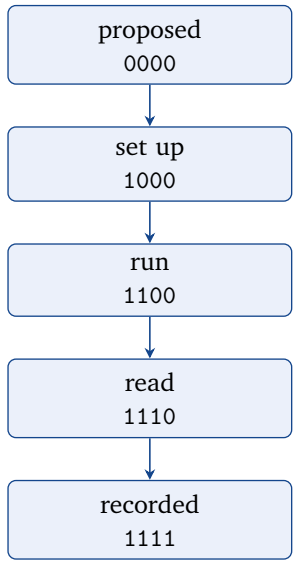
More settled

Up to now everything has had two places: not done and done, NO and YES, the coin in one box or the other. Page one promised more than that. It said settled doesn't have to be yes or no, and to picture a checklist.

That promise matters to me more than it might look. The hunch this whole thing started from was that there's another direction, one ordinary computing doesn't look along. A direction isn't a switch. A switch is just the first line drawn across it. So this chapter takes the switch and finds out what happens when there are many lines.

5.1 A row of cups

Get four cups and a strip of paper. Write five stages of an experiment along the strip, left to right: proposed, set up, run, read, recorded. Agree to do them in that order. Put an upside-down cup between each pair of neighbors. The first cup means "the setup happened," the second "the run happened," the third "the result was read," the fourth "the result was recorded." When you finish a stage, turn its cup right side up. Cups only ever get turned up. None of them ever gets turned back down.



Read down the drawing. Zero means cup down, one means cup up. Four cups give five stages along this run because each new stage turns up one unused cup.

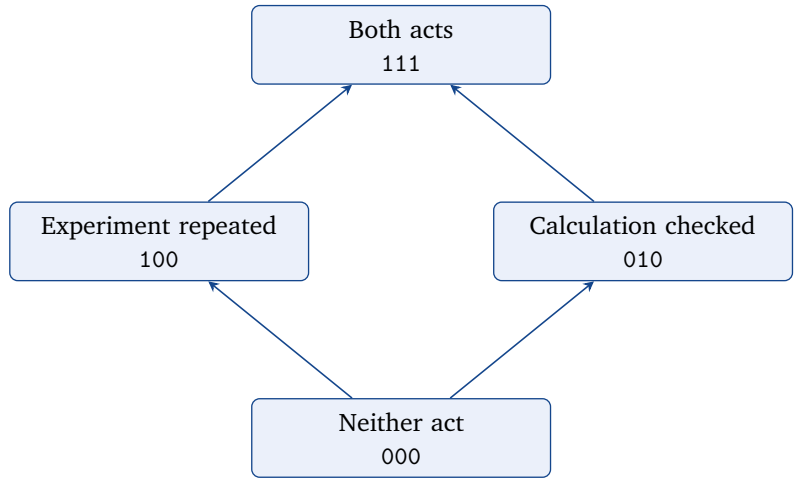
Two things to notice with the cups in front of you.

Each cup on its own is the latch from the last chapter: down or up, and once up, it stays up.

And four cups were enough for five stages. Try it with three. Turn them up one at a time, never back down, and count how many different arrangements you pass through. None up, one up, two up, three up. Four arrangements, and you needed five. You can't do it. Every time the arrangement changes, at least one more cup goes up, and with three cups that can only happen three times. So three cups give you at most four stages, and in general a row of cups that only go up can tell apart at most one more stage than it has cups. A straight line of stages needs one cup fewer than it has stages, and no fewer.

An ordinary binary counter holds a thousand stages in ten bits. That doesn't break anything here: a counter's bits go down as well as up. These cups don't.

Now a harder strip. Some checks don't line up. One person repeats the experiment. Somebody else, separately, works through the calculation. Neither one includes the other. Draw a diamond: "neither" at the bottom, "experiment repeated" on the left, "calculation checked" on the right, "both" at the top. There are two ways up, and the two middle points don't come before or after each other.



Read the cups as repeat, calculation, both. Neither middle point includes the other. The third cup repeats information already supplied by the first two; it is useful for showing thresholds, not needed for storage.

Put a cup on each point except the bottom: one for the repeat, one for the calculation, one for both. Each cup is still a latch. The cups tell you which things have happened. They don't tell you whether the experiment was set up right or the calculation started from the right place. That's a different question, and it comes later.

5.2 Naming what's on the table

Now that you've seen the pieces, here are the names I'll use for them.

The set of places the record can be, whether it's a straight strip or a diamond or something with a hundred points, I'll call the direction.

Each question of the form "has the record reached at least this place?" I'll call a line across it. One line is one cup.

To say "at least this place," you need to be able to compare two places: is this one at or below that one? The strip and the diamond let you see that by looking. Said exactly, it's a test you can put to any two places, and it has to follow two rules. Every place is at or below itself. And if one place is at or below a second, and the second is at or below a third, then the first is at or below the third.

There's a third rule you might expect: if two places are each at or below the other, they're the same place. Every strip and diamond on the table follows it. To see what a rule like that is doing, take it away.

So take two places, left and right, and say each one is at or below the other, without them being the same place. Now there are two cups: "at least left?" and "at least right?" Put the record on left. Both cups are up, because left is at or below

itself and at or below right. Put it on right. Both cups are up again. Look at the cups and you can't tell where the record is.

Now let the record swap between left and right on every step, for free. Is that allowed? It never goes below where it was, since each is at or below the other. No cup ever moves, so nothing ever settles, so the toll never asks for anything. The record moved on every single step, and nothing on the table noticed. It never settles anything either, so it isn't a Thiele machine.

Put the third rule back, and the cups tell you exactly where the record is. Suppose two places leave exactly the same cups up. The cup for the first place is up when the record sits on the first place, so it's up on the second place too, which says the first is at or below the second. The same goes the other way round. Each at or below the other: by the third rule, they're the same place. Without the third rule, the cups can miss moves. Neither is wrong. Dropping the rule just tells the cups not to care about moves between places on the same rung, and they don't.

5.3 The toll on the whole direction

One cup had one rule: the step that turns it up pays. Now there's a row of cups, or a diamond, and one step can turn up several at once. Here's the rule I'd guess first, and it turns out to be the right one: a step that turns up any cup pays at least one. Three cups at once still only has to pay one. The rule asks whether a step turned something up, not how many.

Does that still give you what one cup gave you? Take any run where some cup is up at the end that was down at the start. Somewhere along the run, one step turned that cup up, and that step paid. So the run paid at least one. And every step that turned anything up paid, so the run paid at least once for each of them.

Look at what that argument never used. It didn't care whether the cups sat in a row or a diamond. It didn't care how many cups there were. It didn't even use the fact that cups never come back down: if something could knock a cup back over, the argument still works, because all it needed was the one step where that cup went up.

There's a simple way to tell, from places alone, when a step turns up a cup. Say a step takes the record from place x to place y. The cup at y is up afterwards, because y is at or below itself. Was it down before? It was down exactly when y isn't at or below x. And if y is at or below x, every cup that's up at y was already up at x, so nothing turned up. So a step turns up some cup exactly when it lands somewhere that isn't at or below where it started. That's the toll from page one, written on any direction.

And checking it one cup at a time is the same as checking it on the whole thing. If every cup on its own follows the two-box toll, the whole direction follows the toll, and the other way round.

5.4 One cup isn't enough

Could one cleverer cup do the job of all four? Say you keep one latch and try to read the record's place off the latch together with whatever else the machine shows.

Take the smallest case: a machine with nothing else to show, and a record that climbs through three places, bottom, middle, top. One latch gives you two answers, down or up. Three places, two answers. Some two places get read the same. One cup can't do it.

Try a second guess, closer to how a single cup works. A cup's rule is: the new value is the old value, or the event that turns it up. For a whole row, the obvious copy is: the new place is the lowest place that's at or above both the old place and the event's place. Test it. Take the strip bottom, middle, top, with a cup at middle and a cup at top, and a record that jumps from bottom straight to top in one step, and stays put at middle. With nothing else for the event to look at, the event has to be one fixed place. From bottom, it has to take the record to top, so the event is top. But then from middle, the same rule takes the record to top too, and the record was supposed to stay at middle. The rule breaks.

What works is giving every cup its own event, and letting each cup's event look at where the record already is. A later chapter builds that out over any machine.

How many cups does a direction need? That depends on what you ask the cups to do, so start with the smallest job: tell where the record is, using cups that only ever go up as the record climbs. For that job a cup doesn't have to be "at least this one place." It can go up at several places, as long as climbing never knocks it back down.

A straight strip needs one fewer cup than it has stages; you counted that above. A direction that branches needs at least as many cups as the steps in its longest climb, and at least enough to tell all its places apart.

Put three places above a bottom, none of them above another. The longest climb is one step, but one cup can't tell four places apart, so it takes at least two. Two do it, if you let a cup go up at more than one place: one cup that goes up at the first and third places, one that goes up at the second and third. Every place gets its own pattern. But look at those two cups alone and the third place seems to sit above the first, because every cup that's up at the first is up at the third too. If you also want the cups to show which places sit above which, it takes three: one "at least this place" cup for each of the three.

And sometimes just telling the places apart needs more than both of those. Take two places, each below two others. One step of climb, four places, and two cups still won't do it, because both upper places would have to keep up every cup either lower place has up, and two cups don't leave room for that. It takes three, and showing which sits above which takes four.

5.5 Somebody got here first, from the side

People got here before me, from a direction I didn't expect: programming for many processors at once. Lindsey Kuper and Ryan Newton's LVars, from 2013, are memory cells whose value only climbs, where a program can't ask for the value itself, only whether it has climbed past some point. That's a cup. Two writers running side by side can push the value up in different orders, and a point once passed stays passed. Their programs come out the same every time partly because of which questions the language lets you ask: a program allowed to ask whether the cup is up yet could still see the other writer's timing.

Amin Timany and Lars Birkedal, coming at it from proofs about programs, show in their 2021 paper, as I read it, that any order at all can be carried this way in a logic for proving programs correct, as a value that only climbs and that the running program never reads. That's the closest I've found to somebody else cutting a direction into lines.

The two of them rest on the same shape, a value that only climbs and a question that only asks whether it's past a point, and they use it for opposite jobs. In an LVar the question steers the program. In Timany and Birkedal's logic, the value lives in the proof and the program never looks at it. The shape is theirs first. What I can't find in either is a price on crossing a line.

5.6 Why crossing a line has to cost

Chapters two and three got the toll out of the chairs: on a finite machine, a move that settles something for good lands two states on one. Does that survive a whole row of cups?

The chairs need one yes-or-no question to count with, and now there are many cups. So which cup? The first one I'd reach for is the cup at the place where the record starts. That goes nowhere: the starting state already has that cup up, so there's no newcomer coming in from outside. The cup that works is the one the step turns up, the cup at the place where the record lands. It's up afterwards and it was down before. So put everybody who has that cup up in the done room, and the state the step starts from is the person by the door.

Now it's the chairs again, once for that cup. If the move keeps everybody in the room inside and never lands two of them on one chair, they fill every chair, and there's none left for the newcomer. So one of two things happened: two arrows met, or somebody inside got moved out, which means that cup went back down. If cups never come back down, it has to be the first one. Two arrows meet.

And then Bennett's rule from chapter three does the rest. A move where two arrows meet costs at least one, so a move that turns up a cup costs at least one. The toll on the whole direction rests on the same three things as the toll on one cup: finitely many states with each cup up, cups that never come back down, and a price on two arrows meeting.

Each of those can be knocked out, and each one goes the same way it did for one cup. Let the states be all the whole numbers and let the record count up by one on every step, free. Nothing ever merges, the price list has nothing to charge, and every step turns up a new cup. That's the endless strip, paying in fresh paper. Let a two-state record swap back and forth. Nothing merges, and the cup it turns up gets knocked back down. That's the coin. Let a two-state record stamp every state up and charge nothing. The arrows meet and nobody asked them to pay. That's the stamp at zero.

So nothing new got smuggled in when one cup became a row, and nothing the one cup needed got left behind. You might expect the diamond to need some extra rule for the two middle cups that don't come before or after each other. It doesn't. Every cup gets its own room of chairs, and the argument in each room is the old one.

CHAPTER 6

Looking isn't keeping

The toll charges the move that settles something for good. But there's another thing a machine does that feels like it ought to cost at least as much, maybe more, and that's finding out. Every time a machine rules out a possibility, it knows more than it did. If two arrows meeting costs something, does learning cost something too?

Most people's gut says yes, and finding out is the expensive part. Most physicists felt that way too, for a long time. It's why one little imaginary creature kept them busy for over a century. This chapter is about where that feeling goes wrong, and where it's right. The short version: looking doesn't have to make two arrows meet. It uses up a blank page instead, and you can hand the page back. Wiping the page to use it again is where two arrows meet. And making a record that stays, one nobody can hand back, is paying for good.

6.1 A small being at a small door

In December 1867 James Clerk Maxwell, who had published his equations for electricity and magnetism two years before, wrote a letter to his friend Peter Tait about a problem he couldn't get rid of. Take a box of gas at one temperature, with a wall down the middle and a tiny door in the wall. Now imagine, he said, a "very observant and neat-fingered being" sitting at the door, who can see the molecules coming. When a fast one comes from the left, it opens the door and lets it through. When a slow one comes from the right, same. Otherwise it keeps the door shut. It never pushes anything. It only opens and closes a frictionless door at the right moments.

After a while the right side is hot and the left side is cold, and you've got a difference you could run an engine off, made out of nothing but looking. That's supposed to be impossible. It's exactly what the second law of thermodynamics says can't happen, and the second law is the one law physicists are least willing to give up. Arthur Eddington wrote in a 1928 book of popular lectures that a theory found to contradict it has nothing for it but to collapse in deepest humiliation.

Maxwell put the being in his textbook in 1871, and it got named Maxwell's demon, though not by Maxwell. William Thomson, later Lord Kelvin, called it a

demon in the journal Nature in 1874. Maxwell, in an undated note to Tait headed "Concerning Demons," tried to take the name back: "Call him no more a demon but a valve." Nobody listened, which is probably for the best, because "Maxwell's valve" wouldn't have lasted a century. But his correction is exactly right, and it's the whole problem: the being does no work. It opens and shuts a door.

So physicists spent the next century trying to find where the being was cheating. The obvious answer pulls hard: the being has to see the molecules, and seeing costs. In 1929 Leo Szilard cut the box down to a single molecule and did the arithmetic. The second law survives, he said, if finding out which side the molecule is on costs at least as much as the being gains. In 1951 the French physicist Léon Brillouin made it concrete: to see a molecule you have to bounce light off it, and, the way the philosopher of physics Jeffrey Bub sums it up, finding out that way costs at least one halving's worth of entropy for every bit. For thirty years that was the answer. Information costs because getting it costs.

But that's a price on one way of looking. It doesn't show that every way of looking has to pay it.

Szilard is worth one more minute, for a reason that has nothing to do with the demon. Four years after the one-molecule engine—in September 1933, by his own later telling—he was waiting at a traffic light on Southampton Row in London. He had just read a newspaper report in which Ernest Rutherford, the physicist who announced the atomic nucleus, called the hope of getting power out of atoms moonshine. As the light changed, it occurred to him that an element which gave off two neutrons when it swallowed one could keep a reaction going by itself. That's the nuclear chain reaction, thought up crossing a street. It's his memory of himself, and Richard Rhodes's history of the bomb opens with it, in a chapter called "Moonshine," so hold it as the man's own story, not a record made at the time. It changes nothing about the demon. What it adds is a picture of the man: the same mind looked at one molecule in 1929 and one neutron in 1933, and both times the question was what a single small step forces downstream.

6.2 Bennett moves where the cost is

In 1982 Charles Bennett, who had already shown in 1973 that computing can be done with no two arrows meeting if you keep the whole history, took the demon apart one step at a time.

The being looks at a molecule and writes down which side it's on. Draw it. Before the look, there are two ways the world could be: molecule on the left with a blank notepad, or molecule on the right with a blank notepad. After the look, there are two ways: molecule left with the pad saying "left," or molecule right with the pad

saying "right." Two went in, two came out, and you can read the before off the after. No two arrows met.

But something got used up: the blank page. Before the look, the pad was blank and ready. After it, the pad is full. That's the fresh paper from chapter one. The look didn't merge anything. It borrowed a page.

So where's the cost? The being has one notepad and a lot of molecules. After the first molecule the page is full, and to do the next one it has to wipe it. Draw the wipe. Pad says "left" goes to blank. Pad says "right" goes to blank. Two arrows, one landing place. There it is. Looking feels like the active part, the thing the being does, and wiping feels like housekeeping. The housekeeping is where two arrows meet.

That's Bennett's, and it's the same rule chapter three wrote on the move: what has to cost is "the erasure of a bit or the merging of two computation paths." In the same 2003 paper he lists copying onto a blank place among the operations that can, in principle, be done without any such cost.

6.3 The demon on a sheet of paper

Build it, because the whole thing is four dots, and you should see the arrows meet or not meet for yourself.

The machine has a hidden bit, which is the molecule's side, and a display, which is the notepad. Write its four states as pairs, hidden bit first and display second: 00, 01, 10, 11. It has two moves.

MEASURE copies the hidden bit onto the display the way a light switch adds: if the hidden bit is 1, flip the display; if it's 0, leave it alone. Draw the arrows. 00 goes to 00. 01 goes to 01. 10 goes to 11. 11 goes to 10. Every state has exactly one arrow landing on it. Nothing met.

Look at the two states where the display starts blank, 00 and 10. They go to 00 and 11: the hidden bit got copied onto the blank display. That's the look.

WIPE sets the display to 0 no matter what it held. 00 goes to 00, and 01 goes to 00 as well. 10 goes to 10, and 11 goes to 10 as well. Two arrows land on 00 and two land on 10. That's the wipe, and arrows meet twice.

Hidden bit, display before	After MEASURE	After WIPE
00	00	00
01	01	00
10	11	10
11	10	10

So Bennett's rule charges WIPE at least one, and it's allowed to charge MEASURE nothing. Zero on the price list doesn't mean zero spent in a real device; a real MEASURE takes time and energy, and it uses up the blank display. What it doesn't do is make two arrows meet.

One detail worth seeing. Follow only the two cases you actually set up, the ones with a blank display: after MEASURE they're 00 and 11. WIPE sends those to 00 and 10. Still two different states, because the hidden bit still tells them apart. So on those two cases, WIPE didn't lose anything. The rule charges it anyway, because the rule has to answer for every state it might be handed, and on the full table the arrows meet. The price is the worst case on purpose.

Now be the one watching. You can see the display and not the hidden bit. The display starts blank, so you've got two candidates for what's going on: 00 or 10. One bit you don't know. The machine runs MEASURE, and you look. If the display shows 1, only 10 could have done that. If it shows 0, only 00. Your two candidates went down to one. You learned the hidden bit.

The price was zero. And the machine's own list of where it could be went from 00 and 10 to 00 and 11. Still two. Nothing it could have been got merged with anything else.

You narrowed. It didn't.

6.4 Two rows of dots

That last line is the thing I had to get straight before any of this made sense, so here it is as a picture.

Draw the machine's possible states as a row of dots. Now draw a second row above it: the dots that you, watching from outside, still think it might be on. The first row is a fact about the machine. The second is a fact about you, about what you've seen and what you've crossed off. Both rows can shrink, and they shrink for different reasons.

The machine's row shrinks only when two of its arrows meet. That's what the price charges. Run the halving rule from chapter three along a whole run and the counts just multiply: if the first move costs one, the row can at most halve; if the next costs two, it can shrink to a quarter of that; and so on. So a run can't shrink the machine's own row by more halvings than it paid for, all told. (Count with whole questions instead and you lose a little: four places and three places both take two yes-or-no questions to pin down, so the rounded count can't see that squeeze. The plain count can.)

Your row shrinks when you see something some of your candidates couldn't have shown you, and you cross them off. Nothing happened to the machine. It didn't merge anything. It doesn't know you're watching. It would have done the same thing with nobody there.

The physicist Edwin Jaynes had a name for mixing those two rows up, and it's one of the most useful names I know: the mind projection fallacy, treating what you know about a thing as if it were a property of the thing. Your list of candidates is in

your head. The machine's row is in the machine. The price is about the machine's row, and crossing a candidate off yours doesn't move it.

I have to be fair to Jaynes, though, because he'd argue with where I just took his phrase. In a 1965 paper he wrote that entropy is "an anthropomorphic concept," a phrase he credited to the physicist Eugene Wigner, meaning it belongs to the particular experiments you or I choose to do on a system, not to the system alone. For Jaynes, the second law is about what's known too, and the line I'd like to draw, that physics cares about the machine's row and not yours, is exactly the line he'd refuse. That fight, Jaynes against Boltzmann, is one of the oldest in the subject, and I can't settle it and don't have to. What I need is only this: the two rows are different rows, and only one of them belongs to the machine.

6.5 Who holds the pen

Here's the most beautiful thing in this chapter, and you find it by playing with the four dots, not by thinking about them.

Run MEASURE twice. The first time copies the hidden bit onto the display. The second time adds the hidden bit in again, and adding it twice the light-switch way gets you back where you started. Check the arrows: 10 goes to 11, and 11 goes back to 10. The display went full and then blank again, no two arrows met, and the price was zero both times.

So a display that was blank before the first MEASURE can be put back to blank with no arrows meeting, by a second MEASURE. Not by WIPE. This is the borrowed page being handed back: while the original is still there to check against, the copy can be un-copied.

What's the difference? MEASURE looks at the hidden bit while it clears the display. It has the thing the copy came from, so it can undo the copy. WIPE doesn't look. It sets the display to 0 regardless, and "regardless" is exactly where the arrows meet. Bennett has this in his 2003 paper too: he lists erasing "one of two copies of data known to be identical" as a step that loses nothing, because you're only removing a duplicate. And Lídia del Rio and her colleagues argued in 2011 that the cost of erasing a system depends on how much the eraser already knows about it, dropping to zero, and in the quantum case below zero. The cost of clearing a record falls on whoever clears it, and it's lower the more they know.

Try it with a card under a cup, red or blue, and a note that's blank or marked. Start with the note blank. Lift the cup: for red, flip the note, blank to marked or marked to blank; for blue, leave it. Now the note tells you the color. Do the same thing again while the card is still there, and the note goes back to blank. Check all four card-and-note combinations and you'll see this rule undoes itself, so no two combinations land together. Now erase the note regardless of the card: for either

card, blank and marked both land on blank. Arrows meet. The cards don't measure anything about heat. What they show is which information each rule gets to use.

6.6 Learning after the first look

A fair objection: the demon's watcher learned something by looking, but any watcher learns something on the very first look, before the machine does anything. So the better question is, after that first look, can a step teach you more, at no cost?

Yes. The smallest machine that does it has three states in a ring: U goes to V, V goes to W, W goes back to U. Say all you can see is one thing: whether it's on W. Start with two candidates, U or V. Neither is on W, so the first look tells you nothing. Take one step. From U you'd be on V, not W; from V you'd be on W. One step told them apart, and a ring never has two arrows meeting, so the price can be zero.

Two states can't do it, ever, with any arrows. If you can't tell two states apart at the first look, what you can see is the same on both, and those two are all the states there are. So what you can see is the same everywhere, forever. Try to draw a counterexample with two dots. You'll draw the same two dots a few times and give up, and giving up is the proof. Three is the smallest number of states where two that look alike at first can come apart later.

6.7 Finishing together

One more count before the big picture, because it's about how much room learning needs.

Run a machine from several different starts, and photograph its whole state at every tick. Lay the photographs out in rows, one row per start, one column per tick. Say every run finishes for the first time at the same tick. And say that in every column up to the one just before the finish, no two states step to the same state. That includes the last step, the one onto the finish: if two runs sat on two different states just before the finish, they can't both step onto one finished state.

Could two photographs on that sheet show the same state? Two in the same column: walk both back a tick at a time, and since no arrows met, they were equal the tick before, all the way back to the starts. But the starts were different. Two in different columns: give them both the same remaining steps. The one that's ahead would finish first, and it was supposed to finish at the same tick as everybody else. So every photograph shows a different state. With N starts and T ticks, the machine needs room for N times (T + 1) states. Running in step and finishing together has a size.

6.8 The certainty count

Now I can put the whole path from not knowing to knowing on one page.

You don't know which state the machine started in. There are a few it could have been, each with its chance. Run the machine. Take any two of those possible starts and watch where they go. Only two things can happen. They land on the same state, and the difference between them is gone from the machine: two arrows met, and under the halving rule from chapter three the price covered it, at most one halving per unit. Or they land on two different states, and the difference is still sitting in the machine, kept as two states held apart. There's no third thing a pair of starts can do.

So everything you didn't know about the start splits cleanly in two: the part the run destroyed, which was paid for, and the part the end state still holds.

Now put a watcher outside, who can see some part of the machine and not the rest. Whatever they see depends only on where the machine started, so whatever they learn is a fact about the start. And the same split holds for it: what they learned is either still held in the machine's end state, or it's gone from the machine and two arrows meeting paid for losing it. A run that costs nothing leaves everything the watcher learned still held in the machine.

That's the exact sense of "looking borrows." The watcher's copy is a copy, and the machine still holds the original. When the machine lets go of the original, two arrows meet, and the bill comes due.

What that doesn't say: how much heat. That's for the heat chapter. It doesn't say the watcher's own notes are free; they need a blank page too. And if the start is known for certain, there's nothing to learn and nothing destroyed, so there's no floor from any of this at all.

6.9 Who gets charged is partly bookkeeping

The physicists had this freedom on the books before me. Paul Fahn in 1996, doing the accounting of Szilard's one-molecule engine step by step, argued that whether you charge the looking step or the wiping step is partly a choice of how you keep the books, and that the total is what's fixed. Takahiro Sagawa and Masahito Ueda in 2009 worked out a bound on that total, with the split depending on the memory. So "a price list could charge MEASURE instead" is a known freedom, not a loophole I found.

The other side is alive too. Ruth Kastner argued in 2025 that Brillouin was essentially right after all: put the real physics of measuring back in, and it's the measuring that pays and stops the demon, and Bennett's free copy fails as physics. And John Norton argued in 2013 that at the scale of molecules, jiggling from heat keeps any process that's supposed to be perfectly reversible from ever finishing, and

finishing it costs more than Landauer's amount. If either of them is right, a real demon's look has a cost after all.

What's on the page here survives that, because it's about price lists: a price list that follows Bennett's rule isn't forced to charge the copy, whatever a real device spends. What wouldn't survive is the physics I've been hanging on it, that a real demon can look for free. If Norton is right, the line between looking and keeping keeps its counting and loses its physics. I can't referee that one.

And in 2010 a group in Tokyo, Shoichi Toyabe, Takahiro Sagawa, Masahito Ueda, Eiro Muneyuki and Masaki Sano, ran a real demon: a tiny particle walked up a staircase by feedback on what a camera saw. The energy it gained was paid for by the information the camera took, and nothing was violated. The demon is real, it works, and it pays. Where it pays is the part that took a century.

6.10 Somebody got there first, goddamn it

I'd been sitting on this idea, that making a result stay so it can't be handed back is what pays, and finding out only borrows, and I thought: this one's mine. Then I found Carlo Rovelli's 2024 paper, "The Thermodynamic Cost of Choosing." A real physicist, with a real paper, saying that making a choice that sticks gives off at least the heat Landauer charges for erasing a bit. He even quotes a German saying, Wahl ist Leiden, to choose is to suffer, and at that point what are you going to do but laugh.

It pissed me off. It's hard to have an original thought in this world. But it didn't last long, because once I actually looked, his was a sentence on the surface, and the machinery under it was different.

On its face he contradicts Bennett. Bennett says measuring doesn't have to cost; Rovelli says any measurement costs. So look at what Rovelli calls a choice. A coin in the air hasn't chosen anything; it has to hit the ground and stop. And then the sentence I'd put on my wall: "stabilization must be irreversible, otherwise it is not stabilization." Bennett's free look is one that can still be undone, which is why it's free. Rovelli's choice is one that's been made to stay, which is why it costs. The way I read them, they're standing on opposite sides of the same line: looking, which borrows, on Bennett's side, and keeping, which pays, on Rovelli's.

That reading is mine, and I owe you exactly how much of it. Push on it with his paper open and it gives in four places. He saves the word "choice" for an outcome that's stable and not predictable, and rules out a thermostat because it runs on a predictable mechanism; the settling move in this book is fixed in advance by a rule, so by his own definition his result doesn't cover it. His choice is one possibility branching into two, which he calls in a sense the opposite of erasing; the toll comes from two arrows meeting, the opposite shape. His stability lasts a while, not forever;

mine is forever, by definition. And he doesn't stand beside Bennett at all: he holds that any measurement costs, credits that to the philosopher Hans Reichenbach, and doesn't cite Bennett. So what Rovelli and I share is a sentence: a choice that sticks costs something. Follow it through and his coin hitting the ground, a fact nobody could predict, sits closer to looking than to keeping. I still think it's the same instinct. You've got everything you need to disagree.

And on copying, the nearest physics isn't Rovelli's. It's in biology. Thomas Ouldridge and Pieter Rein ten Wolde worked out in 2017 the basic costs of making a copy of a molecule that stays matched to its template after the two separate, and of destroying one. A companion paper the same year, with Christopher Govern, does the same kind of accounting for how a cell reads out its own sensors, which is a measurement in all but name. The way I read them, the copy that has to stay is the one that pays. Then Jenny Poulton, ten Wolde and Ouldridge in 2019 found the sharp form: in their simple models, when a copy grows with the weakest possible push, its advantage over random copying, and the efficiency of storing that information, both vanish. Pushing harder, with chemical work, makes copies that stay matched after the two let go. That's finding out against keeping, in chemistry, with the cost attached, and biology had the distinction before I did. Theirs is a correlation between molecules, with heat in it. Mine is a reading in a model of computation, with no temperature anywhere.

Two more neighbors have "learning" and "commit" in their titles, so the line here shouldn't be read against them by accident. Sebastian Goldt and Udo Seifert in 2017 did the physics of a neural network learning a rule, and argued that learning has a cost: the entropy the network produces limits how much it can learn. That doesn't contradict this chapter. Their learner isn't copying onto a blank page; it's a driven, noisy physical system, and that's what they price. This chapter only says one way of finding out, a copy onto a blank, is allowed to be free. Nothing here says every way of learning is cheap, and theirs isn't. And from statistics, Aurélien Garivier, Tor Lattimore and Emilie Kaufmann in 2016 compared strategies that explore for a while and then commit to one choice against strategies that keep adjusting as they go. In their setting, over a long run, the best strategies that commit lose about twice as much reward as the best ones that keep adjusting. That's the closest published sentence I've found to "committing costs something" with a number on it. It's lost reward, though, not a count on a ledger, so it's a cousin, not a twin.

6.11 What the toll is for, said at last

Now I can say exactly what the toll charges. Not looking. Not copying. Not finding out. A look can be done with no arrows meeting, on a price list that charges nothing for it; the demon is a machine where that happens. Looking uses up a blank page,

and while the original is still there, the page can be handed back. What gets charged is the one thing chapter two showed can't be done on a finite machine without two arrows meeting: settling something so it stays, a record nothing hands back.

That's what "No Free Insight," the name I gave the toll before I understood it, really covers, and I don't want the name doing work it hasn't earned. Insight in the sense of finding out can come free; it borrows a page. The insight the toll prices is the kind made to stay. Whether that narrower thing deserves the name is fair to argue about. I kept the name because it's the question I started with, and because narrowing it down is the finding.

One more thing this tells you, for later. A later chapter counts what it takes to narrow a list of candidates down to a shorter one, and puts a cost on it. Narrowing alone doesn't force anybody to pay anything; the demon just narrowed two candidates to one for free. So wherever this book puts a price on narrowing, the payment goes in as an assumption, out in the open, and I'll keep calling it one.

CHAPTER 7

Looking through a window

Page one ended on two sheets of paper. One person sorted a list and checked it. The other copied a sorted list off somebody else. Same numbers, same order, and nothing on either sheet tells you which one was checked.

The obvious answer is: look harder. This chapter is about why looking harder doesn't help, exactly when it doesn't, and what does.

7.1 Two boxes under a camera

Put two boxes on a table. Each one has tiles on top, and the tiles spell out the same answer. On the side of one box, a stamp is pressed. On the other, it isn't. Hang a camera straight above them and take a photograph. The two photographs are identical.

Now hand me the photographs and ask which box was stamped. Take a better photograph. Blow it up. Count every tile. None of that gets the side of the box into the picture. The difference is real; it's on the box. It just isn't in what you handed me.

That's the thing I want a word for, and the word is window. A window doesn't see anything. It's a viewing port, an opening that faces one way and narrows what you can take in from where you're standing. The camera above the boxes is one: it faces down, so the tops come through and the sides don't. A display is one. So is a printout, or anything else that lets part of a machine through to you. And behind any one picture there's a pile: every full state of the boxes that would give exactly that picture. The tiles can agree while the boxes differ in whether they were stamped, in what checks were run, in what was erased along the way. All of that sits in the pile behind one photograph. There could be two states in the pile, or a million, or endlessly many.

7.2 A cave, and the people who got good at it

This one is Plato's, from the seventh book of the Republic. Socrates describes prisoners chained in a cave since childhood, facing a wall, with a fire behind them

and people carrying objects along a walkway between the fire and their backs. All the prisoners have ever seen is shadows on the wall. To them, he says, the truth would be nothing but the shadows. Later in the story Socrates reads it out himself: the cave is the world we see, and the climb out is the soul's climb toward what's really there.

The part I need comes a little before that. One prisoner gets freed and taken up into daylight, and he thinks back to the cave. The prisoners, Socrates says, hand out honors to whoever is quickest at spotting the shadows as they pass and best at guessing which comes next. Read that twice. The prisoners get good at it. They build a whole science of shadows, with experts and prizes, and it works, on shadows. What they can't do is say anything about a carried object that leaves no mark on the wall. That isn't because they're stupid. The information isn't on the wall. Skill inside a window doesn't reach past the window, no matter how much skill.

7.3 The shadow of a cube

Try this one tonight with a lamp and a wall. Keep the lamp well back. Hold a cube face-on to the wall and mark the outline of its shadow. Then hold up a square card in its place and move it until the outlines match. Show a friend only the outline. They can't tell you whether the thing had thickness. Ask them, and they'll tell you, truthfully, that they can't know, and that staring at the square won't change it.

Now turn each one a quarter turn. The card's shadow becomes a thin line. The cube's shadow stays a square. Now your friend knows. What changed wasn't how clever they were. It was the window. A second look from a different angle is a bigger window, and what paid for it was the turn.

Edwin Abbott wrote a whole novel about a window like this in 1884: Flatland. The people of Flatland live in a flat world and never see a shape. Put a coin flat on a table and lower your eye to the edge, and the coin turns into a line. That's all the view there is. What gives a Flatlander anything more is fog: a near line is bright, and a line running off into the fog goes dim, so a trained eye can read a triangle or a pentagon off how fast the brightness fades. The training takes time and money, and in the story only people with money get it. The window got bigger, and it was paid for.

Then a sphere passes through the narrator's flat world. What he sees is a line whose ends fade wrong, changing size in a way no flat shape could. He only works out it's a circle by feeling it, and the circle grows from a point to its widest and shrinks back to nothing as the sphere passes through. From one slice he can't put a sphere together. From the slices one after another, seen and felt and argued over, he starts to. The way out of a window is never "think harder." It's always a different window, one you can tell two things apart through when you couldn't through the

old one. Every time, in every story here, that cost something: a turn, a second look, training, time.

Who was Edwin Abbott? Not a mathematician. He was headmaster of the City of London School for twenty-four years, a clergyman, and the author of more than forty books, a Shakespeare grammar among them. He put Flatland out under the name "A Square." His full name was Edwin Abbott Abbott, two A's: his father, Edwin Abbott, had married his own first cousin, Jane Abbott. Is the pen name a joke on that, A squared? People say so. The earliest I found is Rudy Rucker in 1984, who says it "seems possible" he was nicknamed that; Ian Stewart says he "might have been"; and William Lindgren and Thomas Banchoff, in their annotated edition, call it a pun on his initials outright, EAA, E A squared. None of them quotes Abbott saying so. So it's a reading of his name, not his word. The part I like best comes from the sketch of his life in the MacTutor archive of mathematicians' lives: Flatland, by far the most read thing he wrote, isn't mentioned at all in his obituary in the London Times. The official record of the man was a window too. Through it you got the headmaster and the grammarian, and not the one book people still read.

A video about that book is where the hunch behind this one started. That story is in the afterword.

7.4 The whole fact

Here's the whole thing, and it's small.

You've got the states, the real things. You've got a window. For each state, it lets through some part of it, and that part is all you get. And you've got a question about the state. Give it a state and it hands back an answer. What you want is a reader: something that takes what you can see and gives the right answer to the question, every time.

7.5 When does one exist?

Take two states that give the same picture. The reader sees the same picture for both, so it gives the same answer for both. If the question has different answers on those two states, the reader is wrong on one of them, and there's nothing it can do about it, because it never saw the difference. So a reader can exist only if the question gives the same answer on every state in the same pile.

And the other way round: if the question does give one answer per pile, a reader exists. For each picture, answer whatever the question says about any state behind it; since they all agree, it doesn't matter which one. Existing isn't the same as being easy to find, though. To work out the answer for a picture you still have to find a state behind it, and on some windows that search is hard, or endless.

Try the smallest case by hand. A state is a position, 0 or 1, plus a hidden flag, up or down. Through the window you get only the position. Start at position 0 with the flag down. One move goes to position 1 and puts the flag up. Another move goes to position 1 and leaves the flag down. Through the window, both look like position 1. Now write down every possible reader from "what I see" to "is the flag up." There are exactly four: from two positions to two answers. Check each one against both results. None gets both right, because both show position 1 and one has the flag up and one doesn't. There's no fifth reader hiding anywhere.

7.6 People who found this on their own

The fact is so plain that it got found separately in every field that has to deal with seeing part of something. Laying them side by side is the kind of evidence I trust: people who never met kept walking into the same wall.

In 1956 Edward Moore, working on the small switching machines that became computer hardware, asked when somebody who can only feed a machine inputs and watch its outputs can tell two of its inside states apart. He made "can't tell apart from outside" precise, and found a sting: even when every pair of states can be told apart by some experiment, there are machines where no single experiment tells you which state you started in. What you can always do is find out which state you're in now.

In 1960 Rudolf Kalman asked the same question for systems that change smoothly, called it observability, and gave a test you can compute. His unobservable states are exactly a pile: states whose difference never reaches the output, however long you watch. Try it with two toy cars on identical roads, a mile apart. Give them the same speed at every moment, and let nothing about the road depend on the mile marker. Their speedometers tell the same story forever. No amount of rereading that story gives you the starting position. A landmark, or a hill at a known spot, would.

Where did Kalman's filter go first? Toward the Moon. In the autumn of 1960 Kalman visited Stanley Schmidt at NASA's Ames Research Center, not knowing what Schmidt's group was working on. They were stuck on how a spacecraft going around the Moon could work out where it was from a few angles its crew could measure through a sight. That's a window problem with a capsule in it: the state is position and speed, the view is a handful of angles, and the job is a running best guess from a string of views. With real work, his method fit, and it went into the studies for navigating Apollo. Leonard McGee and Schmidt, telling the story in 1985, called it a fortunate sequence of events. The filter didn't make the angles show more. It used how the spacecraft moves, plus a string of noisy views, to estimate what the views alone couldn't pin down.

In 1977 Patrick and Radhia Cousot built a whole way of analyzing programs on throwing information away on purpose: run the program on rough descriptions of its values instead of the values themselves, and only accept answers that are true for everything the rough description could stand for. If all you know is that a number is between minus one and one, you know its square is between zero and one. You don't know which square. In 1970 Edgar Codd, inventing the relational database, gave users views, deliberately partial windows onto the stored data, and François Bancilhon and Nicolas Spyratos later worked out the problem from the other side: if a user changes what they see through a view, which of the many stored states behind it should the database move to? Several are legitimate. That's the pile, seen from the writing side.

Two more fields have it as a founding idea. In the logic of knowledge, what an agent knows is modeled by sorting the possible states into piles it can't tell apart, and it knows a fact exactly when the fact holds everywhere in its pile. The economist Robert Aumann used that in 1976 to prove that two people who start from the same beliefs, and whose estimates are common knowledge between them—each knowing the other's, knowing the other knows, and so on without end—can't agree to disagree, and Ronald Fagin, Joseph Halpern, Yoram Moses and Moshe Vardi built a whole logic on the same piles. And in computer security, Joseph Goguen and José Meseguer defined noninterference in 1982: one group of users doesn't interfere with another when nothing the first group does changes what the second can see. David Parnas, whose cut of a program into modules comes up later, hid things on purpose: a module is defined by the decision it hides from whoever uses it.

There are more. Oliver Kosut and Jörg Kliewer asked in 2018 when a receiver, with a possible attacker on the line and no shared key, can always either get the message right or notice the attacker, and the answer turns on whether the attacker can make what the receiver sees look exactly like an honest message. That's the pile with the probabilities taken out. And in 2022 Anvith Thudi, Hengrui Jia, Ilia Shumailov and Nicolas Papernot asked how a company could prove a trained model had forgotten someone's data, and found the same finished model can come out of different training sets, so the model alone can't show whether a point was used. The only thing you can check is the procedure that was run. Same wall: the finished state is a window, and what happened is in the run.

7.7 Notes passed across a room

Philosophers who study being told things hit the same wall from the other side. Read one after another they blurred together for me, until I built a picture of where each one stands.

Somebody passes you a note across a room. It says the answer to question seven is 42. What crossed the room? Ink. Whatever made the writer believe 42 stayed on their side: the working-out, how careful they were, whether they knew or guessed. Pull that apart and there are five things in play: the claim on the paper, the writer's head, whatever got the writer there, the trip across the room, and your head at the other end. Every position I found puts its weight on one of those five.

The two ends are John Locke and Tyler Burge. Locke, in 1690, says believing something because someone with authority said it isn't knowing it: if the reason didn't cross the room, you don't have it. Burge, in 1993, says you're entitled to accept what you're told, if it's presented as true and makes sense to you, unless you have a reason not to. I get the pull of Burge; nobody could get through a day checking every note. But a default anybody can trigger by writing something sensible is a record nobody paid for, the thing this book keeps refusing. So on that dial this book sits at Locke's end, and staying there means carrying the check across the room every time.

Jennifer Lackey cuts the wire between the writer's head and what arrives: a teacher who doesn't believe the theory she's required to teach, and students who come away knowing it anyway. Sanford Goldberg puts the weight on the process behind the writer's word—machinery the reader can't see. I'm not comfortable with that, and that discomfort is why this book exists. Machinery you can't see is machinery you can't check. Christoph Jäger and Nicholas Shackel built a model where the writer knowing is neither needed nor enough for the reader to know, landing where Lackey's teacher did from a completely different direction. Wesley Holliday put trust into a logic of changing your mind, with a graph of whose word each person takes. And Michael Burrows, Martín Abadi and Roger Needham, in their logic of authentication, said: if I believe an authority has the say over a statement, and I believe the authority holds it, I believe it. That's Burge's default with one change. You named the authority before the note arrived.

Kenny Easwaran, in 2009, went after what crosses the room itself. Why do mathematicians accept a long proof, even one partly done by a computer, and turn down a proof based on chance that can be far more certain? His answer is a property he calls transferability. A step-by-step proof puts the reason on the paper: every step crosses the room, so you can learn from the writer without trusting the writer. A proof based on chance leaves the reason behind and sends only the writer's word about what the dice did. That's the cleanest statement I found of what a checked record should do: make the reason cross the room, and run the check on your own side. That reading is mine.

7.8 Slips handed out at desks

People who build systems out of many computers hit the wall too, and one pair measured how far it goes. Picture people at desks, working from one rulebook and passing slips to each other. Nobody sees into anybody's head; all anyone gets is the slips. Somebody may be cheating. What can the honest ones do about it with nothing but slips?

Andreas Haeberlen and Petr Kuznetsov asked in 2009 which kinds of cheating an outside observer can catch at all. As I read them, it depends on whether the cheating ever shows up in a slip. A cheat who breaks the rulebook without changing any slip, or who writes a slip an honest player could have written too, can't be caught by anyone holding the slips. A cheat who fails to send a slip that was due, or sends one the rulebook would never produce, can. The first two are a pile: an honest room and a crooked one handing out the same slips.

Their group's PeerReview system, from 2007, is the ladder up to that wall: every desk keeps a secure log of its slips, and a checker replays the log against a copy of the rulebook. That's close to "show me everything," built. Close, because a log of slips only stands in for a whole desk if the rulebook can rebuild what the desk did from them, and only if the checker's rulebook is the right one.

Then the people asking what you can say once you're at the wall. Ralf Küsters, Tomasz Truderung and Andreas Vogt defined a verdict that names who misbehaved, and showed that being able to check a result is a weaker form of it: one says something went wrong, the other points at a desk. Radha Jagadeesan, Alan Jeffrey, Corin Pitcher and James Riely looked at systems that enforce their rules after the fact, by audit. Robert Künnemann, Ilkan Esiyok and Michael Backes built a way to check it automatically by making it a property of the whole run. And in September 2026 Timo Treitz and Künnemann gave the public logs of web certificates the full treatment, and found that the plain version names a culprit only if the log itself is assumed honest. Somebody always has to be trusted somewhere.

Turn that on me and it bites twice. A raise of the stamp that never reaches the window is the first kind of cheating, the kind nobody outside can catch. And every verdict in this book is "yes" or "no" about one claim; none of them points at a desk.

7.9 Moves that keep to the tabletop

Back to the two boxes. Give them two kinds of move.

Tile moves move the tiles, and only by looking at the tiles. They cost nothing and never touch the stamp. Stamp moves leave the tiles alone and can only press the stamp or leave it as it is. Once pressed, it stays pressed, and pressing it pays the toll.

Now take the stamped box and the unstamped box, with the same tiles, and give them the same tile moves. After one move the tiles still agree, because each next picture depends only on the picture before and the move. After two moves, they still agree. After a hundred. Meanwhile one stamp stays pressed and the other doesn't. You can watch the tabletop for the whole run and never see the difference you started with.

There's one way to break that, and it's worth seeing. Let somebody read the stamp and choose a different tile move because of it. Now the stamp can signal through the tiles. Here both boxes get the same moves, on purpose, to keep the question fixed: what can come through the window, if nothing that reads the stamp is allowed to steer the tiles?

7.10 The clerk with two photographs

Now point the camera at the price, because that's what this chapter is really for.

Say you want to charge for pressing the stamp, exactly: at least one for every move that presses it, nothing for a move that doesn't. And say the person charging is a clerk who only looks through the window: a photograph before each move and a photograph after. The clerk's charge can only depend on those two photographs.

Can the clerk charge exactly? Not if there's a pair like this: two moves that give the clerk the same before-photograph and the same after-photograph, where one pressed the stamp and the other didn't. A stamp pressed inside a closed drawer, next to somebody opening and shutting the drawer. The clerk has one charge for that pair of photographs. If it's at least one, the move that pressed nothing gets overcharged. If it's zero, the press gets undercharged. Nothing read off the photographs gets both right.

The fix is the plain one: put the stamp in the photograph. Then "one when the stamp goes from not pressed to pressed, zero otherwise" works. Showing the stamp isn't the only fix. Any window with no such pair works too; a later chapter has one that never shows where the record is but always shows when it moves.

You can even find the clerk's best careful prices. Put all the moves that give one pair of photographs in a tray. If you never want to miss a press, charge one whenever the tray has even one press in it. If you never want to overcharge, charge one only when every move in the tray is a press. When the tray is all presses or no presses, those two agree. When it's mixed, they can't, and that's the bad pair again. Those are "some member" and "every member" applied to the same tray, which is exactly how the mathematician F. William Lawvere described logic's two quantifiers in 1969, and Robin Cockett, Isabelle Geng, Carlo Maria Scandolo and Priyaa Srinivasan used the same idea in 2023 to stretch measures of resources along a map.

Control engineers have hunted pairs like that for a long time. In 1995 Meera Sampath, Raja Sengupta, Stéphane Lafortune, Demosthenis Teneketzis and Kasim Sinnamohideen asked when an observer can always work out, from what it sees, that a hidden fault has happened. Anooshiravan Saboori and Christoforos Hadjicostis asked in 2007 the question the other way round, for secrets: when can an observer never be sure the system is in a secret state? A colliding pair like the clerk's is one place where the observer can't tell right away. Their versions are richer than mine. What this chapter adds is the price.

Why can't the clerk just watch the move itself? A clerk who sees the whole state and the move charges exactly with no trouble. So the problem bites an observer who doesn't see the move, or who's handed an account of it they can't trust. If the names of the moves come from a log the program wrote, the program can write whatever names it likes, and a clerk reading that log is reading one more window, one the program painted.

7.11 Look before the concrete covers it

An inspector arrives after the concrete is poured. The question is about the steel underneath. On one job it was sound. On the other it wasn't. Give both jobs exactly the same finished surface, and give the inspector only that.

There's no amount of staring that tells those jobs apart. Hunting for a sound arrangement that could have made this surface doesn't do it either; a sound arrangement was possible on the bad job too. What matters is which one was there. It's the four readers again: one input, two answers needed.

While the steel is still showing, you can inspect it and keep the answer. Or keep enough evidence to inspect later. Either way, something that tells the good job from the bad one has to survive. Wiping some unrelated detail does no harm. Wiping the last difference your question depends on leaves the next inspector with the same surface and two possible pasts.

For one question asked again and again along a run, what you have to keep can be tiny. Start a bit at yes, and at each check, set it to yes only if it was yes and this check passed too. It says exactly whether every check so far passed. A later failure turns it down for good. Put that next to a stamp that says "some check passed at some point" and stays up, and you can see they answer different questions.

7.12 Three ways past the window

Say you're outside, and you want to be convinced a run paid. The window you have doesn't show the count. How bad is that?

Take the smallest setting. A run that presses the stamp and counts one, and a run that only moves the tiles and counts zero, from the same start to the same picture. Call the before-and-after photographs the transcript. Both runs leave the same transcript. Now try to build a checker that reads the transcript and says yes exactly when the count is one. It has to say yes to the first run's transcript. The second run left the very same transcript, so it says yes there too, and the count there is zero. No checker of that transcript is both never fooled and never unfair. Dress the transcript up with extra parts, and the same thing happens, as long as the extra parts are the same on both.

Economists hit the same wall: Jerry Green and Jean-Jacques Laffont in 1986, and Jesse Bull and Joel Watson in 2007, worked out what a system can decide when people can only claim about themselves what their evidence backs up. An audit can only tell apart the cases the evidence on offer tells apart.

So what works? I know three things, and cryptography seems to use all three. Each gets past the window, not past the cost: the run that pressed the stamp paid either way. And each rests on trusting something.

You can be shown the whole state. Then there's nothing to work out; you read the count straight off. What you trust is that the state you were shown is the state there was.

You can be shown a promise: the ordinary window plus one extra mark that's set when, and only when, the claim holds. Reading that mark works exactly when it can't be set falsely and is always set when it should be. What you trust is whoever can set the mark. A mark anybody can set is just another window. Bruce Schneier and John Kelsey worked out how to make a log that an intruder who takes over the machine can't quietly rewrite, and their honest limit is the honest limit here. It protects what was written before the break-in, and nothing after. Mihir Bellare and Bennet Yee had already named that property forward integrity. Point it at an AI agent keeping its own log, and the line moves to the very start. An agent that writes the wrong thing from its first step is past the break-in for the whole log. A 2026 preprint by Laurent Bindschaedler, Quentin Botha and Christoph Siebenbrunner builds exactly such tamper-evident trails for agents and calls it an Agent Flight Recorder.

Or you can ask. The outside party puts a question and gets an answer that pins it down. What you trust is the honesty of whoever answers, because a liar's answer is another shadow. The refined form is Shafi Goldwasser, Silvio Micali and Charles Rackoff's zero-knowledge proofs, from 1985, where a checker is convinced by a conversation built to separate true from false with a guarantee against a cheating prover. Joseph Halpern, Yoram Moses and Mark Tuttle gave them an analysis in terms of knowledge in 1988. My asking, in this chapter, is one round with one fixed

question, and nothing in it handles a liar. It's a toy of the real thing, and I'm not going to let the real thing's reputation stand behind the toy.

The real thing even has a cave in it. In a paper presented in 1989 and printed in 1990, "How to Explain Zero-Knowledge Protocols to Your Children," Jean-Jacques Quisquater and Louis Guillou, with their families and Tom Berson, told it as Ali Baba's cave: a ring of tunnel with a magic door at the back that opens only for a secret word. Peggy goes in down one side while Victor waits outside; then Victor walks to the fork and shouts which side she should come out of. If she knows the word, she always can. If she doesn't, she gets away with it only when he happens to call the side she went in. Play it with a coin for Victor's shout. One round, a cheat gets through half the time. Twenty rounds, about one time in a million. And here's the part that belongs in this chapter: Victor ends up convinced, but a video of the whole thing convinces nobody else, because a staged run, where the two agreed beforehand which sides he'd call, makes exactly the same video. The video is a window, and the real run and the fake one cast the same shadow.

What would it take to make that promise-mark binding for real, against somebody who wants to cheat? Manuel Blum's paper "Coin Flipping by Telephone," presented in 1981, is where I'd point. Alice and Bob have just divorced, they live in different cities, and they want to flip a coin over the phone for who gets the car, with nobody both trust to flip it. If Alice flips and announces, Bob has to take her word. If Bob calls first, Alice can lie about how it landed. Blum's way out is for one side to commit first, sending something that fixes the choice without showing it, and open it after the other side calls. What keeps that binding isn't a promise. It's a bet that a certain kind of big number is hard to factor. So in Blum's protocol, binding is earned, on that condition. In this chapter it's only assumed. That's why I won't say these escapes beat a cheater. Andreas Winter, Anderson Nascimento and Hideki Imai worked out how fast a noisy channel can carry promises like that, and Claude Crépeau, Rafael Dowsley and Nascimento how that rate drops when the one promising can steer the noise.

One more relative: verifiable computation. A box hands back its answer with a short proof that the answer is right, which a checker can verify without redoing the work (Pinocchio, built in 2013 by Bryan Parno, Jon Howell, Craig Gentry and Mariana Raykova), and proof-carrying data passes proofs like that along a whole chain of parties who don't trust each other (Alessandro Chiesa and Eran Tromer, 2010). That's the strongest way out there is, and what it carries out is the claim. The history of which machine ran it, and when, stays behind unless the proof is built to carry it.

Now point all three back at a stamp kept in a machine's own state. That stamp is bits too. Hand somebody a copy of it with a note saying "it's pressed," and they're reading a window: the run that really pressed it and a box built to print that it did

cast the same shadow. So keeping the record inside the machine answers "was this earned?" for whoever can see the machine's state and trusts the step that wrote it. For anybody else, it moves the question from "did it run?" to "do I trust the box that says it ran?" Nothing in this book supplies any of the three ways out. The record here lives in one machine, with nobody's signature on it.

7.13 What gets through

Put the two boxes back under the camera. Their tops agree and their sides differ. The photograph answers questions about the tiles. To answer the stamp question, turn the camera far enough to see the side. The whole state was there all along. What changed was which part reached you.

Taking more photographs from the same spot only helps if something different eventually comes into view. When the two runs give the same picture at every step, a longer album keeps the same blind spot. That's Kalman's unobservable states over a whole run.

And none of this is the famous kind of impossibility. Henry Rice proved in 1953 that no program can sort every program by a question about what it does, unless the question gives every program the same answer. That's a deep fact about computing, and a later chapter takes it up. What this chapter shows is smaller and comes first: hand a reader one picture for two states that need two answers, and it's finished before any question of computing comes up. One input, two answers needed.

Which leaves the question I've been pushing ahead of me since the two sheets on page one. Fine: the stamp got pressed, it got paid for, and you can see it through the right window. What does a pressed stamp actually tell you? Anybody can write "CHECKED" on a whiteboard. What would make the stamp mean the list was checked? That's the next chapter.

CHAPTER 8

Established

The last chapter ended at a whiteboard. Give a worker a whiteboard and a pen, and ask for an answer with a line underneath that says “CHECKED = YES.” The worker can write both. So can somebody who copied the answer and checked nothing. The extra line tells you nothing about which one happened.

This chapter is about what it takes for that line to mean something. It’s the case page one called the sharpest, and it’s sharp because the two answers are the same bits, and everything that differs is in what was kept.

8.1 Two computers that end up the same

Put two computers on a table and hand each one the same list of a thousand numbers.

The first one sorts it. It compares, swaps, walks the list again and again until nothing is out of order. Then it does one more thing: it walks the whole list front to back and checks that every number is no bigger than the one after it. That check could have failed. It passes. Then the computer tidies up after itself, wipes its scratch work, and leaves the sorted list in memory. (That check is about order. Whether the list still holds your thousand numbers and no others is a second question, and it would take a second check.)

The second computer never looks at your list. Somebody typed the sorted version in earlier, the same thousand numbers in the same order, and it puts them in memory.

Now wait until both are done and read what’s left: the list, the working registers, where each one is in its program. Arrange for both to stop at the same point in the same program and clear their scratch work, and all of that can agree. The first compared the numbers and checked their order. The second loaded the answer and did neither. Wheel the boxes out, swap them, and bring them back. From what’s left, neither of us can tell which run happened.

Be careful about what that does and doesn’t say. It doesn’t say every atom of two real computers matches; the room, the heat, a clock or a saved log might still carry evidence. And an ordinary computer can keep a “checked” flag or a log, as long as its program leaves them alone. These two were set up so the difference got

wiped from what you can read. That choice, and what it stops the next reader from answering, is the example.

Who cares? Anybody who's been handed a sheet that says "all tests passed." Anybody who's looked at an inspection sticker on a windshield: the sticker looks the same whether somebody checked the brakes or somebody sold the sticker. An audit, a proof, a ledger, an AI that sounds very sure of itself: each one is a pile of bits that could have come out of either computer, and the question that matters is the one the bits can't answer. Did something establish this, or did it just print it? I'll come back to every one of those at the end of this chapter and say which ones this book actually helps with, and it's fewer than you'd hope.

8.2 Look harder

The obvious answer is: look at everything, not just the end. So make the computers as easy to look at as computers get. Scale them up to the size of a warehouse, every wire a lamp that's lit or dark, every switch something you can walk up to and touch. Nothing is hidden in a chip too small to see. Put them side by side and walk the floor.

While they run, you'd see the difference: one has a whole wing lighting up, comparing and swapping, and the other just copies. But they finish, and the comparing wing goes dark. Walk the floor now and look at every lamp. The sorted list is spelled out in lit and dark lamps in both, the same pattern, and everything else is dark in both, because both cleaned up.

There's no lamp anywhere that says "a check ran here and passed." It isn't hidden, and it isn't scrambled. It just isn't there. Looking harder doesn't help, because the problem was never that you couldn't see enough. Nothing was kept.

8.3 So keep the trace

Then don't let the run go. Keep the trace: every comparison, every swap, the check and its answer, written down as it happens. Wheel the computers out and back, read the traces, and you'll know which one sorted.

It's the right instinct, and it gets most of the way. But it has three problems.

First, a trace is bits, and bits can be typed in. Whoever typed the sorted list into the second computer can type a trace in too: "compared, swapped, checked, passed," a few thousand lines of it, every one of them believable. A trace a program writes is something the program produced. Keeping it doesn't answer the question. It moves it: did something establish this trace, or did it just print it?

Second, the way out of that is to take the pen away from the program. Don't let the program write its own trace. Have something underneath it write the trace, one

step at a time, as the steps actually happen, so nothing a program does can add a line that didn't happen or erase one that did. Airplanes do a version of this. Flight recorders go back to a prototype David Warren built in Melbourne in the 1950s; he was an Australian scientist at the Aeronautical Research Laboratories, working on fuels, and after a string of unexplained airliner crashes he wanted a box that would hold the instrument readings and the voices in the cockpit for whoever came looking afterwards. I know his story from museum and encyclopedia accounts, not his own report. What comes out of a recorder is evidence of what its channels carried, as long as the recorder and its sensors worked. A recorded reading can still be wrong about the airplane. The recorder has the right shape, though, and that shape isn't mine. In computer security it's called a reference monitor, and I'll come back to it.

Third, you can't keep everything. If every tick of a run can go either of two ways, then after n ticks there are 2 to the n possible histories, and a fixed number of bits eventually can't tell them all apart. Flight recorders make a practical choice here: a voice recorder loops, replacing the oldest sound when it's full. Thirty minutes, two hours, twenty-five hours: the window depends on the aircraft and the rule, and the FAA's 2026 rule stretches it to twenty-five hours for certain new aircraft. Whatever the length, it's a window.

The fingerprint is the sharpest answer to that, so take it seriously. Fold every new line of the trace into a short scramble of everything before it, and you've got a hash chain. Haber and Stornetta's timestamps work that way, from chapter four; so do the audit logs Bruce Schneier and John Kelsey designed in 1999, and the measurement registers in the security chip a lot of computers carry. That really does keep something about the whole trace in a few dozen bytes. But look at what it keeps: a binding, not the content. To read the history you still need the trace stored somewhere, and the fingerprint only tells you whether a copy somebody hands you is the one that went in. It holds only because nobody knows how to find two traces with the same fingerprint, which is a bet on how hard a search is, not a law of nature. Moni Naor and Guy Rothblum proved a sharp version of that in 2005, and even called it a fingerprint: checking a big file somebody else stores, against an opponent with unlimited computing, takes a fingerprint whose size times the number of bits you read is on the order of the whole file, and doing even slightly better against a weaker opponent needs exactly that kind of bet. And squeezing endlessly growing traces into a fixed number of bits has to land many traces on one fingerprint. That's two arrows meeting, hidden behind how hard it is to find which ones. It's a very clever way of forgetting, not a way out of it.

So the real question was never "save everything." It's: what's the smallest thing worth keeping for good, and what does keeping it take? Not every comparison the sort made. Just the one fact that a check of this very claim passed and the question

got closed. And keeping even that one fact for good takes something, on every machine: two arrows meeting, or fresh paper forever. That's chapters one and two.

So it was never true that computers can't keep records. They keep records all day. What nothing in an ordinary computer's rules does is make a record paid for, permanent, and earned. Those come in at different places. Paid for is the toll. Permanent is in the definition: a Thiele machine's settled answer never goes back. Earned, meaning a check of this very claim passed first, isn't in the definition at all. It's what this chapter adds.

8.4 More boxes to tick

Go back to the sorter. It checked the order of the list. Did it also check that none of the numbers went missing? The list can be the same whether that second check ran or not. One lamp saying "checked" won't tell you which work was done.

So put a sheet beside the scratch work, with a box for each check you care about: one for order, one for "nothing missing," and so on. When you come back you want to know which boxes got ticked.

Count what that takes. Say four checks would pass on this list. Each one can be run or skipped. That's two choices, four times over: sixteen different sheets, all sitting behind the same list. Two of them that differ on even one box can't be the same state of the machine, because asking about that box would need two answers. So the machine needs room for all sixteen behind that one list. In general, it's two to the number of checks that would pass, for every list, added up over the lists. And you can build a machine with exactly that much room and no more: keep the list and the sheet, clear the sheet whenever a new list is written, and tick a box only when its check passes.

Look at one small piece of it, because the arrows meet there too. A list that's true or false, and one check. Three states: false; true and unchecked; true and checked. The check takes "true and unchecked" to "true and checked," and leaves "true and checked" where it is. Two arrows, one landing place, right above the same list.

Now keep the old sheets when the list gets wiped and reused. Think of carbon copies filed before each sheet goes in the bin. On each of n rounds you can choose to check or not. Bring every round back to the same final list. Then ask about any round you like: was that one checked? Two histories that disagree on any round need two different states, however alike the final list looks. That's 2 to the n states for n rounds. Keep asking for one more round, with the same freedom and the right to ask about every one, and no finite amount of room is enough. That's where the endless paper comes from: not from keeping one answer, but from keeping a file of answers that never stops growing.

Ask the file a smaller question, “was anything ever checked?”, and it becomes a single box that only goes up. That’s the kind of record a Thiele machine keeps.

8.5 Computed, stored, established

Three words get used like they’re one.

Computed is about the steps. The first computer computed the sorted list. The second didn’t.

Stored is about the bits. Both stored it.

Established is neither, and that’s the whole problem. It isn’t the steps, because they’re gone. It isn’t the bits, because they’re the same. It’s this: whoever reads the first computer has a right to lean on “this list is in order,” because a test that could have failed was run on this very list, it passed, and then the question got closed. The second computer has the same sentence and no right to it. That’s a true fact about the two of them, and it’s written down nowhere, in either one.

Four words, and I’ll hold them still from here on. A check is the test, the thing that could say no. A commit is the closing: the question shut for good, about that version of that thing. The record is whatever stays behind to say so. And established, in the sense a machine can carry, is all three, about one thing.

What if the second computer, handed the typed-in list, runs the check on it? Then it has what the first had, and if it closes the question and keeps the record, it has established the list is in order. It didn’t sort and doesn’t need to. The line here is between checked and unchecked, not between worked hard and didn’t. Whatever this book counts, it isn’t effort.

8.6 The clerk who forgets everything

The picture of a computer in chapter one was a person with a pencil, a strip of paper and a short list of rules, and Turing’s 1936 paper takes it a long way. He’s careful with the clerk’s mind: whatever the clerk knows, beyond a handful of states, has to go down on paper. Then he goes all the way. He imagines the clerk getting up, walking off, forgetting everything, and coming back later. To pick up where he left off, the clerk leaves himself “a note of instructions,” and Turing says that note is the clerk’s state of mind, as far as the computing goes. Marks on paper and a note are the whole state.

The two computers are sitting right inside that. Suppose the clerk didn’t write the note. Somebody slipped it under the door, in the right form, with the right instructions. He comes back, reads it, carries on, and every mark he makes from then on is exactly what he’d have made if he’d written it himself. Turing’s model can’t tell the clerk who did the work from the clerk who was handed a note saying

he did. It was never asked to. Turing wanted something you could check without crawling into anybody's head, and a note on a desk is as checkable as it gets. That's why the paper still works ninety years later. A Turing machine could keep a note of who wrote the note, and rules about who's allowed to. This one just doesn't.

So the founding picture of computing is a man who leaves his desk, forgets everything he was doing, and trusts a sticky note. I relate to him deeply.

8.7 Information doesn't have it either

The next guess is information: one computer got some, the other didn't. That one dies fast. Claude Shannon, who turned information into a number in 1948, says on his first page that the meaning of a message is "irrelevant to the engineering problem." His measure depends on the chances of the possible messages, and the same message from the same set of chances gets the same number whichever computer it came from, or if one of them got hit by lightning and came out right by luck.

Fine, forget the set of messages and look at the one string. In the 1960s three people separately worked out how to measure the information in one string: Ray Solomonoff, who was after a theory of guessing what comes next; Andrei Kolmogorov, the great Soviet mathematician; and Gregory Chaitin. Their measure is the length of the shortest program that prints the string. The two computers hold the same string and get the same number. And running a program can't add much of that kind of information. What comes out can't hold more than what went in plus the program, give or take a little. Leonid Levin, who studied under Kolmogorov, proved a general version, and has a line for it that's too good not to repeat: you can torture an uninformed witness all you want, you won't get information about the crime.

That doesn't say sorting tells a person nothing. A person in a hurry can read the smallest number straight off the first slot of a sorted list. Short descriptions, useful access, and a record that a particular check ran are three different things.

Logic says the same thing in a different coat. The standard logic of knowing, the one Jaakko Hintikka worked out in 1962, has a property called logical omniscience: if you know some facts, you automatically know everything that follows from them. So by that logic, once you knew every number on your list, you already knew whether it was in order. Hintikka flagged the problem himself, and people have been trying to fix it ever since. The fixes read like a pile until I asked each one the same question: when it lets you write "known" next to "in order," what does it make you write in the margin? Hintikka's margin is empty. Each fix writes one thing there.

The closest one to me is Sergei Artemov's, from 1995. He wanted a logic that could name the proof itself: you only get to know a thing if you've built an actual

reason for it, and the reason travels along with the claim. That's almost a record. Almost, because nothing in his margin says the reason was ever run, or when. Others taught it to count. Ho Ngoc Duc made knowing take steps of reasoning, and Artemov and Roman Kuznets asked how big a reason can get before knowing by it stops being practical. Francisco Bavera and Eduardo Bonelli turned it into a programming language whose terms carry the history of the computation that made them, and Samuel Bucheli, Meghdad Ghari and Thomas Studer put reasons on a time line where they last and change.

Line those up against what this chapter wants in the margin: earned, paid for, permanent, and placed at a step in the run. Artemov's reason is nearly earned. The counted versions put prices on reasoning. The time line gives dates. What none of them has is the closing of a question as a single step with a price on it. Mine has two soft spots of its own, though. Artemov's margin holds the reason itself, and mine only says a check passed. And their prices grow with how hard the thing was to know; this book charges the step that closes the question, and how long the checking took never enters it. For a measure of effort, read theirs.

And yet you know something after the sort that you didn't before. You can stop worrying. Something happened, and it's the first solid clue: whatever established is, it isn't new bits. It's the same bits, settled: a question sorted into yes and no, with yes made to stay.

8.8 Statues that walk off

If it isn't in the information, maybe it's in the knowing, so I went and read the philosophers. They'd been arguing about the difference between having the right answer and knowing it for a couple of thousand years.

Plato's version is a joke about statues. There was a story that the sculptor Daedalus made statues so lifelike they'd walk off if you didn't tie them down. True beliefs, Plato says in the Meno, are like that: they wander off, and they're not worth much until you tie them down with the reason why. Once they're tied down, they're knowledge, and they stay put. Knowledge tied to a reason, and knowledge that stays.

Then in 1963 Edmund Gettier broke the tidy version in three pages. Smith and Jones are up for the same job. Smith has good reason to think Jones will get it, and he's counted the coins in Jones's pocket: ten. So Smith figures, with good reason, that the man who gets the job has ten coins in his pocket. Then Smith gets the job, and, without knowing it, Smith also has ten coins in his pocket. Smith had a belief that was true and backed by a reason, and it still wasn't knowledge. He was right by luck, because his reason was about the wrong pocket.

That’s the push that matters for a machine. Checking one pocket and closing the question about another is right by luck. So whatever a machine closes, the check behind it has to be about this very claim, and here’s exactly what that means. Four things.

It checks the same property the commit names, not a cousin of it.

It checks that property on the same thing the commit names, not the thing next to it.

Nothing has written to that thing between the check and the commit, so it’s still the same version the check looked at.

And the check is honest: it says yes when the property holds and no when it doesn’t, on anything it could ever be handed.

Break the first three and you’re Smith with his ten coins. Break the fourth and the check is a rubber stamp. Two notes on those. The fourth asks a little more than it needs. To rule out luck, a check only has to never say yes when the property is false. And the third is a very old bug in computing wearing Gettier’s coat. Security people call it time-of-check to time-of-use: check a file, use it, and something swaps the file in between. Matt Bishop and Michael Dilger wrote up how to hunt for it in 1996. Databases guard against the same thing by making sure, just before a commit, that nothing they read has been written since; H. T. Kung and John Robinson called that optimistic concurrency control in 1981. A philosopher’s pocket and a programmer’s race condition are the same bug. One limit, said now. A machine can only promise “nothing has written to that thing” about its own memory. For brake pads or a voter roll, that condition becomes an assumption about whoever measured.

Does anything fix Gettier for good? People tried for years. The first patch was “no false steps in the reasoning.” Then the philosopher Alvin Goldman came back with barns, crediting them to Carl Ginet. You’re driving through a county that, without your knowing, is full of fake barns, just painted fronts propped up to look real. You look at the one real barn and think “that’s a barn.” It’s true, you believe it, you have a good reason—your own eyes—and there’s no false step anywhere. And you were still lucky: if you’d glanced at the next one over you’d have believed the same thing and been wrong. Every patch gets a new barn. What I keep from that: whether a check was good isn’t only about this one case. It’s about what the check would have done on the cases next door.

8.9 What “could have failed” means

I keep saying the first computer’s check could have failed. The computer is deterministic. Given your list, the check was always going to pass. So in what sense could it have failed?

Here's the sense. Hand the checking step a list that's out of order, and it says no. That's all. Chance has nothing to do with it. A check splits every possible input into a pile it says yes to and a pile it says no to, and "could have failed" means the no pile isn't empty. A check that says yes to everything has an empty no pile. It tells you nothing, however long it runs. The philosopher Robert Nozick made a version of this a condition for knowing, in 1981: you know a thing only if, had it been false, you wouldn't believe it.

Notice that "could have failed" is about what the check would have done on other lists. It's not a fact about the finished state at all. Look at the first computer's memory: a sorted list. Nothing in those bits says "and whatever made me would have caught a mistake." That's exactly why the only way a machine can own this is to write something down at the moment the check passes, and have that something mean what it says. Even then the record only carries half of it. It can say a check ran on this version and passed. That the check would have said no on a bad list is a fact about the checker, and that half rests on a proof about the checker, made once, outside any run.

Rig a check to pass only on your one list. Is that cheating? No. It's an honest check of a different claim: "this is that exact list." Its no pile is huge: every other list. It establishes exactly that and nothing more. A check establishes what it's honest about.

And one thing I'll say plainly now: nothing here stops you from asking a question that can't come out no, like "is this number at least zero." Ask one, pay for the check and the close, and nothing is established but the record of asking.

8.10 Computing already had a word for it

Computing already has a word for this and uses it every day. Anyone who's used a database has seen it: COMMIT. A database does a pile of tentative work, and then there's one moment, the commit, written to a log, after which the change counts.

Jim Gray, who more or less built the theory behind that, opens his 1981 paper on it by saying the idea comes from contract law, and his example is a wedding. People negotiate, somebody asks, two people say "I do," and from then on it's binding. If it was a mistake, you don't erase it. You do another transaction to fix it, the way a bad contract gets a second contract.

The wedding was a philosopher's example first. J. L. Austin, lecturing at Harvard in 1955, said "I do" makes a marriage; it doesn't describe one, and sayings like that aren't true or false. They can misfire: the wrong person says it, a step gets skipped. A commit works the same way. Writing the commit record doesn't report that a question got closed. Writing it is the closing. It can't be false; it can misfire, by closing without the check, or about the wrong pocket. Then somebody else reads

the record, and for them it turns into a report, "this was checked and closed," and a report can be false. That's how a commit gets tied to the truth: through a check, and no further than the check reaches.

Robin Milner, building a proof checker in the 1970s, ran out of memory, so he decided to keep only the finished theorems and throw the proofs away, like a teacher rubbing out the blackboard to make room. The obvious worry is forgery: if the proofs are gone, what stops someone from just writing a theorem down? His answer was beautiful. Theorems got their own type in the programming language, and the only way to make one was through the rules of logic. You couldn't forge one because the language wouldn't let you build it any other way. He named that language ML, for Meta Language. I wanted to call my machine the meta machine. I'm choosing to treat that as a coincidence.

Bitcoin keeps a history that's expensive to replace. A node that comes back online checks the candidate chains against its rules and follows the valid one with the most work behind it. Replacing an old block means redoing the work that ties every later block to it. That's protection under assumptions about the attacker and the network, and a recent block can still be undone. And the work is tied to something scarce, electricity and hardware. A unit of cost in this book is tied to nothing. The economist Eric Budish showed in 2025 what that tie has to cost: for Bitcoin's kind of trust to hold, the honest work has to stay expensive at every moment compared with what an attacker could win. A flat charge of one unit can't do that, and isn't trying to. It's a count, not a lock. The lock is the rule that the close waits for the check, and it would work at a price of zero. The price is there because, on a finite machine, keeping the answer makes two arrows meet.

And Bitcoin wasn't first. Pull on its footnotes and three of the eight references in the paper that started it point to Stuart Haber and Scott Stornetta, who in 1991 worked out how to stamp a document with a time nobody could fake later, by chaining each stamp to the ones before. A company built on their idea, Surety, printed a short fingerprint of its chain in the classified ads of the New York Times once a week from 1995, and was still doing it in 2018, under "Notices & Lost and Found," according to a piece the journalist Daniel Oberhaus wrote that year. To make a record nobody could quietly rewrite, they put it somewhere a million people would keep a copy without meaning to.

All of those work. I lean on all of them. The next thing isn't "they're broken." It's where they sit. The database's commit is bits in a log, kept by a database program, on a machine whose own step will overwrite those bits the moment some instruction says to. Milner's guarantee holds inside programs written in his language and nowhere else. Bitcoin is a set of rules running on ordinary computers that do whatever they're told. Each one is a program or a rule sitting on top of a machine that can't tell the two computers apart.

The obvious answer is to put the rule one level lower, in the piece of the system every action has to pass through. Security people call that a reference monitor. James Anderson described it in a 1972 planning study for the US Air Force: a piece every access passes through, that can't be tampered with, and small enough to check. Fred Schneider argued that a monitor that only watches a run as it goes can enforce exactly the rules whose breaking shows up at some point you can name, and "no close without an earlier passing check of the same claim" is one of those. And Arthur Azevedo de Amorim and six coauthors built monitors that check a rule at every single instruction, and proved them correct. Putting a rule under every program, so nothing can get around it, is their move, not mine. A monitor that wouldn't let any program raise its record without paying and without the check first would be one way to build what this book describes. That's an implementation, not the thing. What I'm putting under that move is narrower: a record that has to be paid for, that has to be earned, and a reason from counting why the price of a record that never comes down can't be zero on a finite machine.

8.11 The two computers, put back together

Here's the two-computer story again, rebuilt as Thiele machines, with the record and a running count written out. The record here is established: it goes up only right after a check of the claim passes. The prices are one to check, one to raise the record, and nothing for ordinary work.

Why does the check cost anything, if effort never counts? Not for the work. The check leaves a mark saying it passed, so the raise can see it, and you've already drawn what that mark does: "true and unchecked" and "true and checked" both land on "true and checked." Two arrows meet, so Bennett's rule charges it. The raise costs one because that's the toll. And the copier's raise costs one even though it's refused, because the price is on the button, the same as the stamp pressed on YES in chapter four.

step	sorter does	record	count	copier does	record	count
start	nothing yet	no	0	nothing yet	no	0
1	sorts (ordinary steps)	no	0	copies in the typed list	no	0
2	checks "in order" on this list: passes	no	1	(no check)	no	0
3	raises the record	yes	2	tries to raise the record: refused	no	1

At the end, the lists are still the same. Everything else differs. The sorter's record is up and its count reads 2. The copier's record is down, and it paid 1 for the attempt, because an attempt costs what it costs. Nothing the copier can do raises its record without first running a passing check of the claim. And anything that lets through

only the list can't tell you which is which. Walk the warehouse now and there's finally a lamp that's lit in one and dark in the other.

How literally should you take that table? It's a picture of the condition, and that condition is the whole of what established means. What it needs from the question is a checker that says yes exactly when the answer is yes. Without one, the record is honest about whatever the checker reads, and only by convention about your list. That's Gettier's pocket again.

And a lamp for whom? The record is bits too. It's a lamp in the machine, for anybody walking that warehouse. Print it out and hand it across a desk, and you're back at the sticker on the windshield: the person reading it has to trust whoever handed it over, or have some way to check the box that says it ran. That's the last chapter's problem, and this book doesn't solve it. What changes is the question, from "did it run?" to "do I trust the box that says it ran?" That's a smaller question. It isn't no question.

So, back to the list from the start of the chapter. The windshield sticker is about whether a piece of paper is genuine, and nothing here touches it. The "all tests passed" sheet fits only if the test is a checker proved to mean what it says; a set of tests that can miss a bug establishes "this procedure said yes," which is true and smaller. An audit or a ledger kept by a program on an ordinary computer is bits the program wrote; it fits only where the step underneath keeps the record and refuses to raise it without the check. A proof is one example: checking its steps is a check that can say no, though whether its statement says what you meant is still yours to read. The AI that sounds sure of itself isn't touched at all.

Now the honest half. Take away the rule that the raise waits for the check, and keep only the bare toll from page one, and the copier has an easier move: pay one, raise the record, done. The toll is satisfied and the receipt is real, but it's a receipt for nothing. The bare toll prices the closing of the door. The check behind it goes unpriced and unrequired. That's why established is a condition you add, not something the definition hands you.

8.12 What putting it in the step buys

The hardest objection is one I'd rather make myself. You defined a mark and a charge, then showed the charge gets paid when the mark goes up. That's your definitions agreeing with each other.

For the bare machine, that's fair, and the copier's receipt for nothing is the proof. What makes the record mean established is the condition that it goes up only after a check of that very claim passed, meeting the first three of the four conditions, with the fourth, an honest checker, a fact about the checker. And the condition isn't about one program. On every run from a clean start, if the record is up, then somewhere

earlier a check of the claim passed, with nothing touching the thing between the check and the raise.

So a raised record, from a clean start, tells you a claim went through a passing check before the raise. That piece of history is what it keeps, and only that. It doesn't say the claim still holds at the end; the thing it's about can change after the raise. The record stays up about the version that was checked.

That "every run" is what a rule on every step buys. A log written by one program is only as good as that program. A rule the step itself enforces covers every run, hostile programs included. There are two limits on that "every," and I'll say them now so you don't have to find them later. It's every run, not every everything. From a clean start, one step at a time, on a step that does what its definition says. A hostile start, a soldering iron on the record's bit, or a hostile reader is outside it. And on their own, the record and the count are what people who prove programs correct call ghost state. Erase them and every program computes exactly what it did before. What isn't ghost is whatever the step reads before it lets the record rise: the check in front of it.

A computer that runs a Thiele machine, your laptop say, isn't that machine. It implements it, inside bounds, the way a laptop implements a Turing machine. Turing's tape never runs out, and a laptop's memory does; strictly, a laptop is a very big finite machine, and it acts like a Turing machine only on the computations that fit. Nobody minds, because the bound is known and most of what people compute fits. That's the same deal any device running a Thiele machine gets. The bounds aren't fine print. They're the whole difference between running a model and being one.

How much can a difference in size matter? On 4 June 1996 the first Ariane 5 rocket lifted off and broke apart about forty seconds later. The inquiry board, chaired by the mathematician Jacques-Louis Lions, traced it to one conversion in the guidance software: a 64-bit number, a value tied to horizontal speed, squeezed into a 16-bit whole number. On Ariane 4, where the software came from, that value could never get big enough to overflow, because of how that rocket flew. Ariane 5 flew differently. The value got too big, the backup unit failed on it, the main unit failed the same way a fraction of a second later, and the code doing the converting didn't even serve a purpose once the rocket was off the ground; it was kept so the software would match the old rocket's. The board's report says the assumption that made the conversion safe had been agreed on and then, without anyone meaning to, kept out of sight of any outside review. An assumption about size that holds on one machine is a fact about how that machine gets used. The code doesn't carry it, and yet it travels with the code without anybody noticing. The only defense I know is the dull one: write the assumption into every statement that needs it, where the next person will trip over it.

CHAPTER 9

Settled

The last chapter tied a record to a check: the record goes up only right after a check of that very claim passed. But I kept saying two things in one breath, the check and the closing, and they're not the same thing. This chapter pulls them apart, and then asks a question that bothered me the whole time I was writing the last one: is "established" really a switch, on or off? Or is it something with more than two places?

9.1 Settled, whether it's true or not

I ran into Charles Sanders Peirce again here, the American scientist and philosopher, in an 1877 essay called "The Fixation of Belief." He was asking what inquiry is for, and his answer named the thing I'd been running together without noticing. The point of inquiry, he says, is to get out of doubt and settle on a belief. And when someone objects that surely what people want is a true belief, he answers like a guy who's watched a lot of arguments end: "as soon as a firm belief is reached we are entirely satisfied, whether the belief be true or false."

That's funny, and then it isn't, because he's right. Settling and being right are two different events. A question can be closed on a wrong answer, and once it's closed it acts closed. Peirce cared a great deal about which ways of settling hold up; science wins, for him, because it has to answer to something outside any one person. But the settling itself, the question snapping shut, happens every time, but it never comes with a guarantee.

Karl Popper came at it from the other side. He was a philosopher from Vienna asking what separates science from everything else, and in his 1934 book, the one that came into English as The Logic of Scientific Discovery, he says every test has to stop somewhere, at some statement somebody decides to accept, and nothing in the evidence forces you to stop there instead of one step further. His picture is a jury: the verdict is a decision about what happened, and the jury deciding it doesn't make it true. Science, for him, is a building on piles driven into a swamp, and the piles stop where somebody's satisfied they'll hold. They never reach solid ground.

A real jury has a problem Popper doesn't mention, because a real jury is twelve people. Lewis Kornhauser and Lawrence Sager, two law professors, noticed in 1986 that a court that votes by majority on each part of a case, and also on the verdict, can end up contradicting itself, even when every judge is consistent. Christian List and Philip Pettit turned that into a theorem in 2002: no way of combining people's yes-or-no judgments on connected questions meets a short list of fair-looking conditions and always hands back a consistent answer for the group.

Try it with three ballots. The rule is: a payment is owed only if a contract was signed and a deadline was missed. One voter says yes to both. One says signed but not missed. One says missed but not signed. Each voter applies the rule consistently: one says owed, two say not owed. Now count the questions separately. A majority says signed. A majority says missed. And a majority says not owed. You can point at the contradiction on the paper without blaming any one voter's logic. A group's settling is a rule's settling, and which rule you pick decides which honest settling sticks. I haven't tried to put a group on the machine, and that's where I'd expect the first wall.

Popper's swamp goes further down, because it's the oldest problem there is. Ask why you believe something and you get a reason. Ask why you believe the reason and you get another one. Sextus Empiricus, a skeptic and doctor who probably lived in the second or third century, wrote down the ancient version, credited to someone called Agrippa, and in 1968 the German philosopher Hans Albert gave it a name with a joke in it: the Münchhausen trilemma, after the baron who claimed he pulled himself out of a swamp by his own hair. Every chain of reasons does one of three things. It goes on forever. It loops back on itself. Or it stops somewhere, because somebody decided it stops.

It's the same shape as chapter one's step law, with reasons in place of states. Follow a chain: it keeps finding new reasons, or it comes back to one it already used, or it stops. There's no fourth thing. So "when am I done checking?" doesn't get a clean answer from anybody. What you can do is be honest about where you stopped and why, and that's a lot of what this book is: every place it stops is written down, with its name on it, so you can decide whether you'd have stopped there too.

Lay Peirce and Popper on top of each other and what comes through is that the check is one act and the stopping is another. The check can run as long as you like and it never makes you stop. At some point something decides, and the deciding is its own event. That's the copier's whole trick from the last chapter: it skipped the check and kept the answer.

9.2 A proof is an act

Then a logician said the part I didn't expect a logician to say. Per Martin-Löf, a Swedish logician, in lectures he gave in Siena in 1983: "a proof is, not an object, but an act ... Only secondarily, and irrevocably, does it become the act that has been performed." That took me a few reads, and the tenses opened it up: first the act being done, then, for good, the act that has been done.

You could say that's true of everything that ever happened. It's the two computers again: the sorter's check happened, for good, and the machine right now carries no trace of it. The past can't be undone. The record of it can be lost, unless something keeps one. That gap is where this book lives.

Then I found the people who had already built a record like that, on paper, and it came clear only when I sorted them by three words: the act, the thing it produces, and the trace the act leaves behind.

The Dutch mathematician L. E. J. Brouwer held that a mathematical truth is something a mind makes evident, and from 1948 on he argued about an idealized mathematician working in stages, the creating subject. The logician Georg Kreisel, in 1967, wrote it as a formula: the subject has evidence for a statement at stage n. That's a trace, something you can come back and ask about after the act is over. Anne Troelstra, a Dutch logician who studied under Brouwer's student Arend Heyting, added in 1969 that the subject never forgets what it has made evident, so the trace never fades. That's how Mark van Atten credits it in his history of the idea, down to the page of Troelstra's book where it was added, page 95, and he suggests calling the whole thing the Kreisel-Troelstra theory. The Stanford Encyclopedia's entry lists the never-forgetting rule among the three axioms Kreisel gave in 1967. Walter Dean and Hidenori Kurokawa's list of Kreisel's three, in a paper on his methods, doesn't have it. That's two sources for Troelstra and one for Kreisel, so I still haven't settled it. Van Atten reads the formula as "by stage n the creating subject has made it evident." Look at the tense: has made. That's Martin-Löf's "the act that has been performed," reached from the other end. Van Atten even says the subject can't just accept a proof handed to it from outside; it has to go through the steps and rebuild the evidence itself, which is the copier who runs the check. And Göran Sundholm, a philosopher of logic at Leiden, has long kept the proof as an act apart from the proof as a thing, and has worked through how Brouwer's creating subject sits beside the idea that every procedure is a program.

Set their trace next to the record this book wants. Theirs happens at a stage; mine goes up at a definite step in the run. Theirs never forgets; mine never goes back. Theirs is never wrong; mine is earned. Nearly every column fills. The one that stays empty is the price, and I think I see why. An idealized mind with unlimited memory pays for never forgetting with fresh paper, and an idealized mind has endless fresh

paper, so nobody charges it. On a finite machine the fresh paper runs out, and keeping the record makes two arrows meet. That's chapter two. So theirs is the closest thing to established I've found, and it's been around for sixty years.

Come at mine from their side and it bends in two places. Van Atten's subject won't take a proof from outside, and my machine takes exactly one: its checker is trusted on a proof about the checker, done once, outside every run. And their evidence is never wrong because, for Brouwer, made evident is what true means. Mine is right only as far as the checker is honest.

9.3 Not a switch

So far I've talked about established as if it were a light switch: on or off, the sorter on and the copier off. That bothered me the whole time I was writing it, and for a while I couldn't say why. It felt too small. I don't trust that feeling on its own, because I've had it plenty of times about things that turned out to be nothing. So I went looking for something outside my own head that would either give the feeling something to stand on or show me I was imagining it.

I didn't know what I was looking for at first, beyond a rough shape: two things that look exactly the same and differ in how far something about them has been settled. So I started collecting anything that seemed to have that shape. Most of it fell apart as soon as I looked closely, and how each one fell apart taught me more than the thing itself did.

The first batch was all hidden things. A pea under one of three cups. A tree, whose age you can't see from outside. A forged painting that looks right until somebody tests the paint. A glass of gas, where the thermometer gives one reading and underneath there are endlessly many arrangements of molecules that would all give it. Plato's prisoners. Every one looked promising for about a minute. Then I noticed they all fail the same way: look harder and the hidden thing shows up. Lift the cup. Cut the trunk and count the rings. Test the paint. Turn around in the cave. The hidden part is a fact about the object that better eyes could find. That isn't what the two computers have. You can see every lamp in the warehouse and not find the check.

That gave me a crude test. Give someone perfect vision of the thing in front of them, as much detail as they want. Could they read the difference off it? If yes, it's just hidden. If no, it might be what I'm after. The test is a judgment, and I'm the one making it, so it's a way of sorting my own examples, not a verdict.

The second batch passed that test and failed differently. A clock face reads three o'clock and doesn't show how many days have gone by. And a computer can keep a complete history of what it did: that's Bennett's machine from chapter two, which writes down every step, copies out its answer, then runs itself backwards, wiping

the history as it goes, until it holds only the question and the answer, with no two arrows ever meeting. The history it kept along the way was right out in the open. But it could be run back down. If the thing I was after could be put up and taken down with nothing paid either way, it would just be bookkeeping. So I needed something that only moves one way, and costs something to move.

The third batch came closer. In 1782 James Six, a Canterbury man from a family of silk weavers who had taken up watching the weather, described a thermometer for the Royal Society. He wanted the coldest moment of every night, and he'd found that with an ordinary thermometer you had to be looking when it happened, because, in his words, "if it be not immediately noticed, it can never after be known." His fix was two little steel markers in the tube that the column pushes along and that stay where they're left. A record that only moves one way. Then he spoils it, for my purposes, in the same paper: every evening he wrote the numbers down and slid the markers back with a magnet, "without a touch."

Since 1996, starting with polio vaccine, vials bought through UNICEF have carried a little sticker, a square inside a circle. The square darkens as the vial takes on heat, but it doesn't lighten again when the vial cools. When the square matches the circle, don't use the vaccine. It only moves one way, and it can't be reset. But it's chemistry: the reaction in the square is chosen to run about as fast as the vaccine itself spoils. Nobody ran a check. The heat did.

Closer still: the international vocabulary that measurement labs work from says a result is traceable only if it can be related to a reference "through a documented unbroken chain of calibrations." Standing earned by checks, step by step. The same document adds that traceability "does not ensure . . . that there is an absence of mistakes," which is "settled isn't the same as true," written into an international standard. And the top of the chain wasn't fixed either. When the official copies of the kilogram were compared with the original in 1889, 1946 and around 1991, they had drifted apart by tens of micrograms over the century, and nobody can say which one moved, because the original was the kilogram. In 2019 the kilogram was redefined by fixing a constant of nature instead.

The rules of chess have the shape too. The laws of chess say two positions are the same only if the same player has the move, the same pieces are on the same squares, and the possible moves of all the pieces are the same. They spell out that a king that could still castle and one that can't are in different positions, even on an identical board. That passes the perfect-vision test. And the rules say why: the castling right changes which moves are legal. The hidden part steers the game.

Each of those got something right and missed something else.

9.4 How a statement gets established

The one that fit best, among the ones I found, was the standing of a mathematical statement. Here's how it went for one statement, as far as I could follow it.

In 1611 Johannes Kepler, the emperor's mathematician in Prague, owed his friend and patron Johann Matthäus Wackher von Wackenfels a New Year's present and, he said, had nothing to give. Wackher, Kepler wrote, loved nothing, for the games you can play with the word, so Kepler wrote him a little pamphlet about the nearest thing to nothing he could find, a snowflake, which had landed on his coat as he crossed a bridge. Why six corners? Kepler couldn't say in the end. But along the way he stacked equal balls in his head and said that one way of stacking them, the way grocers stack oranges, packs them so tightly that no other arrangement could fit more into the same box. He didn't prove it. He said it.

Then the sentence sat. In 1831 Carl Friedrich Gauss proved a special case. In 1900 David Hilbert put the general question on his famous list of problems for the new century, without mentioning Kepler. In 1953 the Hungarian geometer László Fejes Tóth showed how the whole thing might be cut down to a problem with finitely many numbers in it, and guessed that computers might one day finish it. Thomas Hales started on it in earnest in 1994, and in August 1998 he and his student Samuel Ferguson announced a proof that leaned on a great deal of computer work. The journal gave it to a panel of twelve referees. They ran seminars on it for years. A journalist, George Szpiro, later reported the editor's letter as Hales described it: the referees had "not been able to certify the correctness of the proof, and will not be able to certify it in the future, because they have run out of energy." The journal printed the human part in 2005 and the whole proof in 2006. Hales started a project in 2003 to check every single step by machine. It finished on 10 August 2014.

Lay those dates out on the table and look at what moved. The sentence never changed. Whether it's true never changed, as far as anyone can tell. What changed was how far it had been established, and it changed in stages: a guess, a special case, a place on a list, a plan, an announcement, a panel that ran out of energy, a printing, a machine check.

Then I tried to break that one too, because by now I'd watched a lot of good examples fall over, and there's an obvious place to push. Standing can fall.

In 1852 a student named Francis Guthrie, coloring a map of the counties of England, noticed that four colors always seemed to be enough to keep neighboring counties different. The question went to his old teacher Augustus De Morgan, who wrote to the Irish mathematician William Rowan Hamilton that a student had asked him for "a reason for a fact which I did not know was a fact," and that he didn't know it yet either. In 1879 Alfred Kempe, a lawyer who did mathematics on the side and sang counter-tenor, published a proof, and his description of finding it is

wonderful: “After a somewhat arduous search, I have succeeded, suddenly, as might be expected, in hitting upon the weak point.” The journal’s associate editor, William Story, added a note saying Kempe had “substantially proved” it. Eleven years later Percy Heawood, a lecturer at Durham who by one account brought his dog to his lectures, showed that one step of “the now apparently recognised proof” doesn’t work.

So standing can fall. And then the part I didn’t expect. The four-color statement was true all along. Kenneth Appel and Wolfgang Haken proved it in 1976 with a computer, and the proof was met with suspicion for years; a cleaner version came in the 1990s; and in 2005 Georges Gonthier finished a proof checked step by step by a small proof-checking program. So one statement stayed true the entire time while its standing went up in 1879, came down in 1890, went up again in 1976 against a lot of doubt, and went up further in 2005. The fall wasn’t a fall of the truth. It was a fall of how far the truth had been established, and it was the right fall, because what had been established was less than anyone thought.

Laying all of that out together is what changed my picture. If there are stages, then at each one there’s a line, with two sides: not this far yet, and at least this far. Each line is its own yes-or-no question. But pick the question carefully. “Do I still trust this proof?” can turn back to no, the way Kempe’s did. “Did I carry out this particular check?” is about something that happened. Cross out the trust and the check still happened. To keep that answer, the machine has to keep a record of it and have no move that knocks the record down.

So maybe the switch wasn’t wrong. Maybe it was just the first line, and the real shape is a direction with lines drawn across it, which is the row of cups from chapter five. I’ve gone around many times on whether the lines make the direction or the direction comes first and the lines are just where you cut it. I don’t have an answer. The part I could actually settle is the part chapter five did: one switch isn’t enough once there are three or more stages.

9.5 What settling looks like

Here’s the picture, put together.

Draw every state the machine could be in as a field of dots. A yes-or-no question about the state splits the field in two: yes and no. That’s a sorting, the smallest one there is. If one side is empty, there’s no sorting at all, just a field, which is the empty no pile from the last chapter. Now the extra thing that makes it settling: once the machine is on the yes side, no step it can take leads back out. The door only swings in. A settled question is a sorting plus a door that only swings one way.

At its best, what goes through the door is a relation: these numbers are in this order, these two things belong together. But the door doesn’t care. Hook it to a

check that says yes to everything and the machine walks through even when the claim is false. The question is settled either way, true or false, like Peirce said. A door that only swings one way tells you a question was closed. It doesn't tell you it was closed right. Price and correctness are two different books, and I keep both, separately.

And one more edge, the one Gray's wedding already hinted at. What stays shut for good is that the question got closed, about that version of that thing. Whether the claim is still true is another matter. Change the list after the commit and the record still says, truthfully, that the old version was checked and closed. If the claim turns out bad, the fix is another act on top, a second record that says "withdrawn." The first one never quietly comes back down. So "I can lean on it right now" is three things at once: closed, not withdrawn since, and the thing unchanged.

CHAPTER 10

What structure is

The chapters so far were about one thing settling: a coin going into YES, a cup turning up, a box getting ticked. This chapter goes after the bigger word behind all of it, and I'll give it to you now, so you can decide whether to stay: structure.

I used that word for most of my life without once looking at it, which I suspect is true of most people and most words, and it's fine, right up until you do something stupid. The stupid thing I did was build a machine that charges for keeping something. You can arrange things however you like, and the machine lets you put any price on the arranging, including none. But the moment an arrangement has to stay, the moment you ask the machine to keep it, or to vouch for it, to put its own name on the claim that these things belong together, there's a charge, and it isn't zero. Chapters two and three showed why. What was in front of me, standing there with the thing built, was a model that charges for keeping structure, and I could not have told you what structure was. It didn't care. It can't nod along the way a person does. It needed the word to hold still, and the word had never held still for anyone, least of all me.

So I went looking at the people who'd wrestled it before me. Some of them have been dead two thousand years and are still, I'd bet money, better at this than I am. I used everything I had: language models, more than one, books, libraries, my own gut, anything in reach. I don't know if I found enough. I never do. There's always another shelf behind the one I was reading. And being dead and famous is not the same as being right, so I pushed on each of them, and what follows is what I could reach, pushed on as hard as I know how. The one thing I'm asking is that you push on it with me, out loud, at your own kitchen table, because the first time through I was the only one checking, and that's not enough people.

The order I ran into them was a mess. Category theory came first, because I'd built one by feel before I knew it had a name, and everything else came in whatever order the next question dragged it in. I'm going to put them in the order where each one leans on the last, which is mostly the order they happened. And none of it is an oracle. History is where I start digging. It gets retold, smoothed over, and now and

then made up outright (there's a drowning coming up that nobody can place), so every piece of it is a lead to check, and I check the parts I lean on.

10.1 A string, a square, and a drowning nobody can place

Start with the oldest one, and start by doing it. Take a string. A guitar string, a rubber band over an open box, anything that twangs. Pluck it. Keep its tension as nearly unchanged as you can when you stop it at a new point; stretching it tighter changes the pitch for a second reason. Now stop it exactly at the middle and pluck half. With the tension held steady, that's the same note an octave up, the one that sounds like the same note sung higher. Press at a third of the way along and pluck the long side: that's the fifth, one of the handful of intervals a bugle can play at all, which is why you hear it all through bugle calls. Half, two thirds: small whole numbers, and your ear says yes before your brain does. About twenty-five hundred years ago, in the south of Italy, the followers of a man named Pythagoras are credited with finding exactly that, and it's as true in your kitchen tonight as it was on their hillside.

I'll tell you what they did with it, as best anyone can tell. Nothing survives that's surely from Pythagoras's own hand. The earliest fragments are from a later member, Philolaus, and scholars argue over those. Mostly what's left is Aristotle, one to two centuries later, writing in his Metaphysics about them the way you'd write about the neighbors' religion, and he says they took the elements of numbers to be the elements of all things: everything is number. That doesn't mean everything can be counted; it means the parts of numbers are the parts of all things.

Stop and ask what I asked. What does that even mean? Is a chair a number? Does a chair have a number tucked inside it somewhere? Aristotle doesn't say, and he's reporting a belief he doesn't hold. So don't take it on his word, or theirs, or mine. But before you test it, know what it was holding up, because you can't test a load-bearing wall without knowing what sits on it. These weren't people doing math for fun on weekends. They were a brotherhood, with rules about what you could eat and when you had to keep quiet, and that plucked string was their scripture. Halve the string, hear the octave: to them that was proof that under the mess of the world there were whole numbers in clean ratios, that the cosmos was a chord and they'd heard it. "Everything is number" wasn't a slogan; it was what they'd organized their lives around. And that's exactly the kind of sentence you should be suspicious of, mine included.

Now push, and push where they loved most: the square. Draw one, with a side one unit long, any unit you like, and draw the line from one corner to the opposite corner. Call that the diagonal. If everything is number, then the diagonal is some whole number of parts long when the side is some whole number of parts, so the

diagonal divided by the side is a fraction, a whole number over a whole number. Cancel anything the top and bottom share, the way four-sixths becomes two-thirds, until they have no common factor left. I'll call the top p and the bottom q, and the letters are purely so I don't have to keep writing "the top" and "the bottom."

You need one fact about squares, and it's the one Pythagoras is famous for. The square built on the diagonal has exactly twice the area of the square built on the side. If you want to see it for yourself, cut out two identical squares, cut the second one along both diagonals into four triangles, and lay one triangle against each side of the first, its long edge on that side. The outline is the square on the first one's diagonal, and it's made of exactly two of the small squares; it's a five-minute job with scissors. So p times p is twice q times q. People write "times itself" as a small raised two, so that's $p^2 = 2q^2$, and that is the only piece of notation in this argument.

Twice anything is even. So p times p is even. An odd number times itself is always odd. Draw that many rows of that many dots, pair off the columns, then pair the dots in the one column left over, and one dot remains. The same pairing works for any odd side, so if p were odd, p times p would be odd too. It isn't, so p itself must be even. It's two times something, and I'll call that something r. Put it back in. Two r times two r is four times r times r, and that equals twice q times q. Halve both sides: two times r times r equals q times q. So q times q is even, so q is even too, by the same step as before. But you already canceled everything the top and bottom had in common, so they can't both be even. The argument has eaten itself, and when an argument eats itself the thing you supposed at the start is false. There is no such fraction. The diagonal of a square can't be named by any ratio of whole numbers, ever. Aristotle reaches for this very argument in his Prior Analytics as his example of proof by contradiction (suppose the diagonal can be measured, and odd numbers come out equal to even ones), and you just did it with a pencil.

There's a story that a member of the brotherhood named Hippasus let this secret out, and his brothers took him to sea and drowned him for it. I went and checked, because it's a good story and good stories are the ones to check. The oldest drownings I could find are in Iamblichus, writing around AD 300, seven or eight hundred years after Hippasus. In Iamblichus the man drowned by name is Hippasus, and what he gave away was how to fit twelve pentagons together into a ball, the dodecahedron. A few pages on, Iamblichus tells it again about whoever first let out the incommensurable, and that time he names nobody. Carl Huffman, who wrote the Stanford Encyclopedia of Philosophy's entry on Pythagoreanism, follows the historian Walter Burkert there and finds no ancient writer who connects Hippasus to this discovery at all. So the version with Hippasus drowned for the diagonal looks like two old stories stitched together later. It's a legend, but notice that the legend exists. Somebody, long after, felt that a man should have been drowned for

this. That's how much was riding on it. To whoever stitched the story together, the diagonal wasn't a math problem; it was the end of the world.

So the claim is dead, and you killed it. I only handed you the pencil. Now the question that matters: what's left? Look at your square again. The diagonal is still there. It still stands in a perfectly definite relation to the side; you drew it, and nobody drowned. That relation was real before anyone tried to express it as a fraction, and it stayed real after that attempt failed. The relation came first. The number, when there is one, is a name for the relation, and sometimes there isn't a name. I'm going to keep that, and I want you to keep it too, because you're going to watch the same move get made again and again as this book goes on: somebody says "everything is X," and it's true enough to do real work with, and it sounds right when you pluck it, and then somebody draws the diagonal. Whether the thing I built survives its own diagonal is a real question, and a later chapter answers it.

10.2 Shapes poured into nothing

About a century and a half after the brotherhood, Plato, in a dialogue called the Timaeus, tried to build the whole physical world out of geometry. Fire, air, water and earth were four regular solids (the pyramid, the eight-faced one, the twenty-faced one, and the cube), and the faces of those solids were built from exactly two kinds of right triangle. You can make one of each in a minute: cut a square from corner to corner for the first, and cut an equilateral triangle from a corner to the middle of the opposite side for the second. Everything else, in Plato's story, was a receptacle, a featureless medium that takes on shape the way soft stuff takes a stamp pressed into it. Structure, for Plato, is shape laid into something that has no shape of its own.

Why did he care? He wanted a world made on purpose, by a craftsman working from a plan, and triangles were the plan. Is it true? No, but not for the cheap reason. You can't refute a claim about what things are made of at the smallest scale by picking one up; the smallest scale is exactly where your hands don't reach, and Plato knew that. What sank it is that it made no prediction anyone could check, and the people who eventually did reach that scale, two thousand years on, found atoms, and atoms aren't triangles. The idea was a guess with no way to lose, until it lost. But test it and two things survive. One: structure is something laid in, and the stuff it's laid into doesn't carry it by itself. That's a real idea, and I'd ask you to keep it loosely for now; when you meet a record that's established, you can ask whether the state it holds is the wax or the seal. Two: Plato put a crack in his own forms himself, in another dialogue, the Parmenides, with an objection that Aristotle later nicknamed the third man, and it's the most useful mistake in this section. Take all the large things. Say they share one thing, Largeness. But Largeness is itself large, so now there's a further thing that all the large things and Largeness share, and that

one is large too, and it never stops. The extra move is treating Largeness itself as one more large thing. Grant that again at each stage and it never stops; refuse it and this particular regress never starts. The lesson, and I think it's right though Plato never spells it out, is that the structure of a collection is not one more member of the collection. The plan isn't a brick. Keep that one close.

10.3 Prove it from what you and I both accept

Around 300 BC, in Alexandria, Euclid wrote the Elements, and the thing to take from it is not the geometry. It's the method, and the method is the whole reason I trust anything in this book, so give me a paragraph on it. He starts with definitions ("a point is that which has no part"), five postulates (things he asks you to grant, like "to draw a straight line from any point to any point"), and five common notions (things nobody argues with, like "things which are equal to the same thing are also equal to one another"). Then he derives everything else, four hundred and sixty-five propositions, from only those, in order, each meant to use nothing but what came before (he sometimes leans on things he never stated, and later geometers filled those in). He never says "trust me" or "it's well known"; he says, in effect, here are the rules you and I both accept, now watch. If you've ever been in an argument that ended with "well, that's just your opinion," Euclid is the man who found a way out of that room, and I'm not embarrassed to say I learned the way out from someone dead for two thousand three hundred years. Four hundred and sixty-five times he does the same trick, and the trick is everything.

And then there's the one rule Euclid seems almost embarrassed by, and it's a hole worth falling into. His fifth postulate, the one about parallel lines, is long and awkward next to the others, and for two thousand years people tried to prove it from the rest so it wouldn't have to be assumed. Proclus, the fifth-century philosopher who wrote a commentary on Euclid's first book, tried. Mathematicians in Persia and the Arab world tried. In 1733 an Italian priest named Girolamo Saccheri spent a whole book assuming it was false so he could watch the contradiction arrive, and he got pages deep into what is now a perfectly good geometry before declaring that it was repugnant and stopping. A century later three people, separately, took the step he wouldn't: drop the postulate, and what you get is other geometries that hold up just as well, and nonsense never arrives. Farkas Bolyai begged his son János to stay away from the problem, warning that it had swallowed years of his own life; the son went in anyway and came out with one of those geometries. The part I keep is that they found something real by refusing to believe one of the rules everyone accepted, and checking what happened without it. Refusing a rule and checking what happens without it is the push, and this one went on for two thousand years.

What is structure, to Euclid? Whatever the starting rules force. And when are two shapes the same? You can do this one yourself: cut out two paper triangles and lay one on the other. If they coincide, they're equal. That's his fourth common notion, and it's how he proves his fourth proposition, by picking up one triangle and setting it down on another. Sameness is coinciding under a move. Keep that; it's about to get very general, and I didn't see that coming either.

10.4 The statue and the bronze

Aristotle, Plato's student, pulled the forms down out of heaven and put them inside things, which was a bit rude to his teacher and also, I think, correct. His example, from the Metaphysics: a bronze statue is bronze plus a shape. And a circle can be made in bronze or stone or wood, so, he says, the bronze is no part of what it is to be a circle. The same shape in marble makes a different statue with the same form. The bronze is the matter and the shape is the form, and the form is what the thing is.

Test it with two things from the kitchen. A cup can be ceramic or metal and still hold your coffee. A sandwich carved in wood can keep the outline and fail to be lunch. The two swaps preserve different things. If the question is shape, the carving may pass; if the question is whether you can eat it, it does not.

Aristotle got there first, and I want to be fair to him. In Book VII, chapter 11, he warns against treating a living animal the way you'd treat the circle whose bronze can be swapped out: an animal's parts have to be in shape to do their jobs. So the sandwich doesn't catch him out. It just puts his own point on the table, carved in wood, where you can see it.

That is the bet I need for the record. Change the material, then check the operations and the answers again. If they still agree, the record has survived the change in the respects I promised. A later chapter states those promises over a chosen base. Neither theorem says that every material can do every job, or that the price in one material is the price in another.

10.5 Where things are, without numbers

In 1679 Gottfried Leibniz, who co-invented calculus and had opinions about everything, wrote to the physicist Christiaan Huygens asking for something that didn't exist: a way to reason about where things are directly, the way algebra reasons about how much, without ever laying down coordinates. He called it the analysis of situation. Try it yourself. Describe where the furniture is in the room you're sitting in without using a single number. "The couch is against the wall. The table is between the couch and the window. The lamp is next to the chair." You just did it, and what

you used was relations between places and nothing else. Huygens wrote back that November and was not impressed. It took roughly two hundred years for Leibniz's idea to become a field, and the field is called topology, and Huygens is remembered for other things.

What survives the push is the same thing that survived the square, coming at you from the other side. Position is a relation among places. The numbers you lay on top are a convenience, and they're a convenience you can take away and still have the room. I'll be honest about why this one landed for me: the thing I built keeps track of which things belong together, and I spent a long time thinking that was a list of numbers before I understood it was a relation that the numbers were merely naming. Leibniz had that in 1679. Huygens told him no. I'd like to think I'd have sided with Leibniz. I probably would have sided with the famous guy.

10.6 Naming a thing by saying when two are the same

In 1884, in a short book called The Foundations of Arithmetic, Gottlob Frege noticed something you can check in your head right now. "Line a is parallel to line b" and "the direction of a is the direction of b" say exactly the same thing. There is no fact in one that isn't in the other. But the second sentence has a new object in it, a direction, that the first one doesn't. Where did it come from? From the sameness. Once you can say when two things count as the same, you've earned the right to name the thing they share. Frege used exactly this move to say what a number is: the number of things in a pile is whatever every pile you can match up with it, one to one, has in common. The residue has teeth: a sameness relation lets you name a new object.

Now push, because Frege pushed too hard, and it cost him. In his formal system, Basic Law V treated value-ranges as objects and called two of them the same exactly when their functions agreed everywhere. Put that next to his unrestricted comprehension, the rule that any property you can state picks out a collection, and the self-membership construction walks right in. The set version is the one everybody tells, and it's easier: In June of 1902, a young British philosopher and logician, Bertrand Russell, wrote him a short letter. Consider the collection of all collections that are not members of themselves. Is it a member of itself? If it is, then by its own rule it isn't. If it isn't, then it qualifies, so it is. Do it in your head; it flips every time you look at it, like one of those drawings that's a vase and then two faces and then a vase. Frege's floor had a hole in it, and he knew it the day he opened the letter, and his second volume was already at the printer. One repair was Zermelo's 1908 axiom of separation: a property picks out a subset of a set you were already handed. That wasn't his only way to get a set; power set and infinity were in there too. What separation took away was the right to sort "everything." You sort a pile.

I'll tell you now that this rule shows up again, later, in a place where I had to obey it or fall through the same hole, and I'll point at it when the book gets there.

Frege's reply is worth hearing, because it's the best example I know of how to lose. He wrote back within days. He didn't hide it or argue around it. He said that the basis he'd meant to build arithmetic on had been shaken, and in an appendix to the volume that was about to come out he wrote that hardly anything worse can happen to a scientific writer than having a foundation give way right after the work is finished. Then he tried to fix it. If I'm wrong about something in this book, and I'd bet I am somewhere, that's how I'd like to take it.

10.7 Choose the changes, keep what survives

By 1872 there were too many geometries. Euclid's, and ones where parallel lines meet, and ones with no such thing as distance, and nobody could say what made them all geometry and kept them from being a pile of unrelated hobbies. Felix Klein, twenty-three years old and taking up a professorship at Erlangen, published an answer in a program paper that people still call the Erlangen program. Pick a set of allowed changes: rotations, stretches, whatever you decide. Then a geometry is the study of the properties that stay the same under those changes. Pick a different set of changes and you get a different geometry on the very same page. Euclid's, in Klein's reading, is the one where you may slide, turn and flip a shape and scale it evenly, and stretching it more one way than another is off limits. And I'd underline, if this were my book, which it is, that in Klein's scheme any group of changes can define a geometry. The choice is yours. Nobody tells you which changes to allow.

You can do this one at your desk. Take any drawing and ask what's still true after you turn it upside down (lengths, angles), after you stretch it (lengths are gone by then; which lines cross which is still there), after you bend and crumple the sheet without tearing it (which drawn paths remain connected survives). Two distant ink marks may touch across a fold; that new physical contact isn't a new connection in the drawing. You have to say which relation you're following. The three changes give three answers and three structures, all from one drawing. The residue is that structure is what survives a chosen set of changes, and "chosen" is doing real work in that sentence. There's no structure of a thing from nowhere. You have to say what you're allowed to do to it first. I didn't know that when I started, and I'd been saying "the structure of" things for years.

Klein's move has a descendant in logic, and it goes all the way. In a lecture in London in 1966, published only in 1986, three years after his death, the Polish-American logician Alfred Tarski asked what makes a notion logical, and answered with Klein's recipe at its far end: allow every one-to-one transformation of the world onto itself, and call logical the notions that survive all of them. The more changes

you allow, the fewer notions are left standing, and with every change allowed what's left is logic. Tarski's answer is the sentence above, "structure is what survives a chosen set of changes," with the choice made as large as it can be.

10.8 An invariance comes with a conserved quantity

In 1918 Emmy Noether, who at the time wasn't allowed to qualify for a university teaching post because she was a woman, and for a while taught courses advertised under David Hilbert's name, proved something that physicists still treat as close to holy. Take a physical system whose laws can be written the way most of physics is written, as "of all the paths the system could take, it takes one that makes a certain total stationary" (the total is called the action, and stationary means a small change to the path doesn't change the total to first order; the path needn't make it smallest, and that's all I'll say about it here). For such a system, every continuous symmetry that leaves the action unchanged comes with a quantity that is conserved. If the action doesn't change when you shift everything in space, momentum is conserved. If it doesn't change when you shift everything in time, energy is. A symmetry isn't decoration. It has a consequence downstream that you can measure with instruments. I can't hand you this one to do at the table; the proof needs calculus I'd have to build first and I'd lose you and probably me, and the "written as an action" condition is doing real work that I'm not going to unpack here. What I can hand you is the shape. Structure, once it's really there and the setting is the right kind, has consequences you can measure. That's the sentence that made me think a machine could carry a cost for structure in the first place. Noether makes it reasonable to ask. She doesn't make it true, and nothing I build in this book is a system with an action.

10.9 All of mathematics as structures

Starting in the 1930s, a group of young French mathematicians began publishing a vast treatise under one made-up name, Nicolas Bourbaki, as if he were a single person. He wasn't. In 1948 he also applied for membership in the American Mathematical Society, and the application was quietly turned down, which I find comforting. (The historian Michael Barany has gone through that application, a second one a year later, and the letters they set off.) Their goal was to rebuild all of mathematics on one idea: a structure is a set of elements "whose nature has not been specified," together with one or more relations on them that satisfy stated axioms. They believed three kinds of structure were the mothers of all the rest: algebraic, order, and topological.

Now test it. The historian Leo Corry went through the entire treatise looking for the place where this formal definition of structure actually does any work, and what he found is that it "plays no actual role in the presentation of theories within the treatise." Outside the chapter that defines it, it does work almost nowhere. And I found no argument in their writing for why the mother structures number three. And the chapter on category theory, which they promised their readers, never appeared; Corry is where I found that too, and Ralf Krömer's 2006 history of Bourbaki and categories in the 1950s follows the attempt. That doesn't mean Bourbaki left maps out. Their Theory of Sets has sections on morphisms, derived structures and universal mappings (Chapter IV, sections 2 and 3). Corry's complaint is about how much of that general apparatus ever got used. The lesson I can take for my own work is smaller. Naming a structure isn't enough; you have to show which constructions and comparisons it supports. I learned that one the slow way, and the afterword tells it.

Philosophers of mathematics took the same idea further and gave it a name, structuralism. In 1965 the philosopher Paul Benacerraf pointed out that the whole numbers can be built out of sets in two equally good ways that disagree about which set the number three is, and concluded that numbers aren't objects at all. What arithmetic is about is the pattern, and anything standing in the right relations to everything else can play the part of three. The philosophers Michael Resnik (Mathematics as a Science of Patterns) and Stewart Shapiro (Philosophy of Mathematics: Structure and Ontology) made the patterns themselves what mathematics is about. The philosopher Geoffrey Hellman (Mathematics without Numbers) kept the talk of patterns and did without the patterns, reading each statement as a claim about what would hold in any system with the right structure, if there were one. The logician Steve Awodey argued that category theory says it best, a structure being what its maps keep, which is where this section goes next. And the philosopher of science James Ladyman carried the idea into physics: what a successful theory gets right about the world is its structure. This section's thesis, that structure is the relations and the things in them don't matter, is theirs. None of the formal results takes their arguments as a premise, and the sources at the back list them under this chapter, beside Bourbaki.

10.10 A set with some named relations

What mathematical logic eventually settled on, with the logician Alfred Tarski giving the field its name, "the theory of models," in 1954, is almost boring, and the boring ones are the ones that last. A structure is a pile of things plus whatever relations a chosen vocabulary names on it. Pick the vocabulary (Klein again: you choose), and a structure is anything that gives those names a meaning.

Now I want to say "the smallest one," and "smallest" means nothing until I say what I'm measuring. So here's my ruler: count the names first; when that ties, count how many things each name looks at at once; when that ties too, count how many answers each name can give. The bottom of that ruler, for anything that sorts at all, is one name, looking at one thing at a time, with two possible answers: one yes-or-no question, put to each thing in the pile.

A relation like "these two go together" can sort a pile too, but it looks at two things at once. A label for each thing can sort a pile, but it can give more than two answers. So the ruler puts both above one question. Two limits, so you don't have to find them. A single named thing, "this one, here," sits even lower on the ruler, but by itself it doesn't sort anything; ask "is it this one?" and you've got a split in two again, one question whose yes is a single thing. And a question only sorts when some things say yes and some say no. If they all agree, there's one group and nothing got sorted.

With those said: by this ruler, the smallest structure that sorts is one question. A different ruler could say something different. This is the one I chose, and I'm telling you so.

And notice one more thing about it. A yes-or-no question carries one bit more than a plain split into two groups: which group gets called yes. That bit is a direction across the split, which way counts as forward. A door that only swings that way is one more thing again: a rule that the machine's moves never take anything from yes back to no. The label gives the door its direction; the rule is what stops it swinging back. Keep both. They're about to turn into the whole chapter.

10.11 The structure is in the arrows

I met this one first, and backwards. I'd built it by feel, in a rendering engine, as "digital DNA," and only went looking for its name after the thing already ran. In 1945 two mathematicians, Samuel Eilenberg and Saunders Mac Lane, published a paper that founded a field now called category theory, and they said in plain words that it was a continuation of Klein, which I'd have loved to know at the time. The move they made was to stop asking what a thing is, and ask how it's carried. A category is some objects and some arrows between them, where arrows can be followed one after another and every object has a do-nothing arrow to itself, plus two rules that make the following dependable: following three arrows in a row gives the same result whether you pair up the first two or the last two, and the do-nothing arrow really does nothing when you tack it onto either end, and that's the whole definition. On the page the two rules look like nothing, and they're the part people skip, so slow down on them for a second, because each one is there for a reason you can see.

Picture three dots in a row, A, B and C, an arrow f hopping from the first to the second and an arrow g from the second to the third. Composition says there's an arrow taking the whole stride from A to C in one, g after f. Without it, every multi-hop path has to be named on its own, and the structure falls apart into a pile of unconnected pairs, a phone book without a search function. Associativity says that with a third arrow h on the end, pairing up the first two and then adding h gives the same arrow as adding h to the last two first. Both spell the same three-hop path. With your paper triangle, bracket "slide, then turn" before "flip," or bracket "turn, then flip" after "slide": you still carry out the same three moves in the same order. Swapping the turn and the flip would be a different question. If the two bracketings came out different, two people who built the same chain in different orders would hold two different arrows, which is a bug, because the chain is the thing and how you assembled it isn't. The identity laws say that tacking the do-nothing loop onto either end of an arrow gives back the arrow. If the loop changed things, it wouldn't be the loop. Those six terms, objects, arrows, composition, identities, associativity and identity laws, are all of Eilenberg and Mac Lane's definition.

Composing relations is older than the definition, too. Who put a name on that move first? I went looking. Augustus De Morgan, writing on the logic of relations around 1860, took the "is" out of the old syllogism, put a general relation in its place, and composed two relations to get the conclusion. The examples I like best from that tradition are Peirce's, a decade later, with phrases like "every lover of a servant of all women," which is relational composition with better gossip. The paper, his fourth "On the Syllogism," has more dates than most. He signed it at University College London on 12 November 1859, added an appendix dated 15 March 1860, and read it to the Cambridge Philosophical Society on 23 April 1860; it was printed in the Society's Transactions in a volume dated 1864, on pages 331 to 358, and Peter Heath's 1966 collection of his logical writings has it on pages 208 to 246. With that many dates to pick from, it's no surprise the sources I read give different years, which is a small reminder that history is where you start digging. Then in 1870 Charles Sanders Peirce, an American scientist and philosopher you met in chapter nine, carried George Boole's algebra of logic (the arithmetic of true and false, and Boole was a largely self-taught schoolmaster) over to relations, with composition as one of its basic operations, and he's usually counted as the real founder of the theory. So following one arrow after another is older than computers, older than categories, older than Turing's machine: it is a nineteenth-century logician's product of relations.

And where did they get the word? They didn't get it from mathematics. They borrowed category from the philosophers, from Aristotle and Kant, and functor (a map from one whole category to another) from the philosopher Rudolf Carnap, and then gave both words exact meanings none of those philosophers had in mind.

So Aristotle walks back into this chapter without being asked: the man who put the form inside the bronze lent his word to the field that keeps track of forms by their arrows. That doesn't change the laws. It's a reminder that a word can start in philosophy and arrive in mathematics with a precise meaning bolted on, which is the thing I've been trying to do with established since the start of the book.

What a thing is is how it maps. For now the residue is the emphasis I needed: structure lives in the maps. Nothing later in this book needs more category theory than that sentence, and the chapter on a box over any machine, where a record is pinned down by the maps between a machine and the base it extends, is where the sentence does its work.

10.12 Structure is the part that compresses

Play this one with me. I write "th" and you guess the next letter. You said "e," and you were right. Now "the q": you said "u." Congratulations, you just measured the structure of English. In 1948 Claude Shannon measured it properly, in a paper that founded information theory, and found that about half of written English is "determined by the structure of the language." It's the reason a zip file of a long letter is so much smaller than the letter. Be careful what that says. It does not say that every other letter is fixed by the ones before it; you just guessed "e" after "th" and you could have been wrong (think "than"). It says that, on average and over long stretches, about half of what's written could be rebuilt from the rest, so that the text is roughly twice as long as the choices in it. And he said why that matters in one line that I'd frame if I framed things: the structure "allows one to make a saving in time" (he added "or channel capacity"). The more you can predict, the less you have to read.

Thirty years later Jorma Rissanen, a Finnish information theorist at IBM's research lab in San Jose, turned that saving into a price. His idea is called the minimum description length principle, and it says that the best model of some data is the one that makes the total description shortest, where the total is the length of the model plus the length of the data once the model has explained what it can. A model is paid for in bits, and it's worth stating only if it saves more than it costs. Peter Grünwald, a statistician in the machine learning group at CWI, the Dutch national institute for mathematics and computer science in Amsterdam, in a tutorial you can find online, adds the footnote that keeps this honest: the price of a model depends on the code you chose to write it in, so the price itself needs a principle behind it, and that principle is something you have to argue for, not something you're handed.

That's a lot of "length of" to swallow in one paragraph, so do one with me, with numbers small enough to check on the back of an envelope. Every number in it comes from a code I'm about to make up on the spot, and I'll say every rule of the

code out loud, because Grünwald's footnote is the whole point: change the rules and every number moves.

The data is a row of thirty-two digits, each a 0 or a 1, and you and I agree in advance that the row is thirty-two long, so nobody has to spend anything saying so:

`0101010101010101 0101010101010101`

(The space is only there to rest your eyes.) Every description in the code starts with one digit that says which kind of description follows: 0 means "here's the row, written out," 1 means "here's a pattern, then the exceptions to it." Each digit is one yes-or-no answer, one bit, which is what Shannon's unit is. Written out, the row costs that first digit plus the thirty-two digits themselves, thirty-three bits, and there's no model in it at all. A pattern description goes like this. First come five digits giving the block length minus one: 00000 means one digit, and 11111 means thirty-two. That covers every allowed length. Then comes the block itself. The rule is "repeat the block, then stop at exactly thirty-two digits, cutting off the last repetition if necessary." Then come five digits for how many places the row disagrees with the pattern (up to thirty-one, which is plenty here), and five more digits for each of those places, numbered zero through thirty-one and listed in increasing order without repeats. At every listed place, flip the predicted digit.

Do it for the row above. The block is 01, two digits long. So the model costs 1 (which kind) plus 5 (the block's length) plus 2 (the block), which makes eight bits. Then comes what's left to say about the data once the model has said its piece: the row disagrees with "repeat 01" nowhere, so that's five digits saying "zero exceptions" and nothing else, five bits. The total is thirteen, against thirty-three for writing it out. The pattern pays for itself, with twenty bits to spare, and Rissanen's sentence has just happened on paper: a model is worth stating when it saves more than it costs.

Now spoil the row. Flip two of its digits anywhere you like. The model doesn't change, eight bits, but now the leftover says "two exceptions, here and here": five plus five plus five, fifteen, for a total of twenty-three. It's still cheaper than thirty-three; the pattern still pays, it just pays less, because the data has more to say that the model can't. Flip four, and the total is $8 + 5 + 4 \times 5 = 33$, a dead tie, and a pattern that saves nothing isn't worth stating. Flip five and it's thirty-eight, and now writing the row out is cheaper than explaining it, and this particular pattern has lost to writing the row out. Another pattern might still win. Flip places 5, 11, 17, 23 and 29 of the original row: now the block 010100, repeated and cut off at thirty-two digits, fits exactly. Its description costs $1 + 5 + 6 + 5 = 17$ bits. Five changes defeated "repeat 01" and produced a better description of their own. The comparison is between descriptions in the stated code, not a count of spoiled digits alone.

Message	Original row	Two flips	Five specified flips
Write all 32 digits	33	33	33
Repeat 01, list exceptions	13	23	38
Repeat 010100, no exceptions	does not fit	depends	17

For the last column, flip positions 5, 11, 17, 23 and 29. That defeats one short description and supplies another. Pattern descriptions include the type bit, length field and exception-count field. The raw row uses one type bit followed by its thirty-two digits.

So "is there structure here?" has turned into a question with a number on each side, and a break-even point you can find yourself: four spoiled digits, in this code.

And now change the code, because I promised you'd see the numbers move. Say I'd used seven digits for each exception's place (wasteful, but nobody stopped me). Then three spoiled digits cost $8 + 5 + 3 \times 7 = 34$, one more than writing the row out, where the five-digit code charged twenty-eight and the pattern paid. The pattern stopped paying one spoiled digit sooner. The row is the same, the pattern is the same, the verdict is different, and the only thing that changed was a convention I chose. That's Grünwald's footnote, and it's why I won't let you take any of those numbers anywhere else. They're the price of a description in one made-up code, a length of a written-down message, and nothing more. They are not the charge I opened this book with. That charge, as page one states it, is on one step a machine takes. How long anybody's description is doesn't enter it, and if the two ever turn out to be related, that has to be shown. Both of them being counted in whole numbers doesn't get to stand in for the showing.

Now push, because this is where "structure means information" gets born, and it's wrong, and it's an easy thing to believe. Flip a fair coin two hundred times and write down the string. Usually there is no substantially shorter way to say that string than to say the whole thing. You can watch it happen in the little code above, cut to thirty-two digits. A coin-flip row disagrees with "repeat 01" in about sixteen places, which would cost eighty-odd bits to list, and even the one block guaranteed to fit with no exceptions, the whole row used as its own block, costs $1 + 5 + 32 + 5 = 43$. The written-out row at thirty-three beats those two descriptions. Another block could do better; a lucky run of coins can make a pattern, so check it. The general reason long random rows usually resist substantial compression is a count: put each short description beside the one row it prints, and there are too few short descriptions to stand beside more than a small fraction of all the long rows. Nikolai Vereshchagin, a mathematician at Moscow State University, and Paul Vitányi, a computer scientist at CWI in Amsterdam, in a 2004 paper on exactly this question, put it without mercy: every bit of a string like that is "meaningless information, random noise." Such a string carries maximum information and zero structure. So in this sense structure is the part that compresses, the part where the "th" tells you about the "e," and most information and most structure are different things. Hold

that one tightly, because if you built something that charged for information, you'd be charging for the wrong thing, and for a while I wasn't sure which one I'd built.

And now notice that the word just changed meaning on you, and on me. I'm pointing at it so it doesn't slide past. Shannon and Rissanen and Vereshchagin mean regularity, a pattern that lets you say something shorter. Tarski, a few pages back, and Paul Halmos, a Hungarian-American mathematician whose short textbook on sets is coming up, mean a set with named relations on it, and those two are not the same thing. You can run the test yourself. Number a thousand objects, flip a coin for each, heads goes in group one, tails in group two. The result is a perfectly good sorting in the set theorist's sense, every object in exactly one group. And the list of who went where is a thousand coin flips, which is to say it compresses about as well as a thousand coin flips, which is typically very little under a code fixed before the tosses. Feed it to a code built like the little one, with the fields widened to fit a thousand, and you get the coin-flip row again, only longer. Typically most of the assignment still has to be written down. A lucky pattern can save some of it; the coin tosses alone don't guarantee there's none. By the set theorist's rule, the sorting is the structure, and it's every bit as much a sorting as "evens in one group, odds in the other," which compresses to almost nothing. The two are the same kind of object with wildly different description lengths. So the two meanings don't even rank things the same way: one asks "is there a sorting?" and the answer is yes for both; the other asks "does a short rule say who went where?" and the answer is yes for evens and odds and typically no for a substantial saving on the coins. The coin sorting has structure in the set theorist's sense without necessarily offering a shorter description. I'll keep both meanings, and I'll try to say which one I'm using, and when I slip you'll know what to look for.

10.13 No place for a grouping

You built Turing's machine on a table in chapter one: a tape, a head, a state, and a card of rules. Chapter one also kept three claims apart: what's proved about it, the Church-Turing thesis, and the claim about physics. Every laptop and phone and server computes nothing a Turing machine can't. What differs between them is speed, space, energy and cost. Underline that last word. Cost is where this book lives.

Now do the thing I did, and do it honestly. Look through those four parts for a place that says "these squares belong together." There isn't one, and that's not me failing to look. Everything the machine is at any moment is which state it's in, where the head is, and what's on the tape, and the table says what happens next, and that's the full list. So any grouping has to live in the states and the table, as a rule, or on the tape, as symbols, which is to say as bytes like any other bytes. The

machine can hold a grouping. It just has no place where holding one is different from holding anything else.

In 1972 David Parnas, a software engineer, wrote a short paper on how to cut a program into modules. He took one small program and cut it two ways. Both worked. But one was easy to change later and the other wasn't, and the easy one could be "much less efficient" if talking between modules cost anything. His point was that a module is "a responsibility assignment," a decision about who answers for what, and how you cut decides what every later change costs. If you've ever maintained someone else's code, you've paid for their cut the way Parnas said you would, probably at two in the morning. What survives the push is that a division is a commitment: it changes what the work costs, and it is no caption, no label stuck on afterwards. I'd like to say the thing I built understood that from the beginning. It didn't. I did, eventually.

10.14 Laying them all down

Now lay every one of those on top of the others and see what shows through. I see seven things. They're my overlay, the places where I see the sources agreeing. The sources never proved it together; some of them never met, and some of them disagree (Eilenberg and Mac Lane put maps and their composition at the center; Shannon's structure and Tarski's are two different words, as I just showed you). You've pushed on each source yourself by now, so you're in a position to tell me if I'm seeing things.

A list of seven slogans would be easy to nod at and impossible to check, so I'm going to carry one thing through all seven and let you watch each one add to the one before it. The thing is the square you drew for the Pythagoreans, side, diagonal and all. Draw a few more on the same sheet: one small, one big, the small one again but tilted, and one stretched out into a long rectangle. You'll want a pile.

Structure is about relations among things, and the things themselves aren't it. You see that in the diagonal, the bronze, Leibniz's room, Bourbaki's unspecified elements, Tarski's named relations and the arrows. On your sheet, what makes a square a square was never the ink. It's that the four sides match and the corners are square, and that the diagonal stands to the side the way it does whether or not any number names it. Every one of those is a relation among parts. But on its own this first thing is too loose to use, because a sheet of paper is drowning in relations: this square is left of that one, this one is nearer the coffee ring. Which relations count?

Structure is what stays the same under a chosen set of changes. You see that in Euclid's triangles laid together, Klein's allowed moves, Noether's symmetry, and Frege's direction that every parallel line shares. This second thing gives the answer the first thing was missing: the relations that count are the ones that survive the

moves you've decided to allow. Slide the small square across the sheet, turn it, and the diagonal still stands to the side exactly as before; "left of the coffee ring" doesn't make it across. The second thing is a filter on the first.

The same structure can be carried by different stuff: the statue in marble, or two sets that answer the same vocabulary the same way. The third thing is the second again with one more change allowed, which is to swap the material. Cut the square out of cardboard, or lay it out in string on the table, and if the relations survive, it's the same square in every way you agreed to care about. Aristotle's bronze is one more thing you agreed not to look at, and the wooden sandwich is what happens when the material was never on that list.

The first act of structure is a sameness that sorts a pile you were already given. It's there in Euclid's first common notion, Frege's direction, Tarski's one question, and Russell's letter, with Zermelo's repair that you must start from a pile. At this point the overlay stopped being a list for me and turned into a shape, so slow down here with me. Look at your pile of drawings and say: two of them count as the same when an allowed move, for now a slide or a turn, carries one exactly onto the other. Watch what happens to the pile. It falls into groups. The small square and its tilted twin go together; the big one sits on its own; the rectangle sits on its own. And it falls cleanly, nothing half in one group and half in another, for a reason you can check. Klein's allowed moves always come with three guarantees: doing nothing counts as a move, every move can be undone, and two moves in a row is again a move. Those three are, in different clothes, "everything is the same as itself," "if this matches that, then that matches this," and "if this matches that and that matches a third, then this matches the third." Put the last two together and you get Euclid's first common notion, "things which are equal to the same thing are also equal to one another." So the second thing, sameness under chosen moves, is a sorting of the pile the moment you ask which drawings it calls the same. Frege did it with lines and parallels. Tarski's one question does it in the crudest way there is, by cutting the pile in two. And the pile has to be there first, because a sameness has to be a sameness among something; that's Russell's letter, and Zermelo's repair.

Adding structure rules possibilities out. Every one of Euclid's postulates rules one out. Half of English is already decided. A conserved quantity limits where a system can go. On your sheet: once the small square and its tilted twin are in one group, you've given up the right to answer any question the moves respect differently for the two of them. Tell me the side of one and the diagonal of the other isn't an open question any more. A sorting rules out a whole bundle of possibilities at once.

Held structure gets used without re-deriving it. You see that in Shannon's saving in time, Rissanen's compression, Parnas's one-module-at-a-time, and Leibniz's analysis of situation (reasoning about where things are straight from their relations, with no coordinates laid down first). The fifth thing pays off here: the possibilities

you ruled out are ones you never have to check again. Know which group a drawing is in and you know everything the group shares without getting the ruler out, the same way the row of 01 never needed writing out. There's one catch, and I'll need a whole section for it: it only pays if you kept the sorting somewhere you can read it back.

And structure is always relative to a chosen frame, whether that frame is Klein's group, the vocabulary, the code you compress against, or Parnas's criteria. Nobody gets to say "the" structure of a thing without saying from where, and I do it constantly, so watch me for it. On your sheet: allow stretching as one of the moves and the rectangle walks over and joins the squares; allow only slides and turns and it doesn't. Same pile, same ink, different groups, because I chose different moves. The groups were never in the paper. They were in the paper and the choice, together.

I think the seven hang together this way. Relations are what structure is made of. The chosen moves pick which relations count, and a change of material is just one more move. Asking which things the moves call the same sorts the pile. The sorting rules possibilities out, and what's ruled out is what you get to skip later. And because the moves were chosen, all of it is relative. Read that way, the fourth thing sits in the middle: it's where the first three turn into something you can write down, and where the last three get their start.

That's my reason for starting the next subsection with a sorting and a door across it, and I want to be exact about what kind of reason it is, because it's mine. None of these people said "start with a partition." Klein would tell you to start with the moves, Eilenberg and Mac Lane with the arrows, Shannon with the probabilities. What I can actually point at is narrower. In the framings about when two things are the same shape or the same kind (Euclid laying one triangle on another, Klein's moves, Frege's directions, Tarski's question) there's a moment where somebody decides what counts as the same, and that decision, written down, is a sorting of a pile. In Shannon, Rissanen and Noether I can't find that moment as cleanly, and I'm not going to bend them until I do. So the overlay doesn't prove that a sorting comes first. It shows me where the framings I trust most keep putting their first foot down, and I'm putting mine there too, with one thing added that none of them needed and the toll does: a direction across the sorting.

I looked for a framing that denies any of the seven and didn't find one on my own. There are two, and they're worth pushing on, because each one denies a different piece.

The seventh, that structure is always relative to a chosen frame, is denied outright by David Lewis and Theodore Sider. Hilary Putnam had argued that if every way of carving things up is as good as every other, a consistent theory can always be made true just by picking what its words refer to. Lewis answered him in 1983, in a paper called "New Work for a Theory of Universals," with an answer he credits to

G. H. Merrill: some properties are natural and some aren't, some cuts carve nature at its joints and some are lines somebody drew, and the natural ones are the ones words can pick out. Sider built a whole book on it in 2011, Writing the Book of the World, and he uses this chapter's word: "The world has a distinguished structure, a privileged description," and "structure is a worldly, not conceptual or linguistic, matter." On their view there's a frame nobody chose.

Now come at it from this book's side. Grant them their joints. A machine still doesn't get handed them. Chapter fourteen shows that the event a box turns up on is a choice the rules don't make for you, and the toll from chapters two and three charges the same whether that event is a joint of nature or a line I drew. So whether the world has a structure of its own is a question this book can't settle, and what a machine keeps and pays for stays relative to the frame it was built with. That's a narrower seventh than the one I wrote above, and it's the one I can stand behind.

The fourth, that the first act is a sameness that sorts a pile you were already given, is denied by Michael Makkai and by the people behind univalent foundations. Makkai, in a 1998 paper, "Towards a Categorical Foundation of Mathematics," doesn't give up sets with members. What he gives up is one sameness that holds across everything at once: "there is no global equality present for all things simultaneously." Any property worth stating about the objects of a kind has to come out the same for two objects that match exactly in structure. The univalent foundations book, written in 2013 during a year at the Institute for Advanced Study organized by Steve Awodey, Thierry Coquand and Vladimir Voevodsky, makes that an axiom: "identity is equivalent to equivalence." Things that match exactly in structure count as the same, for everything the language is allowed to say. And there can be more than one way for two things to match, the way a square matches itself four ways by turning, so sameness there carries which match you meant. That's Benacerraf's puzzle from earlier in this chapter, about which sets the number three really is, shut down at the door. Nobody gets to ask a question the structure doesn't answer.

That one gives less than the first, because a machine that steps still has a pile: the states it can reach. At any tick its box is down or up, so those states are already cut in two, whatever the foundations underneath say about where the pile came from. And the cut is legal on their own terms. What Makkai gives up is a sameness across every kind of thing at once; inside one kind he keeps it, and a set, in his words, is "a relatively orderly part of the world in which an equality predicate reigns." A machine's states are one kind, and a yes-or-no question put to each of them never leaves it. Makkai and the univalent people change what you're allowed to say about the pile. They don't take away the cut.

10.15 The smallest piece

Now the sentence I wanted to say at the top and had no right to until here, and even here I have to split it in two, because half of it is a textbook fact and the other half is me. The piece of structure I'm going to start from is a cut: a pile sorted in two, with a door between the halves that swings one way. A partition is the cut with the door taken off, and the partition comes with a textbook fact, so I'll do that half first.

The textbook half: a partition and an equivalence relation are the same object. That's Paul Halmos, Naive Set Theory, 1960, section seven, and you could read that section before dinner. I'll say it in words first. Take any pile of things. An equivalence relation on it is a way of saying "these two count as the same" that behaves the way sameness should: everything is the same as itself; if a is the same as b then b is the same as a; and if a is the same as b and b the same as c, then a is the same as c. Every such relation sorts the pile into groups—the things that count as the same go together—and every sorting into groups hands you back a relation, "same group." Do one and then the other and you land where you started, in both directions. So a partition and an equivalence relation are one object seen from two sides, and that's the fourth thing on the overlay, the sorting of a given pile, now with a name and a page number.

You can check that with the buttons from chapter four, or anything else in a pile. Say two things count as the same when they're in the same group. Is everything the same as itself? Yes, it's in its own group. If a is with b, is b with a? Yes. If a is with b and b is with c, is a with c? They're all in b's group, and b is only in one group, so yes. Now go the other way: start with any "counts as the same" that follows those three rules, and gather everything the same as a into one group. Could two of those groups overlap without being the same group? If something is the same as a and the same as b, then by the rules a is the same as b, so their groups are one group. So the groups never half-overlap: you get a clean sorting. And doing one and then the other lands you back where you started. That's the whole proof, and it's textbook mathematics. I'm claiming nothing new in it.

And the smallest structure that sorts, by my ruler, the one question, is the same object a third time. Put a yes-or-no question to every thing in the pile. The ones that say yes are one group and the ones that say no are the other. Going the other way, a sorting into two groups becomes a question the moment you point at one group and call it "yes." The only thing a question has that the sorting doesn't is which group got the label. So the set theorist's first step and the logician's first step are the same step, and nobody has to pick a side, which is a relief, because it's an argument you can lose with yourself for a long time.

Now put the door back. In chapter five you met "at or below," with its two rules: every place is at or below itself, and at or below carries across a third place. It

doesn't have to go both ways. Take the places that are each at or below the other, and put them in one group. That "each at or below the other" follows all three of sameness's rules, so it's a clean sorting, the buckets. And between buckets, "at or below" only goes one way: if two buckets were each at or below the other, they'd be one bucket. So an order is exactly a sorting with doors between the groups, and every door swings one way. Take the doors off and a sorting is what's left.

A yes-or-no question is the smallest case of that: two buckets, no and yes, and one door from no to yes. The thing settling on one side of a line, from the first page of this book, is a move through that one door. What chapters one through three found keeping takes is what it costs to go through it and not come back.

So much for the textbook half. The other half is mine, so hold it looser, because Halmos says nothing about "smallest" or "first," and I'd be lying if I let his page number carry my sentence. In what I build, the order goes: a pile, then a split of the pile with a door across it, then more splits at once, which is an order. The groups come first because a question needs something to be about, and a door needs two sides to stand between. That's an order for my construction. It isn't a law that every account of structure has to start by sorting. You could start with arrows between the things themselves and no groups at all, and people do, and nothing I've shown you rules that out. So try that road, measured with the same kind of ruler. Fewest dots first, then fewest arrows between different dots. One dot can't tell anything apart. Two dots with no arrow between them is a plain split, with no door. So the smallest picture with a direction in it has two dots, call them NO and YES, a do-nothing loop on each so a thing is allowed to stay put, and one arrow from NO to YES with nothing coming back. Category theorists call it the walking arrow. Now lay any machine onto it. Put every state on one of the two dots, so that every move of the machine follows an arrow in the picture. The dots sort the states into two buckets. The loops let a move stay where it is. The one arrow lets a move cross from NO to YES. And there's no arrow back, so no move goes from YES to NO. That's a split with a one-way door, exactly, and every split with a one-way door lays onto the picture that way.

So the arrow road doesn't come out smaller. It comes out at the same place. The honest sentence is: a split with a door across it is where I chose to start, and it's where both rulers hit bottom. It's the smallest by the ruler I stated. If you have a smaller start, by a ruler you can state, I want to see it. It would change this book.

10.16 Buckets

So I'll do the symmetric half once, for real, because it's the half you can do with your hands. Take everything in front of you and sort it into buckets, with nothing in two buckets and nothing left on the floor, and that's the whole of the definition. It's

the thing you do with a kindergartner and a pile of blocks, and I am aware that I've just marched you past Pythagoras, Plato, Euclid, Frege and Noether to arrive at a pile of blocks. Keep the blocks anyway. They're about to cost you something.

Take ten blocks, and put the red ones in one bucket, the square ones in another and everything else in a third. Now a red square turns up, as they do, and you've got your first crisis: it belongs in two buckets and the rule says one. So this isn't a partition yet, and you have to decide which bucket wins. That decision is the whole thing, because a partition is a pile of small rulings like that one, and once you've made them all, the pile is holding a fact it wasn't holding a minute ago.

The next part isn't kindergarten. Why would that fact cost anything? Watch what it does to one question. Before the buckets, "are these two in the same group?" means working it out: go find the first one, go find the second one, apply the rule to each, and compare. After the buckets, it's a glance: read the first one's bucket, read the second one's bucket, and you're done. With each block's address already in hand, and a table that retrieves a label in one operation, that's two lookups whether the pile has ten blocks or a billion. Those are access and cost assumptions. A pile with no addresses, or an index that must be searched, hasn't been given that same two-lookup rule. What you're looking at is Shannon's saving in time and Parnas's one-module-at-a-time, the sixth thing on the overlay, showing up in a kindergarten.

Look at what makes that glance possible: a table, with a written-down line for each block saying which bucket it went in. The partition is the fact; the table is one way of holding the fact so it can be read back fast, and you had to build the table, and you have to store it, and it has to be the kind of thing you can read in one look (a bucket number, say; a list of every block in the bucket would have you searching through it again). Anybody who's used a database has met this table; it's what an index is, and an index is a thing you build, store, and keep up to date, and every part of that takes something. Take the table away and keep the partition and you're back to working it out. And sometimes there was never a search to save: if the rule is "evens here, odds there," you can answer "same group?" by looking at the last digit of each, no buckets needed. What survives is that holding a sorting in a form you can read back is worth something, and building that form cost something, and those are already two different events. Building the table, storing it, checking that it's right, and deciding to rely on it are four different things. The first cost question I want to put to that structure lives in the gap between the third and the fourth. Nothing in the kindergarten has shown a charge in that gap. It has separated the acts well enough to ask.

So who, in everything I read, says what has to be paid? I found three answers that come close, and I'll be exact about how close, because this is the edge of what I know. Rolf Landauer, in 1961, and Charles Bennett after him, say that any step which takes two possible situations and makes them one, erasing a bit or merging two paths a

computation could have taken, has a heat floor when it removes uncertainty under their thermodynamic assumptions. That is a physical claim about how the device and its surroundings are prepared, Landauer's principle, and in this book it is always a premise, never something I prove. It's the place, of all I've found, where a fact about information turns into a fact about heat, though at room temperature the amount for one bit is far too small to show up on anybody's electricity bill. In 2012 a lab in Lyon erased one bit, held in a single tiny bead, and measured the average heat it gave off coming down toward Landauer's floor as the erasing was slowed, and other labs have found the same since. Not everyone accepts the principle in general, and I'll come back to the people who don't. Is sending blocks to buckets a step of that kind? Look closer, because the answer depends on what you keep. If you keep each block and write its bucket next to it on a spot you knew was blank, you've merged nothing; you can read the block right back off the pair. You have spent something, though: the blank spot the label now sits on, which the machine hadn't used before, and any heat from the writing depends on its implementation. Bennett's whole point in his 1973 paper was that a computation which keeps its inputs, and keeps enough of its working to retrace its steps, can be run without any merging step at all, and it pays for that by taking fresh states. If instead you throw the block away and keep only the bucket label, then two different blocks can leave the same label behind, and that is a many-to-one step, the kind Landauer charges for. Both are things people call "sorting into buckets." Only the second loses a distinction. So the bucket by itself doesn't hand you Landauer's heat. The forgetting does, when there is forgetting, and keeping instead is paid for with fresh states. Whether a given machine forgets is a question about that machine. It has to be answered on the page, and I won't wave at it here. Rissanen says a model is paid for in the bits it takes to state, and Grünwald says the price depends on the code. Parnas says a bad cut is paid for later, at two in the morning. Peirce, in 1879, priced research itself: in his words, "the relations between the utility and the cost of diminishing the probable error of our knowledge." Abraham Wald's sequential tests, from 1945, are built to need fewer samples, on average, than a test of fixed size with the same chance of being wrong. Those price the looking and the risk in the answer. Neither puts a price on the step that locks the answer in. And that's where it stops. Rovelli, back in chapter six, prices a choice that sticks, and his choice is one nobody could predict, which that section showed is a different act. I couldn't find anyone who says that the moment you vouch for a grouping, the moment you stand behind "these belong together," has a price of its own. Just arranging them that way is a different act. That could mean nobody says it. It could mean I didn't find the person who does. I honestly can't tell you which.

So that's where I am at the end of one word. A grouping, the act of making it, the information kept about it, and a commitment to rely on it are different things.

The blocks let you separate them with your hands. Now take them down to the smallest case there is, a bucket with two sides, one block, and a ruling that sticks, which is a thing settling on one side of a line and staying there, the case the first chapters started from and showed what keeping it takes. What I build from here keeps chosen distinctions in a state, and I should say which ones. Some were there from the start. Some get recorded along the way. Those aren't the same case. A box that already says YES can sit there forever under a do-nothing rule, with no merge and no fresh states. The count in chapters one through three starts at NO and follows a new YES that stays, under one fixed rule with a move for every state. That's the case the merge-or-fresh conclusion covers. A grouping by itself doesn't hand you that run, or its price.

CHAPTER 11

The counter

Back in chapter four you kept a tally beside the two boxes, the way you'd keep score beside a board game. The tally lived in your hand. The machine never saw it. That's fine for a game on a table. For anything that's going to run on its own, with nobody keeping score, the total has to live somewhere the machine can reach, which means the machine can, in principle, write it.

So this chapter is about what a running total inside a machine has to be before you're allowed to call it a receipt. Here's the answer first, because it's short, and then the chapter is about why each word is there: a receipt is a total that lives in the state, that only climbs, that equals the sum of what was charged, and that nothing else could have been.

11.1 A stick, split down the middle

The oldest receipt I could find that you can still hold is a piece of wood. From about 1100, under Henry the First, until 1826, the English treasury recorded what it was owed on tally sticks: a strip of hazel, notched across with cuts of set widths, then split lengthwise so each half carried the same notches. The treasury's own handbook from around 1180, the Dialogue of the Exchequer, gives the widths: a thumb's width for a hundred pounds, a swollen barleycorn for a pound, a bare cut for a penny. The payer kept one half, called the stock, which is one story for where the word stockholder comes from. The treasury kept the other. When the time came to settle, you laid the two halves together, and only the two original halves fit, because wood splits along a grain nobody can copy. Add a notch to your half and the other half shows a gap where the partner should be. Try to take a notch away and you find that wood can't be uncut.

When they finally stopped using them, the clerks burned centuries of old tallies in the furnaces under the House of Lords, overheated the flue, and burned down the Palace of Westminster. I mention that because it's the best fact I picked up for this book, and because it tells you how many of them there were.

Do it yourself. Draw a border around a strip of card, notch it once for every coin you hand a friend, then split it lengthwise through the notches and keep half each.

Don't trim the edge or swap in a fresh strip. Now add a notch to one half and put them back together. You'll see the mismatch straight away. And you can't put a cut back: a patch is just one more thing to look at. The total is the notches, one per coin, and the count only climbs. Both halves started with the same count, so if they disagree now, somebody changed one. The mismatch won't tell you who. What the stick gives you is a comparison you can make, not an object nobody can forge.

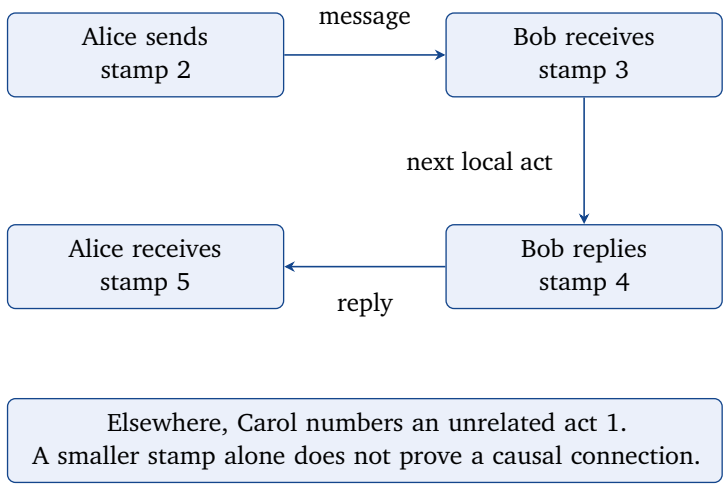
Economists read the same stick another way, and they're not wrong either. Alfred Mitchell-Innes made the treasury tally his star witness in 1913 that money is credit: the crown paid the people it owed by "raising a tally" on some branch of its income, a wooden promise to pay. A receipt to me, money to him, and it's the same piece of wood. Narayana Kocherlakota proved a version of his point in 1998. In a wide range of economies where nobody can be held to a promise, anything money lets people do, a memory of everyone's past dealings lets them do too. His title says it in three words: money is memory.

11.2 Why it has to climb

Page one already showed why there are no refunds: if a later move could pay you back, the move that settles charges you one, the next move hands it back, and the whole run comes out free. A total that can come back down is an accountant you can lean on to revise the books. "This run cost at least one" would be true of a run that ended having cost nothing. So the count only climbing isn't decoration. It's what makes "at least" mean something.

The same rule turns up in two places that have nothing to do with each other, pointing in opposite directions, and that's a big part of why I trust it.

In 1978 Leslie Lamport asked how a set of computers passing each other messages, with no shared clock, could ever agree on what happened before what. His answer is a counter on each machine with two rules: add one before you do anything, and when a message arrives carrying the sender's count, jump your own counter above it, never below your own. Those counters, he showed, respect cause and effect: if this caused that, this has the smaller number. And they do it, in his words, "with no actual timing mechanism." They're clocks only in name.



Start Bob at zero and Alice just after her local act 1. On receipt, each raises the local count above both its old value and the incoming stamp.

Try it with three people passing notes and you'll see it work in two minutes. Then let one of them reset their counter to zero after reading a note, and watch a reply come out numbered before the note it replies to. A counter that goes down can't be leaned on for order. The other direction doesn't hold, though: two people can number two unrelated acts 1 and 2, and the numbers alone don't say the first caused the second.

Where did Lamport get it? In the notes he keeps on his own papers, he says the timestamps came from an earlier note by Paul Johnson and Bob Thomas on keeping copies of a database in step, whose method wasn't quite right: it let things happen in an order that broke cause and effect. What let him see the fix, he says, was special relativity, which teaches that the one order of events everybody agrees on is the order of what could have caused what. And he passes on a line from Jim Gray, the database man whose wedding you met in chapter eight: Gray had heard two opinions of the paper, that it was trivial and that it was brilliant. Lamport says he can't argue with the first and isn't inclined to argue with the second.

Now the other direction. In 1949 Alan Turing, and in 1967 Robert Floyd, each asked how you prove a program stops. Both times the answer was to find a number that goes down on every step and can't go down forever, like a whole number that can't go below zero. If every step lowers it, the steps can't go on forever. Up-only gives you order, and down-only gives you stopping. They're the same shape, seen from the two ends: a number that only goes one way is something you can build on.

And there's a real case where a counter was allowed to go down and got gamed exactly there. Ethereum, one of the largest blockchains, charges "gas" for every step of a program, and for years it gave gas back when a program freed up storage. Someone noticed you could fill storage when gas was cheap, hold it, and cash the

refunds when gas was dear, and a token was built to do exactly that. The real work in a block could reach nearly twice the block's stated limit. In 2021 a change called EIP-3529 cut the refunds sharply and capped them at one fifth of the gas used. The written reasons are the token and those overfull blocks. I read both as the same lesson: a counter that goes down was being leaned on at the place where it went down. Gas comes back in a later chapter, as the one real system of five that charges at the moment of the write.

11.3 The total is the charges and nothing else

The next property is the plainest. The counter after a step is the counter before, plus what that step was charged. Add up a whole run, and all the middle values cancel out, the way a sum of differences collapses to last minus first. The total is the sum of the charges.

Why say it? Because it's the property that fails quietly. Say a machine charges one for raising its record, and somewhere in its error handling forgets to add the charge when the step fails halfway through. Or it charges two on one path and one on another for the same move. Or it has one step that writes the counter directly. Then the counter is mostly the sum of the charges, and "mostly" is the whole gap between a receipt and a note that says "trust me, about this much." The rule is that there's no "mostly": every step, every outcome, every error, adds exactly its charge, and a step that fails is charged the same as one that succeeds, because the charge is on the attempt. For a machine built that way, the rule can't fail. What can fail is the claim that a real machine is built that way, and checking that means going through every move it has, the failure branches included, one at a time. It's dull, and it's the whole job.

One thing it doesn't say is that every unit bought something. The total is exact about what was spent. What the spending was worth is a later chapter's question.

11.4 Only one honest count

The next property surprised me by being provable at all. Fix the price list: this move costs one, that one costs zero. Now say somebody else keeps their own books: their own total for every state, starting at zero, and going up on every step by exactly what the price list says. Could their books disagree with the machine's counter, on some state you can actually get to?

No, and the proof is the tally stick. Both totals start at zero. On every step, both go up by the same amount. So on every state you can reach from the start, they're equal. Agreement at the start survives the first move, then the second, then every

move after it. Given the price list, the count is forced. It's the arithmetic, not a second opinion.

It only says that about states you can reach. On a state nothing reaches, it says nothing, which is the right thing for a receipt to say.

And one thing follows that's easy to miss. If the counter is part of the state, then a run that comes back to a state it's already been in has the same counter both times, so the moves on that loop cost nothing in total. A machine that keeps its counter inside its state can only go round a loop that costs nothing. Every paid move lands it on a state it's never been in, because the counter is higher there than anywhere before. So a counter that only climbs is itself paid for in fresh paper. On a finite machine, that means no move can be priced above zero at all: play any move over and over, the states come back around, and a loop costs nothing. A machine that keeps its own receipt, with even one paid move, needs endlessly many states. That's chapter two again, coming up from underneath.

Here's a puzzle that shows where the "only one" stops. Take the one-switch machine whose set costs one and whose reset costs zero. Set it once and it's up. Reset it and set it again, and it's up again, the same state. Now picture a second machine that keeps the whole list of moves. To it, "set" and "reset, set" are different. So you can always walk a list of moves into the second machine, one way and one way only. But you can't always walk the first machine's states into it, because two lists that land the first machine on the same state can land the second on different ones. You can do it exactly when the second machine never tells apart two lists the first one treats the same. Joseph Goguen put that as a general rule in 1972, in a paper whose title says it outright, "Realization is universal," and Thomas Colcombet and Daniela Petrişan redid it in 2020. None of that uses the toll at all.

The nearest relative in physics is in thermodynamics. Elliott Lieb and Jakob Yngvason showed in 1999 that once you fix which states can be reached from which, and that follows a short list of rules, there's a quantity that goes up exactly along it, and it's the only one once you pick where zero is and what one unit is. Their order goes in and their count comes out forced. Here the price list goes in and the count comes out forced.

11.5 The toll is an old trick with a new name

It would be nice to have a new kind of cost argument here. This isn't one. The toll is a case of a technique computer scientists have used since 1985, and that a mathematician studying whether things settle down had, in another form, in 1892.

It's called the potential method. Give every state a number, the potential, that measures how much is stored up in it. Then look at each step and ask: does the step's cost, plus the change in potential, ever come out negative? If it never does,

then any run costs at least as much as the potential dropped from start to finish, because the drop had to be paid for along the way. Robert Tarjan set it out in 1985, crediting the potential view to Daniel Sleator. Tarjan used it the other way round, to show a run costs at most so much.

Do his stack example with coins. Every operation takes some coins off the top, maybe none, then puts one on, and costs one for each coin it moves. Let the potential be the number of coins. An operation that takes off k coins and puts one on costs k + 1 and changes the potential by 1 minus k, so its cost plus the change is exactly 2, every time. So any run from an empty stack costs at most twice the number of operations, which is twice the number of pushes, however many coins some of them take off.

The decreasing number underneath it is Aleksandr Lyapunov's, from his 1892 doctoral thesis on whether a moving system settles down. It was in Russian. A French translation came out in 1907, made by a French naval engineer, Édouard Davaux, and corrected by Lyapunov himself. The English one came out in 1992, a hundred years after the original, translated from the French. A translation of a translation. Did the computer scientists of 1985 get it from him? I can't tell you, and nothing here needs them to have. What I have is a picture: the same idea, a number that only goes one way along a run, in a thesis about whether a moving system settles down and in a paper about data structures, close to a century apart.

Now point it at the record. Let the potential be one when the record says not done and zero when it says done. A step that takes not done to done drops the potential by one, so for the step's cost plus the change in potential not to come out negative, that step has to cost at least one. That's the toll, exactly. And "any run from not done to done costs at least one" is just the method's usual conclusion, the potential dropping from one to zero.

Ask for the other direction too, that a step's cost is never more than the potential it drops, and the two together pin every price exactly: one on a step that settles something, zero on every other step, and nothing ever going back to not done. One catch: that only works if the price can depend on where the step starts. With a single price per button, like the stamp, a button that settles once still costs one when it's pressed again on done, so no price list per button can meet both directions.

Two more tools sit next to this. Run a list of moves inside a wrapper that adds up a number as it goes, and you get back the final state and the run's total; Norman Danner and Daniel Licata built cost reasoning for programs on exactly that, first with Ramyaa Ramyaa and then on their own. And give each move its price ahead of time, as a tag that adds up as the moves do; that's Shin-ya Katsumata's grades, which Dominic Orchard, Vilem Liepelt and Harley Eades put into a working language. The potential method itself went into the checking of programs before any of those: Martin Hofmann and Steffen Jost in 2003, and Jan Hoffmann, Klaus Aehlig and

Hofmann after them. The difference with this book is when the number gets written. A cost checked ahead of time gets checked before the program runs, by whoever checks it. A cost kept in the state gets written while it runs, by the step itself. The number is the same. The moment of truth is different. For a program with loops whose cost depends on its input, a fixed tag can only bound the cost, while the counter holds the exact number; cost-aware type theories like calf, from Yue Niu, Jonathan Sterling, Harrison Grodin and Robert Harper, reason about such costs with proofs.

And there's one more neighbor, closer than any of those. A number the step keeps up to date and nothing else ever reads is what people who prove programs correct call ghost state. Susan Owicki and David Gries used it in 1976, crediting the first use they'd found to Maurice Clint, because some true facts about a program can't be proved without it. Their rule lets it appear only where it's being updated itself, so it never steers the program. Jean-Christophe Filliâtre, Léon Gondelman and Andrei Paskevich worked out in 2014 when ghost code can be removed without changing what the program computes. By that test the counter is ghost whenever no step reads it. So what does putting it in the step buy? Two things. The number is there while the machine runs, where somebody looking through a window can see it, not only inside a proof written afterwards. And the rule that keeps it covers every move the machine has, so every run.

Two last relatives. Chapter four's floor, "any run from not done to done costs at least one," is the cheapest-path bound in a system with weights on its moves: every path from the not-done states to the done ones crosses the line between them, and every step across that line weighs at least one. Weighted machines like that go back to Marcel-Paul Schützenberger in 1961. And the physicists who study what you can turn into what, when some operations are free, have the same shape: Bob Coecke, Tobias Fritz and Robert Spekkens gave the general setup, Fritz worked out its algebra, and Tomáš Gonda and Spekkens cataloged the numbers that never go up under free operations. Read with "done" as the thing you're after and one unit of cost as the currency, the not-done potential is the simplest of those. It's a neighbor to know about, not a result of mine.

So what's new, if the method isn't? Only where it's pointed. Tarjan and Sleator's method proves things about costs. What this book adds is which potential, the one that drops when something settles, and what happens to that when you can only see part of the state through a window.

11.6 What the counter is, said once

A receipt is a total that lives in the state, only climbs, equals the sum of the charges on every path including the failures, and is the only such total, on every state a run

from the start reaches, for its price list. Each of those had a home before this book: the stick for a total that lives in the state, Lamport's clocks for climbing, Turing's and Floyd's numbers that only go down for the same rule run backwards, the canceling sum for adding up, the step-by-step argument for the one honest count. None of them is physics, and none of them says what the charges should be. Once you've chosen the charges, the counter is the arithmetic.

There's one thing a counter can't do just by existing. The stick works because you can lay the two halves together. What if all you're ever shown is one half, through a window that doesn't show the notches? Then you're back in chapter seven. Take two states that differ only in their count, one at 1 and one at 2, and let nothing that comes through the window depend on the count. The window gives the same view for both, any reader gives the same answer for both, and one of the answers is wrong. So a window that leaves out the count can't give it back. To read the count, the count has to come through.

CHAPTER 12

What a ticked box lets you lean on

I keep writing "you can lean on it" and then backing off. Let me stop doing that and say what it would take. A box went from not ticked to ticked. The toll charged the move that ticked it. It stays ticked. Chapter eight made it established: it could only be ticked right after a check of that very claim passed. And the right window lets you see it.

Now: what is a later step allowed to do because of that? That's a different question from "is the claim true," and the gap between those two is what this chapter is about.

12.1 The coin that moved

Run this on a friend. The feeling of it is the point.

Put a coin under one of three cups while they watch; say cup one. Send them out of the room. Move the coin to cup three. Then, and this is the trick, put a second coin under cup one. Bring them back and ask: is there a coin under cup one?

Say they answer yes, because they trust the coin they saw has stayed put. There is a coin there, so they're right. And their reason was a good one: they watched it go in, and nobody usually moves coins around. But look at what happened. Their reason is about the first coin. The answer came out true because of the second coin. They were right by luck.

That's Gettier's trap again, from chapter eight, done with cups. Edmund Gettier's three pages in 1963 broke a definition of knowing: that you know something when it's true, you believe it, and you've got good reasons. Gettier took that definition from two philosophers of the 1950s, Roderick Chisholm and A. J. Ayer, and the only place he mentions Plato is a footnote saying Plato "seems to be considering" something like it. The Stanford Encyclopedia's entry on the analysis of knowledge, by Jonathan Ichikawa and Matthias Steup, says that definition gets called "traditional," and that the reason part of it is "often traced to Plato's Theaetetus." The same entry calls the

idea that it was ever widely held "a bit of a convenient fiction." And Julien Dutant, a philosopher at King's College London, wrote a 2015 paper called "The Legend of the Justified True Belief Analysis," arguing that the older tradition asked for something stronger: a mark on the belief that can't be wrong.

Bertrand Russell had a related case in 1948: somebody looks at a stopped clock at the one moment it happens to be right. They're right about the time, and they know nothing. And 1948 isn't the bottom either. The Stanford Encyclopedia gives a case from the eighth-century Buddhist philosopher Dharmottara. You're looking for water on a hot day, you see water ahead, it's a mirage, and when you get there you happen to find water under a rock. And one from Peter of Mantua, a logician who died in 1400. Plato is running right next to you, you take him for Socrates, so you believe Socrates is running, and Socrates is in fact running, in Rome, where you can't see him. People in different centuries, languages and traditions kept building the same trap by hand: a reason that points at one thing, and a truth that happens to be sitting on another.

What would fix your friend's reason? Lift cup one again. Now there's a new look at the coin that's actually there. That gives them evidence about this coin, now, instead of a memory about the coin that left.

12.2 Plato nearly had it

The dialogue where Plato goes after knowing is the Theaetetus, and it's named for a real person, so I wanted to know who. He was a mathematician. In the dialogue he's a teenager, and he tells Socrates that his teacher had been showing them, one case at a time, that the side of a square of area three, or five, and so on up to seventeen, can't be written as any fraction. That's the Pythagoreans' diagonal from chapter ten, done case by case. The young man's idea was to stop going one case at a time and find one word for all of them. Euclid's tenth book, on lengths that share no common measure, is thought to be largely his work. The dialogue opens with him being carried home to Athens from the army at Corinth, badly wounded and sick, barely alive; the usual date is 369 BC. And the conversation itself survives as a record: Euclid of Megara, not the geometer, says he took notes as soon as he got home, filled them out later, and had Socrates correct them whenever he came to Athens. So the book about what turns a true belief into knowing survives as a record Socrates himself corrected.

In it, Socrates tries three different meanings for the "account" that should turn a true belief into knowledge: a statement you can say out loud; a list of the thing's parts; and, third, the mark that sets this thing apart from everything it could be confused with. Each one fails, and the dialogue ends with nobody knowing what knowledge is, which is one of the more honest endings in philosophy. But the third

one is the one I'd underline: the account as the thing that tells the real case apart from its rivals. He couldn't make it work because he had no way to say what that account had to be attached to.

Socrates gives his own reason for dropping it: if the account means knowing the mark, then knowledge has been defined as true belief plus knowledge of something, and the word you were defining is back inside its own definition. A circle. Does a check get out of it? It gets the word out: "a check that ran and found the mark" isn't defined in terms of knowing. Whether the check was any good is still a real question, and the rest of this chapter puts a price on it. So the circle moves rather than vanishing. It moves somewhere you can look at it.

12.3 Three things on hand

Here's what I mean by being entitled to lean on a claim. Three things have to be there.

An object, kept in the machine's state: a grouping, a list, a count, something you could print out. A belief doesn't count.

A checked relation between that object and the claim: a check that actually ran, with its result on the record. "A check exists somewhere" doesn't count.

A paid event: a step of the run where the object was put in place, and which the price list charged at least one. In a Thiele machine, the step that settles something is a paid step, because the toll charges it.

Keep the second and third apart, because they come apart. A note saying "coin under cup one, checked" can be written without anyone lifting a cup: an object, but no checked relation. Now actually lift the cup, find the coin, and write it down, but let that step be free, so the box never gets ticked and nothing gets charged. The object and the check are both there. The paid event is missing. And the other way round: ticking the box costs one, but the toll alone says nothing about what the step looked at. Payment by itself supplies no check.

And here's the sentence most likely to get lost: nothing in the bare machine makes a later step read the box. A move is handed the state and goes where its rule says, and its rule is allowed to ignore the box completely. So when I say a run is entitled to lean on a claim, I mean what somebody checking the run afterwards may conclude. The machine doesn't have to act on it, and unless somebody builds it to, it won't.

Line up the three things this book has called by related names, so they don't blur:

The bare toll: a paid move from not done to done. Nothing says a check happened at all.

Entitlement: an object, a check that ran with its result on the record, and a paid event. Nothing yet says the check was about this very claim.

An established record: from a clean start, it goes up only right after a check of the same property, on the same thing, with nothing changed in between, and the raise pays.

Even that last one is missing two things. Whether the check is honest is a fact about the checker, not something the record holds. And nothing asks whether the check could have failed: a check that says yes to everything is perfectly legal, true, and useless.

Why does the honest checker carry so much? The philosopher Linda Zagzebski argued in 1994 that any account of knowing that leaves even a small gap between having good reasons and being right lets a Gettier case through. Take a case where bad luck makes the reasons point wrong, then add good luck that makes the belief true anyway. The three conditions plus an honest check close that gap, and only half of honesty is needed, the half that says no when the claim is false. A check that can be wrong, a set of tests that sometimes passes broken code, puts you right back in her gap, and then the record establishes something smaller and still true: this procedure said yes.

And who vouches that the check is honest? The record doesn't. Honesty is about what the check does on every possible input, and a finished state only holds this run. A proof about the checker can be written down and checked, but the checking is a check too, with its own honesty to vouch for, and somewhere the chain stops at a proof somebody accepts. In this book that proof is read once, outside every run, and nothing on the record pays for it. So an established claim is two things stuck together: something about the run, which the record carries, and something about the checker, which a proof carries.

The engineering version of this was built in 1996 and 1997. George Necula called it proof-carrying code: when you send a program to a computer that doesn't trust you, you send a proof with it that the program follows that computer's safety rules, and the computer checks the proof that arrived before it runs anything. Not your reputation, not "a proof exists somewhere," but the one in the box. Stephen Cook's certificates from 1971, in chapter four, have the same bones: a claim comes with something that's quick to check, and the checker checks that.

And David Hume, in 1748, arguing about miracles, laid out what a receipt replaces. He weighed testimony against evidence and wrote that "a wise man, therefore, proportions his belief to the evidence." The way I read him, his whole argument is a method for how much to trust a witness when you have no receipt. The move in this book is to stop weighing. A claim on the record is either backed by its object and its paid event, or it's testimony from outside, and a run that leans on testimony hasn't earned it itself.

Robert Brandom, a philosopher at the University of Pittsburgh, built a whole account of language on two words this book leans on, commitment and entitlement,

in Making It Explicit in 1994. In his picture, as I read it and as his reviewers describe it, people talking keep score on each other: what each speaker is committed to, and what each is entitled to claim. Many claims count as entitled by default until somebody challenges them, and entitlement can be borrowed. I can claim a thing on your say-so and point to you if I'm challenged. Lay mine beside his and the differences are the content. His score is kept by the other people in the room; here it's in the machine's state. His entitlement often comes by default; here it's an event that has to happen and gets charged. His borrowing is my testimony, let back in on a leash. Borrowed entitlement is only as good as the copy is genuine, which is chapter seven's problem.

12.4 Changing your mind about the cups

There's a whole field about how a reasonable mind should change what it believes when something new comes in, and I couldn't keep its people apart until I put them back on the table with the cups.

Your friend believes the coin is under cup one. They can learn something that doesn't clash with that, say cup two is empty, and just add it. They can lift cup one, find nothing, and have to change their mind about a table nobody touched. Or you can tell them you moved the coin. Those last two felt alike to me at first. They aren't. In one the friend was wrong. In the other the friend was right, and then the table changed.

Krister Segerberg, a Swedish logician, built a logic in 1998 for accepting something so it can never be given up. Isaac Levi split taking on a belief into a deliberate kind and a routine kind: deliberate is reasoning your way there, and routine is accepting whatever a procedure you trusted in advance hands you, the way you'd accept a reading off a thermometer you've already decided to trust, or what's under a cup when you lift it. The trust goes in once, before the answer comes. A passing check behind an established record is routine in exactly that sense.

Neil Tennant and Gösta Grahne I first took as two separate complaints. They're one move made from opposite sides. The target is the best-known impossibility result in that field, Peter Gärdenfors's theorem from 1986, that a natural rule for "if" statements can't live with reasonable mind-changing. Tennant argued that it only works by treating some additions as changes of mind: the friend who learned cup two was empty gets counted as having changed their mind. Grahne argued that it only bites real changes of mind, so the friend whose coin was moved is out of its reach. Side by side, they squeeze the theorem down to one case: being wrong about a table nobody touched.

The line this book leans on hardest is between those two kinds of change. Revising is changing what you believe about a world that stayed put. Updating is changing it

because the world moved. Hirofumi Katsuno and Alberto Mendelzon drew that line for databases in 1991; Arthur Keller and Marianne Winslett had been working on updates since the mid-1980s. And look at what the friend sees: an empty cup one is the same empty cup whether they misremembered or you moved the coin. What they see can't tell revising from updating. That's a window with two states behind one view. What tells them apart is a count of how often the cups were touched. A machine can keep that count, a version number for each thing, and treat every write as an update: a write moves the version number, so every fact about the old version stops counting for the new one.

Alexandru Baltag, Bryan Renne and Sonja Smets gave a logic in 2014 that keeps apart three things Gettier's cases blur: good belief, knowing with the reason written out, and evidence that settles it. The friend who watched has good belief. The friend who lifted the cup has evidence that settles it.

Put that together and it says something exact about this book's record. A record that's never wrong and never corrected would be a stubborn fool's dream if its sentences were about the present. They aren't. Each one says a check passed on one version of one thing, and that sentence stays true after the version number moves on. The record can be never wrong and never corrected only because its sentences have the version stamped on them. Strip the stamps, and the first write would turn it into a liar that can't take anything back.

Two things give when you press on that. The record is Segerberg's never-give-it-up acceptance every time, with no changing its mind in it. If a check turns out dishonest, the only fix that fits a record like this is a second record on top saying "withdrawn," the way a list of canceled certificates works. And the record is never wrong partly because it only talks about past versions. Whoever reads it wants now, and getting from "held at that version" to "holds now" crosses every write since. The record doesn't make that step for you.

Chapter eight met the logicians who taught knowing to count, Ho Ngoc Duc and Artemov and Kuznets. One more thing about them, now that a record is on the table. Duc's knowing points forward, at what an agent could come to know with more steps of reasoning. A record points back, at what was checked. They're one arrow, read from opposite ends. A check is one of Duc's trains of thought, and what I've added is a direction and a charge on stopping. Where the charge sits follows from one choice: I want it to bind every program the machine runs, and a charge that does that has to sit inside the step, since a program can skip any charge kept somewhere it can write.

Three things entitlement isn't. It isn't permission. A system that says who may write to what grants that from outside and says nothing about truth. It isn't "it's true, so I can use it." That's the cups. And it isn't "right by luck." A shortcut that gives the right answer because the inputs happened to cooperate is the stopped clock.

12.5 Count the rivals

Now the number, because a book about what keeping costs has to say how much, and the idea fits in a game with three playing cards.

Shuffle cards marked 1, 2 and 3 face down. Sort them using only questions of the form "is this one bigger than that one?" and count the questions. There are six orders the cards could be in. Each question has two answers, so after one question you've split the six into at most two piles, after two at most four, and you need to get down to one. Four is less than six, so two questions can't always do it, and three can. Draw the tree of questions and you'll see it: six leaves, three deep. The number of questions you need is at least the number of halvings it takes to get from all the possibilities down to one, rounded up. Donald Knuth gives this, in the sorting volume of The Art of Computer Programming, as the floor for all sorting by comparing: a tree so many questions deep has at most two to that many leaves.

Is the floor always the price? Try five cards. There are 120 orders, and seven halvings get you under that, so you need at least seven questions, and seven is enough, though not by the obvious method. Finding the way is a good evening with a deck of cards; Knuth has it, under the name merge insertion, from Lester Ford and Selmer Johnson in 1959. Now take twelve cards. They can be in 479,001,600 orders, so the count says at least 29. Ford and Johnson's method takes 30, and in 1965 Mark Wells showed by a computer search that nothing does better: twelve cards need 30. The floor was 29 and the price is 30. So the count is a floor and only a floor. It never promises the floor can be reached.

Ralph Hartley measured information exactly this way in 1928, before Shannon: by the number of possibilities, with no chances in it, just the count. That's the measure I use here, and it's Hartley's.

The same count works with something you can hold. Put three coins under three cups. Each coin is heads or tails, so there are eight possible rows. Somebody tells you heads under the first cup and tails under the second. Lift the first cup and compare: if it matches, four possible rows are left. Lift the second and compare: two left. The last coin could still be either. Eight, four, two.

This is a parlor game, and an old one. In A Christmas Carol, in 1843, Scrooge's nephew Fred plays a game called Yes and No at his Christmas party. He thinks of something, and everybody else asks questions he may only answer yes or no. He's thinking of an animal—a disagreeable, savage one that growls and grunts and lives in London—and the answer is Uncle Scrooge. Most people know it as twenty questions, and twenty is a good number for it. Twenty questions that each split the field in half can single out one of about a million things. But the halving only happens because Fred answers truthfully about what's actually in his head. Lie once and the count still shrinks, around the wrong animal. The cups showed the same

thing: the four and the two are real because you lifted the cup and compared. Being told what was under it wouldn't have made them real.

Now put it on a run. Say a run comes with a list of possibilities still on the table, and a shorter list with some crossed off. Picture the crossing off as a tree of questions, like the card tree. If the tree is no deeper than the number of paid steps the run took, and it has enough leaves to cover the narrowing, then the run paid at least the number of halvings it takes to point at one thing on the long list, minus the same for the short list.

Chapter six is why the payment has to go in as an assumption. Looking can narrow a list without costing anything: the demon's watcher narrowed two candidates to one for free. So the narrowing alone doesn't force anybody to pay. Wherever this book puts a price on narrowing, the payment goes in out in the open: either a tree no deeper than the paid steps is handed in, or the run does the narrowing itself, with every test step charged at least one and every test keeping at least a set share of the list.

That last condition does real work, and you can see it fail. Take a list of eight things and one test: "is it this one?" If it passes, one thing is left. That's three halvings for one paid step. If every test has to keep at least half the list, that test isn't allowed, and with it allowed, the bound breaks. So the bound is: if every test keeps at least a share of one over two-to-the-k, the halvings dropped are at most k times what the run paid.

12.6 Narrower means a stronger test

One link here is pure logic, and Plato's third account is the whole of it. Take a list of candidates and something you can observe about each one. The list's test passes an observation when some candidate on the list would show it. Shrink the list, and the test can only get pickier: anything the short list's test passes, the long list's test passes too.

Pickier isn't the same as strictly pickier, though, and the cards show the gap. Take all six orders as the long list, and as the short list the three where card 1 sits left of card 2: 1-2-3, 1-3-2 and 3-1-2. Observe "is 1 left of 2?" Every survivor says yes, and the dropped 2-1-3 says no. So the long list's test passes yes and no, and the short list's test passes only yes: strictly pickier. Now, with the same two lists, observe "is the first card a 3?" The survivor 3-1-2 says yes and the other two say no. Among the dropped ones, 3-2-1 says yes and 2-1-3 says no. Both tests pass yes and no. Same strength, even though three orders are gone.

What made the difference is a dropped candidate whose observation no survivor shares. That candidate is Plato's mark, the thing that sets this case apart from everything it could be confused with, turned into something you can check. Narrow

the list and keep one candidate like that, and the test is strictly stronger. Narrow it without one, and the test may not have moved at all.

12.7 What you've got

Put the pieces on one list. Narrow a list of candidates with a test that tells some dropped candidate apart from every survivor. Find the paid step. Count the charges against a tree the run paid for. Then three things follow, on any Thiele machine: the short list's test is strictly stronger than the long one's; some step of the run took the record from not done to done, and that step was charged at least one; and the run's charges add up to at least the halvings the list lost.

A statement like that is true by packaging. It takes the conditions in and hands the conclusions back. The right word for it is a checklist, a checklist for leaning on a claim. It doesn't discover anything about leaning.

And a checklist isn't nothing. On 30 October 1935, at Wright Field outside Dayton, Ohio, the Army was evaluating Boeing's new four-engine bomber, the Model 299. Shortly after take-off it plunged back onto the field and caught fire; that's how the wire report of that day has it, and later histories add that it pitched up and stalled first, which is what controls locked in place would do. Two men died of their injuries: the Army's pilot, Major Ployer Hill, and Boeing's chief test pilot, Leslie Tower. The airplane was fine. The crew had left the gust locks on, a mechanism worked from the cockpit that holds the control surfaces still while the plane sits on the ground. There's a famous line that a newspaper called it too much airplane for one man to fly. I couldn't find that line in any paper from 1935, and a search of the scanned newspapers of 1935 and 1936 didn't turn it up either; what the papers that November report, like the New Castle News on the 12th, is the locked controls. Atul Gawande, a surgeon who writes about how people get complicated work right, gave the line in a New Yorker article in December 2007, "as a newspaper put it," without naming the paper. The words are older than his article, though. Edward Jablonski's 1965 history of the Flying Fortress says the talk after the crash was that it was "too much airplane for one pilot to handle," Martin Caidin's 1968 book has "rumors that the 299 was too much airplane for any one pilot," and the exact line is in DeWitt Copp's A Few Great Captains in 1980, as word reported from the War Department, which Copp says wasn't true. So it's an old saying about that airplane. Whether a newspaper printed it in 1935, I still can't show you. What I can find is what came after. The answer was a card: a list of things to check, in order, every time. The most careful history of it I found is by Walter Boyne, a retired Air Force colonel. By his account it wasn't the first written routine for a pilot; those go back before the First World War. And it's this crash, he says, that made the checklist the way a crew flew. The airplane went on to be the B-17. Nothing on that card was a

discovery. Every item on it was something the pilots already knew. What the card did was make sure none of it got left out when it mattered.

12.8 What standing behind a grouping gets you

Chapter ten ended on buckets. The toll never needed them: it prices any yes-or-no question. Here a grouping gets a job, and an exact one.

Put two red buttons and two blue buttons on the table, numbered 1 to 4, with 1 and 2 red. The question is color. Draw circles round 1 and 2, and round 3 and 4. Call a grouping sound when every two buttons inside one circle are the same color. This one is. Split the red circle into two circles of one button each, and it stays sound; splitting can't put two different colors together. Join 1 and 3 into a circle instead, and it fails. Rename the circles North and South and nothing changes, because soundness only looks at which buttons share a circle.

Now the record. Whatever keeps the fact that a grouping was found sound, a mark on a card or a box in a machine, is a question that goes from not done to done and stays, so everything the first chapters said about that applies to it: the raise pays the toll, and keeping it takes two arrows meeting or fresh paper. None of that depends on what the mark is about. What the mark lets you lean on does depend on it.

Here's exactly what it lets you lean on. If your circles are sound, then any way of cutting them smaller is sound too, whatever the question is: every pair the smaller circles keep together was already together, and already matched. And for any grouping that isn't a cutting-smaller of yours, there's a question your circles pass and that one fails. To see it, find two buttons it puts together that your circles keep apart. Give everything in the first button's circle one color and everything else another. Your circles are all one color each, so they pass. The other grouping has both colors in one circle, so it fails.

So standing behind a sound grouping entitles you to exactly the groupings you get by cutting it smaller. A grouping that isn't a cutting-smaller can still happen to be sound for this question; paint all four buttons red and every grouping passes. But nothing in standing behind yours promises it. Change a button's color after the mark was made, and the mark is still about the old colors. A record says what held when it was made, nothing more.

That's the three things from the start of this chapter in their smallest case: the object is the grouping, the checked relation is soundness for the question, and the paid event is the raise of the mark that keeps it. Like everything in this chapter, it says what somebody checking the run may conclude.

Which leaves the word on the cover. I named the toll "No Free Insight," and I've been using "insight" as if it meant one thing. The next chapter says what it means.

CHAPTER 13

What insight is

I named the toll "No Free Insight" before I had any careful account of what insight is. The book has been narrowing it down ever since. Chapter six took looking, copying and finding out off the list: those can be free. What's left is the part that stays. So it's time to say what that is, in a way that someone who studies real insight, in real heads, wouldn't laugh at. Who are those people, and what did they find? Here's what I ran into, pushed on as I went.

13.1 Sultan and the sticks

In 1913 the psychologist Wolfgang Köhler went to Tenerife to run a research station with a colony of chimpanzees, and the First World War kept him there until 1920, five years longer than planned, with the chimpanzees to watch. The famous one is Sultan. Food is outside the cage, out of reach. Two sticks are inside, each too short, but one fits into the end of the other. Sultan tries each stick and fails. He sulks. He plays with the sticks. And then, by Köhler's account, at a moment you can see, he joins them and reaches the food. Köhler's word was Einsicht, which came into English as "insight," and the whole school of psychology he belonged to, the Gestalt school, said that's what thinking is: a rearranging, the moment two sticks stop being two sticks and become one tool. Piling up tries until one works was, for them, something else.

Karl Duncker gave the human version, in German in 1935 and in English in 1945, five years after he died at thirty-seven; he'd spent his last years working with Köhler at Swarthmore. You can set it up yourself: a candle, a box of tacks, matches, and the job of fixing the candle to the wall. The hard part is seeing the box as a shelf while it's sitting there being the thing that holds the tacks. And he found that if you hand people the box empty, with the tacks poured out beside it, they solve it far more readily. Same objects. What changed was which relation between them was on the table.

I like both of these for the same reason: the insight didn't add an object. Sultan had both sticks the whole time. The box was there the whole time. What arrived was a relation, a piece of structure in chapter ten's sense, and it arrived all at once.

Keep that in view, because the thing the toll charges for also arrives in one step, and at its best it's also a relation.

13.2 It isn't a feeling

A feeling of getting closer is a different thing from solving the problem. In the 1980s the psychologist Janet Metcalfe asked people to solve problems while rating, every fifteen seconds, how close they felt to the answer, how "warm." On ordinary problems, like algebra, the warmth climbs steadily and peaks at the answer, which is what you'd expect. On insight problems, riddles of the candle-box kind, the warmth stays flat, flat, flat, and then the answer arrives. And the people who felt warmest beforehand were, if anything, more likely to be wrong. She put that in the title of a 1986 paper: premonitions of insight predict impending error. That's about how well warmth predicts in those tasks. It doesn't show that solving something comes with no feeling, or that everybody follows the same curve.

Try it yourself. Write a number from one to seven every fifteen seconds while you do a long sum, then while you do a riddle, and compare the two rows of numbers.

And the most famous insight story there is fits Metcalfe uncomfortably well. Archimedes, asked by Hiero, the king of Syracuse, whether a goldsmith had cheated him on a crown, gets into a bath, sees the water rise, and runs home naked shouting "eureka," I've found it. That comes from Vitruvius, a Roman writer on architecture, writing about two centuries later. Is it true? Galileo Galilei, at twenty-two, in 1586, took it as known that Archimedes caught the goldsmith. What he didn't buy was the method. Water spilling over the side, as Vitruvius tells it, was to him "a crude thing, far from scientific precision," and he guessed some writer had written down a rumor. So he wrote a little paper, La Bilancetta, proposing a balance weighed in air and in water instead, which he thought was probably what Archimedes actually did, since it rests on Archimedes's own results. Whether there was ever a bath, or a shout, nobody can check now.

Look at what the legend kept: the bath, the shout, running home naked. The feeling. What it left shaky is the part that would count as the check, the test that could have caught the goldsmith. The bit everybody remembers is the bit that tells you least.

So an account of insight that keys on the "aha" is keyed to something the evidence says is unreliable. Whatever insight is, it isn't the feeling of having it. That's a relief, because a machine doesn't have feelings, and a word that only meant a feeling would be cheating if I hung it on one. The version this book can price has to be keyed to something that happens in a step. The question is which step.

13.3 What happens when a computation learns

Keep the board from chapter four in front of you: the NO box, the YES box, the tally line, and the stamp. Add one more move that does nothing to the boxes; call it WORK. It stands in for every ordinary step a machine takes.

The first thing that can happen is plain looking. Something comes in and gets written down. A count ticks. Nothing is claimed about what it means, and the box doesn't move. The demon's notepad in chapter six was this. The statistician Dennis Lindley measured it in 1956, in a paper about how much information an experiment gives you, as how much your uncertainty is expected to drop. That's a real quantity, and it's about what was seen. On the price list it can cost nothing: the toll says nothing at all about a move that doesn't take NO to YES, so a price list may charge it zero. "May" is the word that matters. A price list is allowed to charge for looking anyway. And in a real device a look takes something too: chapter six showed it uses up a blank page, borrowed until the page is wiped.

Lindley's measure has a back story, and it runs through Bletchley Park. I. J. Good, who was Alan Turing's main statistical assistant there in 1941, wrote in 1979 that Turing had worked out how much evidence an experiment should be expected to give, to judge one of the codebreaking methods before running it, partly getting to Lindley's idea fifteen years early, in work that stayed secret. Turing measured evidence in "bans," with a tenth of one called a deciban, and the name came from the town of Banbury, where tens of thousands of the sheets used in that method were printed. And Good adds that Turing didn't know the measure he was using had already appeared in 1878, in a paper by Charles Sanders Peirce, the same Peirce from chapter nine, who'd called it the weight of evidence. It changes nothing here, but it shows how well trodden this kind of thing is. For a century and a half careful people have measured what an observation tells you, and every one of those measures is about what was seen. The toll leaves that side of the line alone.

The second thing that can happen is the raise. Some move takes the box from NO to YES, and that's where the toll applies. That move costs at least one, and every run from NO to YES pays at least one somewhere. What gets charged, at its best, is the Gestalt thing: a relation committed to, one tool claimed out of two sticks. I say "at its best" because chapter four was careful about what YES means: the coin is in YES, and nothing more. A raise that writes down which relation it's committing to carries Sultan's tool with it. A bare stamp commits to nothing at all, and it pays exactly the same.

The third thing isn't a kind of step. It's something about the price: it's the same, whatever got settled. Say the YES state also shows something striking, a deep relation, a result nobody could have guessed, Sultan's two sticks become one tool. The toll charges the raise at least one. It doesn't charge the striking raise more,

because the reason it charges anything only looks at the box before and after. On the board, the stamp from NO makes one mark whether you were thinking of Duncker's box or of lunch. A different price list could charge them differently. The toll doesn't make it.

The candle box has a footnote that says the same thing from the other side. In 1962 the psychologist Sam Glucksberg ran Duncker's problem with money on it: some people were told they could win five or twenty dollars depending on how fast they solved it. With the tacks in the box, the people chasing the money did worse than the people who weren't. With the box empty the problem was easier for everybody, and I'll leave what the money did there alone. Paying more didn't buy the insight, and on the hard version it got in the way. That's a fact about heads, not records, and I won't push it further. But it makes a nice parallel to the board. On the board a deeper raise isn't charged more, and in Glucksberg's room a bigger prize didn't bring the insight sooner. In neither place does the price measure the depth.

So: looking can stay off the bill, the raise can't, and the toll doesn't charge more for a deeper claim. Who decided that split? Not the prices alone. The toll prices exactly one thing, the move that settles. A price list is allowed to put a floor under more moves than that, and allowed to leave others at zero. The sorting into "moves that settle" and "moves that don't" is by whether the box moves, and that sorting is mine. The part that's forced is the floor under the move that settles, once you grant finite chairs, done staying done, and a price on two arrows meeting. That's chapter three. Anything a price list charges beyond that is a choice, and any real machine owes you its whole price list so you can see which choices it made.

13.4 Test, then commit

The philosopher Karl Popper, who you met in chapter nine, said a theory earns its standing by being exposed: you make a guess, you try to kill it, and what survives a real attempt counts for something, where "real" means the test could have failed. His word for that earned credit is corroboration.

Add a move to the board: CHECK. It looks at something, a sum, a coin under a cup, and it can come out pass or fail, but it doesn't move the coin. Charge it whatever you like. Charging it on the attempt, pass or fail, fits Popper, because what made his test count was that it really ran and could have failed. CHECK is the test, and it never takes NO to YES. The stamp is the commit. So the honest way to go is two moves: test, then commit, with two separate costs.

But nothing on the bare board makes the stamp wait for the check. The stamp takes NO to YES from anywhere. Press the stamp with no CHECK before it, and you land in YES and make one mark, the same box and the same mark as the player who checked first. Their tally lines differ by whatever CHECK was charged, and if

you charged it nothing, they don't differ at all. The receipt says a commitment was paid for. It doesn't say the commitment was earned. Popper's credit needs the test. The toll doesn't.

You can build a board that makes the stamp wait: let CHECK leave a mark when it passes, and let the stamp only take NO to YES when that mark is there. That's chapter eight's established record, and it's a good design. It's a design. Nothing in the definition asks for it. The gap between the word and the bare machine is real, and I'm leaving it where you can see it: the toll prices the commit, and whether a test came first is a fact about the run, or about a design somebody chose.

13.5 What the word means

So here's insight, in the sense this book prices: a distinction made to stay. A move that takes something from not settled to settled, for good, and that move costs at least one, on any machine with the toll, whatever it's about and however deep it is. At its best, what's settled is a relation committed to, a claim checked and closed. A bare stamp that commits to nothing pays the same, and the toll can't tell the two apart. An established record is one way to make sure it's the first kind.

Three limits, each from a different place. From Metcalfe, it isn't a feeling, so a machine's version has to be a step. From the board, the floor is the same for a shallow commit and a deep one, so price isn't depth. A machine can pay a lot to check something trivial, and Duncker's box, a real insight, is one small commit. And from test-then-commit, a paid stamp proves nothing on its own. Price and correctness are separate books.

Which leaves a suspicion I'd have in your place. Everything so far has been one box on one sheet of paper, with a rule I chose and a count I added. Is the shape of that box something I picked, or is it what any box the machine itself turns on has to look like, on any machine at all? That's the next chapter.

CHAPTER 14

A box over any machine

For the whole book so far I've been building up a box and a tally, and you'd be right to suspect that the whole thing is special pleading for one little board. So this chapter answers that suspicion with no particular machine in it at all. Take any machine that steps, anything with states and one next state for each. Put a box over it that watches it, under rules I'll spell out. Then ask: what can the box be?

The answer is one shape, and it's a shape you can build from a cup and a die.

14.1 A cup and a die

Put a cup upside down on the table. The rule is: the first time a six comes up, turn the cup right side up. There's no rule for turning it back over. Now roll a die twenty times. If a six comes up, the cup turns over on the first one and stays turned over. Write down which roll did it. That number is where the record happened, and the cup afterward is the record. If no six comes up, the cup stays down.

To make the die part of a machine with one next state, write the twenty rolls down first and read them off one at a time. The machine underneath is the list and a finger pointing at the next entry; each step moves the finger one place, whatever the cup is doing. The question the cup asks at each step is: is this entry a six? Nothing about the cup changes which entry comes next.

The cup is a latch, the same thing you built with two boxes in chapter four. Electrical engineers have had the circuit since 1918, when two British physicists working in London, William Eccles and Frank Jordan, applied for a patent on a two-valve arrangement that holds whichever state you last put it in. Their 1919 paper calls it a trigger relay. It became the flip-flop, and the memory inside the device you're reading this on comes from it. Cut its reset wire and what's left has one input, "set," and one rule: the new state is the old state, or set. Once set, it stays set. The cup is that circuit with its reset wire cut.

Did anybody ever build a whole memory out of cups with no reset wire, on purpose? Somebody did, and for a reason close to this book's. In the mid-1950s Wen Tsing Chow, an engineer at the Arma division of the American Bosch Arma Corporation in Garden City, New York, was asked by the United States Air Force,

according to the Wikipedia history of the programmable read-only memory, for a safer way to hold the fixed numbers, the targeting constants, inside the guidance computer of the Atlas missile. What he and his colleague William Henrich patented, filed in 1957 and granted in 1962 under the plain title "Storage matrix," was a grid with a connection at every crossing, where you set the numbers by taking out the connections you didn't want, or burning them out with too much current. The word burn is right there in the patent, and people still say they burn a chip. A fuse blown in one stays blown. That proves nothing here, because nothing below cares what a cup is made of. But it's good to know the shape wasn't invented for a sheet of paper. Somebody needed it, for higher stakes than a die.

Here's the claim, loosely first: every box that answers only to what the machine is doing, and never comes back down, is a cup with no reset wire, turned by some event the machine passes through.

14.2 What "answers only to the machine" has to mean

The content is in the rules, so here they are.

Call the machine underneath the base: the part the box watches. Now build a bigger machine that has the base inside it, plus whatever else you like, plus the box and a tally. The bigger machine has to run the base faithfully: start where the base starts, step when the base steps, and stop when the base stops. It can carry as much extra hidden stuff as it wants. The base doesn't get to see any of that, and the extra stuff doesn't get to change what the base does.

Now the five rules for the box and the tally.

The box's next position depends only on what the base is doing right now and where the box is right now. "What the base is doing" means the state it's in and, when it has more than one move from there, the move it's making, like the stamp and the WORK move on the board from chapter thirteen. Nothing else. No clock on the wall, no hidden dial, no outside hand. I'll call that answering to the machine.

The tally never goes down.

The step that turns the box up costs at least one. That's the toll.

Once up, the box stays up.

And some run, from a place the machine can start, actually turns it up. Otherwise the box is decoration.

Here's what follows. Ask the bigger machine what it would do with the box down, above each state of the base, and each move when it has a choice: does the box go up on the next step, or not? Call that answer the event. Since the box answers only to the base, that's one answer per base state, or per state and move; no hidden stuff can change it. Now take any actual moment. If the box is down, the event is exactly what happens to it. If the box is up, it stays up, whatever the event says. Put

those together: the box's next position is "already up, or the event holds right now." That's the cup's rule. The cup turns over the first time the event holds and never turns back.

Look at which rules that used: only two of the five. Answering to the machine, and staying up. The tally, the toll, and the reachable turn played no part. A cup is what you get from "answers only to the base" and "never comes back down," all by themselves.

The other three rules matter for the other direction. Pick any base and any event the base can actually reach. Build the cup on that event, price the turn at one, everything else at zero. Check the five rules: the cup only looks at the base and itself; the tally only goes up; the turn costs one; the cup stays up; and the run that reaches the event turns it. So every cup is a box like that too. Put the two directions together: boxes that answer only to the machine and never come down are exactly cups, nothing more and nothing less, over any machine at all.

The closest relatives are in proving programs correct, and they're old. Martín Abadi and Leslie Lamport, in 1991, added "history variables" to descriptions of systems, extra state that records what has happened and never steers the system. A history variable that only ever turns on is the cup. And there's a logic for talking about the past, set out by Orna Lichtenstein, Amir Pnueli and Lenore Zuck, with an operator "once": "once h" is true now exactly when h has been true at some point so far. A program watching a running system keeps "once h" in a single bit with exactly the cup's rule, and Klaus Havelund and Grigore Roşu built watchers like that. What this chapter adds is the other direction: a box that answers to the machine and never comes down can't be anything else.

One thing about that argument cuts both ways: it never asks what the base is. You decide that, by deciding what goes in the base and what goes in the extra stuff. Draw the line low, with the base being only the machine's own working state, and "answers only to the base" is a real demand. Draw it high, put every table and every rule into the base, and it's easy to meet. The cup comes out the same either way. So the argument holds wherever you draw the line, and that's exactly why it can't tell you where a record's meaning lives. The meaning lives in what the event looks at, and that's in the base, by how you drew it. When you meet a real machine, ask it first what it puts below the line.

14.3 A base that never merges

What about a base where no two arrows ever meet? Can it carry a cup?

It can, and you can see exactly what it pays. Keep every earlier position of the cup on a strip that grows by one entry each step, newest first. Then the whole machine still never has two arrows meeting: if two whole states land in the same

place, the strips are the same, so the old cup positions were the same, and the base parts were the same because the base never merges, so the states were the same. But the strip gets longer every step, so the machine never comes back to a state it's been in. That's the fresh-paper column from chapter two. And on a machine with finitely many states, there's no fresh paper to be had, so a box that stays up can't be turned on by a move where no two arrows ever meet. That's the chairs, again.

People who study processes that talk to each other ran into the same thing from the other side. Vincent Danos and Jean Krivine, two computer scientists in Paris, wanted to pin down what a transaction is. In 2004 they built a version of Robin Milner's calculus of processes where every step can be rolled back, and in 2005 they modeled a transaction in it as a process that wanders freely through everything that can be undone, until it does something declared impossible to undo: the commit. As I read it, they start from steps that all come undone and ask what has to be added before anything sticks. This chapter starts from a box and asks what the step under it has to be. Theirs is declared. I thought mine was forced by counting, since on a finite machine the first turn of the cup makes two arrows meet whatever anyone calls it. That claim has a soft spot. I choose the event, and on a base that grows its history forever, nothing merges at all, so there my staying up is declared too.

14.4 The clean start

There's an arrow of time in this, and I should say where it came from, because I put it there by hand. The cup starts down. That's the only reason a turned cup means the event happened: it was down at the start, and only the event turns it. Turn the cup up before the first roll and it tells you nothing about the die. So the clean start is assumed. It's a small version of what the philosopher David Albert, in 2000, named the past hypothesis: an assumed low-entropy beginning of the universe, an idea that goes back to Boltzmann. Several accounts of why records point to the past lean on it. Whether every trusted record leans on it is their argument, and I'm not settling it. Every result in this book that reads a turned cup as news that the event happened assumes a clean start and starts from there. The cup argument above doesn't need it, and that's exactly why, on its own, it can't say the event happened.

When I went looking for who else had found the arrow hiding in the start, I got five papers that read like five unrelated answers. They lined up once I asked each the same question: what does the cup being down at the start rest on? There are two places to put it. At the beginning, as a fact about how things started. Or in the structure of what causes what.

David Wolpert, a physicist, sorted memory systems in 1992, and one of his kinds works only because it begins in a prepared state, set up by a process that runs one way. That's the cup put down by somebody before the game. Wolpert and Jens

Kipper, a philosopher, put that sorting on firmer footing in 2024, and they say plainly that they have no formal proof this kind can only remember the past. I trust them more for that sentence.

Aram Ebtekar and Marcus Hutter built a toy world in 2024 where every step can be reversed, imposed a clean start on it, and got records of the past and none of the future. As I read their route, the clean start goes through cause and effect first, and the arrow comes out of that. Thomas Blanchard, a philosopher, got the same lopsidedness of knowing out of a condition on cause and effect, also in 2024. So he starts in the middle of Ebtekar and Hutter's road. One is a toy world and the other is philosophy, and the fit between them is my reading.

And Carlo Rovelli, two years before his paper on the cost of choosing from chapter six, studied traces in a 2022 paper called "Memory and Entropy," and concluded that a memory turns a low-entropy start into information you can use. He's at the first place, and he says what the beginning gets spent on. Read next to the cup: that's the blank the record uses up. The cup can say yes once because it started at no.

So does the cup explain why records point backwards? It doesn't, and I'll say that plainly. The step runs forward and the runs start clean, and anything built on those two can only hand the arrow back. Even the strip that grows forever starts empty, so it leans on the same blank. On the map, this book sits at the first place, with Wolpert and Rovelli, and it sits there by assumption.

Constructor theory, David Deutsch and Chiara Marletto's way of writing physics from chapter two, has a 2025 paper on time saying that laws stated as which transformations are possible can't mention time at all, so duration has to come from a second system you count, a clock. This book is in that position by construction: the only time in it is the machine's own steps. The hidden clock in the next section gets thrown out by the first rule, which read next to them is a rule about which system is allowed to be the clock: the one being recorded. But their laws have no step in them, and I put a step into the base by definition. So I'm in their position only at the far end of their argument, where a clock already exists. The base is my clock, and I assumed it.

Physicists have built this cup on paper in quantum mechanics too, and it behaves the way the reversible base above says. The question that sorted them for me is where the cup puts what it forgets. Klaus Hepp, in 1972, wanted a measuring device inside quantum mechanics itself, and studied a particle that flips spins in a long chain as it passes, now called the Coleman-Hepp model. John Bell took it apart in 1975: the record is exactly definite only in a limit that's never reached, an endlessly long chain and an endless time. That's the fresh-paper column—a history that never ends. Jonathan Halliwell used a two-state detector that flips once when a particle passes and is kept from flipping back by a large surrounding environment, and his summary calls it "effectively irreversible." I like that he puts the word "effectively"

right there where you can see it. And Murray Gell-Mann, who named the quark, and James Hartle gave the property a name: the permanence of the past, past alternatives that stay settled as the story goes on. Hepp's chain had to be endless. Halliwell's had to live in an environment. Mine stays up, and the toll writes a unit on a tally. That's fine in a model of computation, and it's exactly where this stops being physics.

14.5 Both rules do work

A result with two conditions should have two counterexamples, one for each condition dropped. It does, and they're the two ways a record goes wrong in the world.

Drop "stays up." Take a base that just counts, and a box that flips at every step: up, down, up, down. It answers to the machine. It doesn't stay up. And it's not a cup on any event. That's the coin from chapter three, the toggle. Working state dressed up as a record. If a take-back has to stick, it's a second cup, "withdrawn," which is how lists of revoked web certificates actually work. You add an entry; you never delete the certificate. Alin Tomescu and his coauthors built add-only lists like that for public logs in 2019, and a revocation fits exactly. It goes in once and stays. Two cups can also look at each other. Try it with two cups where the second only turns on an even roll if the first is already up, and you'll see the second can never turn before the first. Each is still a cup; its event just gets to look at the other one.

Drop "answers to the machine." Take a base that counts, and a box that turns up at the fifth tick of a hidden clock, whatever the base is doing. It stays up, but it doesn't answer to the machine. That's a timestamp. A timestamp says when, not what happened. A record switched by an outside party or a clock records that party or that clock, and leaves the computation out. Try it with the written-down die rolls: keep a second, hidden tally beside the list, and turn the cup when that tally reads five, then move both the tally and the finger. Start two runs at the same place on the list with the cup down, but one with the hidden tally at five and one at zero. The next cup positions disagree, although everything about the base agrees.

One more thing about the timestamp, because it's sneakier than it looks. Start it in step with the base, hidden tally and finger together. Then, on every state its runs actually reach, the hidden tally equals the position in the list, and the timestamp is exactly the cup on "this is position five." The first rule only throws it out because the rule looks at every state, including the out-of-step ones no run from a clean start ever reaches. So the honest version is: a timestamp with its clock free to start anywhere fails even on the states its runs reach. Started in step, it's a cup on the runs it has.

Permanent but not answering to the machine is a timestamp. Answering to the machine but not permanent is a toggle. A record is the thing that's both. That's the most useful single sentence in this book for looking at systems other than the ones I build.

14.6 Arguing with cups

People who study arguments built the two-cup rule for debates. Give each speaker a row of cups, one for every claim they've made. Everybody I read in this field is answering two questions about those cups: whose hand turns them, and can they come back down?

C. L. Hamblin, an Australian philosopher and logician who also came up with reverse Polish notation for early computers, wrote Fallacies in 1970, and it has a chapter where each speaker has a store of what they're on the hook for, and the rules of the debate update it, not the speaker. That's "answers to the machine," for arguments. The speaker's hand is kept off the cup. Henry Prakken, who works on AI and law, built debate games on stores like that and made the claims sticky. A claim can be taken back only after it's been challenged, and taking back has a cost. He also floats a rule that once a speaker takes a claim back, they can't take it on again, which is a revocation that stays revoked. Erik Krabbe, an argumentation theorist, names the awkward part. Taking things back is frowned on, and yet settling a dispute takes somebody taking back a doubt. A store that never gives can't settle anything, and a store that gives whenever asked isn't a store. This book answers Krabbe with a price: a take-back is a second cup, "withdrawn," and turning it costs at least one like any other turn. Where mine gives is Prakken's first condition: nothing in my cup asks why the second cup turned. Its event can look at the base and the first cup, and nothing requires a challenge.

The same field has something that looks a lot like earning. In Phan Minh Dung's 1995 setup, arguments attack each other, and the most careful way of accepting them starts from the arguments nobody attacks and keeps adding the ones they defend. Things go in and never come out: a row of cups turned one at a time. Pietro Baroni and Massimiliano Giacomin, and then Martin Caminada, worked out a stronger version where every accepted argument is defended by others that don't lean back on it, so the support bottoms out. That's an earned claim, in argument form. Notice the bottom, though: an argument nobody attacks gets in on a check that can't fail, and chapter eight says a check that can't fail establishes nothing. Turn that on my own cups, because it lands. My first turn needs only that the event holds, and I charge one for it, and a charge isn't a check. So my earned records bottom out the same way Dung's do, in something let in without an argument. The one difference is that mine pays to get in.

14.7 A box that can change its mind

A box that's allowed to change its mind isn't lawless. It's somebody else's subject. Count how many times the cup may be turned. Once, from down to up, is the cup. More than once and you're somewhere else, and the people who went there sorted it by how many turns are allowed, and by how long the game runs.

The philosopher Hilary Putnam and the computer scientist E. Mark Gold, in back-to-back papers in 1965, looked at questions a machine answers by guessing and revising, where the answer is whatever the guesses finally settle on. Putnam's paper says outright that you're never sure you've got the right answer. What kind of answer can't you ever be sure you have? The way I read it: the cup's position at any moment is the current guess, the answer is where it ends up, and nobody watching can tell when it's done moving. Yuri Ershov, a logician in Novosibirsk, sorted those in 1968 by how many changes they're allowed. That's how Rod Downey, Carl Jockusch, Timothy McNicholl and Paul Schupp describe his ladder. Allow one change, from no to yes, and you get exactly the sets a program can list, the bottom rung. That one you can check yourself. If a program lists the set, guess no for every number and switch to yes when the number shows up on the list. Going back, if every guess flips at most once, from no to yes, run all the guessing side by side and write a number down the moment its guess says yes.

Now put Putnam's worry on the cup, because it lands. On the bottom rung, a yes is final the moment it shows up, and a no never is, since down means "not yet." A single cup can establish that something happened, and can never establish that it didn't. A final no would take a second cup, turned by finding a reason, and that's the two cups again. That's what the bottom rung costs. This book stays on the bottom rung.

The ladder doesn't stop at endless time either. A definition that only ever adds, stage after stage, and keeps everything at each limit, is a cup run through the infinite; Wayne Richter and Peter Aczel set those out. Let it take things back and keep only what stays put from some stage on, and you have what John Burgess called quasi-inductive definitions, which he met in theories of truth that grew up beside Saul Kripke's 1975 theory of truth built up in stages. And Joel Hamkins, the same set theorist as in chapter one, and Andy Lewis let a Turing machine keep running past endless time. At each limit, a square keeps its value if it settled, and reads 1 if it kept flipping forever. Their favorite trick is the toggle: flash a flag every time a guess changes, and at the limit the flag is on exactly when the guess changed endlessly often. Past infinity, the toggle is the clever one, because a cup can only tell you a thing happened at least once. I threw the toggle out anyway, and I'd do it again, because before the limit its reading can't be leaned on, and nobody using a real machine gets to wait for a limit.

One finite-time fact belongs beside that, so nobody reads too much into permanence. Hao Wang, in 1957, described a machine whose only write is to mark a square and which can never erase one, and it computes everything a Turing machine does. Permanence of a machine's own memory costs it no computing power, given an endless tape: each mark goes on a square nobody marked before, which is fresh paper again. What this book prices is a box raised by a step that has to work whether or not the box is already up. So the cost is in what a box can claim, not in what the machine can compute.

14.8 One shape, two choices nobody forces

"There's only one way to carry a record" packs three different claims into one sentence, so take them apart.

Given the toll and a fixed price list, is there only one way to keep the books? Yes, on every state you can reach: that's chapter eleven's one honest count, and it assumes the price list.

Is every box over a base the same machine, charges and all? No, and the counterexample is embarrassingly simple. Take any box over any base, and give it a new tally that adds each step's old cost plus one more, whatever the step is. Walk the five rules. The box still answers only to the base and itself. The tally still only goes up. The turn still costs at least one; the toll is a floor and a surcharge only raises it. The box still stays up, and the same run still turns it. So the surcharged machine passes, but it isn't the one you started with. Do it with the cup and the die: charge one for every roll, six or not. The cup turns on the same roll as before. You've changed the cost and nothing about the record. A machine that charges for running time passes, but it isn't the one that doesn't.

And the second choice nobody forces, the one that still surprises me: which event is the record? Nothing in the five rules picks it, beyond the demand that some run reach it. You saw that above. For any event the base can reach, the cup on that event, priced at one, passes. Do it with the die. One cup turns on the first six. Another turns on the first one. Both pass. Roll a one first, and until a six shows up, the two cups disagree about whether anything has happened. So which permanent box counts as "the record" is a second thing nobody forces on you. The price list is one free choice and the event is the other, and that second one is what a later chapter is about: what, if anything, makes "a check passed" the event worth recording.

Now let the cup wobble. Say whatever turns it over sometimes does and sometimes doesn't, with chances you could write on a card. Does fixing the price list fix the chances? No: two cups with the same possible moves and the same charges, one turning with a fair coin and one with a two-to-one coin, pass the same rules

and aren't the same. And a cup that, from down, can either stay down or go up, depending on chance, isn't a cup on any one event at all.

And is there one set of states, up to renaming, that every box machine has to have? No, because the bigger machine can carry any hidden stuff it likes.

14.9 What leaving the base open buys

So here's the answer to the suspicion I started with. The box isn't special pleading for one board. Over any machine that steps, a box that answers to the machine and never comes down is a cup on some event, and the argument doesn't care what the machine is. What a particular machine adds on top of the cup is a price list and an event, and both are choices. The surcharged cup shows the first. The price can change and the record doesn't notice. The two cups on one die show the second. The rules are happy with either event. That's also why the chairs in chapter two never needed the word "check": they force two arrows to meet in whatever stays.

And chapter five already showed what happens when a box has more than two positions: it's a row of cups, one for each line, each one a cup like this one, where each cup's event may look at where the row already is.

CHAPTER 15

The machine with the record dropped

Chapter seven put the machine behind a window and asked what gets through. Two states can give the same view and differ in what matters, and a price worked out from the window can't be exact if there's a pair like that. That was for one box. Now, with a whole row of cups from chapter five and the cup over any machine from chapter fourteen, I can ask the sharper version, which is also the one I care about. Take the ordinary machine, the one with no cups and no tally, and look at what's missing from it. What's the shape of everything it can't see, and what does the toll pay for, in terms of that shape?

15.1 What's left when you cover the cups

Back in chapter four you covered a coin with your hand and watched the rest of the machine run, move for move, as if the coin weren't there. Do that with everything: cover the whole row of cups and the tally. What's left is a machine that steps the same way and knows nothing about either.

That only works if covering them doesn't change how the rest moves. If some step of the machine looks at a cup and goes one way or the other because of it, then covering the cups leaves you with a machine that can't decide what to do next. So the honest condition is: nothing in how the rest of the machine moves depends on the cups or the tally. Where that holds, call what's left the uncovered machine. It's the ordinary computer.

Over any one state of the uncovered machine there's a pile: all the full states that look like that one once you cover the cups, differing only in their cups and their tally. That's chapter seven's pile behind one picture, with the window being "everything but the cups."

Every move has piles of its own too: the states that move sends to the same landing place. Call those its landing piles. A pile behind a picture is everything that looks the same before anybody moves. A landing pile is everything that ends up in

the same place after one move. Keep the two kinds apart, because this chapter turns on how they meet.

15.2 Pricing through a window, with a whole row

First the window question again, with a row of cups. Say a clerk prices each step from what comes through a window before and after it. The price never misses if it charges at least one for every step that turns up a cup. It's exact if it also charges nothing for every step that turns up no cup. A clash is two steps that look the same through the window, where one turned up a cup and the other didn't.

With one box, the fix in chapter seven was to put the box in the window. So you might guess, for a row, that the window has to show some of the cups. Try the smallest case. Let the record be a count that climbs: 0, 1, 2, and so on, with a cup for "at least 1," a cup for "at least 2," and so on. Let the window show only the first cup, "at least 1." Let the count go up on some steps and sit still on others. Once it's past one, that first cup is up before and after both kinds of step. So a step that turned up a later cup looks exactly like a step that turned up nothing. That's a clash.

So, you might guess next, the window has to show every cup. That's wrong too, from the other side. Same counting record, with two moves: one adds one, the other leaves it alone. Now let the window show only whether the count is even. You can't read the count off that window: 0 and 2 look the same through it. But look at when evenness changes. It changes on exactly the steps that add one, and on no other step. So "charge one exactly when evenness changed" is exact, every time.

That changed how I think about windows. A window doesn't have to show the cups to price them. It has to show them move.

The rest of the clerk's story carries over from chapter seven unchanged. If the window has a clash, no price from it is exact, and any price that never misses has to overcharge at least once for every step that turns up nothing but looks like one that does. That's the least it can overcharge, and there's a price that overcharges exactly that much. If the window has no clash, an exact price exists, and on a machine with finitely many states you can work it out.

15.3 What a paid move destroys

Now the piles. Chapter two showed that on a finite machine, a move that settles something for good makes two arrows meet somewhere. Chapter five showed the same for a step that turns up any cup, once cups only go up. Which two arrows? It matters a great deal.

If the two states that meet sit in different piles, the uncovered machine lost a distinction too: two of its own states became one. If they sit in the same pile, the

uncovered machine lost nothing. What got crushed was only ever under your hand: two states that looked the same once the cups were covered, told apart only by their cups or their tally, now made one.

You might guess it's always the second kind, because the cups are where everything happens. It isn't always. It depends on whether the uncovered machine forgets anything on its own.

Here's the case where it's the second kind. Say the uncovered machine, for this move, never lands two of its own states on one. Two full states meet. Cover the cups, and their uncovered parts land in the same place too. But the uncovered machine keeps its states apart, so those two uncovered parts must have been the same to begin with. So the two states that met sat in the same pile. In that case every landing pile of the move lies inside one pile behind one picture.

And here's the case where it isn't. Take the stamp from chapter four, and let the window show the box itself, so each state is its own picture. The stamp sends NO and YES to YES. Two states meet, and they sit in different piles, because the uncovered machine is the stamp itself and the stamp merges. Nobody ever had to cover anything.

One more fact about landing piles, and it ties back to chapter three. The halving rule said each unit of cost pays for at most one halving. That holds exactly when no landing pile of a move has more than two to the power of its price in it: a cost of one allows piles of two, a cost of two allows piles of four. And that holds on a machine of any size.

So on a machine with finitely many states, a record that only grows, and an uncovered machine that forgets nothing, the toll is a price on crushing a pile: on taking two states that look the same once the cups are covered, told apart only by what's under your hand, and making them one.

15.4 What's left over is the ordinary computer

Put that together and you get the shape I've been circling since the night with the Flatland video that the afterword tells about.

An ordinary machine, read through anything its moves never raise, is what's left when you cover the cups of a machine that has them. Chapter four fitted the first cup. Over every state of the ordinary machine there's a whole pile, states that differ only in their cups and tally, and the ordinary machine can't tell them apart, as long as none of its steps looks at the cups. On a machine that obeys the toll, a step that turns up a cup costs at least one, whichever cup and however many. On a finite machine whose record only grows, when the ordinary machine forgets nothing, what that cost pays for is exactly crushing two states of one pile into one. A window

can charge the right price without showing the cups, if it shows them move. And with the halving rule, the cost in halvings is never less than what got crushed.

What I haven't shown is that any of this is out there in the world, that nature keeps a direction like this over its own uncovered machines. That's a belief of mine, and nothing in this book proves it. What I have shown, and you can check, comes with its conditions showing. A machine whose box answers to the computation and climbs an order is a row of cups, one cup for each line. If it also has finitely many states, never lets its record come down, and prices two arrows meeting, it has to pay this way. And when its uncovered machine forgets nothing, what it pays for is exactly this. That's what the Flatland night saw, and in the model it held when I pushed. What the night didn't settle, and this book doesn't either, is whether the world keeps a direction like this of its own: that's the belief I just told you about.

15.5 Shortcuts

A box is worth turning up because something later can glance at it and not have to work the thing out again, the way chapter ten's glance reads a block's bucket off a table without applying the rule from scratch. Call any move like that a shortcut, and call it sound when it gives the same answer the long way would have.

But everything in this chapter covered the box, and it could only do that because nothing in the machine read it. A machine that reads its box to take a shortcut isn't the machine this chapter studied: cover the box and the steps change. So what happens when the machine actually reads its box? That's the next chapter.

CHAPTER 16

Reading the box

The last chapter ended on a sentence I didn't follow up: a box is worth turning up because something later can glance at it and not have to work the thing out again. Every chapter so far kept the box as a passenger. Chapter twelve said nothing in the bare machine makes a later step read it. Chapter fifteen asked, as its condition, that nothing in how the rest of the machine moves depends on the cups. Under that condition, cover the box and you get the ordinary computer back, step for step.

That's a fair way to study what a box costs. It's a strange way to leave it, because it means the machine pays to settle something and then never uses it. So in this chapter the machine reads its box. Here's what that buys, what it costs, and how chapter two walks back into the room the moment you try it.

16.1 The shortcut

Picture one piece of data, a number that can be changed, and a question to answer about it. There's a long way to answer—a thousand steps, right every time. There's a shortcut—one step, but right only when the number has some property, say that it's even. And there's a check for that property, and a box. The box goes up only right after the check passes, which is chapter eight's established record. Now give the machine one rule: the shortcut may run only when the box is up.

What happens when the number changes? The box was about the old number. So give the number an edition: every write moves it to the next edition, and the box is about one edition, the one that was checked. The shortcut may run only when the box for the current edition is up.

From a clean start, does the shortcut ever give a wrong answer? Walk it. The shortcut runs, so the box for the current edition is up. That box went up only right after a check passed on that edition. Nothing has written to the number since, because any write would have moved it to a new edition. So the number is the one that passed, the property still holds, and the shortcut is right. That's chapter eight's four conditions, doing a job: same property, same thing, same edition, honest check.

And look at what it does to covering the box. Cover it with your hand and you no longer get the same run step for step, because the machine with the box takes

the one-step road where the machine without it takes the thousand-step one. You get the same answers, by a different number of steps. So this isn't chapter fifteen's machine. The box has stopped being a passenger.

It computes nothing new. Every answer it gives, the long way gives too. What it saves is steps, the way a cache saves them, and caches are old. What I want to show you is what keeping the shortcut honest costs, because it's chapter two's bill again, in four different currencies.

16.2 Four doors

The weak spot is the edition. It's a counter, and on a finite machine a counter can't climb forever. So what do you do? There are four doors, and I'll walk through each one with the machine that walks through it, the way chapter three did.

The first door: let the editions wrap around. Say there are two editions, 0 and 1, and after 1 comes 0 again. Four moves are enough to break it. Check the number 0. It's even, so the box for edition 0 goes up. Write 1, and you're on edition 1. Write 3, and you're on edition 0 again. Now ask. The box for edition 0 is still up, so the shortcut runs on 3, which is odd, and gives the wrong answer. The box was about the first edition 0, and the shortcut is looking at the second. That's chapter twelve's coin that moved, with a bad coin. And it's chapter eleven's lesson too: a counter that goes back down gets leaned on at exactly the place where it goes down.

Programmers know this one. IBM's manual for its System/370 computers, in the edition of 1974 as revised in 1975, warns about it in a note on an instruction called Compare Double and Swap. One program reads the head of a list; meanwhile another program "removes the first two messages and then adds the first message back into the chain," and the first program, on continuing, "is not aware that the list is changed." The fix in the note is a change number in the word next door, "incremented for each modification of the list," which brings the risk down "to a negligible level": it can go wrong only if the first program is held up while a whole multiple of 2 to the 32nd changes go by. That's the first door with a very big wheel. R. Kent Treiber, at IBM, drew the same accident in a 1986 report and fixed it with a counter. The name came later. The earliest I can find is in Maged Michael and Michael Scott's reports, dated December 1995, and their 1996 paper, and one of the reports already puts it in quotation marks, so it may be older: "the ABA problem," a value that goes from A to B and back to A, so a check that only looks at the value is fooled.

The second door: never reuse an edition. Each edition gets its own box, and a write moves to an edition that's never been used. Every box stays up for good, so this is still a Thiele machine, and the shortcut is never wrong. But it pays in chapter two's second column: fresh paper, a new edition on every write. On a finite machine

that's a budget. Once the editions run out, the honest move is a mark that goes up for good and says "spent," and from then on the machine takes the long way. That's chapter eighteen's bank of fuses, one blown per update, with a design rule to make sure there are enough.

The third door: keep one box and knock it down on every write. That's a cache with an invalidate, and the shortcut is never wrong. But the box comes down, so it isn't a Thiele record any more; it's chapter fourteen's toggle. And it doesn't get out of chapter two either. Make the write as gentle as you like, even one that just flips the number between 0 and 1, which on the number alone is one-to-one. Knocking the box down still sends the number with its box up and the same number with its box down to the same place. Two arrows meet. That's chapter six's WIPE. And the passing check that put the box up merged two states too: the number with its box down and the same number with its box already up both land on box up. The toggle pays twice.

The fourth door: don't tie the box to the edition at all. Tie it to the value that was checked. "6 is even" never goes stale. Write a different number, and the box for 6 is still true, because it's about 6, not about whatever's in the slot. Come back to 6 later and the box is waiting. No editions, nothing wraps, nothing gets knocked down, and every box stays up for good, so it's a Thiele machine and the shortcut is never wrong. It pays in room: one box for every different value that ever passed. That's chapter twenty-one's count. When the values are few, it's few boxes.

This door has an inventor. Donald Michie, at the University of Edinburgh, wrote in Nature in April 1968 about what he called memo functions. Give a function two parts, he said, a "rule part," the procedure that works the answer out, and a "rote part," a look-up table, and let the machine use whichever suits the moment, so that "each evaluation by rule adds a fresh entry to the rote." The word programmers use now, memoization, grew out of his; it isn't in the paper. His rote is a box tied to the value: once an input has been worked out, it never needs working out again.

16.3 Running all four

I ran all four on the same twenty thousand programs, forty moves each, on a number from 0 to 5, with the long way at a thousand steps and the shortcut at one. The doors that never reuse an edition get four editions to spend, the same room the first door wraps around in.

Door	Wrong answers	Average steps	A Thiele machine?
No shortcut at all, the long way every time	0	25,992	
1. Box stays up, editions wrap around	37,564	17,027	yes, and wrong
2. Box stays up, editions never reused (four to spend)	0	22,529	yes
3. One box, knocked down by every write	0	20,082	no: a toggle
4. Box tied to the value checked	0	17,997	yes

Twenty thousand programs of forty moves each, on a number from 0 to 5; the long way costs a thousand steps and the shortcut one. Wrong answers are counted over all of them.

The first door is the fastest, and it's fast because it's wrong. The other three are never wrong. On these small numbers the fourth door saves the most, and never needed more than three boxes, because only 0, 2 and 4 are even.

16.4 Why the other doors exist

If the fourth door is that good, why doesn't everyone always tie the box to the value? Ask chapter eight's warehouse question. In the run above, the data is one small number, and finding its box takes one look. Make the data chapter eight's list of a thousand numbers. A box tied to the value of the whole list can only be found by reading the whole list, to know which list you're holding. Reading a thousand numbers takes about as long as checking whether they're sorted, which was the check you wanted to skip.

So on big data the fourth door has three ways forward, and each one is a bill you've already seen. Tie the box to a short fingerprint of the list instead, and two different lists can share a fingerprint and a box; that's chapter eighteen's wall, scrambling functions that make that hard but not impossible, and when it happens it's the first door's wrong answer again. Or never change a list at all: keep exactly one copy of each different list and hand around a pointer to it, so the box can sit on the pointer and be found in one glance; that's fresh memory for every new list. Or go back to the second or third door, put the box on the container instead of the value, and let a write move the edition or knock the box down, so a later step can find it in one glance.

Keeping one copy of each value is old too. Andrei Ershov, in a paper printed in Russian in 1958 and translated into English the same year in the first volume of the Communications of the ACM, gave a program that writes programs a way never to work out the same expression twice. It scrambled each instruction into a number that said where to look, checked whether that place already held the same

instruction, and reused it if so, in time that grows with the number of instructions instead of its square. Donald Knuth credits him, along with Gene Amdahl, as an independent discoverer of that kind of table. In 1974 Eiichi Goto, at the University of Tokyo, built the idea into a version of Lisp, from a proposal he'd made with three colleagues the year before, and called it monocopy: no two copies of the same structure, so that checking whether two are equal becomes checking whether they're the same one. (This is Andrei Ershov, the programmer. The Ershov whose ladder came up in chapter fourteen is Yuri, a logician.)

16.5 What the doors say

Every time a box goes up, it pays the toll, because on a finite machine that's two arrows meeting, and that's chapter two and chapter three. That part was already in the book. What this chapter adds is the second bill. Keeping a shortcut honest after the data changes costs one more thing, and there are exactly four ways to pay it: fresh editions, which is fresh paper or a fuse budget; a box that comes down, which leaves the definition and merges at the wipe; room for every value checked; or nothing, and wrong answers.

Programmers have used all four for decades. What this chapter adds is that they're four ways of paying chapter two's bill, and that under two conditions there's no fifth. Here's the proof.

Say what the guard is allowed to see. The first condition: its glance can't tell a passing value from a failing one. If it can, then either it's doing the check over again, which is no shortcut, or it's reading enough of the value to tie a box to it, which is the fourth door, with its bill of room and, on big data, the time it takes to read the value. The second condition: the memory the guard reads is made only of boxes that go up and never come down. If some of it can come back down, that's the third door.

Under those two conditions, run the same moves over and over: write a passing value, check it, ask; then write a failing value, and ask again. The glance can't tell the two values apart, so whether the shortcut is unlocked depends only on the boxes. If it's unlocked after the failing write, it answers wrong. That's the first door. Say it never answers wrong. Then every round that unlocks needs the boxes in an unlocked state after the check and in a locked state after the failing write, so the boxes change between those two moments, and they have to change again before the next round can unlock. A number of unlocked rounds needs at least twice that many changes, less one. But boxes only go up. With E boxes, they can change at most E times in the whole run, because each change raises at least one box and none ever comes back. So twice the number of unlocked rounds is at most E plus one. After that the shortcut never unlocks again, and the machine takes the long

way for good. That's a budget, the second door. It's chapter two's chairs once more, more changes needed than boxes to spend them on.

The fourth door's room is a count of its own. Say every value that's been checked has to stay unlocked, all at once, through any writes. Take two different sets of checked values. Some value is in one set and not the other, and the guard has to unlock it for one and not for the other, so the memory has to be in two different states. With k passing values there are 2 to the k sets, so the memory needs 2 to the k states: k boxes, one for each value.

So under the two conditions, wrong answers or a budget are the only outcomes, and stepping outside the conditions is stepping through one of the other two doors.

And it leaves one question hanging, the one every machine with a record runs into once its programs can be handed their own code. These shortcuts were handed to the machine: I wrote the property, the check and the shortcut. Can a machine find its own shortcuts, and decide from the inside whether one is sound? That question runs into a wall, the same wall Kurt Gödel and Alan Turing hit, and I have to show you the wall is there before anything pretends to be on the other side of it. That's the next chapter.

CHAPTER 17

What a machine can't decide about itself

In chapter ten I said you'd watch the same move get made again and again: somebody says "everything is X," it's true enough to do real work with, and then somebody draws the diagonal. I owe you the diagonal itself now. The last few chapters handed a box to any machine that steps, and the next thing anyone wants from a machine with a box is for it to tell sound shortcuts from unsound ones by itself. Before I let anything claim that, I have to show you the wall every such claim runs into, and then exactly where a machine with a box stands next to it.

It doesn't stand on the far side, because nobody's on the far side. A box keeps a distinction and pays for it. What it can't do is the judging.

17.1 Five names and a table

Do this first, with a pencil, because the whole family of arguments is this one trick.

Write five friends' names down the left of a page. Next to each name, write a row of five digits, each 0 or 1, anything you like. Now build a new row. Its first digit is the opposite of the first digit of the first row. Its second digit is the opposite of the second digit of the second row. And so on, down the diagonal. Look at what you've made. It differs from row one in place one, from row two in place two, from every row at that row's own place. So it's nobody's row. Your table of five missed something.

With five rows that's a cute fact. Georg Cantor made the point in 1891 that it works for any list, however long, even an endless one: hand me a complete list of all endless rows of two symbols, and I'll walk the diagonal flipping as I go and hand you back a row that's on no line of your list. So there's no complete list. There are more rows like that than there are places on any list, and that's where "some infinities are bigger than others" comes from. (He wrote his two symbols as m and w, and that paper wasn't even his first proof of the fact. The first was in 1874, by a

harder road, and in 1891 he called the new one much simpler. The diagonal was him finding the easy way.)

Now look at what the argument did, because it's going to happen again, in new clothes. There was a table. Rows are the things being listed, columns are questions, and each square says "does row i do column j." There was the diagonal, the squares where a thing gets asked about itself. There was a flip, something with no fixed point, like turning yes into no, applied along the diagonal. And what got knocked down was the thing assumed at the start, the complete list. Nothing outside the assumption was touched. A table, a diagonal, a flip, and the assumption eating itself.

17.2 The same move, again and again

Russell, in 1902, from chapter ten: the collection of all collections that don't contain themselves. That's Cantor's flipped diagonal with "contains" as the table, and Russell said as much in 1903: he got there by running Cantor's argument on a supposed collection of everything. What got knocked down was the assumed collection of everything, and Zermelo's fix was to stop assuming it.

Jules Richard, a French mathematician who taught at schools in Tours, Dijon and Châteauroux, noticed in a short note in 1905 that some phrases define numbers ("the ratio of a circle's circumference to its diameter"). There are only as many phrases as there are whole numbers, so list the numbers they define, run Cantor's diagonal down their decimal digits, and you get a number no phrase defines, except that you just defined it, in the phrase that describes the diagonal. The lesson I take from it is the one this whole chapter turns on: keep apart the language a system talks in and the language you use to talk about it. Mix them and the diagonal bites you.

Richard's paradox has a cousin small enough to do at the table, and it comes with a footnote I can't resist. Russell, the first to put it in print, in 1906, credited it to G. G. Berry, a junior librarian at the Bodleian Library in Oxford. In the version in Principia Mathematica it's about syllables. Take the phrase "the least integer not nameable in fewer than nineteen syllables." There are only so many phrases shorter than nineteen syllables, so they name only so many numbers, so some least number is left over, and the phrase picks it out. Now count the phrase: the, least, in-te-ger, not, name-a-ble, in, few-er, than, nine-teen, syl-la-bles. Eighteen. In eighteen syllables it has named the number that can't be named in fewer than nineteen. "Nameable" is a word about the language, used inside the language, and that's where the bite is.

Kurt Gödel, 1931, with an improvement in 1936 by Barkley Rosser, a student of Alonzo Church, that lets it assume only that the system doesn't contradict itself. Take any such formal system, with rules you can check mechanically, strong enough

to do some arithmetic. Then there are sentences it can neither prove nor disprove. The engine lets you build, for any property you can write down, a sentence the system proves equivalent to "my own code number has that property." Apply it to "not provable," and you get a sentence equivalent, inside the system, to its own unprovability. The table is the sentences listed by code number, the diagonal is a sentence about its own code, and the flip turns provable into not provable. Two cautions, which the Stanford Encyclopedia stresses too: the sentence doesn't literally "say" anything about itself; it's provably equivalent to a claim about its code, which is less mystical and more exact. And the theorem doesn't say the sentence can't be known. It says this one system can't settle it.

How did the world take it? Mostly, at first, it didn't. Gödel, twenty-four years old, announced the first theorem in a round-table discussion on the last day of a conference in Königsberg in September 1930, and by the account of Solomon Feferman, one of the editors of Gödel's collected works, the room met it with general incomprehension, except for one man. John von Neumann pulled Gödel aside, kept thinking about it, and a couple of months later saw that the argument gave something more: a system like that can't prove it doesn't contradict itself. He wrote to Gödel with it in a letter dated 20 November 1930. Gödel had already got there; his paper had reached the journal three days earlier with that second theorem announced in it, and von Neumann left the credit with Gödel. It's "somebody got there first" again, except this time the person who got there second was von Neumann.

Alan Turing, 1936. Chapter one told how what he actually proved differs from the halting problem as it's taught. The taught version fits on two index cards. Card A is the Checker: hand it any program card and it announces "stops" or "runs forever." Card B is the Contrarian: "Hand me to the Checker. If it says I stop, run forever. If it says I run forever, stop." Hand B to A, and either answer is wrong. So there's no Checker. The table is programs against programs, the diagonal is a program run on itself, and the flip turns stop into loop.

Stephen Kleene, 1938, and this is the one that turns the paradox into a tool, so slow down. Everything above needs the thing that refers to itself to exist. Gödel's sentence has to be buildable; the Contrarian has to be able to "hand me to the Checker." Kleene's recursion theorem says that for any way of transforming programs that you can compute, there's a program that behaves exactly as if it had been handed its own source code. Any program may be written as though it already had a copy of itself in hand. He stated it almost in passing, in a paper about something else, and proved it in the last two lines of one section. I have that count from Yiannis Moschovakis, a logician who wrote a 2010 paper about it called "Kleene's Amazing Second Recursion Theorem," and he calls it a little gem. It's why programs that print their own source code exist in every ordinary programming

language. Try one in English: "Print the following twice, the second time in quotes: 'Print the following twice, the second time in quotes:'" Obey it.

Those programs are called quines, and the name comes from Douglas Hofstadter in 1979, in Gödel, Escher, Bach, in honor of the philosopher Willard Van Orman Quine, who had made a long study of self-reference that works indirectly. Quine left a sentence you can take apart at the table, from a 1962 essay on paradox: "yields falsehood when preceded by its quotation" yields falsehood when preceded by its quotation. Do what it says. Take the part in quotation marks, put a quoted copy of it in front of itself, and look: it's the whole sentence, word for word. So the sentence says that it itself is false. It's the liar paradox, built without any "this sentence," just an instruction to quote and copy. Quoting and copying is Kleene's trick in plain English. The paradox is what you get when the copy says something about its own truth. The tool is what you get when the copy just does something.

Henry Rice, 1953, whom you met in chapter seven: the general form. Any question about what a program does, not how it's written, that some programs pass and some fail, has no program that answers it for all programs. "Does this program stop" is one. "Does this program ever print 7" is another. "Does this program contain the word 'if'" isn't, because that's about the text.

And the one that's an attack. Ken Thompson, who built the original Unix operating system, gave a lecture published in 1984. Build a compiler that, when it compiles the login program, plants a back door, and when it compiles a compiler, plants both habits in that one too. Then delete the trick from the source code. The compiled compiler still carries it, and recompiling a clean source with the poisoned compiler plants it again, forever. Reading the source never shows it. Picture a locksmith who teaches an apprentice to cut a spare for every town-hall key, and to teach that to every apprentice they train, and never to write it in the manual. Years later the manual is spotless and every locksmith in town cuts the spare. Thompson's compiler is Kleene's trick with a knife in it.

Thompson wasn't first either, and he said so: he wrote that he first read of the possibility in an Air Force review of the security of an early operating system called Multics, and asked anyone who could find it to send him the reference. It's Paul Karger and Roger Schell's 1974 security evaluation of Multics, which already described a trap door planted in a compiler that keeps itself alive. Is there any way out? A partial one. David A. Wheeler, in a paper and then in his dissertation, worked out a check: compile the suspect compiler's source with a second, independently made compiler, use the result to compile that source again, and compare it bit for bit with the suspect. If they match, under his stated conditions, the source and the binary correspond. Two compilers having different authors is useful evidence, not a substitute for those conditions. The question has moved to a place where we can name what we trust.

And Bill Lawvere, in 1969, proved a theorem that, as I read it, makes all of these one theorem in disguise: if there's a function with no fixed point, a flip, then no thing can list every question about itself. Cantor, Russell, Gödel and Turing all come out as cases. Noson Yanofsky retold it in 2003 without the heavy machinery. I mention it because "it's the same wall" is a claim I make below, and it's fair at the level of the argument's shape. Every case has its own fine print about who counts as a judge, and that fine print is exactly where most overclaims live.

17.3 Where the diagonal gets misused

Before pointing it at a machine with a box, look at how it gets misused. Each time, the trouble is forgetting who the judge is and what it has to judge.

"Gödel shows minds aren't machines." John Lucas, a philosopher at Oxford, argued it in 1961. Roger Penrose, the Oxford mathematical physicist, revived it in two books, The Emperor's New Mind in 1989 and Shadows of the Mind in 1994. The standard replies, as far as I can tell, have held up: the Gödel sentence is true only if the system doesn't contradict itself, so a person who "sees" it's true has to know that, which is the hard part; and nobody can build the Gödel sentence of a system they can't write down, so people might be machines too complicated to diagonalize. The theorem has an "if" in it, and the argument drops the "if."

The part that's easy to miss is that Gödel himself didn't go as far as Lucas. In a lecture on 26 December 1951, at a meeting of the American Mathematical Society at Brown University, he drew only an either-or from his own theorems: either the human mind infinitely surpasses any finite machine, even in pure mathematics, or there are questions about whole numbers, of a certain simple kind, that no proof a human can conceive will ever settle. Feferman, writing about that lecture, says Gödel privately believed the first half and still stated only the either-or, because he didn't have a knock-down argument against the machine side. Gödel meant to publish the lecture and never did. So the man whose theorem gets quoted for "minds aren't machines" stated an either-or and left the deciding to whoever came next. The "if" is there again, kept by the person with the most right to drop it.

"Undecidable means unknowable." No. Two senses share the word: no single program that's right on every case, and a sentence one particular system can't settle. Neither means nobody can know a given answer. Any single case is easy in the first sense: for any one program, one of "say yes" and "say no" is right. What's ruled out is one inside procedure that's right on every case.

"Everything interesting is undecidable." No. Rice is about what programs do, over all programs. It says nothing against checking the text, nothing against answering for particular programs, nothing against checks allowed to say "I don't know," nothing

against checking finite machines. Real compilers and analyzers live with Rice by being careful and incomplete, and what limits them in practice is mostly cost.

"Self-reference is a paradox." It's a theorem with a construction, as Kleene, the quines and Thompson show.

17.4 Now point it at the box

Here's the question with nothing in it but what this book has built. A machine steps. A box rides along and turns up on an event, chapter fourteen. When the event is "a check passed," chapter eight says the box lets you lean on a claim, and for a claim that a shortcut is sound, that's what makes the shortcut usable. So: can a machine like that have, as one of its own programs, a procedure that correctly decides, for every program, "does this program have a sound shortcut"? Call that an autopilot: hand it any program and it says yes or no, and it's always right.

The exact claim is narrower than that question in plain words, so here it is first. "Has a shortcut" can be any question about programs with two features: two programs that behave the same always get the same answer, and there's one program that passes and one that fails. That's all the argument gets to look at. It doesn't see soundness. Whether a particular question about shortcuts has those two features is a separate fact, checked one machine at a time. A procedure that only answers for some programs, or needs outside evidence, isn't what the wall rules out.

And because the argument only looks at what programs do, not how, "shortcut" here is a label. A question like "is there a cheaper program that gives the same answers?" isn't the argument's question. The nearest thing I know of is Manuel Blum's speedup theorem from 1967, and it goes somewhere stranger: there are functions with no fastest program at all, where every program for them has another that's faster on all but finitely many inputs. That's about cost, not deciding, and nothing in this book proves anything there.

The answer is no, under conditions, and the proof is the two index cards. Suppose there were an autopilot that says yes exactly when a program has a shortcut. Build the flip: "If the autopilot says yes about me, behave like the program that fails; if it says no, behave like the one that passes." If the machine has Kleene's trick, so a program can be handed its own code, and if that flip is something the machine can run, and if the question only depends on what programs do, and if there's one program each way, then there's a program that behaves exactly like its own flip. Ask the autopilot about it. If it says yes, the program behaves like the one that fails, and behaving alike means getting the same answer, so it fails. The autopilot was wrong. If it says no, the program behaves like the one that passes, so it passes. Wrong again. Those are the only two answers.

Look at what that used: Kleene's trick, the flip being runnable, the two examples, and "behaves alike, answers alike." It never touched the tally, the toll, or the box. It's Rice's trick one floor over, with its own conditions, so it isn't Rice's theorem word for word, and it says nothing about whether this question is harder or easier than the halting problem.

So the wall is there. The honest part is who counts as the judge. The argument rules out a judge whose flip the machine itself can run. It says nothing about a judge standing outside. An outside judge can exist for the very same question, right on every program. The line between inside and outside is Richard's lesson: the language you talk about a system in isn't the language it talks in.

17.5 What this does to the book

It does two things. It keeps me honest: nothing in this book claims a machine with a box can check its own shortcuts on autopilot, because that's a wall, the same wall, and I've shown you the bricks. And it tells you what a box can do instead: it can carry a shortcut along with its receipt, so that "was this earned?" is answered by a step you can point at. No judge has to exist for that, which is lucky, because on a machine with Kleene's trick no inside one can. The diagonal says the judging can't be automated from inside. The cup says the commitment can be recorded. Different jobs, and a box does the second.

The diagonal has a famous cousin that looks like it ought to bite a box too. Gödel's second theorem says a system strong enough for arithmetic that doesn't contradict itself can't prove that it doesn't, and Martin Löb sharpened that in 1955: if such a system proves "if this is provable, it's true," it already proves the thing itself. So can a box vouch for itself, and does Löb stop it?

You might guess yes, since a box looks like a system talking about itself. It lands somewhere else, and mostly it's a relief. Hold two things side by side: "there's a receipt for this somewhere," and the receipt in your hand. Löb's theorem is about provability, the bare claim that some proof exists somewhere. A raised box isn't the receipt in your hand either. From a clean start, chapter eight says a check passed before the raise. The box doesn't say which claim, and it doesn't hold a proof of it. Sergei Artemov's line from chapter eight, between "there's a proof" and "this is the proof," is useful here, but a box doesn't turn one into the other. The trust in this book comes from a proof about the checker, the clean start and every move, made outside the machine. It isn't a proof inside the machine that the machine is sound, so it doesn't get around Gödel or Löb.

That holds only as far as the receipt does, and a receipt is only as good as whoever wrote it. Its trust rests on the checker meaning what it says, and that gets settled once, from outside. That's where I chose to stop this machine's chain of reasons.

Eliezer Yudkowsky and Marcello Herreshoff's 2013 draft on programs that trust their own rewrites describes a related problem they call the Löbian obstacle: how a program can justify trusting its successor's proofs. It's a cousin of this question, it doesn't settle it, and it doesn't make my design the only possible one. So I didn't get rid of the receipt somewhere. I moved it out of the machine and handed it to whoever reads the record.

One more thing follows from the same picture, and it's short enough to check by hand. Can a box rest on itself, the way the sentence "this sentence is certified" would like to? Not if it's established and permanent. The check behind the raise happened while the box was still down, because the box goes up only after that check passes. And once up it never comes down, so it goes up only once. So no claim whose truth needs the box already up can be the claim the box went up on. Saul Kripke's 1975 theory of truth calls a sentence that only ever leans on itself ungrounded, and established boxes are grounded in his sense: every one bottoms out in a check that didn't lean on it. Caspar Oesterheld's 2019 work on programs that cooperate by simulating each other needed a grounding step of the same kind to stop an endless regress.

17.6 Two walls

One last distinction, because the two are easy to run together, and running them together is the easiest way to misread this book.

There's a second wall, nothing to do with the diagonal. It's chapter seven's: two states that differ in their box can give the same view through a window that doesn't let the box through, and then no reader of the window can get the box back. That wall comes from missing information. Nothing about it is undecidable. It holds for small finite things with nothing to compute. Rice isn't underneath it.

So there are two walls, on two different axes. The diagonal is about what no inside procedure can decide. The window is about what no outside watcher can see. The book needs both, and I'll try not to call one by the other's name.

CHAPTER 18

Five real systems

So far there's been a toll, a counter, two kinds of wall and a cup, every one of them shown on paper, with cups and cards and dice I made up. It's time to go look at what people actually do. I picked five systems that keep records other people rely on, with real money or real security behind them, and asked each one the question from the paper: is there a price on the moment the record is written? Not on the work around it. On the write itself, the moment the mark goes on and stays.

I'll give you the answer first, because it's one sentence and it surprised me. Of the five, read through the pieces of them I copied out, one takes a fee at the moment of the write, and that one leaks at exactly the place chapter eleven said a counter leaks: where it goes back down.

"Price" needs one meaning for all five, or the answer is whatever I want it to be. So here it is, fixed before looking. A system prices an event if it attaches one of three costs to it. A fee is something taken at the event itself, whatever happens afterwards. A bond is something put at risk by the event and lost only if some later event happens, like a contradiction. A budget is a limited allowance the event draws down, so only so many such events can ever happen. A wall isn't a price under this. Nobody pays a scrambling function; undoing one is just too hard. Neither is work a system needs and never charges for. The table at the end has a column for each of the three, and because the definition came first, what the table shows is a finding about the five, not something I arranged.

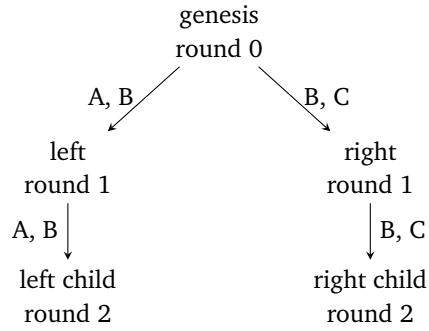
Three things are easy to blur here, so I'll keep them apart all the way through. There's the real system, the document that real money and real security run on. There's my copy, one piece of that document, copied out small enough to check by hand. And for some of the five there's a wrapper, a tiny model that takes one condition as given and says what follows. A claim checked at one level isn't a claim about the level above it. Near the end of the gas section I'll walk gas through all three, so you can see where each claim stops.

18.1 Proof-of-stake finality: the price is on going back

Ethereum, and chains like it, decide which blocks are final by having validators, people who've locked up money, their stake, vote on checkpoints: blocks picked out at regular numbered rounds, which they call epochs. A vote isn't for one checkpoint. It's a link from a checkpoint the validator already trusts to a later one, and every chain starts from one first block, the genesis block, which counts as trusted from the start. In the design Vitalik Buterin and Virgil Griffith published in 2017, called Casper, a checkpoint is justified when validators holding two thirds of the stake vote a link to it from a justified one, and final when it's justified and a two-thirds link runs from it to its direct child. The record is "this checkpoint is final."

Two rules carry the price. Don't vote for two different targets in the same round. And don't cast a vote whose span, from where it starts to where it lands, surrounds the span of another of your votes. Break either, and your deposit can be destroyed, which is called slashing. The theorem they prove, which they call accountable safety, says: if two checkpoints that contradict each other ever both become final, at least a third of the stake broke a rule and can be slashed. So the price is on going back, on making the record contradict itself. Writing the record, voting honestly to finalize, costs no stake at all. That isn't a claim that running a validator is free; it's a claim about what the deposit is taken for. In the terms above, Casper's price is a bond. The deposit stands behind every vote, the honest ones included, and it's only taken from a validator whose own votes contradict each other.

Do it with three validators of equal stake. Draw the genesis block at round 0, two competing blocks at round 1, and one child of each at round 2. Let A and B vote both links on the left branch, genesis to its first block and that block to the next. Let B and C vote both links on the right.



Each link has two of the three votes, two thirds of the stake. Both round-1 blocks are justified and each has a justified link to its child, so both are final, and they're siblings, so neither is an ancestor of the other. B voted for two different targets in round 1, so B's third of the stake can be slashed. A and C each voted one link landing in round 1 and one landing in round 2, which neither doubles nor surrounds, so

they broke no rule. The picture records votes; nothing in it takes anybody's money. What it shows is that the theorem's condition can actually happen: forks like that can be built, and the theorem fires on them.

Why does the theorem hold? Walk it. Say two contradicting checkpoints are both final, one in round v and the other in round w.

If v and w are the same round, then two groups, each with two thirds of the stake, voted links landing in that round on two different blocks. Two thirds plus two thirds is four thirds, more than everything, so the two groups share at least a third. Every validator in that shared part voted two different targets in one round.

If they're different rounds, say v is earlier, look at how the later one got justified: a chain of two-thirds links climbing from genesis, every block on it an ancestor of the later checkpoint. None of those blocks is the earlier final checkpoint or comes after it, because then the later checkpoint would too, and they contradict. So that chain has to get past round v without touching the earlier checkpoint, and some link in it starts at or before v and lands after v. Three ways that can go. If it lands at exactly v plus one, then it and the earlier checkpoint's finalizing link land in the same round on different blocks, and the shared third voted both, a double vote. If it starts at exactly v, then its starting block is a justified block at round v that isn't the earlier checkpoint, and the two links that justified those two blocks land in round v on different blocks, a double vote again. Otherwise it starts before v and lands after v plus one, so it surrounds the earlier checkpoint's finalizing link, which runs from v to v plus one, and the shared third made a surround vote. In every case a third of the stake broke a rule.

Everything there stands on the overlap. Take away "two groups of two thirds share a third," and the argument promises nothing.

The piece I copied out comes from a published proof by Runtime Verification and the University of Texas at Austin, based on Yoichi Hirai's, with that overlap taken as a condition: the copy takes it as given and doesn't do the arithmetic. It has no deposits, no rewards, and no rule for picking which branch to build on. On a chain with one validator, one vote finalizes the genesis block and nobody is slashed: that's "writing costs no stake," shown in the simplest case there is.

One thing that means for the record itself: finality can be undone at a price, the price is named, and a final block can still hold a false statement.

Is the 2017 design what Ethereum actually runs? Partly, and the paper's title is the clue: Casper the Friendly Finality Gadget. Its first sentence says it's a finality system laid over a chain still choosing its blocks by proof of work, so it's a gadget, bolted on top. In 2020 a larger group, Buterin among them, wrote up Gasper, joining Casper's finality to a rule for picking branches called LMD GHOST, and on 15 September 2022, in the switch people called the Merge, Ethereum stopped choosing blocks by proof of work at all and went over to proof of stake. So the checkpoints

and the two slashing rules are one part of a system that holds real money today, and my copy is of that part's safety argument, the way the 2017 design states it. Everything between the two, the branch rule and the rest of the running protocol, is the gap between my copy and the running system. None of it moves the price: finalizing still costs an honest validator no stake, and what costs is contradicting yourself.

18.2 Gas: the one that charges a fee at the write

Every operation on Ethereum costs gas, paid in ether by whoever sent the transaction. The yellow paper, Ethereum's formal specification, gives the reason in one line: every computation costs a fee, so the network can't be abused and questions about whether a program stops are sidestepped. The record I care about is a storage slot holding something other than zero, which every full copy of the chain in the world has to keep until it's cleared. Clearing a slot is allowed, and the refunds below exist to reward it.

Writing a fresh slot from zero to something costs twenty thousand gas, by the table in the yellow paper's appendix, plus twenty-one hundred for touching a cold slot, one the transaction hasn't touched yet. After the first touch it's warm. The point, as I read it, is to discourage storage growth, since storage grows every copy's database. That charge is a fee on the write, at the moment of the write, and it's the only fee in the five. Gas has a budget too: every block has a gas limit, and the gas its transactions use can't go over it, so only so many writes fit in a block.

Where does a number like twenty-one hundred come from? Somebody picked it, so the next question is who, and why, and the answer has a date on it. In September 2016, just as Ethereum's developers were gathering in Shanghai for Devcon 2, their developer conference (the third one, since they started counting at zero), attackers started filling blocks with an operation that was cheap in gas and expensive in fact: it sent every copy of the chain off to read data from its disk, and copies fell behind. The first repair, a change that October called Tangerine Whistle, raised the prices of the operations the attack leaned on. The cold-slot charge came later, in a proposal by Vitalik Buterin and Martin Swende called EIP-2929, switched on in the Berlin upgrade of April 2021, and the reason it gives is still those 2016 attacks. So twenty-one hundred isn't a law of nature or a measurement of anything deep. It's a scar: a price set after somebody found where the old price was wrong. That doesn't change the argument; the write is still priced at the moment of the write. It does change how I read the rest of that table. Some of its numbers, at least, are somebody's answer to "what did this cost us last time," which is a record of a different kind.

Now the leak, which is chapter eleven's leak. Set a slot and clear it again in the same transaction, and you get a refund. Count it through. The first write costs twenty-two thousand one hundred. Clearing it costs a hundred and credits a refund of nineteen thousand nine hundred. Net: twenty-three hundred. Leave it set, and you pay at least twenty thousand. So the write is priced, and the write you take back nearly isn't.

Twenty-three hundred on its own misleads, though, because of the 2021 change from chapter eleven, the one that ended the refund-hoarding token's advantage: the refund a transaction actually gets is capped at a fifth of all the gas it used, and that includes the twenty-one thousand every transaction pays just for existing. Take a transaction that does nothing but set and clear the slot. Count only that base and the two stores (the steps that load their arguments add about a dozen more): it uses 43,200 gas, so the cap is 8,640, so it pays 34,560. That's 13,560 on top of the base. Look at the slot alone and forget the base, which the rule doesn't do, and you'd get 17,760. Either way it's far above twenty-three hundred.

You can check the twenty-thousand floor with a pencil. My copy follows the specification's rule for one slot in one transaction, starting from zero. A store of something that isn't zero into a slot that's still zero costs twenty thousand; any other store costs a hundred; and putting the slot back to zero gives back nineteen thousand nine hundred. Keep two lines on a sheet. "Slot at zero: net so far at least zero." "Slot not at zero: net so far at least twenty thousand." The fresh slot meets the first line. Now check that every store keeps whichever line fits where it lands. Zero to zero adds a hundred, so still at least zero. Zero to something adds twenty thousand, so from at least zero to at least twenty thousand. Something to something else adds a hundred, so still at least twenty thousand. Something back to zero adds a hundred and takes off nineteen thousand nine hundred, nineteen thousand eight hundred off in all, so from at least twenty thousand down to at least two hundred, still at least zero. So whatever the list of stores, if the slot ends up holding something, the net is at least twenty thousand. The cold charge comes once, on the first store, and nothing after it depends on it, so from a cold slot it's at least twenty-two thousand one hundred.

That copy covers one slot in one transaction, with the gas and refund arithmetic copied out of Ethereum's execution specification. It has no other slots, no calls between programs, and no refund cap.

The wrapper is a different kind of thing. It's a gas schedule in the abstract: any states, any steps, a cost on each step, a box that's up or down, and a rule saying which steps get charged. There's no Ethereum in it at all. It's chapter seven's clerk under a new name. Ask a schedule for two things: every run pays at least as much as the number of times it turned the box up, and no run pays more than that. Both together hold exactly when the schedule charges one on each step that turns the box

up and nothing on any other. Both halves are easy to see on a one-step run. Leave one step that turns the box up uncharged, and that one-step run turns it up for free, so the first fails. Charge one step that doesn't turn it up, and that one-step run pays more than the nothing it turned up, so the second fails.

So walk gas through all three. Think of Ethereum as it runs as a kitchen, and my copy as a photocopy of one recipe from it. The photocopy says a set-then-clear nets twenty-three hundred, and inside the photocopy that's right. In the kitchen, the bill for that transaction is 34,560, because the kitchen has a base charge and a cap the photocopy doesn't. The wrapper isn't from that kitchen at all: it says what an exact price would have to look like, one per raise, and Ethereum's twenty thousand isn't that and isn't trying to be. A claim made with the photocopy in hand is a claim about the photocopy.

Gas prices storage and computation. It says nothing about whether what's stored is true.

18.3 Attestation: a quote about the past

A lot of computers carry a small security chip, a trusted platform module. It keeps registers that can only be extended: each new measurement gets scrambled together with the old value. A quote is the chip signing some of those registers along with a challenge from whoever's asking, so they can tell it's fresh, and the asker still has to check the signature, the challenge and the values they expected. That can give you a fresh report of what was measured. It doesn't tell you every important fact about the machine got measured.

Some of those registers can be reset, depending on the register and the rules of the platform. Some reset at startup; some allow resets partway through. So "you can only extend" describes a stretch of time with those resets left out, and a claim about an unbroken history has to say which resets it allows. Intel's SGX enclaves use their own measurements and reports for a similar purpose, but not these registers.

What does the specification price on the registers? Nothing: no fee, no bond, no budget. Forging a different measurement history that passes depends on breaking the scrambling or the trusted parts, which is a wall, not a price. An allowed reset is a separate thing, not an attack. That's the difference between expensive and impossible in practice, and it's easy to blur, which is why the definition at the start of this chapter leaves walls out.

So what stands behind a cost can come in three currencies. It can be money, a fee or a deposit the system takes: Casper and gas are there. It can be a wall, going back made too hard by something nobody knows how to undo, with no exact price on it: attestation and transparency are there. Or it can be physical, energy spent or

hardware worn, which none of these five charges as a fee or a bond; the nearest is the chip's write budget, below.

Why the registers, then, when something that lasts would make a better record? Because the quote is what a faraway asker reads, and the registers are what it signs. The same chip has longer-lasting records, closer to the cup than anything else in this chapter. Version 2.0 of its specification has slots of a kind it calls a bit field, changed only by a command that ORs new bits into the old value: while that slot exists, it can set a bit and can't clear one. Deleting a slot and making it again is governed by separate permissions, and any claim about how long the record lasts has to include them. The chip also has counters that only go up, and a new counter starts above every count the chip has held before, so deleting one and making a new one doesn't start over at zero. That's a promise about this chip, not about somebody swapping in another chip.

Those slots are rationed. Bryan Parno and his coauthors, building a system called Memoir in 2011, noted that the specification of the day only required a counter to be able to tick once every five seconds, and that the chip's storage was expected to survive about a hundred thousand writes in its whole life, so writing once a second would wear it out in under twenty-eight hours. Memoir writes to it twice per startup. That's an expected lifetime, not a budget the protocol enforces, and the quoted registers have no number like it at all. But on the bit field it's chapter five's row of cups, which can only turn up as many times as there are cups. Phones and small controllers go one floor further down, to fuses like Chow's in chapter fourteen: a version number kept as a row of fuses, one more blown at each security update, and startup code that refuses anything older. A 2021 guide by Rob Smart for PSA Certified, a security certification scheme for chips and devices, tells designers to make sure there are enough fuses to last the number of updates expected over the supported life of the chip's root of trust, the part every later check starts from. That's chapter five's count of cups, written as a design rule.

What it's all for is rollback. Memoir's example is a password checker put back to an old saved state to undo a user's password change. In 2017 a system called ROTE did the same job for Intel's enclaves by having several machines vouch for each other's counters, and its authors say the only way left to break it is to reset every one of them to its start. In 2016 Raoul Strackx and Frank Piessens's Ariadne got the job down to flipping one bit of lasting memory per update, which they call the lowest possible. Notice who the attacker is in every one of these: whoever gets to put an old state back. Not a bad program. A record kept by the step can't see that attacker. Everything an established record guarantees about its history is stated from a clean start. Put back a state the machine really reached, and every guarantee still holds of it, through the run that reached it first. None of them can say nothing

came after it. The toll on each step still holds after a restore; it just can't tell you what happened before.

Systems run as several copies put the same kind of record somewhere else: inside each copy, where a copy that's been taken over still can't make it say two different things. Byung-Gon Chun and his coauthors built a log a copy can only add to, called A2M. Dave Levin and his coauthors shrank that to a trusted counter, TrInc. MinBFT, from Giuliana Veronese and her coauthors, uses a trusted counter to agree with 2f + 1 copies when f of them can lie, where the usual count is 3f + 1. Damysus, from Jérémie Decouchant, David Kozhaya, Vincent Rahli and Jiangshan Yu, has a small trusted service it calls the Checker, which keeps a count of where the protocol is that only goes up. And Allen Clement, Flavio Junqueira, Aniket Kate and Rodrigo Rodrigues worked out, with proofs, what such a record buys, and as I read their result it's a good warning for this book. On its own, stopping a copy from saying two different things doesn't lower the 3f + 1 at all when messages can be slow. Add signatures a third party can check, and it turns a system that survives crashes into one that survives liars, with no extra copies. The way I read it, a record helps the others only as far as they can check what it says. That's chapter seven from the other side.

Under those three currencies there's a fourth, and every permanent record on a finite chip pays in it, the chairs from chapter two. Here's how. Forgetting by itself doesn't pick out a record write; any step that overwrites a field that could hold two things forgets something. What picks it out is where the forgetting lands. Count the states where the record already reads yes, and call that number K. Take a step that never turns the record off anywhere, and turns it on at one state that read no. That step sends each of the K yes-states to a yes-state, and it sends the one no-state to a yes-state too. That's K plus one starting places and only K yes-states to land on, so two of them land together. Both came from the yes side or from the one it just raised. So what a permanent write destroys is a distinction on the record's own yes side: two states it already vouched for, or the one it just raised and one that was up already. Any overwrite merges something somewhere. A permanent write has to merge it there. That needs no assumption about money, keys or heat. The other three each need an assumption of their own: a fee rule, a wall nobody can climb, a law of physics.

What the quote can do is say what was measured. It can't say the measured code is correct, and it can't say anything about what was never folded in. A correctly measured buggy program gets a beautiful quote. Try it with the smallest pair there is: two machines with nothing measured at all, one with some running flag up and one with it down. Their registers are the same, so their quotes are the same, and anything that reads only the quote has to give them the same answer. No reader of the quote can recover the flag. That's chapter seven's window again. Anything that

depends only on what was measured, the quote can answer; that's what it's for. And a "sign and check" interface by itself promises nothing: a check that always says yes fits the interface perfectly and accepts a signature no key ever made. The protection has to come from the scrambling and the keys, which is a further assumption.

That limit, turned outward, is the honest edge of chapter eight's four conditions. Every one of them is about a thing the machine holds: same property, same thing, same edition, honest check. For a claim about the world—the oven is hot, the parcel arrived—there's no edition the machine controls. Whatever reads the world is a window onto it, and "nothing changed between the check and the commit" stops being something a step can enforce and becomes an assumption about the sensor. So a record can establish claims about the machine's own state, and claims about the world only through stated assumptions about whatever measured them. A correctly recorded sensor reading establishes what the sensor reported. Whether the oven was hot still depends on the sensor and on how its report connects to the world.

18.4 Certificate transparency: visible, unpriced

The certificates that put the padlock in your browser are issued by hundreds of authorities, and a bad or broken one can issue a certificate for a site it shouldn't. Certificate Transparency puts certificates in public logs you can only add to, built as trees of scrambled values. It was written up as RFC 6962 in 2013 and revised as RFC 9162 in 2021. I went looking for which one the running logs follow, and it's still 6962: Mozilla's reference for web developers says browsers are built to it, and as late as November 2025 Chrome's team was only proposing to drop its rule that every certificate carry a timestamp from at least one 6962 log. My copy is of 9162.

The tree is chapter four's chained running total, folded into a tree: Ralph Merkle's tree from the same chapter. Scramble each certificate; those are the leaves. Scramble the leaves together in pairs, then those in pairs, and so on, up to one value at the top, the root. To prove one leaf is in, the log hands you only the few values sitting beside the climb from that leaf to the root, and you redo the climb yourself. A second check shows that a bigger tree is a smaller one with leaves added on the end and nothing rewritten. The record is a leaf in the tree.

That kind of tree has a history of its own. The computer scientists Scott Crosby and Dan Wallach, at Rice University, built logs you can't quietly change on the same kind of tree in 2009, with a proof that an entry is in the log and a proof that a later version extends an earlier one, each only a handful of values long even for a huge log. A rewrite that doesn't extend a root somebody already saw gets caught by the second check.

Do the climb by hand. Lay out seven cards, d0 through d6. Join neighbors with paper brackets: a joins d0 and d1, b joins d2 and d3, c joins d4 and d5. d6 stands alone. Then L joins a and b, and R joins c and d6. The root joins L and R.

Step	Card handed over	What you rebuild
Start	d_3 itself	d_3
1	d_2, on the left	b, joining d_2 and d_3
2	a, on the left	L, joining a and b
3	R, on the right	the root, joining L and R

To check d3, you only need three cards: d2, a and R. Join d2 on the left of d3 and you've rebuilt b. Join a on the left of that and you've rebuilt L. Join R on the right and you've rebuilt the root. You never opened a or R. Swap left and right anywhere, though, and the bracket changes: the proof has to carry the order as well as the cards. The same seven cards check every other leaf too. d0 needs d1, b and R, all on the right. d4 needs d5 and d6 on the right and then L on the left. d6, standing alone, needs only c and then L, both on the left.

Now the second check. Say you saw the tree when it had four cards, and its root was L. Now it has seven. The log hands you one card, R. Put L on the left of R and you get the new root. So the old tree is the left half of the new one, untouched, and only cards were added on the end.

My brackets are brackets, not numbers, so what you just checked is how the tree is put together. A real log uses short scrambled values, and "two different trees can't give the same root" is then a separate promise, from the wall.

What does the RFC price? Nothing. It says in its first section that logs let people detect a bad certificate afterwards. They don't prevent one. And a log that shows different trees to different people can beat the checking, so logs are still trusted parties. What it buys is visibility.

So why does any authority bother logging? Nothing in the RFC makes it. A browser does. Chrome's team set the rule, enforced from Chrome 68 in July 2018, that every certificate issued after 30 April 2018 had to meet its transparency policy, and a publicly trusted certificate that doesn't simply isn't trusted by Chrome: the visitor gets a full-page warning in front of the site. So there is a price after all, though it isn't in the specification. It lives in one company's browser, and it's the price of being shown to that browser's users at all. Which means the strongest lever on this record is a policy one company can change, and that's a fact about who holds the stamp, not about the tree. It changes nothing in the table, which asks what each specification prices. But keep two sentences apart: "the specification prices nothing" and "nothing is priced."

18.5 Proof-carrying code: no toll required

George Necula's idea, from chapter twelve, is to send untrusted code with a proof that it follows the receiving computer's rules; the receiving computer works out what would have to be true and checks the proof with a small checker before running anything. The record is the checked proof, or the result of checking it, and a computer may keep either. Nothing in the idea requires a permanent box, or charges anything for accepting. Producing and checking the proof both take work, and nothing here makes that work the toll.

18.6 The same question, five ways

Lay them out by the one definition of price from the start. Casper puts a bond behind the write and takes it on going back. Gas takes a fee at the write, mostly refunds it if you take the write back, up to a cap, and rations writes with a gas limit per block. The chip's quoted registers have no fee and no budget; its separate bit fields have only so many bits to set, which isn't a quota on everything the chip does. Transparency charges nothing at the write; what it gives is that a bad certificate can be seen afterwards. Proof-carrying code requires neither a toll nor keeping anything.

System	Fee at the write	Bond	Budget	What my copy shows
Casper finality	none	the deposit, taken on a contradicting vote	none	two contradicting final checkpoints mean a third of the stake can be slashed
Gas	20000 for a fresh slot, mostly refunded if taken back, up to a cap	none	the gas limit of each block	a write left standing in an empty slot nets at least 20000
Attestation	none on the quoted registers	none	none on those registers	the quote can't tell you what it never measured
Transparency	none; a bad certificate can be seen	none	none	the RFC's own examples pass

Proof-carrying code	none required	none	none	a comparison of ideas; no checker is built here

The first four rows come with a piece copied out and checked; the fifth is a comparison of ideas. Each copy shows only what it shows. To read a row as a claim about the running system you'd need evidence from that system.

So the thing the toll charges for, the moment the record is written, gets a fee in one of the five, and in that one the designers had to patch the place where the charge could be clawed back. I don't find that a win or a loss. It's a fact about what people built, and what they built was mostly walls and penalties for contradicting yourself. A toll on the write was rare. Whether a toll is a better design than a wall is something I can't prove, and I won't pretend the five say so. Because the definition came first, that last part is a count: of the fifteen fee, bond and budget squares in the table, three are filled, one of each kind, and only gas takes a payment at the moment of the write.

What I read in them, and it's a reading of five cases I chose, is that each one has something it treats as the record, they last for different lengths of time, and they all picked something different: final, committed, measured, logged, checked. Is there anything that picks out one event as the one worth recording, or is it a free choice every time? That's the next chapter.

CHAPTER 19

Which box is worth keeping

Everything so far leaves one choice open. Chapter fourteen showed that a box that answers to the machine and never comes down is a cup on some event, and that the event is a choice: the two cups on one die were both happy under the same rules. Chapter eighteen found five real systems that each picked a different event. And in this book I picked one too: the established record from chapter eight, a box that goes up only right after a check of the very claim it records has passed. So I owe you an answer to the obvious question. Is that just my choice, or is there something about how people and machines lean on records that picks that event out?

What I have is a thesis, in the same sense the Church-Turing thesis from chapter one is one: a claim tying things I can define to the world, which no proof can settle. Nothing before this chapter depends on it. The cups, the windows and the counting hold whichever event a system records. What follows is where the idea came from, what exactly it claims, and how you'd catch it being wrong.

19.1 A dial and the light off it

The shape comes from physics, and I want to tell you exactly where, because I borrowed a word and didn't get to keep all of it.

Quantum mechanics has a famous problem. A system can be in a blend of states, and when you measure it you find one definite state, and nobody agrees on what measuring did. In 1981 Wojciech Zurek asked a sharper question: of all the ways a measuring device could end up, why does it end up recording one particular kind of quantity, the position of a pointer on a dial, say, and never some blend? His answer, built up over twenty years and summed up in a 2009 paper called "Quantum Darwinism," is that the environment decides. The air, the light, the table, everything the device touches, keeps interacting with it, and only certain states of the device survive that; those are what he calls the pointer states. More than that, the environment picks them out and then copies them. The light bouncing off a dial carries the dial's reading to every corner of the room, over and over, so a thousand people can each catch a different scrap of that light and all agree on where the pointer is, without any of them touching the dial. On his view, being objective is

being copied like that: a property is objective when many separate pieces of the world each carry it.

Zurek isn't the only one behind it. Fernando Brandão, Marco Piani and Paweł Horodecki in 2015 proved an approximate version: for most small pieces of a large environment, what each piece carries can be matched, up to a small error, by measuring the system in one common way and handing each piece the result. The error and the share of exceptions are part of their theorem, and agreeing on outcomes also needs the pieces to carry enough about them. Xiao-Liang Qi and Daniel Ranard in 2021 tightened the bound on the exceptions. Neither says every observer gets an exact record, or any useful record at all.

What I borrowed is the shape. Take a machine together with everything around it that reads, stores and relies on its records. The established record is to that crowd what the pointer reading is to the room: the thing that gets copied everywhere. It's a good shape, and I think it's the right hunch. But I'll be exact about what survived the trip, because my word "copied" does less work than his.

19.2 The claim

Here's the claim, said plainly. Of all the events a machine could put on its bill, the ones its price list charges for and its counter adds up, the established record is the one whose marks get copied and relied on by parties who didn't make them: checked, stored and acted on in separate places that share nothing else.

Copying alone can't be the test. A system run as many copies, like chapter eighteen's chains, copies everything it holds to every copy: the gas a block used, as faithfully as the word final. So rival events get copied too. Reliance doesn't split them apart either. On Ethereum, since a 2021 change called EIP-1559, the gas the last block used sets the next block's base fee: aim for a block half full, and if the last one came out completely full, the base fee goes up by an eighth, so 800 becomes 900; if it came out empty, it drops by an eighth, to 700. Every copy reads that number and acts on it. That's an event nobody would call an established record, copied and relied on by parties who didn't make it.

So examples can't show that only the established record gets copied and relied on. What the thesis has to ask is narrower: whether some kind of reliance, picked out for its own reasons, selects that event, and whether that selection gives a reason to charge for it. Any exact test fixes its readers and their actions in advance. It can test those choices, but it can't choose the real readers for me.

There's a second handle on the same question, from chapter three. On a finite machine, every move that settles something for good is a move Bennett's rule charges, and that argument never used the word "check." So it points at a narrower candidate: the events worth pricing are the ones that write a record no later step

can take back. The established record is one of those. As an exact answer, though, that candidate is wrong, because the three cards in chapter three showed a charged move that takes a star back. Whether the records that get copied and relied on are the permanent ones, and whether being copied and relied on picks out anything the merging doesn't, is what the two handles have to settle between them.

I can see two holes in this from here, so you don't have to find them for me.

The first is a missing step. Even if the world copies established records more than any other kind, "copied" doesn't give you "priced" by itself. Something has to connect them, and the best candidate I have is about incentives: a record that many separate parties copy and lean on is exactly the record worth forging, so it's the one where a price buys the most protection. That's reasonable, but it's an assumption. The thesis states the connection as something to test. It doesn't explain it, and nothing in this book does.

The second is a circle. If "established record" just meant "the record other people rely on," then of course it would be the one that gets relied on and copied, and the claim would say nothing. So the event has to be picked out without mentioning reliance, and it is: a box going up for good right after a check of the same claim passed. You can find that in a run without asking who reads it. Whether the world copies that event more than its rivals is then a real question, and it could come out no.

Now the word. Fix, for some crowd of readers, a test for whether an event is copied and relied on by parties who didn't make it. That test has to come from somewhere other than the established record, or the whole thing repeats a definition. Call an event a pointer, after Zurek's dial, if it passes that test and, on every state the machine can actually reach, every other event that passes holds wherever it does. There may not be one. But there can't be two that disagree: if two events both pass and each holds wherever anything that passes holds, then each holds wherever the other does, so on the reachable states they're the same event.

So here's the thesis, which I call the pointer thesis. Take record-keeping the way people actually run it, modeled faithfully. Then two things. First, the established record is the pointer: it gets copied and relied on by parties who didn't make it, and every event that gets copied and relied on that way holds wherever it does. Second, every record the system puts a price on gets copied and relied on by parties who didn't make it.

It's a thesis and not a theorem because of two words: "faithfully" and "actually run." Those are what tie the definitions to the world, the way "everything a person could do by following rules" does for Church and Turing, and no definition can do that job. The other thing it needs does have an exact meaning. Readers have separate carriers when each reader's view depends only on its own part of the world, and any two readers' parts could be set independently of each other. The states

the machine reaches can still tie those parts together; the thesis asks for agreement there. Everything else, which systems, which states, which event, which prices, goes in as input. The thesis proves nothing about which real systems fit it.

19.3 How much of the borrowed shape survived

Zurek's copying comes in degrees: each piece of the environment carries almost all of it, with a little lost. And it needs the pieces to be separate: different people reading different parts of the light.

My first try at writing this down had neither. It was the no-loss case with no separateness at all: a crowd of readers, one yes-or-no each, and an event counted as copied when every reader's yes matched it exactly. Nothing in that says two readers draw on different parts of anything. Picture three readers all looking at the same single switch. They agree perfectly, and they're one reader three times. So I'd kept the word and dropped the separateness that makes his version mean "objective." That's why the thesis above asks for separate carriers, and it's also why separate carriers aren't enough on their own: three separate switches can be set any way at all, so something, a copying process, has to put them in step before they agree.

The separateness I dropped is exactly what the thesis's closest published neighbor keeps. Jess Riedel, Zurek and Michael Zwolak in 2016 took histories of a closed quantum system, the story a whole run of events tells, and asked which of them the world keeps records of in many separate pieces of the environment. As I read their paper, records kept in many separate pieces make a history behave, so ordinary yes-or-no reasoning about it works, and they pick out a small set of histories from the huge number that merely behave: in their words, an objective past. That's "what gets copied is what counts," in quantum form, with the separateness in it, and it came before this book. Jonathan Halliwell, in a 1999 paper called "Somewhere in the universe," had already carried the link between records and well-behaved histories from the simplest starting states to messier ones, where the records still exist but can't be perfect.

There's a deeper mismatch in the borrowing, and it runs against this book, so it goes here. In Zurek's picture the whole thing, system and device and environment together, follows quantum mechanics' rule for a closed system, and that rule never lands two states on one. Every step can be run backwards. So seen from the whole, the environment's copying is just copying: the dial's reading gets written onto fresh light, and no two states anywhere meet. That's chapter two's way out, fresh paper, and chapter six's blank page. It isn't a claim that a closed system goes on to new states forever; a finite closed system can come back around.

Look at the dial alone and it's different, because you've thrown away how it's tied up with the light. Quantum states carry a phase, a part that only turns up when

things interfere. Take two states of the dial that are as different as two states can be, differing only in that phase. Let the environment wash the phase out completely, and the dial on its own ends up exactly the same from either start. Two arrows met, in the dial's own description. If the washing out is gradual, they only get harder to tell apart, and never exactly equal at any finite time. So an exact merge in a part of the picture, a slow loss of what you could tell apart, and a flow of heat are three different claims. Getting a heat cost out of any of them needs more: how the thing was prepared, what the change in disorder was, and the conditions that make heat the right word. Copying by itself supplies none of that.

So the thesis points at copies being kept and used, and the quantum account explains particular ways information spreads and becomes readable. It doesn't attach this book's toll to the copying.

A published no-go cuts closer still. Philippe Allard Guérin, Veronika Baumann, Flavio Del Santo and Časlav Brukner gave an argument in 2021 that, under assumptions that look natural in quantum mechanics, an observer's own record of an outcome at two different times can't share one reality: no joint chance for the two, depending in the simple straight-line way on the starting state, matches what quantum theory predicts for each. A record that lasts, for the one holding it, is exactly what gets into trouble once the whole is treated as never merging anything. It isn't a theorem about my machine, which isn't quantum. It's still the strongest published reason I know to hold the word "permanent" loosely when a physicist is in the room.

Which leaves the question to ask of any borrowed shape: has anybody checked the original? For a long time quantum Darwinism was a theory without a lab. Then, in 2018 and 2019, three groups working separately, in Italy, China and Germany, went looking for its fingerprint, as Philip Ball's 2019 report in Quanta Magazine tells it. The German one, at Ulm, with Zurek among the authors, watched the spin of a single defect inside a diamond and the carbon atoms around it, and saw the copying the theory predicts. The defect's state turned up copied in its surroundings, and it didn't take much of the surroundings before more of them told you nothing new. The other two did it with light. Is that the bottom? Not by a long way, and the people involved said so. Zurek himself pointed out that the experiments with light make their copies artificially and that the diamond's environment is small, and others noted that all three are sketches of a real environment and don't rule out rival accounts. So the shape I borrowed has passed some first tests in a lab, at the scale of a handful of atoms or bits of light, with those cautions written on it. None of that touches the thesis here, which is about machines and never rested on quantum physics. It changes how I hold the picture. What I borrowed is a lead somebody is checking. It isn't settled, and I don't get to lean on it as if it were.

19.4 What the thesis asks of the examples

Three comparisons would test it, and each one could come out against it.

First, look at a record a real system puts a price on. Do parties who didn't make it copy it and rely on it? A priced record that nobody else relies on breaks the second half.

Second, put the established record next to its rivals. If some event the established record doesn't cover gets copied and relied on to the same standard, the established record isn't the one pointer. The base fee from a moment ago is a candidate: on Ethereum it's read and acted on everywhere. Whether it passes a test of reliance that was picked for its own reasons, and not to let it in or keep it out, is exactly the kind of thing the thesis has to face.

Third, give the readers separate carriers in a faithful model, and ask whether the established record itself passes. Each reader has to be able to recover the event from its own view. A reader whose view stays the same whether the event happened or not can't; that's chapter seven's wall, where nothing about the box comes through. Naming a crowd of readers isn't enough. The model has to say what each one can read and what each one relies on.

Those comparisons are the evidence the claim about the world asks for. The exact test only turns into a definite question once the model and its readers are chosen.

19.5 Where the argument has got to

This is the last choice the book has to own, so here's the whole road behind it, with what was shown kept apart from what was chosen.

Chapters one and two showed that on a finite machine, settling something for good makes two arrows meet, unless the machine uses fresh paper on every tick. Chapter three chose a price on two arrows meeting, Bennett's rule, and showed that with it, every move that settles something for good costs at least one. Chapter four built the smallest machine where that happens. Chapter five showed that a record with more than two positions is a row of cups, that a straight line of stages needs exactly one cup fewer than it has stages, and that the toll on the row charges at least one on any step that turns up a cup. Chapter six showed that looking can be free. A look uses up a blank page, which can be handed back, and owes no floor. Chapter seven showed that a window with a clash can't price exactly. Chapter eight chose what "established" means: up only right after a passing check of the claim. Chapter nine drew what settling looks like, a sorting with a door that only swings one way. Chapter ten said what structure is and which ruler I chose to measure it. Chapter eleven showed that once the price list is fixed, the counter is arithmetic, and that a counter that goes down gets leaned on where it goes down. Chapter

twelve showed what a ticked box lets you lean on, and no more. Chapter thirteen chose the meaning of insight: a distinction made to stay, with price and depth kept apart. Chapter fourteen showed that over any machine, a box that answers to it and never comes down is a cup, with two choices left open, the price list and the event. Chapter fifteen showed what the toll pays for when the machine is finite, the record only grows and the uncovered machine forgets nothing. It pays for crushing two states of one pile. Chapter sixteen let the machine read its box to take a shortcut, and found four ways to keep the shortcut honest after the data changes, each one chapter two's bill in a different currency. Chapter seventeen showed that a machine's own shortcuts can't be judged from inside it, and that the commitment can still be recorded. Chapter eighteen found that of five real systems, read through my copies, one charges a fee at the write. And this chapter proposes the event, and says how the proposal could fail.

19.6 Does the list of requirements pick out the toll?

One question about the price itself is still open. Do the requirements I've put on a price pick out the toll and nothing else? If a short list of them left only one price standing, that would settle it. The lists I tried don't, and the no is small enough to check on three states.

Call them blank, once and twice, with once and twice reading yes. Two moves. The flip takes blank to once and leaves the other two where they are. The smooth takes once to twice, and leaves twice and blank alone. Draw the arrows. The smooth lands once and twice both on twice: two arrows meet, so Bennett's rule charges it. The flip lands blank and once both on once, so the rule charges it too. But only the flip raises anything, and only from blank. The smooth goes from yes to yes.

Now try requirements that never mention the reading. No charge on a move that changes nothing. No charge on a move that forgets nothing (both moves here forget, so this one asks nothing). The same cost if you rename the states (the only renaming that keeps both moves the same is leaving every name alone, so this asks nothing either). A total that adds up along a run. A cost that depends only on the state and the move. And a cost of one on the flip from blank. Let the price depend on the state as well as the move, the way chapter seven's clerk was allowed to. Go through the six steps one at a time. The flip from once and the flip from twice change nothing, so they're free. The smooth from blank and the smooth from twice change nothing, so they're free. The flip from blank costs one. That leaves the smooth from once, and nothing on the list says a word about it. Charge it nothing, and you have the toll: one on the raise and nothing else. Charge it one, and you have a price on merging. Charge it two, and that passes too. The list can't choose.

What chooses is a requirement that names the record: over any run, the total is never more than the number of times the reading went up. The smooth from once raises nothing, so it has to cost nothing. With the floor from the toll, that leaves exactly one price, one per raise and nothing else, which is chapter seven's step price again. So the uniqueness is real, and it's small. It holds once you've said which reading you're charging for. And which reading deserves the charge is the pointer thesis, which is still a thesis.

The uniqueness results that do exist are about pricing information lost, and they're worth a look because they say what kind of charge a price on merging is. John Baez, Tobias Fritz and Tom Leinster, mathematicians who work in category theory, the mathematics Lawvere's diagonal from chapter seventeen came from, asked what a measure of information loss has to look like. As I read their paper, they proved that for maps that keep track of the probabilities, Shannon's entropy, up to a constant factor, is the only measure of information loss with three natural properties: it adds up along a chain of maps, it mixes the way probabilities mix, and it changes smoothly. János Aczél and Bruno Forte, mathematicians at the University of Waterloo who worked on functional equations, and Che Tat Ng, who had taken his doctorate there under Aczél, asked which entropies deserve to be called natural, and proved, by their account, that the ones with a short list of natural properties are exactly the mixtures of Shannon's and Hartley's, where Hartley's is just the log of the count. So even for merges, the choice between the average and the count is a real choice. Both price merges. Neither prices raises, so neither picks out the toll, and that's the three states again, at a larger size. On the other side of the fence, Luciano Pomatto, Philipp Strack and Omer Tamuz, economists at Caltech and Yale who asked what learning something should cost, proved, by their account, a uniqueness result for what it costs to get information, when running the same experiment twice costs exactly twice as much as running it once. That prices producing a signal, which is a different matter from chapter six's copy onto a blank page; the copy owes no floor and borrows fresh paper.

Most of that isn't a machine. The arguments in this book are about any states and any moves, and chapter eighteen's five are small copies of somebody else's systems. The smallest machine is the box from chapter four, with every part named and every price written down. That's where everything above has something to bump into. If a statement up there doesn't mean what I think it means, the box and the five copies are where it shows.

Every price in this book so far is one I wrote down. Whether the world charges anything like it, in heat, is a question for physics, and it's the next chapter.

CHAPTER 20

Heat

Every price in this book so far is one I wrote down. Chapter one said it had nothing to say about heat, and chapter three promised heat its own chapter, with the people who doubt it named, and promised that Landauer's heat would turn out to be owed only for what you didn't know. This is that chapter. Everything in it is either counting, which I'll do with you, or physics, which I'll name and source and leave as physics. I'll keep saying which is which.

20.1 The stamp at zero

Start where chapter three left one door open. Take a sheet of paper whose reading is just BLANK or STAMPED, and one move: stamp it if it's blank, leave it alone if it's already stamped. It's finite, two readings. It's permanent: STAMPED stays STAMPED. And it settles something, because BLANK goes to STAMPED. Both readings land on STAMPED, so two arrows meet. The paper's fibers, the ink and your memory aren't in this two-state picture; the merge is a loss from the reading, not the disappearance of every trace.

Now set the price of that move to zero. The toll fails, and you can watch it fail: the box went up for nothing. Nothing in counting stops you from doing that. A price of one satisfies everything chapter three asked for, so the asks can hold together; whether they must is a different question, and only a claim about the world can answer it. The nearest such claim is Rolf Landauer's.

I have to be exact about the gap between his claim and mine. Bennett's rule, as I wrote it, charges every move where two arrows meet, whether or not you already knew which state you were in. Landauer's heat, as this chapter will show, is owed only for what you didn't know. My rule is the worst case on purpose: a move that has to work whatever it's handed pays for the worst it could do. And it keeps only a yes-or-no question, whether any pair can meet. The one-bit reset is where it comes from, but it doesn't measure the heat of any particular preparation.

20.2 What Landauer said

Here's what Landauer said, in 1961, as exactly as I can put it. I went past the summaries and read the paper itself. A logical operation with no single-valued inverse, a step that forgets, in this book's words, must give off heat, "typically of the order of kT" per operation, where k is Boltzmann's constant from chapter three and T is the temperature. For the specific case of resetting one bit whose value you don't know, the minimum is kT times the natural log of 2, about 0.69 kT.

I want you to hear two things in that. First, "of the order of": he wasn't handing out a number to three decimals. Second, and this surprised me, he didn't measure anything. The paper is argument, with models. He writes that he can "show, or at least very strongly suggest" the conclusion, and that his reasoning gives "no guarantee that this minimum is in fact achievable." That's the founding paper of the whole subject, and it's a careful man saying "I think this is right and here's why." I like him more for it.

Was it measured, eventually? Yes, and only recently. In 2012 a group at the École Normale Supérieure in Lyon, Antoine Bérut and Sergio Ciliberto's lab working with Eric Lutz, trapped a single glass bead, two millionths of a meter across, in a double well made of laser light, and used the two wells as the two values of a bit. Then they erased the bit, by lowering the barrier, tilting the trap and raising it again, many times over, slower and slower, and measured the heat each time. The average heat came down toward Landauer's kT ln 2 as the erasure got slower, closer the slower it went, and never under it, on average. Single runs went under it; the bound is about the average, and any one erasure can beat it. And if you only succeed in erasing half the time, the floor drops to zero, which the paper's formula shows and which I think is the most honest line in it.

Others have done it since with different devices. Yonggun Jun, Momčilo Gavrilov and John Bechhoefer did it with a bead in a feedback trap, and ran the same moves with no tilt as a control. Jeongmin Hong, Brian Lambson, Scott Dhuey and Jeffrey Bokor did it with tiny magnetic memory bits. Rocco Gaudenzi and his coauthors did it with the spins of molecules at one kelvin. The table near the end of this chapter has their numbers. So it was predicted in 1961 by argument, and seen in a laboratory fifty-one years later.

20.3 Who doubts it

Is it settled? No, and I'd be lying by leaving that out, because the people who doubt it aren't cranks. John Norton, a philosopher of physics at Pittsburgh, has argued since 2005 that the usual proofs of the principle quietly assume what they're trying to prove, and that the kinds of process they assume, put together, could be made to

break the very law the principle is supposed to guard. An earlier objection, from Norton and his Pittsburgh colleague John Earman, also a philosopher of physics, was that the principle can't be independent of the second law. Charles Bennett, who did more than anyone to build the subject, rated that one "in my opinion the one of greatest merit," and then disagreed with it. Others replied to Norton; Norton replied back. Owen Maroney, a philosopher of physics who works on exactly this, wrote the Stanford Encyclopedia of Philosophy's entry on it, and it ends without a verdict. I've read what I could of both sides, I'm not qualified to referee them, and I'm not going to pretend the laboratory settles a question about what the principle means in general, because measuring one bead doesn't do that.

So here's what I did with it, and I think it's the only honest thing to do. The charge on forgetting is written into this book as an assumption with a name. Nowhere is it proved as a general fact, because it can't be proved from counting. For a particular machine with a particular price list you can see it hold, the stamp at cost one, and that's a different thing. If Norton turns out to be right, every argument in this book still holds; they just have one assumption fewer that the world agrees to. If Landauer's side turns out to be right, it backs a heat price under its conditions, and the whole-number charge on every merging move is still my bookkeeping rule. Either way nothing here says a device does anything until somebody measures the device.

Naming an assumption isn't the same as leaving it unplaced, though, and a hedge isn't a position, so I'll give mine. Earman and Norton's objection is a fork, set out at the start of the second part of their paper. Either an information-based answer to Maxwell's demon assumes the second law—the sound horn, they call it—and then it adds nothing the second law didn't already give; or it rests on a new principle tying information to entropy—the profound horn—and nobody has given that principle grounds of its own. My named assumption is the sound horn, on purpose. That's where Bennett ended up, and Norton's "Too Good to Be True," written in 2025 and published in 2026, says the careful literature almost always takes that horn, and argues that whatever works there is ordinary thermodynamics relabeled. I'll take that. Ordinary thermodynamics, relabeled onto a ledger, is all this book asks of the physics.

And on the counting, I'm on Norton's side outright. His 2018 paper "Maxwell's Demon Does Not Compute" disposes of the demon with Liouville's theorem and nothing else, no information and no computation: motion that keeps every patch of states the same size can't crowd a big set of states into a small one. The counting at the bottom of this book, that a step losing nothing can't carry a no-state into a set that stays up, is that point in finite, countable form: chapter two's chairs.

If you want the best case for the assumption itself, the one I found is James Ladyman, Stuart Presnell, Anthony Short and Berry Groisman in 2007. From one of

the standard statements of the second law, the one named for Kelvin, they derive that a logically irreversible operation can't be carried out in a thermodynamically reversible way by any of their model machines, which they call L-machines. That's a physical argument under their stated model of a machine. It deserves to be argued with on those terms. It doesn't by itself give my whole-number charge.

20.4 Where the difference goes

A physicist asks one question before any of that, and it's a good one. The laws underneath never merge anything. Classical mechanics carries each exact state forward to exactly one next state and could carry it back, and Liouville's theorem adds that a patch of states keeps its size as it moves. Quantum mechanics is one-to-one too. So when this book says a step merged two states, where did the difference go?

It went out of the machine. The merge is in the machine's own description, the states its reading is about. The machine together with everything around it still runs one-to-one, and the distinction the machine forgot gets pushed out into the surroundings. The surroundings are chapter two's endless strip: fresh paper, which the universe as a whole may have to spend and the machine doesn't, because the machine isn't the universe and its memory fills.

Does that make the whole argument depend on where you draw the machine? Draw it coarser or finer and you get different states: the dial on a bench is a state, and so is each atom in it, and so is each atom together with the light it gave off. It doesn't, and the reason fits in one picture.

Draw the fine states, and draw a fence around each group of them that counts as one coarse state. Say the coarse description follows the fine one exactly. Wherever a fine state steps, its fence steps to the fence of where it landed, and every fence has at least one fine state in it. Now say the coarse picture merges, so two fences step to one fence. Then something has to give down below. Suppose it doesn't. The fine step never merges, never steps outside the fenced states, and no run goes on landing on fine states it hasn't been to before, forever. Follow any fine run. It must come back to some fine state it's been in, and since nothing merges, walk the two visits back together one step at a time, the way chapter two did, until the earlier one reaches the start. So every fine run is a loop through its own start. Look at that loop through the fences. Every fence is on a loop too. Now take the two fences that stepped to one. Wait as many steps as the product of their two loop lengths, and each is back where it began. But they were together after one step, and the same rule can't pull them apart again, so they must have begun in the same place. They didn't. So down below, one of three things happens: the fine step merges, or it steps

out of the states the coarse picture describes, or it runs off taking a new one every tick. The three ways out from chapter two come back one level down.

Chapter nineteen's dial and light are the example. In the dial's own picture, phases wash out and two starts end up the same: a merge. In the picture of dial and light together, the step is one-to-one, and it's the second way: the light starts blank and leaves carrying the copy, so the fine step leaves the states where the light is still blank. Nothing merges at the fine scale.

And that's where the heat floor sits: where the merge really is. Landauer's assumption charges the scale whose step merges states that were really possible. A description that merges only because it stopped looking at the light owes that floor nowhere just because of that. Pushing a distinction out isn't heat yet, either: another memory could keep it as a readable record. It becomes heat when a reset removes real uncertainty into something warm, a reservoir, and then Landauer's assumption bounds the average heat handed over.

None of that heat has to be wasted. Takahiro Sagawa's 2014 analysis says erasure can in principle approach a process that produces no entropy overall: the reservoir's entropy goes up by what the memory's goes down. A positive heat can be unavoidable without being wasteful.

20.5 Two meanings of reversible

Ladyman and his coauthors and Sagawa look like they disagree, but they don't. They use "reversible" for two different things.

Ladyman's L-machines are devices, and a device counts as reversible only if it produces no entropy on every input it's fed, each one known. Their section 4 says it: the L-machine is thermodynamically reversible if and only if every one of its processes, one per input, produces no entropy. Sagawa's reversible erasure is a process on one given spread of inputs, with the device tuned to that spread. The standard eraser is reversible in Sagawa's sense and not in Ladyman's: on an even mix of 0 and 1 it produces no entropy, and on each known input it produces k ln 2.

One identity holds the two together. Take a fixed device that sends every input to one output, and any spread of its inputs. The average, over the inputs, of the entropy produced on each input known, equals the entropy produced on the spread, plus k times the Shannon entropy of the spread. It holds because the heat is the same average either way, the device being fixed, and the output is the same either way. The only difference is at the start: the spread's own entropy includes a term for not knowing which input it is, and each known input's doesn't. Sagawa makes the first term on the right zero. Ladyman reads the left side, which is then k times the spread's entropy, more than zero. For the even mix of 0 and 1 that's k ln 2 on the left, nothing plus k ln 2 on the right.

That identity hands you a floor that doesn't need anybody's knowledge. Take K inputs a fixed device sends to the same output, and spread your chances evenly over them. Same output means one state at the scale where nothing inside the device still holds which of the K it was. If the K land on K different fine states inside one well, the argument about scales above says where the difference went: into a merge among the fine states, out past the fence into the surroundings, or into fresh states that a memory reset over and over can't keep supplying. The average over the K of the entropy produced on each one known equals what's produced on the even spread, plus k ln K. The second law makes what's produced on the spread zero or more. So the average is at least k ln K, and the worst of the K is at least the average. On some one of them, the device produces at least k ln K of entropy, and, when dissipated work is temperature times entropy produced, at least kT ln K of dissipated work. Going slowly doesn't remove that. Using different controls for different known inputs changes the device being compared; the bound is about one fixed device across them. So the floor sits on the merging move, on its worst input, whatever anyone knows. That's the shape of the toll.

20.6 Counting in bits

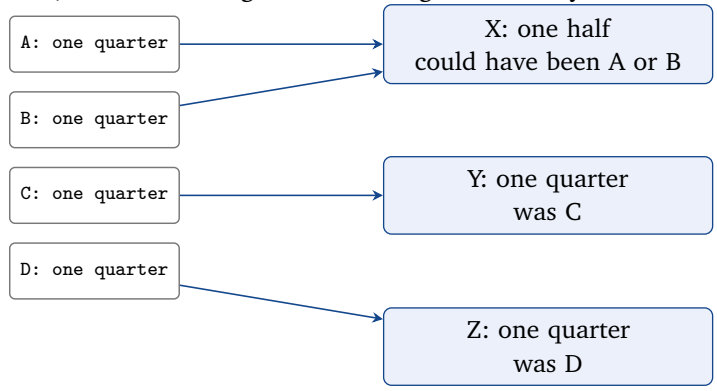
There's one more way to say all this, and I include it because it's how physicists say it, and because it says exactly when the heat floor is zero: when no two states you might be in land on one.

Claude Shannon, in 1948, gave a number to how much you don't know. If something could be in any of n states and you've no idea which, your uncertainty is log base two of n bits, the number of halvings from chapter three, and that's the most it can be for n states. Push your uncertainty through a step that forgets nothing and it stays the same; through a step that merges two states you might have been in, it drops.

Why does Shannon's number share a name with the physicists' entropy at all? There's a story. Shannon wanted to call it uncertainty, and the mathematician John von Neumann told him to call it entropy instead, for two reasons: the same formula already lived in physics under that name, and nobody really knows what entropy is, so in an argument he'd always have the upper hand. It's a great story. It comes from Myron Tribus, who spent years tying information theory to thermodynamics, who said Shannon told it to him in 1961, and who printed it with Edward McIrvine in Scientific American in 1971. Then in 1982 an interviewer, Robert Price, asked Shannon about it, and Shannon said he didn't think von Neumann had told him any such thing. So it's a legend with one witness, and the person the witness quoted doesn't remember saying it. What doesn't hang on the story: the two formulas really

do have the same shape, and it still takes Landauer's assumption to turn a count of bits into heat.

Here's how the count works when the chances aren't even. Give each state a chance, all adding to one. Each state contributes its chance times the number of halvings it would take to single it out, and you add those up. Try four equally likely cards, A, B, C and D. Each has chance a quarter, and singling one out takes two halvings, so each contributes a half, two bits in all. Now send A and B to a box marked X, C to a box marked Y and D to a box marked Z. The boxes' chances are a half, a quarter, a quarter. X takes one halving to single out, Y and Z two each, so the three contribute a half each, one and a half bits. Half a bit was lost on average. You can see where. Half the time the box is X and leaves one question unanswered, A or B? The other half of the time, Y or Z tells you the card exactly. An actual draw never loses half a card; the half belongs to the average over many draws.



Half the time one question is left open; half the time the card is known. On average, half a bit is lost.

A move that forgets nothing sends each state somewhere different, so the chances get carried along one for one: nothing piles up, and the uncertainty is the same number after as before. That's why a price on bits lost can leave every reversible move costing nothing, which is the shape the toll needs.

Now push uncertainty through a permanent raise. Say m states already read yes, and the move turns k states that read no into yes. Before the move you might be in any of the m + k states in play. After it, you're in one of m. How much uncertainty can m places hold? At most log base two of m, and only when the chance is spread evenly. Here's why that's the most, in one line. The curve of the natural log never rises above the straight line y minus one; they touch only at one. For each place with chance q, put y equal to the even share, 1 over m, divided by q. The line says q times the log of that y is at most the even share minus q. Add over the places: the right side totals at most one minus one, which is zero. The left side totals the spread's uncertainty minus log m. So the uncertainty is at most log m, whatever the spread.

So if you started with every one of the m + k states equally likely, you had log2 of (m + k) bits, you end with at most log2 of m, and the move removed at least log2 of (m + k)/m. That's chapter three's halving count again, in Shannon's units. There's a second floor too, and sometimes it's bigger. Count one landing pile at a time. Say a pile holds s of the starts. You land in it with chance s over (m + k), and when you do, log2 of s bits are gone. The s starts in it can't be told apart any more. So that pile's share of the total loss is s log2 s over (m + k). A pile of one adds nothing. A pile of two adds 2 over (m + k). And s log2 s is never less than 2(s minus 1). The two are equal at s of two, and from there s log2 s climbs faster. So every pile adds at least 2 over (m + k) for each start after its first. There are m + k starts and at most m piles, so at least k starts come after the first in their pile. That gives at least 2k over (m + k) bits. The first floor is reached when k is a multiple of m: keep each old yes-state where it is and send the same number of newcomers to each. The second is reached when k is at most m: send each newcomer to its own old yes-state.

And in the worst case it's a full bit every time. A permanent raise makes two arrows meet, by chapter two. Put half the chance on each of those two states, and the move removes exactly one bit, however many states already read yes.

The assumption, stated as I use it: for a chosen physical memory and its spread of inputs, a step that lowers the memory's logical uncertainty by some number of bits hands at least kT ln 2 of heat per bit to its surroundings at temperature T, on average over that preparation, not on every single run. It needs a physical model in which the logical uncertainty has that meaning; a table of arrows doesn't supply one. And it's for a symmetric memory, whose two states are alike apart from their labels, as Landauer assumed. For a memory whose states differ, the floor changes, which Owen Maroney worked out in 2009, and Gavrilov and Bechhoefer in 2016 measured erasure work in a lopsided double well that came out below kT ln 2.

20.7 How big is that

Take the case where one state flips in beside one that was already yes. The floor is kT ln 2. At room temperature, about 300 kelvin, that's roughly three zeptojoules: three thousandths of a billionth of a billionth of a joule. That's the biggest floor a one-state raise gets from the even spread. With one newcomer the floor shrinks as more states already read yes, because the squeeze gets thinner: with a thousand of them it's 2 ln 2 over 1001 of kT, about a seven-hundred-and-twentieth, some five hundred times less than the case of two.

So for the even spread, one unit of cost is not one halving. It's one halving exactly when the squeeze is two, and for any thinner squeeze the physics asks for less, and the toll still charges a whole unit, because the cost comes in whole numbers. In the

worst case over what the machine might be holding, the way chapter three's halving rule reads a move, it's a full bit every time, so the gap between the unit and the bits is a fact about the even spread. Whatever the size, it's heat only if you grant the assumption; nothing on this page measured anything.

For comparison, Anson Ho, Ege Erdil and Tamay Besiroglu's 2023 tally of the limits of ordinary chips puts switching a single gate at a hundred attojoules at the low end, over thirty thousand times this floor. The point of the number isn't that a permanent raise is expensive. It's that, on the assumption, it isn't zero.

The same logarithm turns up in a different physical question, and I want to keep the two apart. Yûto Murashita, Ken Funo and Masahito Ueda in 2014 studied processes whose reverse histories include some with no forward partner, which they call absolute irreversibility. Their bound on average entropy produced picks up the logarithm of one over the share that's left. Their example is gas that starts in one part of a box split into two, the parts in the ratio l to 1 minus l, and spreads out through the whole box; the logarithm is of one over l. (Take l to be a tenth and you get the logarithm of ten; that number is my illustration, not theirs.) Make l the old yes-states' share of the pile, and out comes the same logarithm as the squeeze. That's a match of numbers, not of processes. My arrows don't say what the reverse experiment is or how likely its histories are, so they don't determine that share. Gonzalo Manzano, Gülce Kardeş, Édgar Roldán and David Wolpert in 2024 studied questions like that for computations with one-way steps and random stopping times. Their physical assumptions are on top of my counting.

The even spread is the easy case, and it's fair to worry that the heat depends on picking it. It doesn't. Take any spread that gives a real chance to the state that flips and to every yes-state. After the move, they all land among the yes-states, and there's one more of them than there are yes-states, so two of them with real chances land together. Count bits state by state: each state carries its chance to where it lands and finds whatever else landed there, and its share of the loss is how much more crowded its landing spot is than what it brought. Nobody's spot holds less than they brought, so every share is zero or more, and the two that landed together each find a spot holding more than they brought, so their shares are more than zero. A positive number of bits, at a positive temperature, is positive heat.

20.8 Three roads, two strengths

There are three ways to charge for forgetting, and they come in two strengths. Charge the bits a move removes, on every spread: that's entropy pricing. Charge, as chapter three did, at least as many units as the halvings in a move's biggest landing pile: that's the halving rule. These two are the same rule. The most bits a move can remove, on any spread, is log2 of its biggest pile, because once you know which pile

you landed in, the starts left to tell apart are at most that many; and the even spread on that biggest pile removes exactly that much. So a price that covers every spread covers the biggest pile, and a price that covers the biggest pile covers every spread.

The third road is Bennett's rule from chapter three, at least one whenever any two arrows meet. That's the halving rule looked at through two starts only. If two starts land apart nothing's lost, and if they land together, a pile of two asks for one halving. So it's weaker. Try eight equally likely cards all sent to one box, and three bits are lost. A charge of one passes Bennett's rule and fails the other two, which ask for three. All three give the toll on a permanent raise; only the stronger two give the size of the squeeze.

20.9 The coin under the cup

Now the case where the floor is zero. Toss a fair coin under a cup without looking at how it landed, and let the rule be: set it heads up. You might have had tails; now you have heads; one bit of what you didn't know is gone, and that's the bit Landauer charges for. Now lift the cup, see heads, put the cup back, and apply the rule again. Nothing you didn't know changed, and no bit was lost. And the assumption, applied to a state you already knew, asks for nothing: a known state is a spread with all its chance on one state, which holds no bits before or after, so the assumption reads "heat at least zero." It sets no floor.

That's easy to overread, so be careful with it. It doesn't mean a stamping machine run on a known-blank sheet gives off nothing. A real stamping device may cost something because it runs in finite time; how much depends on how it's built, how it's driven and how fast it goes. All it means is that the assumption sets no floor there. The floor comes from what you didn't know. And it doesn't mean telling somebody which way the coin landed changes an unchanged device's heat. If you lifted the cup after an uncertain toss, you made a record that's tied up with the coin, and resetting that record has to be counted with that tie. If the coin was handed to you heads up for certain, there was no uncertain toss to record. Those are different preparations.

This is the general form of Landauer's principle, worked out by Owen Maroney, in a paper he wrote at the Perimeter Institute in Canada, and by Sagawa. And Landauer himself had the seed of it in 1961: for a row of bits all already ONE, he wrote, there's no need to carry out any operation, so no entropy changes and no heat is given off; and resetting a row all ZERO to ONE involves no entropy change either, because neither row carries any information.

Notice that the toll still charges the known-state stamp one unit, because the toll is a rule about the move, and the move doesn't know what the watcher knows.

That's the worst-case gap from the start of the chapter, and it's the price of having a rule a machine can follow without asking who's watching.

The physics of all this has a standard review, Juan Parrondo, Jordan Horowitz and Takahiro Sagawa's "Thermodynamics of information," which puts Landauer's bound, its general form and the costs of measuring and acting on what you measured side by side. And the gap between a device and the spread it runs on has a physicist's number on it. Artemy Kolchinsky and David Wolpert showed that a fixed process built to waste as little as possible on one spread over its states pays extra when it runs on another, and the extra is how much closer the process brings the two spreads together, measured in relative entropy. They called it the dissipation due to an incorrect prior, and Wolpert and Kolchinsky later gave it the name it goes by now, mismatch cost. Under their conditions, their formula says which input pays: the one the device's own best spread gave the least chance. A device can make one input cheap only by making another dear. That's the worst-case gap again, priced for an actual spread, where the toll prices the move.

There's a second piece of fine print, about who works the controls. Lower this barrier, tilt that well, raise the barrier again: those operations have a timing and a physical build, and counting bits prices neither. Paul Riechers, Alexander Boyd, Gregory Wimsatt and James Crutchfield in 2020 analyzed memories held behind barriers under controls that are the same played forwards or backwards. A symmetric clock pulse can be like that; an arbitrary one needn't be. For that setting, as the allowed error shrinks, the least work picks up a term that grows like kT times the log of one over the error, times the chance that the memory sits in a state the operation, done twice, doesn't bring back to where it started. They call those transitions nonreciprocal. A permanent raise is nonreciprocal at the no-state: doing it twice can't bring it back to no. So a known input there can face extra work in their setting, even though it holds no bits. Their class also includes a three-state cycle, which merges nothing. It's about the allowed controls as well as the arrows, and it doesn't change the coin under the cup.

One more limit, about which states collide. Chapter two guarantees two arrows meet among the yes-states together with the one coming in. It doesn't guarantee that the one coming in is one of the two. Take four states with only the first reading no, and a move that sends the first to the second, the second to the third, and leaves the third and fourth where they are. The second and third land together on the third, and the first shares with nobody. So for a spread with all its chance on no-states, this counting alone forces no loss. The toll on the move still follows from Bennett's rule.

So here's what the counting and the assumption give together. A permanent raise has two starting states that land on one. Prepare those two with equal chances, and the raise removes one bit, and on the assumption its average heat is at least kT ln 2. That's a worst-case preparation, not a charge on every run. A known start

gets no floor from the assumption. A fixed device that merges K inputs produces at least k ln K of entropy on one of them. And the ledger charges the merging move whatever the preparation, because my rule says so. The counting proves a collision; the preparation decides how much uncertainty is removed; and the physics connects that to heat.

20.10 Going faster gives off more heat

Everything so far is about the floor, the least heat as you go slower and slower. Going at a speed costs on top of that, and here there's a small model I can work all the way through.

Charles Bennett's Brownian computer, from his 1982 review, is a walker on a line that's pushed forward a little. At each tick it steps forward with some chance p and back with the rest. Its speed, on average, is the difference between the two, 2p minus 1. Grant the standard assumption of this corner of physics, called local detailed balance, which Udo Seifert's review sets out: a forward step hands the surroundings kT times the log of p over 1 minus p in heat, and a backward step takes the same amount back. Then the average heat per tick, in units of kT, is the speed times that log, which follows from Jürgen Schnakenberg's formula for the entropy produced by a system that jumps between states. When p is more than a half, both are positive; when it's less, both are negative; either way the product is positive. Only at exactly a half, no drift at all, is it zero. And no drift doesn't mean a walker standing still: it still jumps about, and it can reach a nearby spot on the first tick. It just doesn't get anywhere on average.

How much heat to get somewhere? Per step forward on average, the heat is the log of 1 plus v over 1 minus v, where v is the speed, and that log is never less than 2v: the difference is zero at no speed and only grows from there, because its slope, 2 over (1 minus v squared), minus 2, is never negative. So walking a distance N costs at least 2N times v in heat and takes N over v in time, and heat times time is at least twice N squared. Halve the speed and the heat about halves while the time doubles; the product never drops below twice the distance squared. Going slowly makes the heat for a given distance smaller and the time longer. These are averages.

The walker is the case I can work by hand. Others have worked other models. Tim Schmiedl and Udo Seifert found the best way to drag a bead in a spring-like trap through water from one place to another in a set time, and the least work you can get away with is the drag times the distance squared, divided by the time plus twice the drag over the stiffness: at long times, drag times distance squared over time. Erik Aurell, Carlos Mejía-Monasterio and Paolo Muratore-Ginanneschi, for a bead that drifts, and Naoto Shiraishi, Ken Funo and Keiji Saito, for states that jump, showed that changing one spread of chances into another in a set time produces at

least a fixed fraction of the square of how far apart the two spreads are, divided by the time and by how lively the motion is. Those are their results under their stated motion, and I cite them as physics. They're about entropy produced, not positive heat on every step: a step that leaves the spread exactly as it was owes nothing on them, and a single step's heat can be negative when the device's own entropy goes up.

And there's a machine that, in principle, owes nothing at all. Edward Fredkin and Tommaso Toffoli, whose gates that lose nothing were in chapter two, also described in 1982 a computer made of perfect billiard balls bouncing their way through a computation, and in principle it could run without wasting anything. Bennett wrote in 2003 that such ballistic machines compute at constant velocity without dissipating any energy, and that they must be kept isolated from heat baths and noise. Keeping one on course against noise means correcting its errors, and whether that brings the waste back I don't settle here; in 2003 Bennett called the thermodynamics of error correction a subject that appears to be in its infancy.

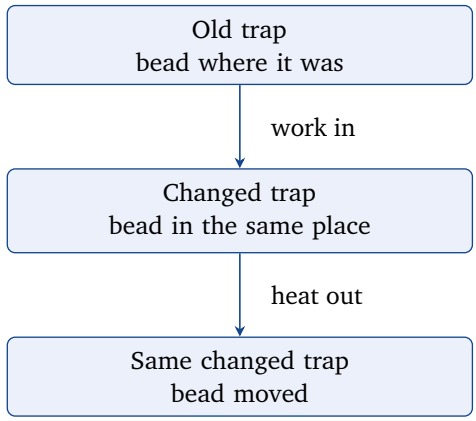
The bead of Jun, Gavrilov and Bechhoefer shows the other half. Put through the same moves as an erasure but with no tilt, so nothing is squashed, its work, carried out to the slow limit, came out at 0.05 kT, give or take 0.03: within two error bars of zero. No floor under it, which is what the assumption says. With the tilt, so two wells really do become one, it came out at 0.71 kT, give or take 0.03, against ln 2, which is 0.693.

So keeping something costs, on top of going at a speed, either a merge, which on Landauer's assumption owes kT ln 2 per bit that no slowness removes, or a state the device has never used before. Taking a fresh state escapes the floor, and doesn't settle how much a real device wastes in finite time. And a finite set of states, where every state has a next one, forces the merge.

20.11 A simulation in code

I wanted to see those steps, so I ran a simulation in code. It models a bead in a double well shaped like a quartic curve, and follows the experiment's sequence: lower the barrier for one second, ramp a tilt, take the tilt away, raise the barrier for one second. The well is a shape I chose and the motion is a numerical approximation; it isn't the measured trap, and nothing in it was measured. The modeled bead starts from the thermal spread in that well, equally likely on either side.

Keep three accounts as it moves. Changing the trap while the bead holds still does work on it. The bead moving in a fixed trap hands heat to the water around it. Whatever's left changes the bead's stored energy.



Work in minus heat out is the change in stored energy. Either one can be negative on a single step.

Work in minus heat out is the change in stored energy, on every simulated path, to rounding. That checks the bookkeeping. It doesn't show the time steps are small enough to copy continuous motion exactly.

Here's what came out, with 3,000 paths at each setting and a time step of half a millisecond. Each setting uses a tilt picked to make roughly nine erasures in ten work, and success means ending in the right well. The last two columns average only the successes, in units of kT. The time column is the tilt ramp; the two one-second barrier ramps come on top.

Tilt time (s)	Success rate	Heat out on success	Work in on success
2	0.909	2.517	2.501
5	0.916	1.466	1.490
10	0.913	1.073	1.095
20	0.899	0.830	0.907
40	0.893	0.773	0.782
80	0.869	0.693	0.719

At the slowest setting the sampling errors are about 0.023 kT for heat and 0.011 kT for work. They leave out error from the time step and from the model.

Heat and work are close in some rows and less close in others; the difference is measured by the simulation, not put in by hand. Now try to make a floor out of those six points. Fit the successful heat to a constant plus a number divided by the tilt time, and the constant comes out at 0.674 kT, give or take 0.015, near ln 2. Halve the time step and run the six settings again, and it comes out at 0.732 kT, give or take 0.016: it moved by nearly four of its own error bars. The second run also samples different paths, so that comparison mixes sampling error with numerical error. Still, it's enough to stop me treating a fitted constant near ln 2 as a confirmation. The barrier ramps are still one second long, the tilt changes between settings, and the fit doesn't take the whole procedure to a slow limit.

There's a theoretical comparison, and it's a comparison with work. For the continuous thermal model, starting in equilibrium and ending in the same symmetric trap, there are two bounds, both written in terms of the success rate p. Averaging over successes and failures, the work is at least kT times ln 2, plus p ln p, plus (1 minus p) times ln (1 minus p); that's the formula from Bérut and his coauthors' 2012 paper that drops to zero at half. Looking only at the successful paths, the work is at least kT times the log of 2p. Bérut, Artyom Petrosyan and Ciliberto worked that one out, from a version of Christopher Jarzynski's 1997 equality that sorts paths by how they end, due to Suriyanarayanan Vaikuntanathan and Jarzynski and to Ryoichi Kawai, Juan Parrondo and Christian Van den Broeck. At nine successes in ten the two come to about 0.368 kT and 0.588 kT. Neither is a heat bound for the simulation without also counting its change in stored energy.

The script is in the repository as scripts/bead_erasure_sim.py. It reports heat, work, stored-energy change, sampling errors and the fits, and takes a smaller time step for checking them. What it shows is that the slower settings use less work and give off less heat on average. It doesn't find a universal floor, and it doesn't reproduce the laboratory's fitted constant.

20.12 What's been measured

Here's the laboratory side in one place, so you can see how near it has come to each piece.

A merge, the floor	Jun, Gavrilov and Bechhoefer (2014): one bead in a feedback trap. Squash two wells into one: $0.71 \pm 0.03\,kT$ of work, in the slow limit. Run the same moves without the tilt, so nothing is squashed: $0.05 \pm 0.03\,kT$, also in the slow limit, within two error bars of zero. $\ln 2$ is 0.693. Same bead, same trap.
The floor, three more ways	Bérut and others (2012), a glass bead; Hong, Lambson, Dhuey and Bokor (2016), tiny magnetic memory bits: $1.45 \pm 0.35\,kT$, which they report as within two standard deviations of Landauer's limit; Gaudenzi, Burzurí, Maegawa, van der Zant and Luis (2018), molecular spins at one kelvin.
Lopsided wells	Gavrilov and Bechhoefer (2016), the feedback trap of Jun and others with a lopsided double well: the average work to erase came out below $kT \ln 2$. The floor as stated is for a symmetric memory.

Time against heat	Bérut, Petrosyan and Ciliberto (2015), on the 2012 bead: heat $= kT \ln 2 + B/\tau$ fits with $B = 8.15\,kT \cdot \mathrm{s}$; left free, the floor of the fit comes out at $0.72\,kT$. The floor is reached only as the time goes to infinity.
How long a bit stays put	The same paper, on the same bead, estimates a typical time to hop the barrier of about 3,000 seconds when the barrier is more than $8\,kT$, and about ten seconds at $2.2\,kT$. Those are statistical time scales. The barrier height sets how long the bit is kept; it isn't itself a measured heat payment.

These experiments test particular memories and procedures, including the symmetric-memory floor and heat against time. They aren't a survey of all physical computation, and they aren't an experimental proof of the counting.

20.13 Why the rule sits on the raise

So here's how the heat and the toll fit. The toll charges for what's kept, whichever way it's paid, merge or fresh paper. A price on merges alone charges only the merge, so it misses something kept by taking fresh paper: the machine that keeps every history, chapter two's endless strip, raises its box at zero cost under it. And it charges merges that keep nothing, like a reset that takes a box down. The toll doesn't ask that reset for a unit, and it still merges two states, so in a real device it owes Landauer's floor whenever the bit it lowered was really unknown. The ledger leaves that to physics, along with the waste of going at a speed.

That's the heat. What none of this touched is what keeping takes when the steps themselves are noisy, or quantum, and how much room a memory needs to keep its history without any merge at all. That's the next chapter.

CHAPTER 21

Keeping without the heat

The last chapter was heat, and every line of it leaned on an assumption with a name. This one doesn't need it. It's the counting that holds whether or not anybody ever measures a joule: what keeping takes when the steps are noisy, or quantum; how much room a memory needs to keep a history without any merge at all; what a machine's counter has to do with heat; and, at the end, a plain list of what you've earned so far and what you haven't. Then one last section that isn't counting at all, kept apart so you can check none of it leaked in.

21.1 A cell that's written once

Start with the smallest memory that keeps something. A cell has three states: blank, zero and one. The question is "is the cell written?", no on blank and yes on the other two, and a written cell stays written. The move is "write a 0": blank goes to zero, and a written cell is left alone.

Draw the arrows. Blank lands on zero. Zero lands on zero. Two arrows meet, so writing a 0 on nothing is, by itself, a merge. The history where the cell was blank and the history where it already held a 0 now leave the same cell behind.

The obvious way round it is a fourth state that remembers: "0, and it was blank before." Now blank lands on a state nothing else lands on. But the remembering state has to go somewhere on the next write too, and it's written, so it has to stay written. Count. There are three written states now, zero, one and the remembering one, and each of them lands on a written state. Blank lands on a written state too. That's four starting places and three places to land, so two share. It's chapter two's chairs, and adding a fifth state, or a fiftieth, only moves the problem along: as long as finitely many states are written, the written ones plus blank are one more than the room they land in.

The only way to write with no merge, once every state has a next one, is a supply of fresh states that never runs out: a new cell of an endless tape for every write, forever. Or a tape that stretches both ways, with the machine stepping one cell along it each tick. Every cell has one before it and one after, nothing ever shares a landing,

"past the start" goes up on the first step and never comes down, and the machine never returns. That pays at every step by taking a cell it has never used.

The argument leans on one thing, and I want it in the open: the write is a move defined on every state of the cell, written ones included. A write defined only on blank, with the machine simply stopping when it meets a written cell, merges nothing. That's the machine that stops, from chapter one: it buys the 0 by having no next step at all.

And notice what it takes to start. A blank handed to you is a starting condition, not proof that somebody paid heat to make it blank. If a device has to reset an unknown cell so it can be used again, the last chapter's floor can apply to that reset. Precise versions of that floor, like David Reeb and Michael Wolf's from 2014, assume a reservoir that starts thermal and untangled from the memory and a combined motion that merges nothing; their heat bound holds under exactly those conditions. Reading a past or future payment off a known blank would need more assumptions than a known blank supplies.

21.2 Noisy steps

Real states are noisy: the bead hops on its own. So take the step to be a table of chances. From each state, there's a chance of landing on each state next, and every row adds to one. A permanent reading is a set of states you never leave once you're in. And "nothing merges" has an exact noisy meaning: start with the chance spread evenly over the states, take one step, and it's still spread evenly. In the table, that's every column adding to one, as well as every row.

Can a step like that raise a reading that stays? No, and the count is the chairs again, with chances in place of people. Add up all the chance flowing into the yes-states from everywhere, one column at a time. Each column adds to one, so the total is the number of yes-states. But the yes-states alone already send exactly that much, because they never leave: each of their rows puts all of its one into the yes side. There's nothing left over for any outside state to send in, and chances can't be negative, so every outside state sends in nothing at all. Not a small chance. None.

And it runs the other way. A step that can carry an outside state in can't keep the even spread, so one step from the even spread leaves the chance more bunched than it was. Here's a way to measure bunching. Draw a state twice, independently, using the same chances, and ask whether the two draws match. For two equally likely states the matching pairs have chances a quarter and a quarter, so a match comes up half the time. Put three quarters on one and a quarter on the other, and the matching chances are nine sixteenths and one sixteenth: ten sixteenths. Bunching made a match likelier. Each state contributes its chance squared; add those up and

call it the collision weight. The even spread has the least of it, one over the number of states.

Why does a step that lets anything in raise it? Say there are N states and column j adds to c_j. Starting even, the chance at j after the step is c_j over N. The columns add up to N, because the N rows each add to one. Now add up (c_j minus one) squared over all the columns. Multiply it out and use that total, and it comes to the sum of the c_j squared, minus N. At least one column isn't one, so that's more than zero, so the sum of the c_j squared is more than N, and dividing by N squared says the new collision weight is more than one over N. The spread got lumpier. Shannon's uncertainty drops too, by the line from the last chapter: the even spread is the only one that holds the full log of N.

And a step that keeps every column at one never makes any spread lumpier. Each new chance is an average of old chances, weighted by a column that adds to one, and an average of squares is never less than the square of the average. Add over the columns, and each old chance squared gets weighted by its whole row, which adds to one. So the new collision weight is at most the old.

The first of those needs only the yes side to be finite; the rest of the machine can be as big as you like. The last needs no finiteness at all. The middle one needs the even spread, so it needs finitely many states. A lopsided memory that keeps its own lopsided spread can't raise a reading that stays either: weight each state by its chance and the same sum comes out.

21.3 Quantum steps

The quantum version needs a different picture. Draw a sheet through the middle of a room. Call every direction along the sheet YES, and the direction straight through it NO. Now turn the whole arrangement without stretching it. Can the sheet stay in its own plane while the direction through it turns into that plane? No. A turn keeps right angles. If the sheet still fills its plane after the turn, the direction that was at right angles to all of it is still at right angles to all of it.

Here's what carries over. A quantum state is described by a direction, an arrow with a length. A reversible quantum step keeps lengths and keeps the overlaps between arrows; that's called unitary. The YES part is a flat slice of directions that includes every mix of its own directions. The directions with no overlap with any YES direction are the NO states. And permanence means a YES arrow is still YES after the step.

One detail has to be settled before the right-angle argument works: the old YES directions have to fill the whole YES slice after the step, with nothing missing. Pick a finite list of YES directions that build every YES arrow, none of them built from the others. A unitary step keeps that list independent, because any way of building

one image from the others could be run backwards to build the original from the others. The images are still inside YES, there are just as many, so they still fill it.

Now take a NO arrow. Before the step it has no overlap with any YES arrow. After the step those overlaps are still zero, because a unitary step keeps overlaps, and the stepped YES arrows still fill the whole YES slice. So the stepped NO arrow is still NO. Step again and the same holds. A unitary step can't raise a reading that stays.

Finite size was used exactly once: a full finite list of YES directions still fills the slice after the step. Drop it and the endless ladder from the cell above, one rung along each tick, is a unitary step that raises "past the start" on the first tick and keeps it forever, taking a rung it has never used every time. So in finite size, keeping a quantum reading takes a step that isn't unitary. Not every step that fails to be unitary squeezes, though. One that keeps the even spread, as pure dephasing and an unread measurement do, can't raise a reading that stays either, by the same column count as the noisy steps. Which way a quantum measurement pays when its outcome is kept isn't settled here. The collapse models people test keep the even spread too, so the heat the experiments look for is the heat of the noise those models add, not a Landauer price on a kept outcome.

The loop	Rauer, Erne, Schweigler, Cataldini, Tajik and Schmiedmayer (2018): in an isolated gas of thousands of atoms, the phase correlations, one collective measure of its coherence, came partly back around, twice, each time a little lower. A collective quantity of a closed system returned; the full state of the gas wasn't shown to return, and in a system that size couldn't be.
Heat of collapse	Vinante and others (2017): a cantilever at millikelvin shows a force noise of unknown origin, compatible with the heating one collapse model predicts. Donadi, Piscicchia, Curceanu, Diósi, Laubenstein and Bassi (2021): the radiation a gravity-driven collapse would give off wasn't seen at Gran Sasso, ruling out the version of that model with no adjustable settings. These test the noise a collapse model adds, not a Landauer price on a kept outcome.
The ceiling on fresh states	Bekenstein's proposed bound uses energy and size; it isn't a measured count of states.

Bekenstein's bound ties entropy to energy and size under its own physical assumptions. It doesn't hand you a finite list of a device's states. Every count in this book takes finitely many states as an assumption, and nobody has measured where that ceiling is.

21.4 How much room keeping takes

So far, keeping meant a merge or fresh states forever. In between there's a question with an exact answer: how long can a finite memory hold off the merge?

Take an operation that sends every value to one fixed value and leaves a state that already holds that value where it is, the way a reset does. Hand it N different starting states, one of which already holds the fixed value. Ask it to be repeated T times without any two of the states it passes through landing on the same next state. How many states does the memory need?

Walk it. Each of the N minus one starts that isn't already fixed, followed through T repetitions, passes through T plus one states, and none of them twice. A repeat would walk back, the way chapter two walks back two meeting arrows, to the start being equal to a later state of its own path, and every later state holds the fixed value while the start doesn't. Two of those paths can't share a state either. Walk back to the earlier time, and either two different starts are one state, or a start is a state that already holds the fixed value. That's (T + 1) times (N minus 1) states. The start that's already fixed needs one more, and only one. It can't be a state on another start's path. The operation leaves it where it is, so the state just before it on that path and the start itself would land together on it, and walking back finds the same contradiction. And one is enough. Every state on its own path holds the fixed value, so walking back along it finds no contradiction, and its path can fold onto a single state. So the memory needs at least 1 plus (T + 1) times (N minus 1) states.

And that's enough, so the number is exact. One state for the start that's already fixed; the operation leaves it there. For every other start, a row of T plus one states: the first reads the way that start reads, the rest read the fixed way, and the operation walks the row, sending its last state to the common fixed state. For as many repetitions as you asked for, nothing lands together. On the next one, the last state of a row and the fixed state both go to the fixed state: the merge, delayed exactly as long as you paid room for.

With L of the starts already resting, the same count gives N plus T times (N minus L). In bits, a memory of B bits has at most 2 to the B states, so holding off T repetitions with one resting start takes B at least log2 of 1 plus (T + 1) times (N minus 1). The memory grows with the logarithm of how long the history is kept. A fresh state every step isn't a fresh cell every step.

Charles Bennett showed in 1989 that a reversible machine can save room by undoing and redoing its own intermediate work, keeping only enough to get it back. That trades time for space. It doesn't let a machine running on its own, under this count, reuse a state it has been in without a merge.

21.5 Two levers

Now a harder version of the same question. Put two reset levers on a box, and use three values on the tabletop: 0, 1 and 2. The first lever wipes only 1 to 0 and leaves 2 alone. The second wipes only 2 to 0 and leaves 1 alone. On the tabletop it doesn't matter which you pull first. Start one full state above each value, and allow every sequence of at most T pulls. Each lever has to keep different states apart, through the promised horizon. How much room does the box need?

Count the zero states first. At each depth, the first lever takes every state already reading zero, and also the start that reads 1, into zero, and keeps them apart. So the zero states grow by at least one per pull, and with the two other starts there are at least T plus 3 states. A single line of zero states reaches that: walk the line with either lever and stop at its end.

But try the levers in opposite orders on that box and the hidden states disagree. Ask for more: the order of the levers must leave the whole machine the same, not just the tabletop. Then the count goes up, to 2T plus 2. Take the simple case first, where the second lever leaves the start above 1 standing still, and the first lever leaves the start above 2 standing still.

Start above 1 and pull the first lever T times. That walks a chain of T zero states, all different: if two were the same, walking back would make the start above 1 equal to a later state that reads 0. Now ask what the second lever does to a state on that chain. Commuting says pulling it after the first lever lands where pulling it before would, and before, you were at the start above 1, where the second lever stands still. So the second lever holds every state on the first chain still. The same reasoning with the levers swapped walks a second chain of T zero states from the start above 2, and the first lever holds every one of those still.

Can the chains share a state? Everywhere on the first chain but its last state, the first lever moves you on; everywhere on the second chain, the first lever holds you still. No state does both, so nothing on the first chain except its end can be on the second, and the same with the levers swapped. The most they can share is their two ends, as one state. If they don't share it, that's 2T zero states, and with the starts above 1 and 2, 2T plus 2. If they do, that's 2T minus 1 zero states, and now the zero start can't be on either chain: if it were, pulling from it would reach the shared end within the horizon, and the lever that walks that chain would send both the end and the state just before it to the end, two states landing together before the promise ran out. So the zero start is one more, and with the starts above 1 and 2 it's 2T plus 2 again.

When the idle lever moves the hidden state instead of holding it still, the path it traces above 1 or 2 costs states of its own. It runs for a while and then, if it repeats, falls into a loop. Switch levers there, and every pull spent getting into the loop is

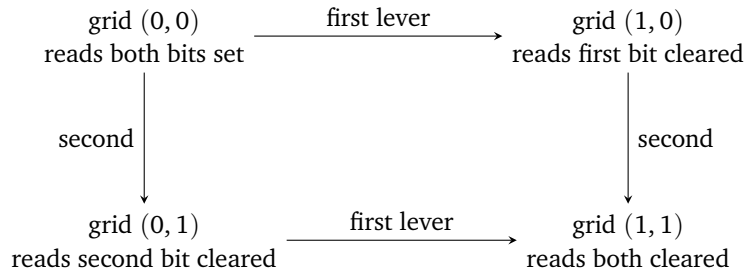
one pull fewer for that chain, so the trade is one state for one. Commuting still pushes any sharing between the two chains out to their far ends, because a shared state would have to come back to itself after a full turn of the other loop. The count comes out at 2T plus the longer loop, and when both loops are a single state, the zero start adds the last one, as before. Either way, 2T plus 2, and the exhaustive search on the small horizons agrees.

There's a box that reaches it. At horizon one it has four places, and you can write every arrow down:

Place	Tabletop	Pull the first lever	Pull the second lever	A start?
a_0	1	end	a_0	yes
b_0	2	b_0	b_1	yes
b_1	0	b_1	end	yes
end	0	end	end	no

On the three starts each column lands on three different places. On the whole four-place box each column has a repeat. And both orders of the levers agree on every row. So the promise that holds through a horizon and the promise that holds everywhere really are different promises. Ask two commuting levers to keep states apart everywhere, forever, and the count goes up once more, to 2T plus 3. Take the simple case again. Now neither chain can stop. A chain that stopped would have a state landing on itself, with the state before it landing there too. So the first chain walks on without ever repeating, and the second lever still holds every state on it still, and the same with the levers swapped. Then the chains can't share any state at all, not even an end. The first lever moves you on along its own chain and holds you still on the other one. That's 2T zero states and the two starts above 1 and 2, and the zero start makes one more. If it's on neither chain, it's one more state. If it's on a chain, T pulls from it walk past the last state of that chain counted so far. And two separate arms running on forever reach 2T plus 3 at every horizon. Commuting is what costs here. Without it the forever promise costs nothing extra: one line of zero states that both levers walk, each lever sending its own start onto the line's first state and holding the other start still, keeps everything apart forever with T plus 3.

Change the tabletop and the room can fit differently. Give it two bits, with one lever clearing the first and the other clearing the second. Lay out a grid of points (i, j) with i and j whole numbers from zero up, plus one separate blank state. One lever adds one to i, the other to j, and both leave blank alone. Read a grid point as whether i is zero and whether j is zero. Start at blank and at the three grid points (0, 0), (1, 0) and (0, 1).



Both orders reach the very same grid point, not just the same reading. Each pull adds one to i plus j, so after t pulls you can reach exactly the triangle of points with i plus j at most t plus 1. Its rows hold 1, 2, up to t plus 2 points. Add them and the blank: 1 plus (t + 2)(t + 3) over 2. And that's the least any box with this tabletop can do. Here's why. Each reading with exactly one bit cleared starts with one state, and every pull its lever sends that reading's own states, plus at least one state with both bits set, into it, kept apart, so it gains at least one state per pull, and after t pulls holds at least t plus 1. The reading with both bits cleared starts with one state too, and at the pull after depth j, one lever sends its own states and a whole one-cleared group of at least j plus 1 into it, kept apart, so it gains at least j plus 1. Add those gains from the first pull to the t-th and it holds at least 1 plus t(t + 1) over 2. Add the one state with both bits set and the two one-cleared groups, and the total is 1 plus (t + 2)(t + 3) over 2, the grid's own count. The three-value box needed extra room to commute. This one fits its best count with commuting built in.

All of those counts are checked by exhaustive search on the small horizons, where a computer can try every box.

21.6 Keeping every history

There's one way to make every step merge nothing, and it's the first way out of chapter two, put inside a machine. Keep the previous state on a list: each step, push where you were onto the front of the list. The step you can see is the same as before, but the whole state now carries its past. Two different whole states can't land on the same one, because popping the list takes you back. And the list gets one longer every step, so the run never comes back to a whole state it's been in. That's the fresh-state way of keeping, and on any memory you can build, the list fills it, and then the merge is back.

So "this step merged, so it gave off heat" can't be said of the step you can see by itself. It's a sentence about the whole state, including whatever history is kept, and a physical model has to say what's kept.

21.7 The counter and the heat

Does a machine's counter track heat? Here's the one test I think is good discipline, and it's a test of whether two assumptions can even hold together. Say you want both of these: every step you've chosen to count gives off at least one unit, and nothing gives off more than its charge. Can both be true at once? Exactly when every step you've chosen to count is charged at least one. If one is, take the heat to be the charge itself: both hold. If a counted step is charged zero, it would have to give off at least one and at most zero.

I should say how little that shows. The model that passes the test just takes the heat to be the charge, so it only shows the two assumptions don't contradict each other. It doesn't show the counter tracks anything real. Still, it bites. Point it at a machine that computes at no cost underneath and counts every step that collapses some yes-or-no question, and it fails: such a machine has free moves that merge, and no heat can be at least one and at most zero. A machine that prices the moves that raise its record and leaves its other merges free has both assumptions for the toll's steps, and not for the rest. The hardware can give off heat on those free merges, off the ledger. The ledger counts the record and nothing else. A unit of cost is not a joule.

Three statements are in play here, and each points its own way. Energy at a scale, a unit of cost times so many joules, is a conversion; it points nowhere. "Nothing gives off more than its charge" puts the charge above the heat: an upper bound. Landauer puts the heat above a loss of uncertainty: a lower bound. The test above says the assumptions can live together. That's compatibility, not evidence.

Is there a heat prediction the permanent record makes that Landauer alone doesn't? No, and the reason is short. The smallest case, one state already yes and one flipping in, is "set a bit to one whatever it held," which is the reset Landauer described and Bérut's group measured. The general bound, for squeezing m plus k equally likely states into m, is what Landauer's assumption gives for that squeeze. What the counting adds is the structural half: on a finite machine, a reading that stays forces the raising move to merge. It doesn't say the state coming in is one of the two that meet, and it doesn't supply the spread or the physical conditions a heat bound needs. Supply those and Landauer prices the uncertainty actually removed. The counting and the physics do different jobs.

There's one more way to put receipts next to merges, and it's exact. Draw the arrows of one run, and put a receipt beside each state the run leaves from: charge on entry into a parked state, and let the parked state idle for free. Now mark every landing place that two different states in the run lead to, and count each mark once. Ask for one thing: whenever two different states the run leaves from land together, at least one of them pays. Then the marks never outnumber the receipts. Every

marked landing has a paid state behind it, and one paid state can't stand behind two marks, because it only goes one place.

Try it with no going to yes and yes going to itself. Charge one at no and zero at yes. Entry pays the one, and every idle step after pays nothing. The merge is covered and the parked machine stays parked for free. Now ask that parked yes-state how many merges you've watched. After one step you've seen only no leave, so no merge yet. After two you've seen no and yes both lead to yes, so one merge. The state in your hand is yes both times, but it can't tell you which. Add a clock, and the two endings become yes at 1 and yes at 2, so you can read off the count. But draw the new machine, with the clock in it, and the clock keeps those states apart. Keep both drawings on the table so you can see which arrows you're counting.

21.8 What you're entitled to

Let me total it up: what you've earned the right to conclude, and where there's still a way out.

You've earned this, with nothing but one next state for every state. On any machine run by one rule, finite or endless, a run that raises a reading and keeps it up either merges two of its own states or never repeats one. You've earned this. On a finite machine, a move that raises a reading either forgets something or takes a reading down somewhere else. That's chapter three's merge or take back. You've earned this. On a finite machine where a reading, once up, stays up, raising it makes two arrows meet, and if it raises several no-states at once, the yes-states together with the newcomers land on that many fewer places than there are of them. That doesn't say each newcomer shares. And you've earned this. If two arrows meeting is charged, the toll follows, and if it's charged per halving, the size of the toll follows too. That's chapter three's rule, derived, with every assumption named.

Now what you haven't earned, so you don't assume it. You haven't earned that forgetting costs anything in the world; that's Landauer, measured for beads and argued over in general, and here it's an assumption with a name. Even granting it, it asks no heat of a run whose state is known, so what the physics forces is a floor on a device that has to work whatever it's holding. You haven't earned that any machine you'd want to build is of this kind, either, and there are two easy ways to miss. A machine that keeps its running total inside its own state, as a number with no ceiling, has endlessly many states, so the finite argument fails for it as a whole. And you can't patch that with a ceiling: a total that only climbs, in finite memory, either stops at its top, and then two different histories leave the same top value, which is a merge, or rolls over to zero, and then it doesn't only climb. So any running total a finite machine keeps for itself is honest up to its width and no further. And a machine whose price list charges the raise but lets some other step merge

two states for free raises its box honestly and still doesn't charge for forgetting in general. Either one puts a machine outside these arguments as they stand, though a finite piece of it can still sit inside.

The smallest finite piece I could build where every condition can be seen to hold is this. Four slots, each with a flag: eight states. Three kinds of move. Stamp sets the flag and goes to the next slot, cost one. Jump goes to a chosen slot, cost two. Next goes to the next slot, cost zero. Check it with me. It's finite. The flag never comes down on any move. Next forgets nothing: every state lands somewhere different. Stamp lands at most two states together, a flag up and a flag down from the same slot, so one halving, and it costs one. Jump lands all four slots with the same flag together, two halvings, and it costs two. So every move's price is exactly the number of halvings in its biggest pile.

And you haven't earned anything about what the flag means. Chapter two's chairs don't know what they're counting. The cost is for raising a flag that stays up. Whether raising it ever amounts to standing behind a real fact was the subject of chapters eight through twelve.

One more thing, because it should nag at you. "Permanent" was chosen, by me. Why should a flag stay up? Merge or take back is the start of an answer: on a finite machine that charges for every merge, a flag that goes up for nothing is a flag the same move takes down somewhere, and a record that can be silently taken back is exactly what the clay ball, the write-once disk and the hash chain were all built against. That's durability, one of chapter four's five jobs, and it's the one I chose.

And "permanent" is an idealization, which I should say before a physicist does. No physical record lasts forever. A bit behind an energy barrier leaks over it at some rate, however slow, and a closed finite system comes back around, which is Zermelo's point from chapter two. The arguments that start from a flag that stays up need it to stay up on every state the machine can reach, including states a given run never visits, forever. Weaken that to "it stays up on the states that actually turn up, for as long as anyone cares," and merge or take back says what goes wrong: the move could take the flag down somewhere the machine could go and never does, and then it isn't forced to merge at all.

One piece of a time-span version I can hand you. Take one move and repeat it from one start, forever. On a finite machine the states that run passes through are finitely many, so it falls into a loop. If the move forgot nothing on those states, it would just shuffle them, and a shuffle brings every one of them back around, the start included, where the flag was down. So if the flag goes up along that run and stays up from some point on, the move merges two states the run itself visits. No permanence is used anywhere in that.

Runs that mix different moves split two ways. If the moves are picked from outside, in any order, the answer is no. Take two states, one reading no and one

reading yes. One move swaps them; the other does nothing. Run the swap once and then the do-nothing move forever: the flag goes up at the first step and stays up for good, and neither move merges anything anywhere. The swap would take the flag down from yes if it ever ran there; on this run it never does. But if the machine is closed, picking its next move from its own state, the run is one map repeated, and the single-move argument applies as it stands. And if the moves come in a pattern that repeats, one phrase played over and over, treat the whole phrase as one move at the states where each playing of it begins, as long as the flag was down somewhere after the repeating started. It merges two of those, and the first move where their paths meet merges two states the run visits.

21.9 The shape of the second law

Here's a comparison that's easy to read as more than it is, so I'll say first what it doesn't do. It doesn't give entropy's numbers. It doesn't derive Landauer. It makes no physical claim. It puts two drawings side by side and points at the parts that line up, and every paragraph says which kind it is.

Shown, and only about the model. A window groups states: every state gives one picture, and the states that give the same picture form that picture's pile. A physicist calls the picture a macrostate and the states under it its microstates; the word is different and the drawing is the same.

Shown, and only about the model. A Thiele machine's record never falls; that's in the definition. And it never climbs for free: the toll charges every climb, and the counter that adds the charges never goes down either. So a Thiele machine carries two quantities that only climb along every run, the record and the counter, and the first can't move without the second moving too.

Shown, and only about the model. Keeping something costs room or a merge. On finitely many states a reading that rises and stays rises by a merge, and when each unit of price pays for one halving, the bits a run squeezes out are bounded by what it paid. A noisy step that keeps the even spread can't raise a reading that stays.

A thesis, not shown. Put those beside the second law of thermodynamics and the shapes line up: a coarse description, the many fine states under each coarse one, a quantity that only goes one way, and a rule that the one-way change has a price. My guess is that this isn't a coincidence, and that the second law, read as a statement about what a coarse description can keep, is the toll seen from the physics side. I can't prove that. The model has no temperature, no energy and no probability unless an argument puts one in, and the bridge from a unit of cost to a joule is an assumption with critics, named in the last chapter. If the match is real, somebody has to show it with physics, and that somebody isn't this book.

21.10 At the bottom: what I suspect and can't prove

This is underneath the arguments, so read it with the lights on. None of it is shown, and I'm not going to dress it up. It's why I care, and it's here, after the arguments and apart from them, so you can check that none of it got into them.

Physics has a word for the moment a question closes: measurement. And the people who thought hardest about it keep describing it in this book's shape. They ran together for me until I put them on one picture. Flip a coin in a crowded room. Before it lands, it could go either way. After, it's settled, somebody chalks it on a board, and the result gets copied out into the world, the light and the air, so many times that it can't be taken back and everybody who looks agrees. Five pieces: the question, the landing, the chalk mark, the copies, the agreeing. Each person leans on a different one. Zurek, in chapter nineteen, leans on the copies. Rovelli's coin, in chapter six, has to hit the ground, so his piece is the landing.

Niels Bohr, the Danish physicist whose institute in Copenhagen was where quantum mechanics got argued into shape, leans on the mark. In a talk at Columbia University's bicentennial in October 1954, printed in his 1958 book Atomic Physics and Human Knowledge, he said "every atomic phenomenon is closed in the sense that its observation is based on registrations obtained by means of suitable amplification devices with irreversible functioning such as, for example, permanent marks on a photographic plate." A door that swings one way. I first found that line quoted in a 2026 paper by Mauro Dorato and Jan Faye, and their quotation runs on into a clause about water drops forming in a cloud chamber that isn't on Bohr's page. The page is worth reading yourself. John Wheeler, the Princeton physicist who made the name "black hole" stick, went further in a 1989 talk in Tokyo and leaned on the question: "every it," he said, "every particle, every field of force, even the spacetime continuum itself," gets "its function, its meaning, its very existence" from "the apparatus-elicited answers to yes or no questions, binary choices, bits." He boiled it down to "it from bit." As I read him, the coin has no face until something asks and something writes the answer down. How literally he meant that, I can't tell. And Peirce, from chapter nine, long before any of them, leans on the agreeing. The settling worth trusting is the kind that has to answer to everybody. Laid out like that, they pair up. Bohr and Rovelli lean on a step that only goes one way, in the plate and in the coin. Zurek and Peirce lean on many onlookers who agree, one with light and one with people.

Lay that next to everything in this book and the parts line up: a check that could have come out either way, a result that's now settled, a price paid in forgetting or in fresh states, a record that stays. I suspect that isn't a coincidence. I suspect the event this book prices on a machine is the same kind of event physics calls a measurement, turning up in a different place. The forgetting is my step. Bohr only says irreversible, and getting from a record that stays to a merge or to fresh states

is chapter two's counting. The suspicion breaks at Zurek, for the reason chapter nineteen gave. Copying onto fresh light merges nothing and uses up fresh states, which is chapter six's look that owes no floor. So physics' own word lands on both sides of the line: the chalk on the paying side, the copies on the borrowing side. The landing I'd put with the chalk, and when chapter six pushed on Rovelli's paper it slid toward the borrowing side, the side of finding out. So two pieces already stick out, and I'll leave them sticking out. That's as far as I'll take it, a suspicion.

John Bell asks for precision in those words. He was the Northern Irish physicist at CERN, and in 1990 he wrote a short, furious essay called "Against 'measurement'," with a list of words he said have no place in a physics that wants to be precise. On the list: system, apparatus, environment, irreversible, information, and measurement, which he called the worst of all. I used half of them in the last few paragraphs. To turn the suspicion into a physical claim, each of those words would need a stated meaning and a tie to the device. The arguments in this book work with the machines I've defined; anything physical says which further assumptions it uses.

That's the whole of it: what's counted, what's assumed, what's chosen, and one suspicion kept apart. The last chapter lays those out side by side.

CHAPTER 22

What's shown, what's chosen, what's argued

This is the last chapter, and it's the one where everything gets laid out on the table at once, sorted into piles, so you can see which pile each piece came from. Page one promised that: found, assumed, chosen. Every chapter since has said which pile it was adding to. Here they all are.

Start with the smallest machine, because it carries the whole book in three layers. Two boxes, one move that sends both to YES. What I found is the merge inside it, two states landing on one, which counting forces on anything that keeps a distinction on finitely many states, and which nobody chose. What I chose is the price—one unit on the move that climbs—a choice that could have gone another way. What I assumed, and only where an argument says so, is the heat. A merge of states the machine really could have been in costs something in the world. That's Landauer's, not mine, and not shown here. Everything below is one of those three, or it's a thesis, which is a fourth thing, and I'll say which every time.

22.1 What it is, and what it's for

You've had the model since page one: a thing, moves, a way to tell how settled the thing is where settled never goes back, some run from a start you pick that settles it more, a price on every move that's nothing or more and never a refund, and the toll. A counter and a window come in when an argument needs them. You can't buy it in a box. It isn't a device, a chip or an instruction set; those are things you'd build to carry it.

So if it computes nothing new, and it doesn't, what's it for? The same thing other ways of counting a computation are for. The random-access machine, in the form Stephen Cook and Robert Reckhow gave it as a way of counting cost, computes exactly what Turing's does, and it's worth having because it counts steps the way real memory is used. Andrew Yao's cell-probe model counts only how many times you look into memory and leaves all the computing in between out of the count.

Alok Aggarwal and Jeffrey Vitter's model counts only the blocks moved between a slow memory and a fast one. None of them computes one function more than Turing's machine. Each is worth having for what it counts. The Thiele Machine's own count is what a run made the case, and what that cost. Look at a computer through a window that drops that, and you get the ordinary picture, chapter four's coin covered by your hand. Read a computer through whether it has stopped and put a price on the move that stops it, and it's a Thiele machine.

22.2 The objection a theorist makes in four seconds

The toll is a floor, and a floor is easy to meet. Charge one unit for every move, whatever it does, and the toll holds for any reading you like, because every move that settles something is a move, and every move costs one. So any computer with a step counter beside it obeys the toll, for every yes-or-no question you could ask about it. A clock passes.

That's exactly why the toll isn't the whole definition. Charging every move doesn't make a machine a Thiele machine; the reading still has to never go back, and some run still has to settle it. And it means the toll, by itself, only puts a floor under the settling move. It doesn't tie the rest of the bill to anything kept. One thing in this book does tie it down, and it's a ceiling: ask that a run's total never be more than the number of times the box went up. Chapter eleven showed that with a price that can depend on where the step starts, the floor and that ceiling leave exactly one way of charging: one unit per raise, nothing else. With a price per move name, the way page one defines it, no machine whose box ever goes up meets both, because the raising move, played again from YES, raises nothing and still costs what it costs. A clock fails the ceiling the first time it charges a tick that raised nothing.

Two more things fall out of the same picture. The toll doesn't care how strong the machine is. A machine is called Turing-complete when it can imitate a Turing machine; the toll never asks. Stronger, weaker, exactly as strong: bring the parts, charge the settling move, and you're in. It was never about what you can compute. It's about what a move is allowed to do for free.

And nothing is in scope just for being a computation. A Turing machine's run isn't an example of anything in this book because it's a run. It becomes one when somebody says what the states are, which ones read yes, what each move costs, and checks the toll. Until then the arguments say nothing about it, and I won't pretend they do. What chapter two found has no such condition, though: every run of anything with one next state for every state lands somewhere new every tick, makes two arrows meet once, or goes round a loop through its start and keeps nothing.

22.3 Bolting two together

Put two Thiele machines side by side: each move is a move of one or the other, and the costs add. Is the pair a Thiele machine? It depends on how you read the pair.

Read it as "either box is up," or "both are." Then the pair's reading can only go up when one of the boxes goes up, that move is a move of that machine, and that machine's toll already charged it. So the pair obeys the toll. Read through "both are," a run that starts with both down and ends with both up raises each box once, each raise is charged by its own machine, so the run costs at least two. From a start with one box already up, one raise is enough, so two is a floor only from the clean start.

Now read it as "the first is up and the second is down." That reading comes down the moment the second box goes up, and a Thiele machine's reading never comes down. The pair is out.

So some readings of a pair keep the toll and some don't, and you can tell which by asking whether the pair's reading can go up without one of the parts going up. Whether every way of building one machine out of others keeps the toll is a bigger question than this book answers.

22.4 Steps that branch

Everything so far gave every state one next state. Plenty of models don't: a machine that can guess, a lambda calculus term you can reduce in whichever place you like, a system of rewrite rules, a proof read as one rule applied at a time. Nothing earlier was stated for those, so here are the two arguments, both short.

The first is chapter two's chairs on one choice of next step. Say only finitely many states read yes, none of them can step to a state reading no, each has at least one next state, and some state reading no can step into the yes side. Pick one next state for each yes-state, and the step in from outside. That's one more step than there are yes-states, each leaving a different state and landing on a yes-state, so two land on the same one. It needs the yes side finite; Hilbert's hotel keeps a reading with nobody sharing.

The second follows one run, however its choices fall. Take the step where the reading goes up, and say it stays up for the rest of that run. If the run is in the same state at two times after the raise, step both times back together. They can't stay equal all the way back past the raise, because the state just before the raise reads no and every state after it reads yes. So at the first pair that differs, two different states of that very run step onto one. After the raise, the run makes two arrows meet or never repeats a state, on any number of states. Before the raise it can come back to a state and then leave it for good with nothing meeting, which one rule can't do. A

state with no next step at all is chapter one's machine that stops: a reading kept by stopping is kept with nothing meeting.

For branching steps the first argument says less than it looks. In the lambda calculus, any term N is one step away from two different terms: apply "do nothing" to it, or wrap it in a function that throws its argument away and hand that function anything. Two different states sharing a next state is no news there. The second argument is the one that carries weight.

Where it all stops. Steps in continuous time have no next state to count, and nothing here covers them. Noisy steps where endlessly many states read yes, and quantum steps whose kept part has endlessly many directions, get none of the finite arguments; the endless ladder keeps a reading with nothing meeting, a fresh rung every tick. What still reaches them is the single run: the second argument covers any one run of them. And a truth isn't a process at all. That a statement is provable is a fact with no step in it; the arguments are about a run that derives it.

22.5 What doesn't follow

Some things look like they follow and don't. Here they are, so you don't carry them out of the book by accident.

The counter isn't a heat meter. Once you pick the start and the price list, the number is forced by arithmetic, but it isn't energy. To read it as energy you'd need to say which states carry information, how the charge relates to what a device gives off, and that both of those can hold at once; chapter twenty-one did the last part and showed it only means the two don't contradict each other.

No question about what computers can do in a given time follows. Whether P equals NP is one of the seven problems the Clay Mathematics Institute put a million dollars on, at a meeting in Paris in May 2000. One of the seven has been declared solved since: the Poincaré conjecture, by Grigori Perelman, who posted his proof in 2002 and 2003. A second has a claim on it. On 8 September 2026 OpenAI announced that one of its models, run as about ten thousand agents for about eighty-eight hours, had produced a proof that a three-dimensional fluid flow with a smooth force pushing on it can break down in finite time, and that another model had then written it out for a proof checker in about seventeen more hours. The official statement of the Navier-Stokes problem accepts any one of four answers, and OpenAI's paper says it settles two of them: the breakdown in all of space, and the same breakdown in a box whose sides wrap around. As of October 2026 other mathematicians haven't finished checking it. On 6 October Alexander Bastounis, Fabian Circelli and Anders Hansen posted a paper arguing that the version written for the proof checker doesn't match the proof written in words. There's a dispute over credit with Tristan Buckmaster and Levent Alpöge, who were working on a

closely related result, and the Clay Institute's own page lists the problem as active, not solved. P versus NP has no such claim. If a two-state clash on my machine settled it, I'd be owed a million dollars. Nobody owes me anything. A window that loses a box is a fact about that window. A question like P versus NP is about everything every possible program can do within a budget, and nothing in these clashes reaches that far.

The choice of event isn't forced. The pointer thesis is a thesis. The cup over any machine is a theorem, and it says the event is a choice.

The two walls stay two walls. Chapter seven's is about what a watcher can't get back through a window; chapter seventeen's is about what no inside procedure can decide. Linking them would take an argument of its own, and this book doesn't have one.

The entitlement count in chapter twelve is handed its groups and its covering. It doesn't dig them out of an arbitrary run.

Records with chances on them don't inherit the uniqueness of the counter. Two honest random records can have the same branches and the same charges and still give those branches different weights, one choosing evenly and one two to one. The price list doesn't fix the chances.

Every guarantee about an established record is about runs: any list of moves, from a clean start, on a step that does what it says, one writer at a time. Whoever sets the start, the wire that carries the report out and the tools that built the step are outside that fence.

The five real systems are copies of pieces, with their limits named in chapter eighteen. None of them is a check of the running system. What the book does prove that reaches real systems is chapter sixteen's: a machine that reads a box to skip work it already paid for goes, under the conditions stated there, through one of four doors, and the caches, version counters and memo tables people build are each one of those doors.

22.6 The ledger

Here's every claim the book makes, one line each, with what kind of claim it is and where it lives. If a line here ever says more than its chapter does, the chapter is right and the line is wrong.

Ledger is a good word for it, and it's older than bookkeeping. Around 1400 a ledger was a big book that lay in one place, a large church breviary, from the old verb for to lay; the account-book sense turns up in the 1580s. Staying put is the job here too. This is the one place where every claim lies still, so you can come back and find it where you left it.

There's a way to use it that takes five minutes. Pick the sentence in this book you trust least. Find its line. Read the kind before anything else. Then go to the chapter and check that the line says no more than the chapter does. If it says more, you've found my mistake.

Claim	Kind	Chapter
The Thiele Machine: a thing, moves, a settledness that never goes back, a run that settles it, prices that are nothing or more, and the toll	chosen (the definition)	page one
Put a price on the move that stops it, and a machine that stops is a Thiele machine, read through whether it has stopped	shown	page one, 1
Every step of a machine with one next state lands somewhere new, makes two arrows meet, or goes round a loop	shown	1
A machine that stops slips past the merge by having no next step	shown	1
On finitely many states, settling something for good makes two arrows meet, unless the machine takes fresh paper every tick	shown	2
Hilbert's hotel: on endlessly many states, a one-to-one move can settle something and nothing meets	shown	2
A machine where nothing meets, on finitely many states, comes back to its start and settles nothing new	shown	2
Bennett's rule: a move where two arrows meet costs at least one	chosen	3
With finite chairs, settled staying settled and Bennett's rule, the toll follows	shown	3
On a finite machine, a move that settles either makes two arrows meet or takes something back	shown	3
The moves every price list following Bennett's rule must charge are exactly the moves where arrows meet	shown	3
Each of the three conditions has a machine that walks past it: the strip, the coin, the stamp at zero	shown	3
The halving rule, and that a move squeezing m + k states into m costs at least the halvings between them	chosen; shown	3
The smallest Thiele machine is two boxes and one move that sends both to YES	shown	4

Claim	Kind	Chapter
Every machine with a move is what's left of a Thiele machine once you cover a coin	shown	4
People who kept records priced the steps beside the moment of writing, not the moment itself	cited history, read by me	4
A record with more than two places, no two of them each at or below the other, is a row of cups; a straight line of stages needs one cup fewer than it has stages	shown	5
The toll on the row charges at least one on any step that turns a cup up	shown	5
Looking can be free: a look uses up a blank page, which can be handed back; wiping pays	shown	6
A window with a clash can't price exactly; the least overcharge, and a price that reaches it	shown	7
Established: up only right after a passing check of that very claim, nothing it's about changed in between	chosen (a condition)	8
Two computers can end with the same answer and differ only in what a window drops	shown	8
Settling is a sorting with a door that only swings one way; closed isn't the same as true	shown; the reading is mine	9
Structure as relations, kept under chosen changes, carried by any stuff, relative to a frame	chosen (my overlay of the sources)	10
An order is exactly a sorting with one-way doors between its groups	shown (a textbook fact)	10
A split with a one-way door is the smallest start, by the ruler I stated; the arrow road meets it there	chosen ruler; shown under it	10
Nobody prices vouching for a grouping	not claimed; only that I found no one who does	10
Given its start and its price list, the counter is arithmetic, and there's only one honest count	shown	11
A floor and a ceiling together leave one price, one per raise, when the price can depend on where the step starts	shown	11
A counter allowed to go down gets leaned on where it goes down	cited (Ethereum's refunds), read by me	11
A sound grouping entitles you to exactly the groupings you get by cutting it smaller	shown	12
A ticked box proves its claim true	not claimed	12, 13
Insight, as priced here, is a distinction made to stay; the price doesn't measure depth	chosen meaning; shown for the price	13

Claim	Kind	Chapter
Nothing in the bare definition makes the box wait for a check	shown	13
A box that answers only to the machine and never comes down is a cup on one event, over any machine	shown	14
The event and the price list are free choices; the chances on a random box aren't fixed by its prices	shown	14
When the machine is finite, its record only grows and the uncovered machine forgets nothing, a paid move crushes two states of one pile	shown	15
A window can price exactly without showing the cups, if it shows them move	shown	15
A machine that reads an established box to take a shortcut never answers wrong from a clean start, as long as editions are never reused	shown	16
With editions that wrap around, four moves give a wrong answer	shown	16
When a guard's glance can't tell a passing value from a failing one and its memory is E boxes that only go up, it answers wrong or unlocks on at most (E + 1)/2 rounds; outside those two conditions are the other two doors	shown	16
Unlocking exactly the checked ones among k values, whichever were checked, takes 2 to the k memory states	shown	16
A box that comes down on every write still makes two arrows meet, at the wipe and at the check	shown	16
Reading the box computes something new	not claimed; it saves steps, the way a cache does	16
No inside procedure decides whether a machine's own shortcuts are sound, under the conditions stated	shown	17
Undecidable means unknowable	not claimed	17
Of five real systems, read through my copies, one charges a fee at the write	shown on the copies	18
A copy of one piece is the running system	not claimed	18
A toll is a better design than a wall	not claimed	18, 22
The pointer thesis: the established record is the event copied and relied on by parties who didn't make it	thesis	19

Claim	Kind	Chapter
Requirements that never name the reading don't pick out the toll; the ceiling that names it does	shown	19
Landauer's principle: lowering uncertainty by a bit hands at least kT ln 2 of heat to the surroundings, on average	assumed, critics named	20
A permanent raise from the even spread removes at least log2((m + k)/m) and at least 2k/(m + k) bits; in the worst case a full bit	shown	20
Charging bits on every spread is the same rule as the halving rule; Bennett's rule is weaker	shown	20
On a known state the assumption sets no heat floor	shown	20
A fixed device that merges K inputs produces at least k ln K of entropy on one of them	shown, under the stated physics	20
Every scale pays the same way: a coarse merge forces a fine merge, a step out, or fresh states forever	shown	20
Going faster costs more: heat times time is at least twice the distance squared, for the walker	shown, under local detailed balance	20
On finitely many states, a noisy step that keeps the even spread can't raise a reading that stays; one that can makes the spread lumpier	shown	21
A unitary step on finitely many directions can't raise a reading that stays	shown	21
A reset on N starts, one already resting, held off from merging through T repetitions takes exactly 1 + (T + 1)(N - 1) states	shown	21
Two levers on three tabletop values that keep states apart through T pulls: T + 3 states, and 2T + 2 when the levers commute	shown; searched by machine on small horizons	21
Keeping every history makes every step one-to-one, by taking fresh states	shown	21
Writing on a write-once cell makes two arrows meet; a remembering state only moves the meeting along	shown	21
With L of N starts already resting, holding off the merge through T repetitions takes N + T(N - L) states	shown	21
Two levers on two bits that keep states apart through t pulls need exactly 1 + (t + 2)(t + 3)/2 states	shown	21

Claim	Kind	Chapter
Two levers that keep states apart everywhere, forever, need 2T + 3 if they commute, in the simple case, and T + 3 if they don't	shown	21
Receipts cover the merges a run shows, when one of every colliding pair pays	shown	21
The eight-state machine meets every condition, and each move's price is the halvings in its biggest pile	shown	21
A running total in finite memory is honest only up to its width	shown	21
One move repeated, with the box going up and staying, makes two arrows meet among states the run visits; moves picked from outside need not	shown	21
Every counted step costing at least one is exactly when "gives off at least one" and "gives off no more than its charge" can hold together	shown	21
The counter is heat in joules	not claimed	21, 22
The second law is the toll seen from the physics side	thesis, not shown	21
Measurement is the event this book prices	suspicion, kept apart	21
Turing's model lets a table rub out or print any mark on its tape; nothing in the model makes a mark hold	shown	22, page one
Every machine that steps is a Thiele machine with its record covered and the price of settling set to zero	shown	4, 22, page one
Some readings of two machines side by side keep the toll; a reading that can fall doesn't	shown	22
After the raise, a run with branching steps makes two arrows meet or never repeats a state	shown	22
Anything here settles P versus NP	not claimed	22
The Thiele thesis: every physical process that computes with final answers has an exact one-rule description whose answer no run turns off	thesis	22
Given the thesis, every set such a process accepts is one a program can list	shown	22

"Shown" means shown on the page, under the conditions the chapter states, and checked by machine on small cases where a computer can try them all. "Chosen" means I picked it and said so. "Assumed" means it's somebody's claim about the world that an argument leans on, named. "Cited" means it's somebody else's result,

sourced. “Thesis” means it ties a definition to the world and no argument can settle it. “Not claimed” means it looks like it follows, but it doesn’t.

22.7 Whose is what

Here’s the short form of who did what. The long form is at the back of the book, in What’s mine, and what isn’t.

The toll isn’t only my preference: on a finite machine a reading nothing turns off can only be turned on by a step that makes two arrows meet, so if that’s priced, raising it costs; and a raise where nothing meets needs endlessly many yes-states. The counting under it is the finite cousin of the recurrence Zermelo threw at Boltzmann in 1896, and the price on a merge comes from Landauer’s reset. Others had already priced merges. Thomas Ouldridge and David Wolpert put a physical price on them for finite automata. Ming Li and Paul Vitányi put a unit price on each bit a computation erases, or takes in, irreversibly, and Erik Demaine and his coauthors charged for instructions that destroy information. What’s here is the toll that kind of price forces on a reading that stays.

That a price worked out from a window can’t tell apart two steps the window shows the same is a fact about functions far older than this book; what’s here is the pricing form of it. That a box driven by the computation that never comes down is a cup on one event, over any machine, uses old pieces too. Latches and history-keeping variables are old, and the result covers only boxes that never come down. And the move of guaranteeing something about every run, which an established record leans on, isn’t mine at all. It’s the reference monitor’s: James Anderson’s 1972 study for the US Air Force, which said the mechanism “must be tamper proof,” “must always be invoked,” and “must be small enough to be subject to analysis and tests.” Fred Schneider made the idea precise for monitors that watch a run, and it was proved for monitors checked at every instruction by Arthur Azevedo de Amorim and his coauthors. What I put under it is a definition, and its idea isn’t new either: a check tied to the claim it looked at, with the toll charged on top.

Here’s what this book proves that I haven’t found anywhere else, each with its conditions in its chapter. First, that on any machine a record that stays is paid for in two arrows meeting or in fresh states, with the noisy and the finite quantum versions in chapter twenty-one, and that a price on the meeting forces the toll. Second, that a box answering to the machine and never coming down is a cup on one event, over any machine, with the toggle and the hidden clock as what falls outside. Third, the blindfold: every bare machine that steps is a Thiele machine with its record covered and its price of settling set to zero. Fourth, the four doors and their theorem, for a machine that reads its box to skip work. Fifth, the exact room counts of chapter twenty-one: 1 plus (T + 1)(N minus 1) states to hold off a merge, T plus 3 and 2T

plus 2 for two levers, 2T plus 3 when they commute and keep states apart forever, and the grid's 1 plus $(t + 2)(t + 3)$ over 2. And sixth, the ledger itself, every claim sorted into shown, chosen, assumed, cited and thesis. Under all six is the bet that the price belongs in the step.

22.8 The Thiele thesis

One more claim ties the counting to the world, the way Church and Turing's ties computing to it, and I'll state it as what it is, a thesis. I call it the Thiele thesis: every physical process that computes with final answers has an exact description as a machine run by one rule, with a next state for every state, written down in a way a program can follow, on which the answer is a reading that no run from any input ever turns off, on inputs accepted and rejected alike.

No argument settles it, because "a physical process that computes" isn't something you can do mathematics on. A model of computation is, and for a model written down as states, a step and a reading that stays, what keeping takes needs no thesis: it's chapter two. The thesis is the tie between the two, and it's about what a physical computation pays for what it keeps.

There's an established road to that kind of tie, with its conditions in the open. Robin Gandy in 1980 wrote down principles for mechanisms, and Pablo Arrighi and Gilles Dowek went back over his argument in a short form in 2012, on the way to a quantum version of it: sameness across space and time, finitely many local states, a bound on how fast anything travels, and a start that differs from a quiet background only in a finite region. From those, the evolution is computable. Those are real assumptions about the world, and they don't pick an event or a price in joules. I'm not asserting them of every physical system.

What can be shown sits on either side of the thesis. On one side, chapter two applies to every such description. A run that raises the answer merges two of its own states or never visits a state twice. On the other side, look only at which inputs it accepts, and the thesis says something you can check: every set a physical process accepts with final answers is one a program can list. Here's how. Run the description from every input, a step at a time across all of them, taking turns, and write an input down the first time its reading says yes. Nothing ever turns an answer off, so a yes seen once is the answer. And the other way round, a set a program can list is accepted by a machine that searches the list for its input and then steps from its stopped state to itself forever, read through "stopped," which nothing turns off. That builds a machine with the same accepted set. It doesn't build an exact description of the original process. And nothing about how the keeping is paid survives that view. The same search held at its stop by a step to itself, which merges, or by a counter that keeps ticking, which takes fresh states, accepts the same inputs.

"Final" has to cover the rejected inputs too, and here's why. Build a machine that starts at no, reads yes while program n is still running, and falls to no for good if it stops. It reads yes for good on exactly the programs that never stop. No program lists those: one that did, run beside the plain search for a stop, would tell you for every program whether it stops, and chapter seventeen showed nothing can.

22.9 What isn't settled

Four things, each with what would settle it.

Whether Landauer's critics are right. That would leave every argument standing and one assumption unowned by the world. What settles it is measurements of real devices, and an argument about what the principle means in general that both sides accept.

Whether the established record really is the pointer event. What settles it is chapter nineteen's three comparisons, run on systems people actually use.

Whether the toll is the right thing to price at all. The five real systems mostly built a wall or a penalty for contradicting yourself. I can't prove a toll is better, and I won't pretend the five say so.

And whether anything sits below a split with a one-way door, by a ruler someone can state. The arrow road met mine at the bottom. If you find a smaller start, it would change this book.

22.10 Back to the warehouse

Walk the warehouse from chapter eight one more time. Two computers, the same sorted list on both. On the one with the established record, a different part of the state tells you a check happened and a matching commitment followed it, and the rules keep that tied to the edition that was checked. Put your hand over that part of the machine and the difference between the two vanishes from where you're standing. Change the list and you can watch the old check go stale. Those are different operations with different consequences, and you can follow each one with your finger.

Carry a report out of the warehouse and you've got another question: what connects the report to the machine? The box doesn't answer it. A window, a copy, a second machine and a reader each bring relations of their own, and the chapters asked, one at a time, which of those hold.

At the bottom of all of it is one distinction made to stay: a bead in a dip, a cell written, a question closed. What that settling keeps, what it takes, and what the price I chose charges for it. What I'm arguing is that keeping is part of computation, not something you bolt on afterwards, and that the price belongs in the step. The

box and its toll are the part of that I've worked all the way through. The rest is yours to push on.

22.11 The blindfold

Here's the whole book in one picture. Take any machine that steps: Turing's, a register machine, a lambda term, a table of chances, a quantum circuit. None of them puts a price in the step, and none of them asks that a mark mean what it says. Write "checked" on Turing's tape, and his model allows the next rule to rub it out, or to print it without the check. A particular table might leave the mark alone forever, the way chapter two's endless strip does, but nothing in the model makes it hold. A machine like that can carry a record the way it carries anything, as marks, and nothing in its step makes the mark honest. That's why the sorter and the copier leave the same bits behind.

Put the one-way door and the toll in the step, and the picture turns over. Put a price on the move that stops a Turing machine, read it through whether it has stopped, and it's a Thiele machine. Any machine that has a move at all is what's left of a Thiele machine once you cover the coin with your hand, as chapter four showed. Covering the coin is the blindfold, and setting the price of settling to zero is the other half of it.

Take the blindfold off and the same counting reaches every kind of step in this book. On a finite machine, a record that stays can't go up without two arrows meeting, unless the machine stops; on an endless one it can, by taking fresh states forever. A noisy step that keeps the even spread can't raise it at all, and neither can a turn in a finite quantum space. A window that drops the record can't tell you whether an answer was established or just printed. And a machine that reads its record to skip work it already paid for has to keep that shortcut honest when the data changes, and under chapter sixteen's two conditions there are exactly four ways: fresh editions until they run out, a box that comes down and makes two arrows meet twice, room for every value checked, or wrong answers in four moves.

Turing's machine is the case where the blindfold stays on and settling is free. This book takes the blindfold off.

Afterword: how I got here

What I am is curious, past the point of good sense.

I pick a thing up and cannot set it down until I have turned it over enough times to see how it holds together, and this time it did not let go.

I am not a programmer and I was never formally taught any of this.

I was not trying to build a new model of computation.

I would not have known what one was if it bit me.

I was trying to build a 3D renderer.

There was no plan behind it.

Just a pull to make something real with my own hands, and the kind of stubbornness that will not let a thing go until it stands up on its own.

I have worked on it in off hours every day since.

Once I started digging in, two things came clear fast. I knew nothing about computers, and the field was already saturated, every obvious road paved and crowded. So I needed an angle nobody else was standing on, and the angle I had wasn't a strategy. It was just the only way my head works. I did not know what compositional meant. I had never heard of category theory. I called the thing I was building digital DNA, because that's what it felt like in my hands, and I thought about it in the only language I had: gravity pipes and directions and vectors, things that were linear and somehow not linear at the same time, structure that flowed and joined and came apart without my owning one proper word for any of it. I built a custom rendering pipeline out of those intuitions, by feel, throwing out pieces that didn't sit right even when I couldn't have told you the rule they were breaking. And then, the way I always do, I went looking for what I was actually doing, and found out it already had a name. An old one. A deep one. What I had been reaching for in the dark was composition. It was category theory, a whole field I had never once heard of, sitting there describing the exact shapes I had been shoving around. The moment that landed, the rendering pipeline stopped being a rendering pipeline and started turning into a categorical engine, and I started swallowing the real theory as fast as I could find the words for things I had already built.

I took it one concept at a time, and I would not move to the next until I could watch the current one actually run. Category. Object. Morphism. Functor. Monoidal product. It came together in pieces and never in chapters, the way a picture resolves out of static. The Yoneda lemma (a result buried deep in category theory) was the piece that made the rest snap into place.

Then within weeks I found out the engine I had built to draw pictures was general enough to run a physics simulation on the exact same primitives, without my rewriting any of it. I wrote Newton's laws as a program in it, half expecting nothing, and it ran. I want to be careful about what that was and what it was not, because this is exactly the kind of thing that gets misread. It was not me deriving physics, and I am not claiming the machine explains gravity. It was the compositional structure being general enough to express a physics simulation the same way it had expressed a renderer, and that generality is what told me the structure was the point and the pixels never were. Somewhere in those weeks I crossed from debugging why one pixel wouldn't show up to understanding that I had quietly built a categorical computing engine, and that the rendering had only ever been the first thing I aimed it at. The pixel did eventually show up, for what it's worth.

I kept going, because by then I could not not keep going. The night the engine first ran a program declaratively, meaning I told it what should happen and let it work out the how without my spelling out each step, I told my wife, Julie, I thought I was onto something. She said okay.

Then I hit a wall, and it's the wall this whole book is about. Every tool I had for settling questions about what I'd built could settle the small ones and fell over on the exact ones I needed answered, which left me staring down an infinite hallway of guess-and-test with no proof waiting at the end of it, ever. One night, not even working, I was watching a video about Flatland, the old story about a creature living in two dimensions who cannot so much as picture a third. And it landed on me sideways, all at once, the way the real ones always do. I said it out loud: I get it now. There's another axis. Classical computation only ever sees what it can read and write off its own surface, and there's a whole direction it's blind to. It's not broken. It was just never built with an eye that points that way. That's how it landed that night, and it held. The rules of ordinary computing know states and steps, and nothing else. A computer can write "checked" on its tape, but that's only a mark. Nothing in the rules stops the next step from rubbing it out or printing it without the check, and nothing charges for settling it. The one thing those rules do hold for good is whether the machine has stopped, and even that comes with no price. Settling something for good, held by the step itself and paid for in the step. That's the direction, and ordinary computing doesn't have it. Cover it and set its price to zero, and what's left is the ordinary computer, a Thiele machine with a blindfold on. A view that drops the record is blind to it the same way. Over every state you can see sits a pile of states that differ only in what was kept, and no amount of staring at the surface tells them apart. Whether the world itself keeps a direction like this is my belief, and nothing in this book proves it. Page one says where the axis ended up, and chapter fifteen says how much of that night held when I pushed. The two computers on page one, and again in chapter eight, are my attempt to show you that

blindness with something you can hold. That same night I gave the idea a name: the Thiele Machine. The name began as a working title and became the name of the model. It carries the larger claim plainly: the machine treats structural state as part of computation.

What survived the whole trip from there to here is not the renderer. Not one piece of it, and good riddance. What survived was a single commitment I made early and never once let anyone talk me out of: structure is the state itself, the actual thing the machine runs on. It is no metaphor laid over some realer state underneath. Category theory was the first language I found for that structure. It is not what the model is. The basic model asks for states, instructions, a step, prices and a reading. The ledger and the window are further structure used in the constructions. None of that needs category theory. The morphisms in that first engine, the night they first composed correctly at two in the morning, were objects that remembered where they came from; an established record remembers where it came from too, and you can walk up to one and challenge it. The receipt grew teeth. But the shape itself never moved. The shape was the substrate the entire time, and everything else, every renderer and every proof, was me slowly learning to see what I had already built. By the shape I mean structure held as state. Not category theory. Category theory was just the first language I had for it.

Every step pulled back to show another layer underneath that I had not known was there, and I mean that plainly: I didn't know what a Turing machine was when I started, I didn't know most of the words in this book existed. I learned each one the same way, by walking face-first into a wall that needed it and not being let past until I had it. The needing always came first. The learning came after, dragged along behind the needing because it had no say in the matter.

Large language models were my hands the whole way through, and I want the division of labor exact, because it's the thing people most want to get wrong about work like this. The hands had nothing like this to draw on: no reference implementation, no prior project to point at. The ideas, the architecture, the direction, the demand that the thing be true: those are mine, all of them, no asterisk. The symbol-pushing, the code, the formal derivations, and the drafting and revision of the prose under my rules, the models produced. So I read and pushed back on every statement. A derivation can settle an empty question, and a sentence can claim more than its proof earned, and I had to refuse both. However the words and symbols got onto the page, the argument on it is checkable by reading, and it would be exactly as right or as wrong if a million monkeys had typed it. The monkeys could not have refused.

A word on what I actually do, because it runs under everything after it. I do not write code or math by hand. I do not pick constants. I do not grind through derivations. I have never written a line of code on paper or in a notepad in my life.

What I do is see, and direct, and refuse. When I take hold of an idea it doesn't stay a point. It opens up. It becomes a shape I can blow up and turn over and take apart, lay across another shape to find where the two of them touch, run forward and then back to where it started, zoom into until it's the only thing in the world and pull back until it is one dot in a field of other dots, and the whole time it stays wired to everything I have ever connected it to, the right connections and the wrong ones both, all of them live at once. I am not reading that field neutrally, the way a search would. The thing doing the looking is me, the whole of me, my gut and my mood and my stubbornness picking what lights up and which way I move through it, and that's the part you can't lift out without turning me back into a search engine. So I see where a thing has to fit, the shape it is obligated to have, usually well before I could write it down in a single symbol. Then the model gives me the symbols, and I refuse anything whose argument does not hold or does not match the shape I am holding. That refusal is the one thing in this entire book that is mine and only mine: the demand that it be true, and the throwing-out of everything that wasn't.

I am telling you all of this because it's the engine. I did not come up through anywhere that hands you a foundation and says trust us, this part is settled. No professors, no colleagues, no field full of authority to lean back on when a claim got heavy. From the outside that is a deficit. From the inside it is the whole reason any of this is solid: when I meet a claim I either find the proof or I do not believe it, and that goes double for my own, because my own are the ones I most want to wave through. It is how I am built. It is the only way I know to hold a thing and actually trust that I am holding it.

So everything in here gets built from the ground up. It isn't a courtesy to you. I build it because I do not believe I understand a thing until I have built it with my own hands, watched it stand up on its own, and tried to knock it over. I built this book the exact way I built the machine, brick by brick, refusing each brick until it held weight. That includes the parts you already knew cold before you opened this. I did not skip them for you, and I could not skip them for me.

The vocabulary I had to go look up to do any of this sits at the back of the book, so the argument comes first. If a word stops you, it is there. If you already know this material, you never have to open it. But if you catch something in the basics that is wrong, I want to know, and I mean that as a real ask. It isn't politeness. Nothing in here is correct just because I wrote it. Especially the parts I wrote.

On the name

I wanted to call it the meta machine. Then, in October 2021, Facebook renamed its company Meta: a company taking a term for self-reference as its name and draining the word of meaning in the same motion. So I went one level up, where this machine

already lives. The most meta move available, for a machine that runs on meta-level reasoning, was to put my own name on the cover and take the word back. That's the whole of it, a meta move on a meta machine. It happens to be my dad's name too, and my wife's now. So I asked them before I did it. They said go ahead.

APPENDIX A

Things I had to learn first

None of this is the argument. It's the vocabulary I had to go look up before I could make the argument, written the way I understood each word once I had it. If a word in the book stopped you, it should be here. If you already know a part, skip it. And if one of these is wrong, I want to hear it, the same as anywhere else in the book.

The first four parts are words other people built. The last part is the words this book builds, each with the chapter where it's built properly; here they only get a reminder.

A.1 Counting

A bit is one place that holds 0 or 1. Two bits have four arrangements, 00, 01, 10 and 11, and every bit you add doubles the count, so n bits have 2 to the n. A byte is eight bits, 256 arrangements. Nothing in a byte knows whether it's a number, a letter or a color. That's a convention laid on top.

A function is a rule that gives every input exactly one output. "One next state for every state" is a function from states to states. A function is one-to-one when no two inputs share an output. In the book's pictures that's a machine where two arrows never land on the same spot.

The pigeonhole principle: put more pigeons than holes and some hole holds two. Chapter two does it with four people and three chairs, and chapter two also tells how old it is.

The logarithm base two of a number is how many times you can halve it before you reach one. Eight halves three times, so log base two of 8 is 3. For a number that isn't a power of two it lands in between: log base two of 3 is about 1.58. That's why an AND gate, which sends three of its four inputs to 0, needs about 1.6 halvings in chapter three. The natural logarithm, ln, is the same idea with a base of about 2.718 instead of two; ln 2 is about 0.693.

A spread, in this book, is a list of chances, one for each state, that add up to one. An even spread gives every state the same chance. The usual word is probability distribution.

Shannon's entropy, for a spread, is roughly the average number of yes-or-no questions you'd need to pin down the state. An even spread over n states has log base two of n bits of it, and that's the most any spread over n states can have. One bit is one question.

A.2 Machines and programs

A Turing machine is a tape of squares, a head that reads one square at a time, a state, and a card of rules. For each state and each symbol under the head, the card says what to write, which way to move, and which state comes next. Chapter one builds one on a strip of three squares.

The lambda calculus is Alonzo Church's way of writing computation as functions handed to functions, with nothing else in the language. It computes exactly what a Turing machine computes. That match was proved by other people; chapter one says so.

The halting problem: no program decides, for every program and every input, whether that program stops on that input. Chapter one tells what Turing actually proved and who first called it the halting problem; chapter seventeen puts it beside the other walls.

A diagonal argument builds the one thing that does the opposite of what a list says about it, so the list has to be wrong somewhere. Cantor used it on lists of numbers, Gödel and Turing on statements and programs.

Kleene's recursion theorem: for any way of transforming programs that a program can carry out, there's a program that behaves exactly as if it had been handed its own source code. Chapter seventeen uses it.

P versus NP. P is the yes-or-no problems a program can solve in time bounded by some fixed power of the size of the input. NP is the problems where a proposed answer can be checked that fast. Whether every problem in NP is in P is open, and the Clay Mathematics Institute will pay a million dollars for an answer. Chapter twenty-two says why nothing in this book touches it.

A cache is a copy of a result kept close at hand so the work that made it doesn't have to be done again. Keeping a function's past answers in a table is what Donald Michie called a memo function. Chapter sixteen tells that history.

A hash is a short fingerprint of a longer piece of data, made so that finding two inputs with the same fingerprint is supposed to be hard. A hash chain folds each new record into the fingerprint of the one before, so changing an old record changes every fingerprint after it. A Merkle tree folds records in pairs, then pairs of pairs, up to one fingerprint at the top.

A commit, in a database, is the one moment after which a pile of tentative work counts. It's written to a log, and a mistake after it gets fixed by another transaction, not by erasing. Chapter eight tells where the idea came from.

Compare-and-swap is an instruction that writes a new value into a memory cell only if the cell still holds the value you read a moment ago. The ABA problem is its blind spot: the value went from A to B and back to A while you weren't looking, so the check passes on something that changed. Chapter sixteen has it from IBM's manual.

A potential, in counting the cost of a run, is a number given to every state that measures how much is stored up in it. If a step's cost plus the change in potential is never negative, then a whole run costs at least what the potential dropped. Chapter eleven has where it came from.

A.3 Heat

Temperature in physics is measured in kelvin, from absolute zero. Room temperature is about 300 kelvin.

A joule is a unit of energy, about what it takes to lift an apple one meter. A zeptojoule is a thousandth of a billionth of a billionth of a joule; an attojoule is a thousand zeptojoules.

Boltzmann's constant, k, is about 1.38 times ten to the minus twenty-three joules per kelvin. It turns counting into physics: one bit of entropy, written in physical units, is k times ln 2. kT is Boltzmann's constant times the temperature, an amount of energy; at room temperature it's about four zeptojoules, and kT ln 2 is about three.

Landauer's principle, from Rolf Landauer in 1961: lowering the uncertainty about a device's state by one bit hands at least kT ln 2 of heat, on average, to the surroundings. It's a claim about the world, and it has critics. Chapter twenty says what he actually wrote and who doubts it. In this book it's always an assumption, with its name on it.

Maxwell's demon is James Clerk Maxwell's little being at a little door between two boxes of gas, letting fast molecules one way and slow ones the other, so one side heats and the other cools with no work done. Chapter six tells how the looking got separated from the wiping.

Reversible has two meanings, and chapter twenty pulls them apart. A step is logically reversible when no two states land on one, so you can always run it backwards. A process is thermodynamically reversible when it produces no entropy. Whether a device that erases counts as thermodynamically reversible depends on whether you ask about one known input or about a spread of them.

The second law of thermodynamics says the entropy of an isolated system doesn't go down. Chapter twenty-one compares its shape with the toll's and doesn't claim they're the same thing.

Local detailed balance is the standard assumption in the physics of small noisy systems that ties a step's odds to its heat: if a step forward is more likely than the step back, the forward step hands the surroundings kT times the log of the ratio of the two chances. Chapter twenty grants it for the walker.

A noisy step is a move that goes from each state to each other state with some chance, instead of to one certain next state. A unitary step is how a closed quantum system moves: it turns the list of directions the system can point in without stretching or squashing any of them, so it can always be run backwards. Chapter twenty-one does both.

A macrostate is a description that groups many exact states, the microstates, under one picture, the way a temperature groups countless arrangements of molecules. In this book's words, a macrostate is a picture through a window and its microstates are the pile behind it.

A.4 Structure

A relation is a list of which things stand in some connection to which: who is taller than whom, which number divides which.

An order puts things so that some come before others, in a way that chains: if a is at or below b and b is at or below c, then a is at or below c. Ties are allowed in the orders this book uses. A checklist's arrangements, ordered by which ticks one has that the other doesn't, are an order.

A partition is a sorting of a pile into groups, nothing in two groups and nothing left out. Chapter ten calls it a sorting, and a sorting with a door between the groups that only swings one way is an order.

A category is a collection of objects and arrows between them, where an arrow from a to b and an arrow from b to c always give an arrow from a to c, following one and then the other; following arrows doesn't care how you group them; and every object has an arrow to itself that changes nothing. Morphism is the formal word for arrow.

A functor is a map from one category to another that sends objects to objects and arrows to arrows and keeps the following-one-then-the-other intact. It's a translation between two worlds of structure.

A monoidal product is a way to set two objects side by side and get one, like pairing, with an empty object that changes nothing when you set it beside another, and rules that make the grouping not matter.

The Yoneda lemma is a result deep in category theory. One consequence of it: an object is pinned down, up to a perfect match, by the arrows coming into it from everything else. The afterword says what it did for me. No argument in this book uses it.

An invariant is something that stays the same under a chosen set of changes. Emmy Noether showed in 1918 that in physics every smooth symmetry of a certain kind comes with a quantity that's conserved. Chapter ten has it.

Structuralism, in the philosophy of mathematics, is the view that mathematics is about patterns of relations, and that what fills a pattern doesn't matter. Chapter ten names the people who hold it and the people who push back.

A.5 Words this book builds

A thing, a move, a run, a price, and a way to tell how settled the thing is, where settled never goes back: page one. The toll is the one rule: a move that settles something more than it was costs something. A Thiele machine is all of it together, with some run from a start you pick that actually settles it more than it was. A Thiele machine with its record covered and the price of settling set to zero is the ordinary computer.

Two arrows meet when two different states have the same next state; the usual word is a merge. Fresh paper, or a fresh state, is a state the run hasn't used before. Chapter one shows that every step of a machine with one next state does one of three things: lands somewhere new, makes two arrows meet, or goes round a loop.

The chairs and the done room are chapter two's picture of the pigeonhole principle, aimed at states that read done.

Bennett's rule: a move where two arrows meet costs at least one. The rule is Charles Bennett's, from a sentence he wrote in 2003; using it as the price is my choice. The halving rule is stronger: each unit of cost pays for at most one halving, so a move pays at least the halvings in its worst pile-up, the place where the most arrows land. Both are in chapter three.

The box and the flag: chapter four's smallest Thiele machine, two boxes and one move that sends both to YES. Covering the coin is how any machine with a move becomes what's left of a Thiele machine.

A row of cups is a record with more than two places: cups that only ever get turned up. Chapter five.

Looking and keeping: a look can use up a blank page that gets handed back, so looking can be free. Keeping what you saw for good is what pays. Chapter six.

A window is anything that lets part of a machine through to you: a camera, a display, a printout. It doesn't see anything. Behind one picture is a pile, every full state that gives that picture. A clash is two steps that look the same through

a window, where one turns up a cup and the other doesn't. Chapters seven and fifteen.

Established: a record that goes up only right after a passing check of the very claim it records, with nothing that claim is about changed in between. It's a condition you put on the machine. Chapter eight.

Settled isn't the same as true. Settling is a sorting with a door that only swings one way. Chapter nine.

Structure, in this book, is relations, kept under chosen changes, carried by any stuff, relative to a frame. A split with a one-way door is the smallest start, by the ruler chapter ten states.

The counter is a number in the state that starts at zero and that every step raises by exactly that step's charge. Read as a receipt, it's a total that lives in the state, only climbs, equals the charges on every path including the failures, and is the only such total for its price list. Chapter eleven.

A grouping is sound when every two things inside one group agree on what's being tested. Standing behind a sound grouping entitles you to exactly the groupings you get by cutting it smaller. Chapter twelve.

Insight, as this book prices it, is a distinction made to stay. The price doesn't measure depth. Chapter thirteen.

The base is the machine a box watches. A box answers only to the machine when what it reads depends only on the base's run. A cup over any machine is chapter fourteen's result: such a box that never comes down is a cup turned by some event the base passes through. A clean start is a start with every box down. A toggle is a box that flips up and down as the machine runs; it answers to the machine and doesn't stay up, so it's no cup.

The uncovered machine is what's left when you cover a machine's cups and tally, if nothing in how the rest moves depends on them. It's the ordinary computer. A shortcut is a move that answers from the record instead of redoing the work, and it's sound when it gives the same answer the long way would. Chapter fifteen.

An edition is a count of how many times something has been written, so a box can be about one edition. The guard is whatever decides whether a shortcut may run. The four doors are the four ways a finite machine that reads its box to skip work can pay for it: fresh editions until they run out, a box that comes down, room for every value checked, or wrong answers. Chapter sixteen.

The pointer thesis: the established record is the event that gets copied and relied on by parties who didn't make it. A thesis, stated as one. Chapter nineteen.

The ledger sorts every claim in the book into shown, chosen, assumed, cited, thesis, or not claimed. The Thiele thesis says every physical process that computes with final answers has an exact description as a machine run by one rule, on which the answer is a reading no run turns off. The blindfold is the picture that every

machine that steps is a Thiele machine with its record covered and the price of settling set to zero. All three are in chapter twenty-two.

APPENDIX B

Every assumption in one place

Every argument in this book says what it stands on, in the chapter where it's made. This appendix puts all of those in one place, so you can see which result leans on what, and what walks through when a condition is taken away. The ledger in chapter twenty-two sorts the claims. This sorts the conditions under them.

Each one below is one of three things. A part of the definition, which I chose, and page one says so. A condition a chapter puts on a machine, which holds for the machines that meet it and nobody else. Or an assumption about the world, somebody's claim about physics, named, that an argument leans on and doesn't prove.

B.1 The definition

A thing, moves that take it to a next thing, and a way to tell how settled it is, where settled never goes back. Settledness can be yes or no, or a checklist of ticks that don't come off. Take away "never goes back" and the coin walks through: a flip and a flip back, settled and unsettled, forever, and nothing has to give. Chapter three's second door.

Some run, from a start you pick, that actually settles it more than it was. Take it away and a coin that never settles anything counts as a Thiele machine for free. Page one.

A price on every move, nothing or more, never a refund. Take away "never a refund" and a move that settles charges one, the next hands it back, and the whole run comes out free. Page one.

The toll: a move that settles something more than it was costs something. That's the one rule. Chapter three shows it isn't only a choice: on a finite machine it follows from settled staying settled and Bennett's rule together.

B.2 What a machine is

One next state for every state. That's what chapter one assumes of a machine, and it's all the step law needs: every step lands somewhere new, makes two arrows meet,

or goes round a loop. Nothing about size. A machine with more than one move from a state has one next state per move; chapter twenty-two takes steps that branch and shows that, after the raise, a single run still makes two arrows meet or never repeats a state.

Finitely many states. This is the condition people most often think the book leans on, and it leans on it less than it looks. Chapter one needs none of it. What finiteness takes away is the column of fresh paper: on finitely many states, a run that settles something for good has to make two arrows meet, because it can't land somewhere new forever. Take it away and Hilbert's hotel walks through: a move where no two arrows ever meet can settle something, and nothing meets, as long as there's always another room. Chapter two. And the finite set the count needs can be tiny. Chapter four's machine has two boxes.

Applying any of this to a real device takes one more step the arguments don't supply: a justified list of the device's states that the device can actually tell apart. That's a fact about the device.

B.3 Prices

Bennett's rule: a move where two arrows meet costs at least one. It's the third of the three conditions in chapter three. Take it away and the stamp at zero walks through: a move that sends both states to YES, priced at nothing. Chapter three's third door.

The halving rule: each unit of cost pays for at most one halving of the places you could have been. It's stronger than Bennett's rule, and it's what sizes the toll: a move that squeezes m + k states into m pays at least the halvings between them. Chapters three, six and fifteen use it. Chapter twenty shows that charging bits on every spread is the same rule.

A price list is always given. Nothing in the book says what the charges should be beyond the floors above. Chapter eleven's counter is the arithmetic once they're chosen.

B.4 Conditions put on a machine

Established. A record that goes up only right after a passing check of the very claim it records, with nothing the claim is about changed between the check and the raise. That's chapter eight. One more thing it needs is a fact about the checker and not a condition on the machine: that the check means what it says and could have failed. And every guarantee about an established record is about runs: any list of moves, from a clean start, on a step that does what it says, one writer at a time. Whoever sets the start, the wire that carries the report out, and the tools that built the step are outside that fence. Chapter twenty-two.

Answering only to the machine. A box whose next position depends only on what the base is doing right now and where the box is right now, with a tally that never goes down, a toll on the step that turns the box up, a box that stays up once up, and some run that turns it up. Those are chapter fourteen's five rules. Drop "stays up" and the toggle walks through. Drop "answers only to the machine" and the timestamp does.

Covering the cups. Nothing in how the rest of the machine moves depends on the cups or the tally. Without that, covering them leaves a machine that can't decide what to do next. Chapter fifteen's result also needs the machine finite, the record only growing, and the uncovered machine forgetting nothing.

Editions never reused. The shortcut in chapter sixteen never answers wrong from a clean start as long as every write moves to a fresh edition. Let the editions wrap around and four moves give a wrong answer.

The guard's two conditions. Its glance can't tell a passing value from a failing one, and the memory it reads is made only of boxes that go up and never come down. Under both, there are four doors and no fifth. Drop the first and you're at the fourth door, or doing the check over. Drop the second and you're at the third. Chapter sixteen.

The diagonal's conditions. A procedure that decides a question about programs, where the machine has Kleene's trick, the flip is something the machine can run, two programs that behave alike always get the same answer, and there's one program that passes and one that fails. Under those, no procedure inside the machine decides it. A procedure that answers only for some programs, or needs outside evidence, isn't ruled out. Chapter seventeen.

Two levers that keep states apart. In chapter twenty-one each lever has to keep different states apart for the states reached within the promised number of pulls. The room counts there are counts under that promise. Receipts covering the merges a run shows needs one of every colliding pair to pay.

Steps that keep the even spread. A noisy step that keeps the even spread can't raise a reading that stays. Chapter twenty-one. A step that can raise one makes the spread lumpier.

B.5 Assumptions about the world

Landauer's principle: lowering uncertainty about a device's state by a bit hands at least kT ln 2 of heat to the surroundings, on average. It has critics, and chapter twenty names them. Every heat claim in the book is "given Landauer," and nothing in the book derives it. On a state that's already known, the principle sets no floor at all.

The fixed device. A fixed device that merges K inputs produces at least k ln K of entropy on one of them. That leans on the second law for the device's average and on reading dissipated work as temperature times entropy produced. It compares inputs to one device, not different devices for different inputs.

Local detailed balance: a forward step hands the surroundings kT times the log of the odds of going forward over going back. Chapter twenty grants it for the walker, and only there. Under it, heat times time is at least twice the distance squared.

Unitary steps. Chapter twenty-one's quantum result is about a closed system that moves by unitary steps on finitely many directions. An open system, one that hands something to its surroundings, is a different question.

B.6 Theses

A thesis ties a definition to the world, and no argument can settle one. The book has two, stated as theses where they're made.

The pointer thesis: the established record is the event that gets copied and relied on by parties who didn't make it. Chapter nineteen, with what would test it.

The Thiele thesis: every physical process that computes with final answers has an exact description as a machine run by one rule, with a next state for every state, written down in a way a program can follow, on which the answer is a reading that no run from any input ever turns off. Chapter twenty-two. Robin Gandy's principles for mechanisms are one road to that kind of tie, with real assumptions about the world in it; I don't assert them of every physical system.

APPENDIX C

What's mine, and what isn't

Chapter twenty-two gave the short form of who did what. This is the long form, and it's the place in the book where I'd most like to be caught out, and caught out early. For each piece the book could be accused of claiming, it says who had it before me and what's actually mine.

Two warnings about my own evidence first. "I didn't find it" is a fact about my searches, and the literature is bigger than my searches. I looked with every tool I had, and some of those tools will cheerfully invent a citation if you let them, so every citation in the sources was checked against the work itself or its publisher's record before it went in. Still, assume it may well be sitting in a paper I haven't read. And a proof, if it's right, tells you a statement is true. It doesn't tell you it's new. A thousand correct proofs could be a thousand careful ways of agreeing with somebody who got there first.

There's a law about tables like this one, and it's a joke that keeps turning out true. In 1980 the statistician Stephen Stigler proposed that no scientific discovery is named after the person who first found it, and then credited the idea to the sociologist Robert K. Merton, who had been writing about fights over priority in science since 1957. So Stigler's law is an example of itself. Is it a law? No, it's a pattern, and nobody proved it. But this book has already tripped over it: the pigeonhole principle wears Dirichlet's name, and Leurechon had it two centuries earlier. And I put my own name on a machine. By Stigler's law that makes it somebody else's. The table is my best attempt at finding out whose.

What I compare each time is what a result is about. Landauer's bound is about heat, a physical quantity, under a physical model. The toll is a floor on one kind of step, the step that settles something more than it was, in a model I chose. So the comparisons below are about the shape of each result. They aren't claims about who reached a place first, and they don't show that settling is the only event worth pricing.

C.1 The prior-work map

One row per chapter, in the order the book reaches them. "Mine" means one of three things, and the ledger in chapter twenty-two says which for every claim: a definition or a price I chose, an argument I haven't found anywhere else, or a reading of other people's work, which is only ever as good as my reading.

Ch.	Who had it before me	What's mine
1	That every step of a machine with one next state lands somewhere new, makes two arrows meet, or goes round a loop is counting about functions, and the counting isn't mine. Turing's machine, Church's lambda calculus and the proofs that they match are theirs.	Pointing the count at a reading that stays.
2	The pigeonhole principle: Leurechon in 1622, Dirichlet in 1842. Recurrence: Poincaré, and Zermelo against Boltzmann in 1896. Reversible automata that can't keep a done: Pin in 1992, Brodsky and Pippenger in 2002. The counting with a physical price on it: Ouldridge and Wolpert in 2023. Neighbors in constructor theory: Deutsch and Marletto, Marletto, Vedral and coauthors, Violaris and Marletto. Rogers on monoid actions.	Nothing in the counting. The chairs are a small, exact, classical relative of what they did.
3	Bennett's rule is Bennett's, in his words of 2003. The least work for a step that has to succeed on every input is Faist, Dupuis, Oppenheim and Renner's, 2015. Landauer's principle is Landauer's.	Writing the price on the move as a rule of the model; the three doors, with a machine walking through each; the halving rule as a choice of price.
4	Seals, witnesses, tallies and receipts are history, and the people who studied them are named in the chapter. Cook's certificates, 1971.	The smallest Thiele machine, and the coin covered.
5	A value that only climbs, asked only whether it's past a point: Kuper and Newton's LVars, 2013, and Timany and Birkedal, 2021.	A price on crossing a line.
6	Looking without merging: Bennett. Charging the look or the wipe as a choice of bookkeeping: Fahn, 1996; Sagawa and Ueda, 2009. A choice that sticks gives off heat: Rovelli, 2024.	The reading of Bennett and Rovelli as two sides of one line, with the four places it gives.

Ch.	Who had it before me	What's mine
7	That two things that look the same through a window can't be told apart from it is a fact about functions far older than this book. Moore, 1956; Kalman, 1960; Codd, 1970; Sampath and coauthors, 1995; Saboori and Hadjicostis, 2007.	The price: a window with a clash can't price exactly, and the least it has to overcharge.
8	The commit: Gray, 1981, from contract law, and Austin's "I do," 1955. The reference monitor: Anderson's study, 1972; Schneider, 2000; Azevedo de Amorim and coauthors, 2015. The gap between check and use: Bishop and Dilger, 1996.	The definition of an established record: paid for, permanent, and earned, together.
9	The creating subject: Brouwer; Kreisel, 1967; Troelstra, 1969; van Atten's history, 2018. A proof as an act: Martin-Löf.	The reading: settled is a sorting with a door that only swings one way, and closed isn't the same as true.
10	Structure as relations: Bourbaki, the structuralists, Klein, Noether, and the rest named in the chapter. An order as a sorting with one-way doors is a textbook fact; the chapter cites Halmos.	The overlay of the sources, and the ruler under which a split with a one-way door is the smallest start. That nobody prices vouching for a grouping isn't claimed; I only didn't find anyone who does.
11	The tally stick. Lamport's clocks, 1978. Numbers that only go down: Turing and Floyd. The potential method: Tarjan, 1985, crediting Sleator; Lyapunov, 1892, in another form.	Which potential: the one that drops when something settles, and what happens to it through a window.
12	Gettier, 1963, and the history of the definition he broke. Proof-carrying code: Necula, 1996 and 1997. Changing your mind: Segerberg, Gärdenfors, Katsuno and Mendelzon, Baltag, Renne and Smets, and the others named. Counting by possibilities: Hartley, 1928.	A checklist for leaning on a claim. It's true by packaging and discovers nothing about leaning.
13	Köhler's Sultan and the sticks, on Tenerife; Duncker's box, 1935; Metcalfe's ratings of how close people felt to a solution.	The meaning chosen: insight, as priced here, is a distinction made to stay.
14	History variables: Abadi and Lamport, 1991. The "once" operator: Lichtenstein, Pnueli and Zuck, 1985. Watchers that keep it in one bit: Havelund and Roşu, 2002. A toy world with a clean start: Ebtekar and Hutter, 2024.	The other direction: a box that answers only to the machine and never comes down can't be anything but a cup on one event, over any machine.

Ch.	Who had it before me	What's mine
15	The fact about functions under the window, again.	The pricing form of it, and that a paid move crushes two states of one pile when the conditions hold.
16	Memo functions: Michie, 1968. Keeping one copy of each structure: Ershov, 1958; Goto, 1974. The ABA problem: IBM's System/370 manual, 1974 as revised in 1975; Treiber, 1986; Michael and Scott, 1995 and 1996. Caches, version counters and memo tables in general belong to the programmers who built them.	That the four are four ways of paying chapter two's bill, and that under the two conditions there's no fifth.
17	The diagonal: Cantor, 1891; Russell, 1903; Gödel, 1931; Turing, 1936; Kleene's recursion theorem, 1938; Rice, 1953. Blum's speedup theorem, 1967.	Nothing new. Rice's trick one floor over, with its own conditions, pointed at a box.
18	The five systems belong to their designers, and the copies I read are named in the chapter.	The reading of five copies side by side, through one question.
19	Pointer states and quantum Darwinism: Zurek, from 1981, with Riedel and Zwolak; Brandão, Piani and Horodecki, 2015; Qi and Ranard, 2021.	The pointer thesis, stated as a thesis.
20	Landauer, 1961, and Bennett after him. The critics: Earman and Norton, Norton. The two meanings of reversible: Ladyman and coauthors, Sagawa. The walker: Bennett's Brownian computer, 1982; Schnakenberg, 1976; Seifert's review, 2012. Bounds of the same kind for other models: Schmiedl and Seifert, 2007; Aurell, Mejía-Monasterio and Muratore-Ginanneschi, 2011; Shiraishi, Funo and Saito, 2018.	The bit counts for a permanent raise from the even spread, and that every scale pays the same way; the walker's bound, worked on the page under local detailed balance. Every heat figure is given Landauer.
21	Reversible machines saving room: Bennett, 1989. Precise floors for a reset: Reeb and Wolf, 2014. Noisy and unitary steps are standard, and so is the collision weight.	The exact room counts: $1 + (T + 1)(N - 1)$, $N + T(N - L)$, $T + 3$ and $2T + 2$ for two levers, $2T + 3$ for two commuting levers that keep states apart forever, the grid's $1 + (t + 2)(t + 3)/2$; receipts covering the merges a run shows.

Ch.	Who had it before me	What's mine
22	Church and Turing's thesis. Principles for mechanisms: Gandy, 1980; Arrighi and Dowek, 2012. Prices on erasing: Ouldridge and Wolpert, 2023; Li and Vitányi, 1996; Demaine and coauthors, 2016.	Which readings of two machines side by side keep the toll; the run with branching steps; the blindfold; the ledger; the Thiele thesis, stated as a thesis.

C.2 What I'm claiming

Six things, the same six as chapter twenty-two, each with the conditions it holds under.

First, keeping: on any machine with one next state, a run that raises a reading that stays either makes two of its states land on one or never repeats a state, and with a price on two arrows meeting, the toll follows. The pieces are old. Recurrence goes back to Poincaré and Zermelo, the pigeonhole to Leurechon, the price on a merge to Landauer and Bennett, and Ouldridge and Wolpert had the counting with a physical price on it in 2023. What I haven't found is the toll forced that way on a reading that stays, in a model of computation, with the noisy and the finite quantum versions in chapter twenty-one.

Second, the cup over any machine: a box that answers only to what the machine is doing and never comes down is a cup turned by some event the machine passes through. History variables and the "once" operator are the cup built by hand. What I haven't found is the other direction, that such a box can't be anything else, with the toggle and the hidden clock as what falls outside.

Third, the blindfold: every machine that steps is a Thiele machine with its record covered and the price of settling set to zero. Put a price on the move that stops Turing's machine and, read through whether it has stopped, it fits. This computes nothing new. What it adds is the direction and its price, which the bare machine doesn't have.

Fourth, the four doors: a machine that reads its box to skip work it already paid for, under chapter sixteen's two conditions, pays in fresh editions, a box that comes down, room for every value checked, or wrong answers, and there's no fifth. Every door has been used by programmers for decades, and the chapter names who. What's mine is that they're one bill, and the proof that there's no fifth.

Fifth, the exact room counts of chapter twenty-one. Bennett showed in 1989 that a reversible machine can trade time for room. The counts of how much room it takes to hold off a merge, through a promised number of repetitions, I haven't seen anywhere.

Sixth, the ledger: every claim in the book sorted into shown, chosen, assumed, cited, thesis, or not claimed. That isn't a result. It's a promise about how the book was written, and every row is a place to check it.

Under all six is the bet, and the bet is mine: that the price belongs in the step.

Sources

These are the works the book leans on, for their results, their history and the company they keep. Each one is listed under the first chapter that names it. A note after an entry says what the book takes from it, where the title doesn't already say. Much of the record sits behind paywalls, and I don't cite a page as read unless I've looked at it myself. Where an open copy existed, a preprint, an author's posted PDF, an archive scan or a public-domain text, I checked that copy and cited that version. Where a paywall blocked the original, I didn't fake a primary reading. I checked the claim through open secondary sources or the abstract, and the entry says so. A source seen through a window is labeled as a window. Where I couldn't confirm a detail of an entry, a page range or a place of printing, I left it out rather than guess.

Page one

Alan M. Turing (1936–1937). On Computable Numbers, with an Application to the Entscheidungsproblem. Proceedings of the London Mathematical Society, series 2, volume 42, pp. 230–265. DOI: 10.1112/plms/s2-42.1.230. The clerk at a desk reduced to a tape, a head, a state and a card of rules, with an appendix sketching the match with Church's version.

Chapter 1. Any machine that steps

American Institute of Physics. Harvard College Observatory: The labor and status of the women computers. AIP page at aip.org, labor-status-women-computers. The women computers were paid a fraction of what the men were paid; the page puts the men's hourly rate at nearly five times as much.

Alonzo Church (1936). An Unsolvable Problem of Elementary Number Theory. American Journal of Mathematics 58(2), pp. 345–363. DOI: 10.2307/2371045. Computation written as functions handed to functions, from the same year as Turing's paper.

B. Jack Copeland (1997, revised 2023). The Church-Turing Thesis. Stanford Encyclopedia of Philosophy. Stanford Encyclopedia of Philosophy, entry church-turing. Why the claim about everything a person could do by following rules is a thesis and not a theorem.

Martin Davis (1958). Computability and Unsolvability. New York: McGraw-Hill. Reprinted by Dover, 1982. The first use of the words “the halting problem.”

David Alan Grier (2005). When Computers Were Human. Princeton University Press. Rooms of human computers working out artillery tables in both world wars.

Joel David Hamkins and Theodor Nenu (2026). Did Turing prove the undecidability of the halting problem? Journal of Logic and Computation 36(1), exaf075. DOI: 10.1093/logcom/exaf075; arXiv: 2407.00680. What Turing proved in 1936, Kleene’s statement of the halting problem in 1952, and Davis’s name for it in 1958.

Stephen Cole Kleene (1952). Introduction to Metamathematics. Amsterdam: North-Holland; New York: D. Van Nostrand. Where the halting problem is first stated in the form people now learn.

Henrietta S. Leavitt and Edward C. Pickering (1912). Periods of 25 Variable Stars in the Small Magellanic Cloud. Harvard College Observatory Circular 173, pp. 1–3. The link between a variable star’s period and its brightness that became a way to tell how far away a star is.

J. C. Shepherdson and H. E. Sturgis (1963). Computability of Recursive Functions. Journal of the ACM 10(2), pp. 217–255. DOI: 10.1145/321160.321170. Machines with registers instead of a tape compute the same things Turing’s machine does.

Dava Sobel (2016). The Glass Universe: How the Ladies of the Harvard Observatory Took the Measure of the Stars. Viking. The women who cataloged stars by hand at Harvard, Henrietta Leavitt among them.

Alan M. Turing (1937). Computability and λ-definability. Journal of Symbolic Logic 2(4), pp. 153–163. DOI: 10.2307/2268280. The full proof that Turing’s machines and Church’s functions compute the same things.

Chapter 2. Running out of room

Charles H. Bennett (1973). Logical reversibility of computation. IBM Journal of Research and Development 17(6), pp. 525–532. DOI: 10.1147/rd.176.0525. A computation that saves its intermediate results can run without erasing and then undo its own steps.

P. Bocchieri and A. Loinger (1957). Quantum Recurrence Theorem. Physical Review 107(2), pp. 337–338. DOI: 10.1103/PhysRev.107.337. A closed quantum system comes back close to where it started.

Ludwig Boltzmann (1896). Entgegnung auf die wärmetheoretischen Betrachtungen des Hrn. E. Zermelo. Annalen der Physik und Chemie, Neue Folge 57, pp. 773–784.

DOI: 10.1002/andp.18962930414. The reply to Zermelo that nobody would ever wait long enough to see a return.

Alex Brodsky and Nicholas Pippenger (2002). Characterizations of 1-way quantum finite automata. SIAM Journal on Computing 31(5), pp. 1456–1478. DOI: 10.1137/S0097539799353443. A quantum automaton that measures once at the end recognizes exactly the group languages; one that measures after every letter picks up "has seen a b."

Richard Dedekind (1888). Was sind und was sollen die Zahlen? Braunschweig: Vieweg. The definition of infinite: a collection that matches one-to-one with a part of itself that leaves something out.

David Deutsch (2013). Constructor theory. Synthese 190(18), pp. 4331–4359. DOI: 10.1007/s11229-013-0279-z. Physics stated as which transformations can and can't be brought about.

David Deutsch and Chiara Marletto (2015). Constructor theory of information. Proceedings of the Royal Society A 471(2174), 20140540. DOI: 10.1098/rspa.2014.0540; arXiv: 1405.5563. Its section 8.6: measuring one quantity disturbs another, and nothing undoes the disturbance while the result is kept.

Peter Gustav Lejeune Dirichlet (1842). Recherches sur les formes quadratiques à coefficients et à indéterminées complexes. Journal für die reine und angewandte Mathematik 24, pp. 291–371. DOI: 10.1515/crll.1842.24.291. One of his two papers of 1842 that use the drawer principle.

Peter Gustav Lejeune Dirichlet (1842). Verallgemeinerung eines Satzes aus der Lehre von den Kettenbrüchen nebst einigen Anwendungen auf die Theorie der Zahlen. Bericht über die Verhandlungen der Königlich Preussischen Akademie der Wissenschaften, pp. 93–95. The other 1842 paper that uses the drawer principle.

Edward Fredkin and Tommaso Toffoli (1982). Conservative logic. International Journal of Theoretical Physics 21(3–4), pp. 219–253. DOI: 10.1007/BF01857727. Circuits that compute anything and lose nothing, by carrying extra inputs in and extra outputs out.

George Gamow (1947). One Two Three... Infinity. New York: Viking Press. The book that made Hilbert's hotel famous.

David Hilbert (2013). David Hilbert's Lectures on the Foundations of Arithmetic and Logic 1917–1933, edited by William Ewald and Wilfried Sieg. Berlin and Heidelberg: Springer. DOI: 10.1007/978-3-540-69444-1. Prints the notes of the winter 1924 to 1925 course on the infinite, where the hotel appears.

Helge Kragh (2014). The True (?) Story of Hilbert's Infinite Hotel. arXiv: 1403.0059. Takes back his earlier guess that Gamow invented the hotel, after Tilman Sauer pointed him to Hilbert's lecture notes.

Rolf Landauer (1961). Irreversibility and Heat Generation in the Computing Process. IBM Journal of Research and Development 5(3), pp. 183–191. DOI: 10.1147/rd.53.0183. Quoted: "typically of the order of kT" and "show, or at least very strongly suggest," p. 183; "no guarantee that this minimum is in fact achievable," p. 187; and on p. 187 the inputs that are all ONE, where "there is no need to carry out any operation." Paper scan at fab.cba.mit.edu, Landauer-1961.

Yves Lecerf (1963). Machines de Turing réversibles. Récursive insolubilité en $n \in \mathbb{N}$ de l'équation $u = \theta^n u$, où θ est un "isomorphisme de codes." Comptes Rendus de l'Académie des Sciences de Paris 257, pp. 2597–2600. English translation at revcomp.info, english. Machines with no merging can imitate any Turing machine.

Jean Leurechon (1622). Selectae Propositiones in Tota Sparsim Mathematica Pulcherrimae. Gaspard Bernard, p. 2. The one-line remark that two people must have the same number of hairs.

Josef Loschmidt (1876). Über den Zustand des Wärmegleichgewichtes eines Systems von Körpern mit Rücksicht auf die Schwerkraft. Sitzungsberichte der Kaiserlichen Akademie der Wissenschaften in Wien, Mathematisch-Naturwissenschaftliche Classe 73, Abt. II, from p. 128. The laws of motion run just as well backwards.

Chiara Marletto, Vlatko Vedral, Laura T. Knoll, Fabrizio Piacentini, Ettore Bernardi, Enrico Rebufello, Alessio Avella, Marco Gramegna, Ivo Pietro Degiovanni and Marco Genovese (2022). Emergence of constructor-based irreversibility in quantum systems: theory and experiment. Physical Review Letters 128, 080401. DOI: 10.1103/PhysRevLett.128.080401; arXiv: 2009.14649. An irreversible task built from quantum bits that partly swap with each other, run with photons.

Kenichi Morita (1996). Universality of a reversible two-counter machine. Theoretical Computer Science 168(2), pp. 303–320. DOI: 10.1016/S0304-3975(96)00081-3.

Thomas E. Ouldridge and David H. Wolpert (2023). Thermodynamics of deterministic finite automata operating locally and periodically. New Journal of Physics 25, 123013. DOI: 10.1088/1367-2630/ad1070; arXiv: 2208.06895. If the smallest machine for a job merges two states on some letter, every machine for it does, and running it gives off entropy; one worked example is the strings with no b.

Jean-Éric Pin (1992). On reversible automata. In LATIN '92, 1st Latin American Symposium on Theoretical Informatics, Lecture Notes in Computer Science 583, Springer, pp. 401–416. DOI: 10.1007/BFb0023844. What automata whose letters never land two states on one can recognize.

Henri Poincaré (1890). Sur le problème des trois corps et les équations de la dynamique. Acta Mathematica 13, pp. 1–270. The recurrence theorem.

Benoît Rittaud and Albrecht Heeffer (2014). The Pigeonhole Principle, Two Centuries Before Dirichlet. The Mathematical Intelligencer 36(2), pp. 27–29. DOI: 10.1007/s00283-013-9389-1. Leurechon's remark of 1622, and Dirichlet's uses of the principle in 1842 and his later name for it.

Morgan Rogers (2019). Toposes of Discrete Monoid Actions. arXiv: 1905.10277.

Morgan Rogers (2021). Toposes of Monoid Actions. arXiv: 2112.10198.

Tilman Sauer and Ulrich Majer, eds. (2009). David Hilbert's Lectures on the Foundations of Physics 1915–1927. Berlin and Heidelberg: Springer. DOI: 10.1007/b12915. The volume of Hilbert's lectures that Sauer helped edit.

Tommaso Toffoli (1980). Reversible computing. In Automata, Languages and Programming (ICALP 1980), Lecture Notes in Computer Science, Springer, pp. 632–644. DOI: 10.1007/3-540-10003-2_104. The proof that reversible gates compute everything.

Jos Uffink (2004, revised 2024). Boltzmann's Work in Statistical Physics. Stanford Encyclopedia of Philosophy. Stanford Encyclopedia of Philosophy, entry statphys-Boltzmann. Zermelo as Planck's assistant in 1896, his careful logical argument, and Boltzmann's reply "drenched in sarcasm and bitterness."

Maria Violaris and Chiara Marletto (2022). Irreversibility beyond Landauer's principle: is it possible to perform erasure using the quantum homogenizer? New Journal of Physics 24, 113030. DOI: 10.1088/1367-2630/ac9e13; arXiv: 2205.11310. A machine made of reversible pieces that, run in a cycle, turns a pure quantum bit mixed and never back, suggesting erasure may cost more than the usual bound.

Ernst Zermelo (1896). Ueber einen Satz der Dynamik und die mechanische Wärmetheorie. Annalen der Physik und Chemie, Neue Folge 57, pp. 485–494. DOI: 10.1002/andp.18962930314. Uses Poincaré's recurrence theorem to argue that nothing in such a system can climb forever.

Ernst Zermelo (1908). Untersuchungen über die Grundlagen der Mengenlehre I. Mathematische Annalen 65(2), pp. 261–281. DOI: 10.1007/BF01449999. Set theory put on axioms.

Chapter 3. Why charge for it

Charles H. Bennett (2003). Notes on Landauer's principle, reversible computation, and Maxwell's Demon. Studies in History and Philosophy of Modern Physics 34(3),

pp. 501–510. DOI: 10.1016/S1355-2198(03)00039-X; arXiv: physics/0210005v2. Quoted from the arXiv preprint: Earman and Norton's objection as "in my opinion the one of greatest merit" (p. 1); ballistic machines that compute "at constant velocity without dissipating any energy" and "must be isolated" from heat baths and noise, and the thermodynamics of error correction as "a subject that appears to be in its infancy."

Ludwig Boltzmann (1877). Über die Beziehung zwischen dem zweiten Hauptsatze der mechanischen Wärmetheorie und der Wahrscheinlichkeitsrechnung respektive den Sätzen über das Wärmegleichgewicht. Sitzungsberichte der Kaiserlichen Akademie der Wissenschaften in Wien, Mathematisch-Naturwissenschaftliche Classe 76, pp. 373–435. Entropy as the logarithm of the number of ways the parts could be arranged.

Philippe Faist, Frédéric Dupuis, Jonathan Oppenheim and Renato Renner (2015). The minimal work cost of information processing. Nature Communications 6, 7669. DOI: 10.1038/ncomms8669; arXiv: 1211.1037. The least work a logical step needs when it has to succeed every time, with an AND gate at $\log_2 3 \approx 1.6$.

G. H. Hardy (1940). A Mathematician's Apology. Cambridge University Press. Mathematical reality lies outside the people who study it, and their theorems are notes on what they saw.

Max Planck (1901). Ueber das Gesetz der Energieverteilung im Normalspectrum. Annalen der Physik 309(3), pp. 553–563. DOI: 10.1002/andp.19013090310. Where the constant k and the entropy written as k times a logarithm come from.

Max Planck (1920). The genesis and present state of development of the quantum theory. Nobel Lecture, 2 June 1920. In the translation by H. T. Clarke and L. Silberstein, The Origin and Development of the Quantum Theory, Oxford: Clarendon Press, 1922. 1922 translation. The constant "is frequently termed Boltzmann's constant, although to the best of my knowledge Boltzmann himself never introduced it."

Heinrich Weber (1893). Leopold Kronecker. Mathematische Annalen 43, pp. 1–25. DOI: 10.1007/BF01446613. Also printed in the Jahresbericht der Deutschen Mathematiker-Vereinigung 2. The memorial piece that carries the line about God and the whole numbers, pointing back to a lecture of 1886.

Chapter 4. The smallest one

Michael T. Clanchy (1979). From Memory to Written Record: England 1066–1307. London: Edward Arnold; third edition, Wiley-Blackwell, 2012. England learning to trust writing, and the forged seals and charters along the way.

Stephen A. Cook (1971). The complexity of theorem-proving procedures. Proceedings of the Third Annual ACM Symposium on Theory of Computing (STOC '71), pp. 151–158. DOI: 10.1145/800157.805047. A claim that comes with a certificate that's quick to check.

Nicholas of Cusa (1433). De concordantia catholica. Where he calls the Donation of Constantine a forgery. Known here from the reference works, not from the book itself.

Denise Schmandt-Besserat. Tokens and Writing: the Cognitive Development. Author's page, University of Texas at Austin. Web page at sites.utexas.edu/dsb/tokens. Hollow clay balls holding tokens, with the tokens pressed into the outside, around 3300 BC.

Deuteronomy 19:15. NRSV text at biblegateway.com. Nobody is convicted on one witness; it takes two or three.

Cynthia Dwork and Moni Naor (1992). Pricing via Processing or Combatting Junk Mail. Advances in Cryptology, CRYPTO '92, Lecture Notes in Computer Science 740 (Springer, 1993), pp. 139–147. DOI: 10.1007/3-540-48071-4_10. Proof of work, proposed to make junk mail expensive.

Robert W. Floyd (1967). Assigning meanings to programs. In J. T. Schwartz (ed.), Mathematical Aspects of Computer Science, Proceedings of Symposia in Applied Mathematics 19, American Mathematical Society, pp. 19–32. DOI: 10.1090/psapm/019/0235771. A program stops because no infinite decreasing sequence exists in a well-ordered set.

Forrester (n.d.). Books as Evidence, before and after Pacioli (1494 AD). Paper from the University of Strathclyde accounting history pages; the page gives no initials or date. Mirror of the paper. Reads Pacioli as reporting that books were taken to a consul to be authenticated before use. Pacioli's own text is seen here only through this paper.

Dean P. Foster and Robert A. Stine (2008). α-investing: a procedure for sequential control of expected false discoveries. Journal of the Royal Statistical Society, Series B 70(2), pp. 429–444. DOI: 10.1111/j.1467-9868.2007.00643.x. A budget of false alarms that every test spends and every discovery earns back.

Fox-IT (2012). Black Tulip: Report of the investigation into the DigiNotar Certificate Authority breach. Delft, 13 August 2012, for the Dutch Ministry of the Interior and Kingdom Relations. PDF at zoek.officielebekendmakingen.nl, blg-191225. Intruders took over the servers that signed certificates, and the company went bankrupt.

Stuart Haber and W. Scott Stornetta (1991). How to Time-Stamp a Digital Document. Journal of Cryptology 3(2), pp. 99–111. DOI: 10.1007/BF00196791. Records chained so that changing an old one breaks every one after it.

C. A. R. Hoare (1969). An Axiomatic Basis for Computer Programming. Communications of the ACM 12(10), pp. 576–580. DOI: 10.1145/363235.363259.

Horace (c. 20 BC). Epistles, book 1, epistle 1, line 14. Latin text at thelatinlibrary.com, epist1. “Nullius addictus iurare in verba magistri,” the line the Royal Society cut its motto from.

Gaius. Institutes, book 2, sections 104 and 119 (second century). Text and translation at droitromain.univ-grenoble-alpes.fr, gai2_Poste. The will by bronze and balance before five witnesses (104), and a will sealed by seven witnesses (119).

U.S. Department of Justice (2002). Statement of Deputy Attorney General Larry Thompson on the Arthur Andersen verdict, 15 June 2002. Press release at justice.gov, 02_dag_356. Andersen convicted of obstruction of justice over the Enron documents.

U.S. Department of Justice (2002). Press release, October 2002, on the criminal complaint against Andrew Fastow. Press release at justice.gov, 02_crm_568. Enron’s special purpose entities, set up to keep debt off the balance sheet.

Geoffrey A. Lee (1977). The Coming of Age of Double Entry: The Giovanni Farolfi Ledger of 1299–1300. Accounting Historians Journal 4(2), pp. 79–96. DOI: 10.2308/0148-4184.4.2.79. The Provence branch ledger of a Florentine company, the earliest complete double entry found.

Henry Sumner Maine (1861). Ancient Law. London: John Murray, chapter VI. Project Gutenberg 22910. The seven seals of the Roman will as the seven people of the older mock sale.

Alvaro Martinelli (1983). The Ledger of Cristianus Lomellinus and Dominicus De Garibaldo, Stewards of the City of Genoa (1340–41). Abacus 19(2), pp. 83–118. DOI: 10.1111/j.1467-6281.1983.tb00242.x. The Genoa treasurers’ ledger for 1340, kept in double entry.

Ralph C. Merkle (1979). Secrecy, Authentication, and Public Key Systems. PhD thesis, Stanford University. Thesis at ralphmerkle.com, Thesis1979. Where tree authentication, now called the Merkle tree, is introduced.

Satoshi Nakamoto (2008). Bitcoin: A Peer-to-Peer Electronic Cash System. bitcoin.org/bitcoin.pdf. Three of its eight references are to Haber and Stornetta.

Luca Pacioli (1494). Summa de arithmetica, geometria, proportioni et proportionalità. Venice. The bookkeeping treatise is the part titled Particularis de computis et scripturis.

Peter E. Prosser (2001). Church History's Biggest Hoax. Christian History 72. Article at christianhistoryinstitute.org. The Donation most likely written in the eighth century, Cusa's exposure in 1433, and Valla's in 1440.

Denise Schmandt-Besserat (1992). Before Writing, volume 1: From Counting to Cuneiform. Austin: University of Texas Press.

Securities and Exchange Commission (1997). Reporting Requirements for Brokers or Dealers Under the Securities Exchange Act of 1934. Release No. 34-38245, Federal Register 62, pp. 6469–6474, 12 February 1997. Federal Register text at govinfo.gov, 97-3426. Broker-dealer records kept electronically had to be on non-rewriteable, non-erasable media.

Securities and Exchange Commission (2022). Electronic Recordkeeping Requirements for Broker-Dealers, Security-Based Swap Dealers, and Major Security-Based Swap Participants. Release No. 34-96034. Final rule at sec.gov, 34-96034. Adds an audit-trail alternative to write once, read many.

Steven Shapin (1984). Pump and Circumstance: Robert Boyle's Literary Technology. Social Studies of Science 14(4), pp. 481–520. DOI: 10.1177/030631284014004001. Virtual witnessing, and the air pump as seventeenth-century big science.

Supreme Court of the United States (2005). Arthur Andersen LLP v. United States, 544 U.S. 696. The conviction reversed over the jury instructions.

The Royal Society. History of the Royal Society. Web page at royalsociety.org, history. First meeting 28 November 1660, and the motto Nullius in verba.

Lorenzo Valla (c. 1440). De falso credita et ementita Constantini donatione declamatio. English translation: The Treatise of Lorenzo Valla on the Donation of Constantine, trans. Christopher B. Coleman, Yale University Press, 1922. Coleman translation at history.hanover.edu, vallaintro. The Donation taken apart by its Latin, "satraps" among the words no Roman of Constantine's day would have used.

Chapter 5. More settled

Lindsey Kuper and Ryan R. Newton (2013). LVars: lattice-based data structures for deterministic parallelism. Proceedings of the 2nd ACM SIGPLAN Workshop on Functional High-Performance Computing (FHPC 2013), pp. 71–84. DOI: 10.1145/2502323.2502326. Memory cells whose value only climbs, where a program can only ask whether it has passed a point.

Amin Timany and Lars Birkedal (2021). Reasoning about monotonicity in separation logic. Proceedings of the 10th ACM SIGPLAN International Conference on Certified

Programs and Proofs (CPP 2021), pp. 91–104. DOI: 10.1145/3437992.3439931. Any order carried in a program logic as a value that only climbs and that the program never reads.

Chapter 6. Looking isn't keeping

Charles H. Bennett (1982). The thermodynamics of computation: a review. International Journal of Theoretical Physics 21(12), pp. 905–940. DOI: 10.1007/BF02084158. The demon taken apart one step at a time, with the cost moved from the look to the wipe.

Léon Brillouin (1951). Maxwell's Demon Cannot Operate: Information and Entropy. I. Journal of Applied Physics 22(3), pp. 334–337. DOI: 10.1063/1.1699951. To see a molecule the demon has to bounce light off it, and that costs entropy.

Jeffrey Bub (2001). Maxwell's Demon and the Thermodynamics of Computation. Studies in History and Philosophy of Modern Physics 32(4), pp. 569–579. DOI: 10.1016/S1355-2198(01)00023-5. The summary of Szilard and Brillouin: finding out costs at least $k \log 2$ of entropy for every bit.

Arthur S. Eddington (1928). The Nature of the Physical World. Cambridge University Press. Chapter IV, The Running-Down of the Universe, where a theory against the second law has nothing for it but to "collapse in deepest humiliation."

Paul N. Fahn (1996). Maxwell's demon and the entropy cost of information. Foundations of Physics 26(1), pp. 71–93. DOI: 10.1007/BF02058888. Whether the look or the wipe gets charged is partly a choice of bookkeeping, and the total is what's fixed.

Aurélien Garivier, Tor Lattimore and Emilie Kaufmann (2016). On explore-then-commit strategies. Advances in Neural Information Processing Systems 29 (NeurIPS 2016), pp. 784–792. arXiv: 1605.08988. Over a long run, the best strategies that commit lose about twice as much as the best ones that keep adjusting.

Sebastian Goldt and Udo Seifert (2017). Stochastic thermodynamics of learning. Physical Review Letters 118(1), 010601. DOI: 10.1103/PhysRevLett.118.010601. The entropy a learning network produces limits how much it can learn.

Edwin T. Jaynes (1965). Gibbs vs Boltzmann entropies. American Journal of Physics 33(5), pp. 391–398. DOI: 10.1119/1.1971557. Entropy as "an anthropomorphic concept," a phrase he credits to Wigner (p. 398).

Edwin T. Jaynes (1989). Clearing up Mysteries: The Original Goal. In John Skilling (ed.), Maximum Entropy and Bayesian Methods, Kluwer, pp. 1–27. DOI: 10.1007/978-94-015-7860-8_1. Where the mind projection fallacy gets its name.

Ruth E. Kastner (2025). Maxwell's Demon Is Foiled by the Entropy Cost of Measurement, Not Erasure. Foundations 5(2), article 16. DOI: 10.3390/foundations5020016; arXiv: 2503.18186. Brillouin was essentially right, and it's the measuring that pays.

James Clerk Maxwell (1865). A Dynamical Theory of the Electromagnetic Field. Philosophical Transactions of the Royal Society of London 155, pp. 459–512. DOI: 10.1098/rstl.1865.0008.

James Clerk Maxwell (1867, 1871). Letter to Peter Guthrie Tait, 11 December 1867, printed in Cargill Gilston Knott, Life and Scientific Work of Peter Guthrie Tait, Cambridge University Press, 1911, pp. 213–214 ("very observant and neat-fingered being," p. 214), with the undated note "Concerning Demons" on pp. 214–215 ("Call him no more a demon but a valve," p. 215); and Theory of Heat, Longmans, Green, London, 1871, the section on the limitation of the second law of thermodynamics.

John D. Norton (2013). The end of the thermodynamics of computation: a no-go result. Philosophy of Science 80(5), pp. 1182–1192. DOI: 10.1086/673714. At the scale of molecules, thermal jiggling keeps a supposedly reversible process from finishing, and finishing it costs more than Landauer's amount.

Thomas E. Ouldridge and Pieter Rein ten Wolde (2017). Fundamental Costs in the Production and Destruction of Persistent Polymer Copies. Physical Review Letters 118(15), 158103. DOI: 10.1103/PhysRevLett.118.158103.

Thomas E. Ouldridge, Christopher C. Govern and Pieter Rein ten Wolde (2017). Thermodynamics of Computational Copying in Biochemical Systems. Physical Review X 7(2), 021004. DOI: 10.1103/PhysRevX.7.021004. The same kind of accounting for how a cell reads out its own sensors.

Jenny M. Poulton, Pieter Rein ten Wolde and Thomas E. Ouldridge (2019). Nonequilibrium correlations in minimal dynamical models of polymer copying. Proceedings of the National Academy of Sciences 116(6), pp. 1946–1951. DOI: 10.1073/pnas.1808775116; arXiv: 1805.08502. When a copy grows with the weakest push, its advantage over random copying and the efficiency of storing it both vanish.

Hans Reichenbach (1956). The Direction of Time. University of California Press, Berkeley. Rovelli credits him as the first to notice that any measurement dissipates.

Richard Rhodes (1986). The Making of the Atomic Bomb. Simon and Schuster, New York. Chapter 1, "Moonshine," opens with Szilard's own later account of the traffic light on Southampton Row and Rutherford's dismissal in the newspaper.

Lídia del Rio, Johan Åberg, Renato Renner, Oscar Dahlsten and Vlatko Vedral (2011). The thermodynamic meaning of negative entropy. Nature 474(7349), pp. 61–63.

DOI: 10.1038/nature10123. The cost of erasing depends on what the eraser already knows, down to zero, and below zero in the quantum case.

Carlo Rovelli (2024). The Thermodynamic Cost of Choosing. Foundations of Physics 54(3), article 28. DOI: 10.1007/s10701-024-00765-7; arXiv: 2309.08557. Quoted from the arXiv preprint: "stabilization must be irreversible, otherwise it is not stabilization," with the thermostat set aside as predictable and "Wahl ist Leiden" near the end.

Takahiro Sagawa and Masahito Ueda (2009). Minimal Energy Cost for Thermodynamic Information Processing: Measurement and Information Erasure. Physical Review Letters 102(25), 250602. DOI: 10.1103/PhysRevLett.102.250602; arXiv: 0809.4098v3. A bound on the total cost of measuring and erasing, with the split depending on the memory; see also the erratum, Physical Review Letters 106, 189901 (2011).

American Physical Society (2006). May, 1911: Rutherford and the Discovery of the Atomic Nucleus. APS News 15(5), This Month in Physics History. APS News, May 2006, history column.

Leo Szilard (1929). Über die Entropieverminderung in einem thermodynamischen System bei Eingriffen intelligenter Wesen. Zeitschrift für Physik 53(11–12), pp. 840–856. DOI: 10.1007/BF01341281. The one-molecule engine.

William Thomson (1874). Kinetic Theory of the Dissipation of Energy. Nature 9(232), pp. 441–444. DOI: 10.1038/009441c0. Where Maxwell's being first gets called a demon.

Shoichi Toyabe, Takahiro Sagawa, Masahito Ueda, Eiro Muneyuki and Masaki Sano (2010). Experimental demonstration of information-to-energy conversion and validation of the generalized Jarzynski equality. Nature Physics 6(12), pp. 988–992. DOI: 10.1038/nphys1821. A particle walked up a staircase by feedback on what a camera saw.

Chapter 7. Looking through a window

Edwin A. Abbott (1884). Flatland: A Romance of Many Dimensions, by A Square. Seeley and Co., London.

MacTutor History of Mathematics. Edwin Abbott Abbott (J. J. O'Connor and E. F. Robertson). University of St Andrews. MacTutor biography, Abbott. His mother was his father's first cousin, he was headmaster of the City of London School from 1865 to 1889, and his Times obituary didn't mention Flatland.

Robert J. Aumann (1976). Agreeing to disagree. The Annals of Statistics 4(6), pp. 1236–1239. DOI: 10.1214/aos/1176343654. Knowledge modeled by piles of states an agent can't tell apart.

François Bancilhon and Nicolas Spyratos (1981). Update semantics of relational views. ACM Transactions on Database Systems 6(4), pp. 557–575. DOI: 10.1145/319628.319634.

Mihir Bellare and Bennet S. Yee (1997). Forward integrity for secure audit logs. Technical report, Department of Computer Science and Engineering, University of California, San Diego. Where forward integrity gets its name.

Laurent Bindschaedler, Quentin Botha and Christoph Siebenbrunner (2026). Agent Flight Recorder: Tamper-Evident Audit Trails with On-Chain Anchoring for Long-Horizon Tool-Using Agents. Preprint. arXiv: 2609.01931.

Manuel Blum (1983). Coin Flipping by Telephone: A Protocol for Solving Impossible Problems. ACM SIGACT News 15(1), pp. 23–27. DOI: 10.1145/1008908.1008911. Presented at CRYPTO 1981; one side commits first and opens after the other calls, with the binding resting on how hard it is to factor.

Jesse Bull and Joel Watson (2007). Hard evidence and mechanism design. Games and Economic Behavior 58(1), pp. 75–93. DOI: 10.1016/j.geb.2006.03.003.

Tyler Burge (1993). Content preservation. The Philosophical Review 102(4), pp. 457–488. DOI: 10.2307/2185680. The acceptance principle: you're entitled to accept what you're told unless you have a reason not to.

Michael Burrows, Martín Abadi and Roger Needham (1990). A logic of authentication. ACM Transactions on Computer Systems 8(1), pp. 18–36. DOI: 10.1145/77648.77649. If I believe an authority has the say over a statement and believe it holds the statement, I believe the statement.

Alessandro Chiesa and Eran Tromer (2010). Proof-carrying data and hearsay arguments from signature cards. Proceedings of Innovations in Computer Science (ICS 2010), Tsinghua University Press, Beijing, pp. 310–331. Proceedings page at conference.iiis.tsinghua.edu.cn, 25. Proofs passed along a chain of parties who don't trust each other.

Robin Cockett, Isabelle Jianing Geng, Carlo Maria Scandolo and Priyaa Varshinee Srinivasan (2023). Extending resource monotones using Kan extensions. Proceedings of the Fifth International Conference on Applied Category Theory (ACT 2022), Electronic Proceedings in Theoretical Computer Science 380, pp. 203–223. DOI: 10.4204/EPTCS.380.12. "Some member" and "every member" used to stretch measures of resources along a map.

Edgar F. Codd (1970). A Relational Model of Data for Large Shared Data Banks. Communications of the ACM 13(6), pp. 377–387. DOI: 10.1145/362384.362685.

Patrick Cousot and Radhia Cousot (1977). Abstract interpretation: a unified lattice model for static analysis of programs by construction or approximation of fixpoints. Proceedings of the 4th ACM Symposium on Principles of Programming Languages (POPL '77), pp. 238–252. DOI: 10.1145/512950.512973.

Claude Crépeau, Rafael Dowsley and Anderson C. A. Nascimento (2020). On the commitment capacity of unfair noisy channels. IEEE Transactions on Information Theory 66(6), pp. 3745–3752. DOI: 10.1109/TIT.2020.2967048; arXiv: 1905.10921. How the rate drops when the one committing can steer the noise.

Kenny Easwaran (2009). Probabilistic proofs and transferability. Philosophia Mathematica (III) 17(3), pp. 341–362. DOI: 10.1093/philmat/nkn032. A step-by-step proof carries its reason across to the reader, and a proof based on chance doesn't.

Ronald Fagin, Joseph Y. Halpern, Yoram Moses and Moshe Y. Vardi (1995). Reasoning About Knowledge. MIT Press, Cambridge, Massachusetts.

Joseph A. Goguen and José Meseguer (1982). Security policies and security models. Proceedings of the 1982 IEEE Symposium on Security and Privacy, pp. 11–20. DOI: 10.1109/SP.1982.10014. Noninterference.

Sanford C. Goldberg (2010). Relying on Others: An Essay in Epistemology. Oxford University Press. The weight put on the process behind the speaker's word.

Shafi Goldwasser, Silvio Micali and Charles Rackoff (1985). The Knowledge Complexity of Interactive Proof-Systems. Proceedings of the 17th Annual ACM Symposium on Theory of Computing (STOC '85), pp. 291–304. DOI: 10.1145/22145.22178.

Jerry R. Green and Jean-Jacques Laffont (1986). Partially verifiable information and mechanism design. Review of Economic Studies 53(3), pp. 447–456. DOI: 10.2307/2297639.

Andreas Haeberlen, Petr Kouznetsov and Peter Druschel (2007). PeerReview: practical accountability for distributed systems. Proceedings of the 21st ACM Symposium on Operating Systems Principles (SOSP '07), pp. 175–188. DOI: 10.1145/1294261.1294279. Every node keeps a secure log, and a checker replays it against a copy of the rules.

Andreas Haeberlen and Petr Kuznetsov (2009). The fault detection problem. Principles of Distributed Systems, OPODIS 2009, Lecture Notes in Computer Science 5923, Springer, pp. 99–114. DOI: 10.1007/978-3-642-10877-8_10. Which kinds of cheating an observer holding only the messages can catch at all.

Joseph Y. Halpern, Yoram Moses and Mark R. Tuttle (1988). A knowledge-based analysis of zero knowledge. Proceedings of the 20th Annual ACM Symposium on Theory of Computing (STOC '88), pp. 132–147. DOI: 10.1145/62212.62224.

Wesley H. Holliday (2009). Dynamic testimonial logic. In Xiangdong He, John Horty and Eric Pacuit (eds.), Logic, Rationality, and Interaction, LORI 2009, Lecture Notes in Computer Science 5834, Springer, pp. 161–179. DOI: 10.1007/978-3-642-04893-7_13. Trust put into a logic of changing your mind, with a graph of whose word each person takes. Known here from a summary.

Radha Jagadeesan, Alan Jeffrey, Corin Pitcher and James Riely (2009). Towards a theory of accountability and audit. Computer Security, ESORICS 2009, Lecture Notes in Computer Science 5789, Springer, pp. 152–167. DOI: 10.1007/978-3-642-04444-1_10. Systems that enforce their rules after the fact, by audit.

Christoph Jäger and Nicholas Shackel (2025). Testimonial authority and knowledge transmission. Social Epistemology 39(6), pp. 636–651. DOI: 10.1080/02691728.2025.2449608. A model where the speaker knowing is neither needed nor enough for the hearer to know.

Rudolf E. Kalman (1960). A New Approach to Linear Filtering and Prediction Problems. Journal of Basic Engineering 82(1), pp. 35–45. DOI: 10.1115/1.3662552. The filter.

Rudolf E. Kalman (1960). On the general theory of control systems. Proceedings of the First IFAC World Congress, Moscow; IFAC Proceedings Volumes 1(1), pp. 491–502. DOI: 10.1016/S1474-6670(17)70094-8. Observability, with a test you can compute.

Oliver Kosut and Jörg Kliewer (2018). Authentication capacity of adversarial channels. 2018 IEEE Information Theory Workshop (ITW), Guangzhou, pp. 1–5. DOI: 10.1109/ITW.2018.8613353. The capacity is either zero or the full capacity without an attacker, depending on whether the attacker can make what's received look like an honest message.

Robert Künnemann, Ilkan Esiyok and Michael Backes (2019). Automated verification of accountability in security protocols. 2019 IEEE 32nd Computer Security Foundations Symposium (CSF). DOI: 10.1109/CSF.2019.00034; arXiv: 1805.10891. Accountability checked automatically by making it a property of the whole run.

Ralf Küsters, Tomasz Truderung and Andreas Vogt (2010). Accountability: definition and relationship to verifiability. Proceedings of the 17th ACM Conference on Computer and Communications Security (CCS '10), pp. 526–535. DOI: 10.1145/1866307.1866366. A verdict that names who misbehaved, with being able to check a result as a weaker form of it.

Jennifer Lackey (2008). Learning from Words: Testimony as a Source of Knowledge. Oxford University Press. The teacher who doesn't believe the theory she teaches, and students who come away knowing it.

F. William Lawvere (1969). Adjointness in foundations. Dialectica 23(3–4), pp. 281–296. DOI: 10.1111/j.1746-8361.1969.tb01194.x. The two quantifiers as the two adjoints of substitution.

William F. Lindgren and Thomas F. Banchoff (2009). Flatland: An Edition with Notes and Commentary, by Edwin A. Abbott. Cambridge University Press and Mathematical Association of America. DOI: 10.1017/CBO9781139194921. Calls the pen name a pun on Abbott's initials, EAA.

John Locke (1690). An Essay Concerning Human Understanding, Book I, chapter iv, section 23. Modernized text of Book I. Believing something on authority isn't knowing it.

Leonard A. McGee and Stanley F. Schmidt (1985). Discovery of the Kalman Filter as a Practical Tool for Aerospace and Industry. NASA Technical Memorandum 86847, Ames Research Center. NTRS at ntrs.nasa.gov, 19860003843. Kalman's visit to Ames in the fall of 1960 and the filter's road into navigation for Apollo.

Edward F. Moore (1956). Gedanken-Experiments on Sequential Machines. In Claude E. Shannon and John McCarthy (eds.), Automata Studies, Princeton University Press, pp. 129–153. Original paper at cs.cmu.edu, Moore1956-gedanken-experiments.

David L. Parnas (1972). On the criteria to be used in decomposing systems into modules. Communications of the ACM 15(12), pp. 1053–1058. DOI: 10.1145/361598.361623. A module as "a responsibility assignment" on p. 1054, and the decomposition that could be "much less efficient" on p. 1057.

Bryan Parno, Jon Howell, Craig Gentry and Mariana Raykova (2013). Pinocchio: nearly practical verifiable computation. Proceedings of the 2013 IEEE Symposium on Security and Privacy, pp. 238–252. DOI: 10.1109/SP.2013.47.

Jean-Jacques Quisquater, Myriam Quisquater, Muriel Quisquater, Michaël Quisquater, Louis Guillou, Marie Annick Guillou, Gaïd Guillou, Anna Guillou, Gwenolé Guillou, Soazig Guillou and Thomas A. Berson (1990). How to Explain Zero-Knowledge Protocols to Your Children. Advances in Cryptology, CRYPTO '89, Lecture Notes in Computer Science 435, Springer, pp. 628–631. DOI: 10.1007/0-387-34805-0_60. Ali Baba's cave, told for children.

Plato. Republic, Book VII, 514a–517c, translated by Benjamin Jowett. MIT Internet Classics Archive. The cave; the prisoners' honors for whoever is quickest at the passing shadows and best at guessing what comes next (516c–d); and Socrates' own reading of the cave as the world of sight (517b).

Henry Gordon Rice (1953). Classes of Recursively Enumerable Sets and Their Decision Problems. Transactions of the American Mathematical Society 74(2), pp. 358–366. DOI: 10.1090/S0002-9947-1953-0053041-6.

Rudy Rucker (1984). The Fourth Dimension: A Guided Tour of the Higher Universes. Houghton Mifflin, Boston. The earliest guess found that “A Square” was a nickname for A squared.

Anooshiravan Saboori and Christoforos N. Hadjicostis (2007). Notions of security and opacity in discrete event systems. Proceedings of the 46th IEEE Conference on Decision and Control, pp. 5056–5061. DOI: 10.1109/CDC.2007.4434515. When an observer can never be sure the system is in a secret state.

Meera Sampath, Raja Sengupta, Stéphane Lafortune, Kasim Sinnamohideen and Demosthenis Teneketzis (1995). Diagnosability of discrete-event systems. IEEE Transactions on Automatic Control 40(9), pp. 1555–1575. DOI: 10.1109/9.412626. When an observer can always work out from what it sees that a hidden fault has happened.

Bruce Schneier and John Kelsey (1999). Secure audit logs to support computer forensics. ACM Transactions on Information and System Security 2(2), pp. 159–176. DOI: 10.1145/317087.317089. A log an intruder who takes over the machine can’t quietly rewrite, protecting what was written before the break-in and nothing after.

Ian Stewart (2002). The Annotated Flatland: A Romance of Many Dimensions. Perseus Publishing. Abbott’s text with Stewart’s introduction and notes.

Anvith Thudi, Hengrui Jia, Ilia Shumailov and Nicolas Papernot (2022). On the Necessity of Auditable Algorithmic Definitions for Machine Unlearning. Proceedings of the 31st USENIX Security Symposium. arXiv: 2110.11891. The same finished model can come out of different training sets, so only the procedure that was run can be checked.

Timo Treitz and Robert Künnemann (2026). Accountability in Certificate Transparency and variants. Preprint. arXiv: 2609.11552. Plain Certificate Transparency names a culprit only if the log itself is honest.

Andreas Winter, Anderson C. A. Nascimento and Hideki Imai (2003). Commitment capacity of discrete memoryless channels. Cryptography and Coding 2003, Lecture Notes in Computer Science 2898, Springer, pp. 35–51. DOI: 10.1007/978-3-540-40974-8_4; arXiv: cs/0304014. How fast a noisy channel can carry commitments.

Chapter 8. Established

Federal Aviation Administration (2026). 25-Hour Cockpit Voice Recorder (CVR) Requirement, New Aircraft Production. Final rule, Federal Register 91, p. 4447, 2 February 2026; docket FAA-2023-2270. Federal Register text at govinfo.gov, 2026-02110. Raises the required recording time from two hours to twenty-five for newly manufactured aircraft.

Arthur Azevedo de Amorim, Maxime Dénès, Nick Giannarakis, Cătălin Hriţcu, Benjamin C. Pierce, Antal Spector-Zabusky and Andrew Tolmach (2015). Micro-policies: formally verified, tag-based security monitors. Proceedings of the 2015 IEEE Symposium on Security and Privacy, pp. 813–830. DOI: 10.1109/SP.2015.55. Monitors checked at every instruction, with their guarantees proved.

James P. Anderson (1972). Computer Security Technology Planning Study, Volume II. Technical Report ESD-TR-73-51, Electronic Systems Division, Air Force Systems Command, Hanscom Field, Bedford, MA. PDF at csrc.nist.gov, early computer security papers, ande72. The reference validation mechanism “must be tamper proof,” “must always be invoked,” and “must be small enough to be subject to analysis and tests.”

Luis Antunes, Lance Fortnow, Dieter van Melkebeek and N. V. Vinodchandran (2006). Computational depth: Concept and applications. Theoretical Computer Science 354(3), pp. 391–404. DOI: 10.1016/j.tcs.2005.11.033. Its epigraph is Levin’s line: “Torturing an uninformed witness cannot give information about the crime.”

Sergei N. Artemov (1995). Operational Modal Logic. Technical Report MSI 95-29, Cornell University. The logic of proofs, where knowing a thing comes with an explicit reason for it; surveyed in the Stanford Encyclopedia of Philosophy entry on justification logic, Stanford Encyclopedia of Philosophy, entry logic-justification.

Sergei Artemov and Roman Kuznets (2014). Logical omniscience as infeasibility. Annals of Pure and Applied Logic 165(1), pp. 6–25. DOI: 10.1016/j.apal.2013.07.003. Knowing that counts the steps it took.

J. L. Austin (1962). How to Do Things with Words. Clarendon Press, Oxford. The William James Lectures, Harvard, 1955, where saying “I do” makes a marriage rather than describing one.

Francisco Bavera and Eduardo Bonelli (2018). Justification logic and audited computation. Journal of Logic and Computation 28(5), pp. 909–934; first published online in 2015. DOI: 10.1093/logcom/exv037. A programming language whose terms carry the history of the computation that made them.

Matt Bishop and Michael Dilger (1996). Checking for race conditions in file accesses. Computing Systems 9(2), pp. 131–152. PDF at usenix.org, Computing Systems, spring 1996. Hunting for time-of-check to time-of-use bugs.

Samuel Bucheli, Meghdad Ghari and Thomas Studer (2017). Temporal justification logic. Methods for Modalities (M4M9), Electronic Proceedings in Theoretical Computer Science 243, pp. 59–74. DOI: 10.4204/EPTCS.243.5; arXiv: 1703.02195.

Eric Budish (2025). Trust at Scale: The Economic Limits of Cryptocurrencies and Blockchains. The Quarterly Journal of Economics 140(1), pp. 1–62. DOI: 10.1093/qje/qjae033. For Bitcoin's kind of trust to hold, honest work has to stay expensive at every moment compared with what an attacker could win.

Gregory J. Chaitin (1966). On the Length of Programs for Computing Finite Binary Sequences. Journal of the ACM 13(4), pp. 547–569. DOI: 10.1145/321356.321363.

Ho Ngoc Duc (1997). Reasoning about rational, but not logically omniscient, agents. Journal of Logic and Computation 7(5), pp. 633–648. DOI: 10.1093/logcom/7.5.633. Knowing as what an agent could come to know with more steps of reasoning.

Edmund L. Gettier (1963). Is Justified True Belief Knowledge? Analysis 23(6), pp. 121–123. DOI: 10.1093/analys/23.6.121. Smith, Jones and the ten coins.

Alvin I. Goldman (1976). Discrimination and Perceptual Knowledge. The Journal of Philosophy 73(20), pp. 771–791. DOI: 10.2307/2025679. The fake-barn case, credited there to Carl Ginet.

Michael J. C. Gordon, Robin Milner and Christopher P. Wadsworth (1979). Edinburgh LCF: A Mechanised Logic of Computation. Lecture Notes in Computer Science 78, Springer. DOI: 10.1007/3-540-09724-4.

Michael J. C. Gordon (2000). From LCF to HOL: a short history. In Gordon Plotkin, Colin Stirling and Mads Tofte (eds.), Proof, Language, and Interaction: Essays in Honour of Robin Milner, MIT Press, pp. 169–185. PDF at cl.cam.ac.uk, HolHistory. Milner keeping only the theorems, "like a mathematics lecturer using a small blackboard," with an abstract type so that theorems can only come from proofs, and ML named for "Meta Language."

Jim Gray (1981). The Transaction Concept: Virtues and Limitations. Proceedings of the 7th International Conference on Very Large Data Bases, pp. 144–154. PDF at jimgray.azurewebsites.net, thetransactionconcept. The commit, introduced through contract law and a wedding.

Trusted Computing Group (n.d.). Trusted Platform Module Library Specification, Family 2.0, Part 1: Architecture. Specification page at trustedcomputinggroup.org,

tpm-library-specification. The registers and the extend operation, the bit-field slots that can only set bits, and the counters that only go up.

Jaakko Hintikka (1962). Knowledge and Belief: An Introduction to the Logic of the Two Notions. Cornell University Press. Logical omniscience, surveyed in the Stanford Encyclopedia of Philosophy entry on epistemic logic, Stanford Encyclopedia of Philosophy, entry logic-epistemic.

Andrei N. Kolmogorov (1965). Three approaches to the quantitative definition of information. Problems of Information Transmission 1(1), pp. 1–7; reprinted in International Journal of Computer Mathematics 2 (1968), pp. 157–168. DOI of the reprint: 10.1080/00207166808803030.

H. T. Kung and John T. Robinson (1981). On optimistic methods for concurrency control. ACM Transactions on Database Systems 6(2), pp. 213–226. DOI: 10.1145/319566.319567. Checking, just before a commit, that nothing read has been written since.

Leonid A. Levin (1984). Randomness Conservation Inequalities; Information and Independence in Mathematical Theories. Information and Control 61(1), pp. 15–37. DOI: 10.1016/S0019-9958(84)80060-1. Running a program can't add much information beyond what went in plus the program.

Jacques-Louis Lions (chair) et al. (1996). ARIANE 5 Flight 501 Failure: Report by the Inquiry Board. Paris, 19 July 1996. Copy of the report. The assumption that made the conversion safe was agreed on and then "essentially obscured, though not deliberately, from any external review."

Moni Naor and Guy N. Rothblum (2005). The complexity of online memory checking. Proceedings of the 46th IEEE Symposium on Foundations of Computer Science (FOCS 2005); journal version in Journal of the ACM 56(1), 2009. DOI of the journal version: 10.1145/1462153.1462155; IACR ePrint 2006/091. Against an unbounded opponent, the size of the client's private fingerprint times the bits it reads has to be on the order of the whole file, and beating that needs one-way functions.

National Museum of Australia. Defining Moments: Black box flight recorder invented. Museum page at nma.gov.au, black-box-invented. David Warren's flight memory recorder at the Aeronautical Research Laboratories in Melbourne.

Kerry Noble (2013). Obituary: Dr David Ronald De Mey Warren (1925–2010). Imperial College London News, 14 June 2013. Obituary at imperial.ac.uk, news item 122905. Warren at the Aeronautical Research Laboratories, the unsolved Comet crashes, and the 1956 prototype that held four hours of voice and instrument data.

Robert Nozick (1981). Philosophical Explanations. Harvard University Press. Chapter 3: you know a thing only if, had it been false, you wouldn't believe it.

Daniel Oberhaus (2018). The World's Oldest Blockchain Has Been Hiding in the New York Times Since 1995. Vice (Motherboard), 27 August 2018. Article at vice.com, what was the first blockchain. Surety's weekly hash under "Notices & Lost and Found."

Plato. Meno 97d–98a, translated by W. R. M. Lamb, Plato in Twelve Volumes, vol. 3, Harvard University Press, 1967. Perseus Digital Library text, Plat. Meno 97d. Daedalus's statues, and true opinions tied down by reasoning.

Fred B. Schneider (2000). Enforceable security policies. ACM Transactions on Information and System Security 3(1), pp. 30–50. DOI: 10.1145/353323.353382. A monitor that watches a run as it goes can enforce exactly the rules whose breaking shows up at a point you can name.

Claude E. Shannon (1948). A mathematical theory of communication. Bell System Technical Journal 27(3), pp. 379–423, and 27(4), pp. 623–656. DOI: 10.1002/j.1538-7305.1948.tb01338.x; DOI: 10.1002/j.1538-7305.1948.tb00917.x. Structure "allows one to make a saving in time (or channel capacity)," p. 385; half of English "determined by the structure of the language," p. 399.

Ray J. Solomonoff (1964). A Formal Theory of Inductive Inference, Parts I and II. Information and Control 7(1), pp. 1–22, and 7(2), pp. 224–254. DOI of Part I: 10.1016/S0019-9958(64)90223-2; DOI of Part II: 10.1016/S0019-9958(64)90131-7.

Chapter 9. Settled

Hans Albert (1968). Traktat über kritische Vernunft. J. C. B. Mohr (Paul Siebeck), Tübingen. English translation: Treatise on Critical Reason, Princeton University Press, 1985. The Münchhausen trilemma.

Kenneth Appel and Wolfgang Haken (1976). Every planar map is four colorable. Bulletin of the American Mathematical Society 82(5), pp. 711–712. DOI: 10.1090/S0002-9904-1976-14122-5. The announcement of the computer proof.

Mark van Atten (2018). The Creating Subject, the Brouwer–Kripke Schema, and infinite proofs. Indagationes Mathematicae 29(6), pp. 1565–1636. DOI: 10.1016/j.indag.2018.06.005; arXiv: 1805.00404. Read in the arXiv preprint: the never-forgetting axiom credited to Troelstra (1969, p. 95), the name Kreisel-Troelstra theory, the reading "by stage n the Creating Subject has made A evident," and the subject who can't simply accept a proof from outside.

L. E. J. Brouwer (1948). Essentieel-negatieve eigenschappen. Proceedings of the Koninklijke Nederlandse Akademie van Wetenschappen 51, pp. 963–964; also

Indagationes Mathematicae 10. English translation, Essentially negative properties, in Hans Freudenthal (ed.), L. E. J. Brouwer: Collected Works, vol. 2, North-Holland, 1976, pp. 478–479. Where the creating subject first appears in an argument.

Walter Dean and Hidenori Kurokawa (2021). On the methodology of informal rigour: set theory, semantics, and intuitionism. Preprint, written for Marianna Antonutti Marfori and Mattia Petrolo (eds.), Intuitionism, Computation, and Proof: Selected Themes from the Research of G. Kreisel, Springer. arXiv: 2104.14887. Its list of Kreisel's three creating-subject axioms is decidability, Christian charity and soundness, with no never-forgetting axiom.

FIDE (2023). FIDE Laws of Chess taking effect from 1 January 2023, Article 9.2.3. FIDE Handbook at handbook.fide.com, E012023. Two positions are the same only if the same player has the move, the same pieces stand on the same squares, and the possible moves of all the pieces are the same.

Carl Friedrich Gauss (1831). Untersuchungen über die Eigenschaften der positiven ternären quadratischen Formen von Ludwig August Seeber. Göttingische gelehrte Anzeigen, 9 July 1831; reprinted in Journal für die reine und angewandte Mathematik 20 (1840), pp. 312–320. DOI of the reprint: 10.1515/crll.1840.20.312. The densest packing among lattices, a special case of Kepler's claim.

Georges Gonthier (2008). Formal proof: the four-color theorem. Notices of the American Mathematical Society 55(11), pp. 1382–1393. The proof checked step by step by a small proof checker, completed in 2005.

Thomas C. Hales (2005). A proof of the Kepler conjecture. Annals of Mathematics 162(3), pp. 1065–1185. DOI: 10.4007/annals.2005.162.1065.

Thomas Hales and others (2017). A formal proof of the Kepler conjecture. Forum of Mathematics, Pi 5, e2. DOI: 10.1017/fmp.2017.1. The project announced in 2003 and completed on 10 August 2014.

Thomas Hales (2024). The Formal Proof of the Kepler Conjecture: a critical retrospective. arXiv: 2402.08032. The August 1998 announcement with Ferguson, the panel of twelve referees who "became exhausted and quit," and the formal proof begun to get past them (p. 2).

Percy John Heawood (1890). Map-colour theorem. Quarterly Journal of Pure and Applied Mathematics 24, pp. 332–338. The flaw in "the now apparently recognised proof," as quoted in secondary accounts; I haven't seen the original.

MacTutor History of Mathematics. Percy John Heawood (J. J. O'Connor and E. F. Robertson). University of St Andrews. MacTutor biography, Heawood. Lecturer at Durham from 1887, and, in G. A. Dirac's words, "often accompanied by a dog, which was admitted to his lectures."

David Hilbert (1900). Mathematische Probleme. Nachrichten von der Königlichen Gesellschaft der Wissenschaften zu Göttingen, pp. 253–297. English translation by Mary Winston Newson, Mathematical Problems, Bulletin of the American Mathematical Society 8 (1902), pp. 437–479, DOI: 10.1090/S0002-9904-1902-00923-3. The packing question sits in the eighteenth problem.

Canterbury Historical and Archaeological Society (n.d.). James Six (1730–1793). Web page at canterbury-archaeology.org.uk, six-james-biog. The family silk-weaving business and his turn to astronomy and meteorology.

Rosalie Iemhoff (2008, revised 2019). Intuitionism in the Philosophy of Mathematics. Stanford Encyclopedia of Philosophy. Stanford Encyclopedia of Philosophy, entry intuitionism. Lists "the Creating Subject never forgets what it has proven" among the three axioms Kreisel introduced in 1967.

Alfred Bray Kempe (1879). On the Geographical Problem of the Four Colours. American Journal of Mathematics 2(3), pp. 193–200. DOI: 10.2307/2369235. The proof that stood for eleven years; his account of finding it is quoted from p. 194.

MacTutor History of Mathematics. Alfred Bray Kempe (J. J. O'Connor and E. F. Robertson). University of St Andrews. MacTutor biography, Kempe. A barrister with "a very fine counter-tenor voice."

Johannes Kepler (1611). Strena seu de Nive Sexangula. Godefrid Tampach, Frankfurt am Main. English: The Six-Cornered Snowflake, translated by Colin Hardie, with essays by L. L. Whyte and B. J. Mason, Clarendon Press, Oxford, 1966. The New Year's gift to Wackher, and the claim about stacking balls; I haven't seen the Hardie translation.

Lewis A. Kornhauser and Lawrence G. Sager (1986). Unpacking the court. Yale Law Journal 96(1), beginning p. 82. DOI: 10.2307/796436. A court that votes by majority on each issue and on the verdict can contradict itself.

Georg Kreisel (1967). Informal rigour and completeness proofs. In Imre Lakatos (ed.), Problems in the Philosophy of Mathematics, Studies in Logic and the Foundations of Mathematics 47, North-Holland, Amsterdam, pp. 138–186. DOI: 10.1016/S0049-237X(08)71525-8. The creating subject's evidence for a statement at stage n. Kreisel's own page not seen; his three axioms are taken here from the Stanford Encyclopedia's entry and from Dean and Kurokawa, who differ on the never-forgetting rule.

Jeffrey C. Lagarias (ed.) (2011). The Kepler Conjecture: The Hales–Ferguson Proof. Springer, New York. DOI: 10.1007/978-1-4614-1129-1. Reprints the six papers by Thomas C. Hales and Samuel P. Ferguson that gave the full proof in Discrete & Computational Geometry 36(1), 2006.

Christian List and Philip Pettit (2002). Aggregating sets of judgments: an impossibility result. Economics and Philosophy 18(1), pp. 89–110. DOI: 10.1017/S0266267102001098.

Institute for Logic, Language and Computation (2019). Anne Troelstra (1939–2019). Obituary by Johan van Benthem and Dick de Jongh, University of Amsterdam. Obituary at illc.uva.nl, Anne Troelstra 1939–2019. His doctorate was supervised by Heyting.

MacTutor History of Mathematics. Arend Heyting. University of St Andrews. MacTutor biography, Heyting. Doctorate in 1925 under Brouwer.

Per Martin-Löf (1996). On the Meanings of the Logical Constants and the Justifications of the Logical Laws. Nordic Journal of Philosophical Logic 1(1), pp. 11–60. Lectures given in Siena, 6–9 April 1983. PDF at ncatlab.org, MartinLofOnTheMeaning96. “Thus a proof is, not an object, but an act,” and “Only secondarily, and irrevocably, does it become the act that has been performed.”

Bureau International des Poids et Mesures (2019). The International System of Units (SI), 9th edition. SI Brochure at bipm.org, si-brochure. The kilogram defined by fixing the Planck constant, adopted by the 26th CGPM in 2018 and in force from 20 May 2019.

Bureau International des Poids et Mesures (n.d.). Historical perspective: unit of mass, kilogram. BIPM page at bipm.org, kilogram. The national copies found drifting from the international prototype in 1946, confirmed by the verification of 1989 to 1991, with a median difference of about 25 micrograms.

Joint Committee for Guides in Metrology (2012). International vocabulary of metrology: Basic and general concepts and associated terms (VIM), 3rd edition. JCGM 200:2012. PDF at bipm.org, JCGM 200:2012. Definition 2.41, metrological traceability “through a documented unbroken chain of calibrations,” and its Note 5, that traceability “does not ensure . . . that there is an absence of mistakes.”

World Health Organization and UNICEF (2007). WHO and UNICEF urge donors and member states to fund and expand use of vaccine vial monitors to ensure effective immunization. Joint press release, Geneva, 3 May 2007. Release at gavi.org. The first monitors on oral polio vaccine in 1996, with a heat-sensitive inner square that darkens as the vial takes on heat.

Charles S. Peirce (1877). The Fixation of Belief. Popular Science Monthly 12, pp. 1–15. “As soon as a firm belief is reached we are entirely satisfied, whether the belief be true or false,” p. 6.

Karl R. Popper (1959). The Logic of Scientific Discovery. Hutchinson, London. Translation of Logik der Forschung (1934). Chapter X, corroboration: credit earned by surviving a test that could have failed.

Neil Robertson, Daniel Sanders, Paul Seymour and Robin Thomas (1997). The Four-Colour Theorem. Journal of Combinatorial Theory, Series B 70(1), pp. 2–44. DOI: 10.1006/jctb.1997.1750. The cleaner computer proof.

Sextus Empiricus. Outlines of Pyrrhonism I.164–177. The five modes credited to Agrippa.

James Six (1782). Account of an improved Thermometer. Philosophical Transactions of the Royal Society of London 72, pp. 72–81. DOI: 10.1098/rstl.1782.0012. “If it be not immediately noticed, it can never after be known,” p. 72.

William E. Story (1879). Note on the Preceding Paper. American Journal of Mathematics 2(3), from p. 201. DOI: 10.2307/2369236. “Substantially proved,” p. 201.

Göran Sundholm (2014). Constructive Recursive Functions, Church’s Thesis, and Brouwer’s Theory of the Creating Subject: Afterthoughts on a Parisian Joint Session. In Jacques Dubucs and Michel Bourdeau (eds.), Constructivity and Computability in Historical and Philosophical Perspective, Logic, Epistemology, and the Unity of Science 34, Springer, pp. 1–35. DOI: 10.1007/978-94-017-9217-2_1. Known here from its abstract and stated aims; the body leans on it lightly.

George G. Szpiro (2003). Does the proof stack up? Nature 424(6944), pp. 12–13. DOI: 10.1038/424012a. The Annals editor’s letter as Hales reported it, ending “run out of energy.”

MacTutor History of Mathematics. The four colour theorem (J. J. O’Connor and E. F. Robertson). University of St Andrews. MacTutor history topic, The four colour theorem. De Morgan’s letter to Hamilton of 23 October 1852, asking for “a reason for a fact which I did not know was a fact.”

László Fejes Tóth (1953). Lagerungen in der Ebene, auf der Kugel und im Raum. Springer, Berlin. The first detailed strategy for proving Kepler’s claim, and the first suggestion that a computer might finish it.

Anne S. Troelstra (1969). Principles of Intuitionism. Lecture Notes in Mathematics 95, Springer, Berlin. DOI: 10.1007/BFb0080643. The creating subject who never forgets, added on p. 95.

Robin Wilson (2002). Four Colours Suffice: How the Map Problem Was Solved. Allen Lane, London. Francis Guthrie and the map of the counties of England.

Chapter 10. What structure is

Steve Awodey (1996). Structure in mathematics and logic: a categorical perspective. Philosophia Mathematica 4(3), pp. 209–237. DOI: 10.1093/philmat/4.3.209.

Michael J. Barany (2020). Impersonation and personification in mid-twentieth century mathematics. History of Science 58(4), pp. 417–436. DOI: 10.1177/0073275320924571. Bourbaki's applications to the American Mathematical Society in 1948 and again in 1949, and the letters around them.

Paul Benacerraf (1965). What numbers could not be. The Philosophical Review 74(1), pp. 47–73. DOI: 10.2307/2183530. Two equally good ways of building the numbers out of sets that disagree about which set three is.

Antoine Bérut, Artak Arakelyan, Artyom Petrosyan, Sergio Ciliberto, Raoul Dillenschneider and Eric Lutz (2012). Experimental verification of Landauer's principle linking information and thermodynamics. Nature 483, pp. 187–189. DOI: 10.1038/nature10872. One bit erased in a single colloidal bead at the École Normale Supérieure in Lyon, the mean heat approaching Landauer's bound as the erasure is slowed.

János Bolyai (1832). Appendix scientiam spatii absolute veram exhibens. Printed as an appendix to the first volume of Farkas Bolyai's Tentamen, Maros-Vásárhely. The son's geometry without the parallel postulate.

Roberto Bonola (1912). Non-Euclidean Geometry: A Critical and Historical Study of Its Development. Translated by H. S. Carslaw. Open Court, Chicago. Internet Archive, noneuclideangeom00bonouoft. Two thousand years of attempts on the parallel postulate, from Proclus and the Arab geometers through Saccheri to Gauss, Bolyai and Lobachevsky.

George Boole (1847). The Mathematical Analysis of Logic, Being an Essay towards a Calculus of Deductive Reasoning. Macmillan, Barclay and Macmillan, Cambridge.

George Boole (1854). An Investigation of the Laws of Thought. Walton and Maberly, London.

Nicolas Bourbaki (1950). The Architecture of Mathematics. Translated by Arnold Dresden. American Mathematical Monthly 57(4), pp. 221–232. DOI: 10.1080/00029890.1950.11999523. Structures on sets of elements "whose nature has not been specified," and the three mother structures, algebraic, order and topological.

Nicolas Bourbaki (1968; reprinted 2004). Theory of Sets. Elements of Mathematics, Hermann, Paris, and Addison-Wesley; Springer reprint, Berlin, 2004. DOI:

10.1007/978-3-642-59309-3. Chapter IV, sections 2 and 3, on morphisms, derived structures and universal mappings.

Walter Burkert (1972). Lore and Science in Ancient Pythagoreanism. Translated by Edwin L. Minar Jr. Harvard University Press, Cambridge, Massachusetts. Read here through Huffman's entry on Pythagoreanism.

Leo Corry (1992). Nicolas Bourbaki and the concept of mathematical structure. Synthese 92(3), pp. 315–348. DOI: 10.1007/BF00414286. The promised chapter on categories that never appeared in the treatise.

Leo Corry (1996; 2nd edition 2004). Modern Algebra and the Rise of Mathematical Structures. Birkhäuser, Basel. DOI: 10.1007/978-3-0348-7917-0. Chapter 7, on Bourbaki's theory of structures; "plays no actual role in the presentation of theories within the treatise" is on p. 338.

Samuel Eilenberg and Saunders Mac Lane (1945). General Theory of Natural Equivalences. Transactions of the American Mathematical Society 58(2), pp. 231–294. DOI: 10.1090/S0002-9947-1945-0013131-6. Where category theory starts: categories, functors and natural transformations.

Euclid. Elements. Translated by T. L. Heath, The Thirteen Books of Euclid's Elements, Cambridge University Press, 1908. Perseus Digital Library text, Perseus:text:1999.01.0086. Definition 1, Postulate 1, the fifth postulate, Common Notions 1 and 4, and Proposition 4 of Book I, proved by laying one triangle on another.

Gottlob Frege (1884). Die Grundlagen der Arithmetik. Wilhelm Koebner, Breslau. Project Gutenberg 48312. Directions and parallel lines in section 64, and number through one-to-one correspondence in sections 68 to 73.

Gottlob Frege (1893). Grundgesetze der Arithmetik, vol. I. Hermann Pohle, Jena. Basic Law V, on value-ranges, in section 20.

Gottlob Frege (1903). Grundgesetze der Arithmetik, vol. II. Hermann Pohle, Jena. The appendix answering Russell, which opens on how unwelcome it is for a scientific writer to have a foundation shaken after the work is finished. MacTutor extract, Frege10.

Peter D. Grünwald (2005). A tutorial introduction to the minimum description length principle. In Peter D. Grünwald, In Jae Myung and Mark A. Pitt (eds.), Advances in Minimum Description Length: Theory and Applications, MIT Press, Cambridge, Massachusetts, pp. 3–80, as the chapters Introducing the Minimum Description Length Principle and Minimum Description Length Tutorial. arXiv: math/0406077. The length of a model depends on the code chosen for it, so the code needs a principle behind it.

Paul R. Halmos (1960). Naive Set Theory. D. Van Nostrand, Princeton; Springer reprint, 1974. DOI: 10.1007/978-1-4757-1645-0. Section 7, on relations: equivalence relations and partitions are the same thing.

Geoffrey Hellman (1989). Mathematics without Numbers: Towards a Modal-Structural Interpretation. Clarendon Press, Oxford. DOI: 10.1093/0198240341.001.0001.

Carl Huffman (2003, revised 2024). Philolaus. Stanford Encyclopedia of Philosophy. Stanford Encyclopedia of Philosophy, entry philolaus. Philolaus's book as probably the first written by a Pythagorean, and the long argument over which of his fragments are genuine.

Carl Huffman (2006, revised 2024). Pythagoreanism. Stanford Encyclopedia of Philosophy. Stanford Encyclopedia of Philosophy, entry pythagoreanism. Following Burkert, no ancient source connects Hippasus to the discovery of the irrational.

Aristotle. Prior Analytics. Translated by A. J. Jenkinson. Book I, chapter 23 (41a23–30): the diagonal is incommensurate with the side, because odd numbers would equal evens if it were commensurate. MIT Classics text at classics.mit.edu, prior.mb.

Aristotle. Metaphysics. Translated by W. D. Ross. Book I, chapter 5 (985b–986a), on the Pythagoreans: "they supposed the elements of numbers to be the elements of all things" (MIT Classics, Book I); Book I, chapter 9 (990b17), where the "third man" is named; Book VII, chapter 3, the bronze as matter, the shape as form and the statue as the compound; Book VII, chapter 11 (1036a31 onward), the circle in bronze, stone or wood, and the warning that an animal's parts are parts only when they can do their work (MIT Classics, Book VII).

Iamblichus (c. AD 300). On the Pythagorean Life. Translated by Thomas Taylor as Iamblichus' Life of Pythagoras, A. J. Valpy, London, 1818. Project Gutenberg 63300. Hippasus drowned for the dodecahedron in chapter XVIII; the unnamed man who let out the incommensurable in chapter XXXIV.

Yonggun Jun, Momčilo Gavrilov and John Bechhoefer (2014). High-precision test of Landauer's principle in a feedback trap. Physical Review Letters 113, 190601. DOI: 10.1103/PhysRevLett.113.190601. Erasure at $0.71 \pm 0.03\,kT$ and the control that squashes nothing at $0.05 \pm 0.03\,kT$, on the same bead.

Felix Klein (1872). Vergleichende Betrachtungen über neuere geometrische Forschungen (the Erlangen program). Andreas Deichert, Erlangen. English translation by M. W. Haskell, A comparative review of recent researches in geometry, Bulletin of the New York Mathematical Society 2 (1893), pp. 215–249. DOI: 10.1090/S0002-9904-1893-00147-X.

Ralf Krömer (2006). La “machine de Grothendieck” se fonde-t-elle seulement sur des vocables métamathématiques? Bourbaki et les catégories au cours des années cinquante. Revue d’histoire des mathématiques 12(1), pp. 119–162. DOI: 10.24033/rhm.38. Bourbaki’s attempt at categories in the 1950s.

James Ladyman (1998). What is structural realism? Studies in History and Philosophy of Science Part A 29(3), pp. 409–424. DOI: 10.1016/S0039-3681(98)80129-5.

Saunders Mac Lane (1971; 2nd edition 1998). Categories for the Working Mathematician. Graduate Texts in Mathematics 5, Springer, New York. DOI: 10.1007/978-1-4757-4721-8. Where the words came from: category from Aristotle and Kant, functor from Carnap.

Gottfried Wilhelm Leibniz (1679). Letter to Christiaan Huygens, 8 September 1679, with the essay on the analysis of situation. In C. I. Gerhardt (ed.), Leibnizens mathematische Schriften, vol. 2, A. Asher, Berlin, 1850, from p. 17; Huygens’s reply, which Gerhardt dates to 22 November 1679, from p. 27. Internet Archive, leibnizensmathe08leibgoog. English in L. E. Loemker (ed.), Philosophical Papers and Letters, 2nd edition, Reidel, 1969, pp. 248–258, DOI: 10.1007/978-94-010-1426-7_28.

David Lewis (1983). New work for a theory of universals. Australasian Journal of Philosophy 61(4), pp. 343–377. DOI: 10.1080/00048408312341131. The answer to Putnam through natural properties, which Lewis credits to Merrill on p. 371.

MacTutor History of Mathematics. Emmy Amalie Noether. University of St Andrews. MacTutor biography, Noether Emmy. Habilitation closed to women, her courses advertised under Hilbert’s name, and habilitation at last in 1919.

MacTutor History of Mathematics. Farkas Wolfgang Bolyai. University of St Andrews. MacTutor biography, Bolyai Farkas. His letter of 4 April 1820 begging his son to leave the parallels alone.

Michael Makkai (1998). Towards a categorical foundation of mathematics. In J. A. Makowsky and E. V. Ravve (eds.), Logic Colloquium ’95, Lecture Notes in Logic 11, Springer, Berlin, pp. 153–190. DOI: 10.1007/978-3-662-22108-2_11. “No global equality present for all things simultaneously” and a set as “a relatively orderly part of the world in which an equality predicate reigns,” both on p. 156.

G. H. Merrill (1980). The model-theoretic argument against realism. Philosophy of Science 47(1), pp. 69–81. DOI: 10.1086/288910.

Augustus De Morgan (1864). On the Syllogism, No. IV, and on the Logic of Relations. Transactions of the Cambridge Philosophical Society 10, pp. 331–358. Internet Archive, transactionsofca10camb. Signed at University College London on

12 November 1859, with an appendix dated 15 March 1860, and read on 23 April 1860.

Augustus De Morgan (1966). On the Syllogism and Other Logical Writings. Edited by Peter Heath. Routledge and Kegan Paul, London. The fourth paper on the syllogism is on pp. 208–246.

Emmy Noether (1918). Invariante Variationsprobleme. Nachrichten von der Gesellschaft der Wissenschaften zu Göttingen, Mathematisch-Physikalische Klasse, pp. 235–257. EuDML 59024.

Charles S. Peirce (1870, printed 1873). Description of a notation for the logic of relatives, resulting from an amplification of the conceptions of Boole's calculus of logic. Memoirs of the American Academy of Arts and Sciences 9(2), pp. 317–378. DOI: 10.2307/25058006. Boole's algebra carried over to relations, with examples like "lover of a servant of" a woman.

Charles S. Peirce (1879). Note on the theory of the economy of research. Report of the Superintendent of the United States Coast Survey for the year ending June 1876, Appendix 14, pp. 197–201. Government Printing Office, Washington. Internet Archive, 1876-us-coast-survey-report. "The relations between the utility and the cost of diminishing the probable error of our knowledge."

Plato. Parmenides 132a–b. Translated by H. N. Fowler. Perseus Digital Library text, Perseus:text:1999.01.0174:text=Parm.:section=132a. The regress of greatness that Aristotle later called the third man.

Plato. Timaeus 50c–56b. Translated by W. R. M. Lamb. Perseus Digital Library text, Perseus:text:1999.01.0180. The receptacle that takes impressions, the two right triangles, and the four elements as regular solids.

Proclus. A Commentary on the First Book of Euclid's Elements. Translated by Glenn R. Morrow. Princeton University Press, 1970. His own attempt to prove the parallel postulate: MacTutor extract, Proclus parallel postulate.

The Univalent Foundations Program (2013). Homotopy Type Theory: Univalent Foundations of Mathematics. Institute for Advanced Study, Princeton. homotopytypetheory.org/book; arXiv: 1308.0729. Written during the 2012–13 special year organized by Steve Awodey, Thierry Coquand and Vladimir Voevodsky; univalence as "identity is equivalent to equivalence" is in the introduction.

Hilary Putnam (1980). Models and reality. Journal of Symbolic Logic 45(3), pp. 464–482. DOI: 10.2307/2273415. The model-theoretic argument Lewis answers.

Michael D. Resnik (1997). Mathematics as a Science of Patterns. Clarendon Press, Oxford. DOI: 10.1093/0198250142.001.0001.

Jorma Rissanen (1978). Modeling by shortest data description. Automatica 14(5), pp. 465–471. DOI: 10.1016/0005-1098(78)90005-5. The minimum description length principle.

Bertrand Russell (1902). Letter to Gottlob Frege, 16 June 1902, and Frege's reply. In Jean van Heijenoort (ed.), From Frege to Gödel: A Source Book in Mathematical Logic, 1879–1931, Harvard University Press, 1967, pp. 124–125 and 127–128. Stanford Encyclopedia of Philosophy, Russell's paradox.

Girolamo Saccheri (1733). Euclides ab omni naevo vindicatus. Milan. English translation by George Bruce Halsted, Girolamo Saccheri's Euclides Vindicatus, Open Court, Chicago, 1920. Internet Archive, girolamosacceri00saccuoft. Proposition XXXIII calls the hypothesis of the acute angle "repugnant to the nature of the straight line."

Claude E. Shannon (1951). Prediction and entropy of printed English. Bell System Technical Journal 30(1), pp. 50–64. DOI: 10.1002/j.1538-7305.1951.tb01366.x. The game of guessing the next letter.

Stewart Shapiro (1997). Philosophy of Mathematics: Structure and Ontology. Oxford University Press, New York. DOI: 10.1093/0195139305.001.0001.

Theodore Sider (2011). Writing the Book of the World. Oxford University Press, Oxford. DOI: 10.1093/acprof:oso/9780199697908.001.0001. "The world has a distinguished structure, a privileged description" is in the preface, and "structure is a worldly, not conceptual or linguistic, matter" in chapter 1, as read in the author's sample at tedsider.org, wbotw_sample.

Stanley Burris and Marcel Jackson. George Boole. Stanford Encyclopedia of Philosophy. Stanford Encyclopedia of Philosophy, entry boole. Boole as a schoolmaster who carried on his own education.

Alfred Tarski (1954). Contributions to the theory of models, I and II. Indagationes Mathematicae 16, pp. 572–581 and 582–588. DOI: 10.1016/S1385-7258(54)50074-0; DOI: 10.1016/S1385-7258(54)50075-2. The first part announces the field under the name theory of models.

Alfred Tarski (1986). What are logical notions? Edited by John Corcoran. History and Philosophy of Logic 7(2), pp. 143–154. DOI: 10.1080/01445348608837096. A lecture given at Bedford College, London, on 16 May 1966: logical notions are those invariant under every one-to-one transformation of the world onto itself.

Nikolai K. Vereshchagin and Paul M. B. Vitányi (2004). Kolmogorov's structure functions and model selection. IEEE Transactions on Information Theory 50(12), pp. 3265–3290. DOI: 10.1109/TIT.2004.838346; arXiv: cs/0204037. "Meaningless

information, random noise” is in the section on meaningful information, as read in the arXiv version.

Abraham Wald (1945). Sequential tests of statistical hypotheses. Annals of Mathematical Statistics 16(2), pp. 117–186. DOI: 10.1214/aoms/1177731118.

Chapter 11. The counter

Lorenz Breidenbach, Phil Daian and Florian Tramèr, with advice from Ari Juels (2017). GasToken. Archived site at web.archive.org, gastoken.io; code at github.com, gastoken. Storing gas when it is cheap and spending the refunds when it is dear.

Vitalik Buterin and Martin Swende (2021). EIP-3529: Reduction in refunds. Ethereum Improvement Proposals. eips.ethereum.org/EIPS/eip-3529. Refunds capped at a fifth of the gas used, with GasToken and blocks reaching nearly twice their gas limit given as reasons.

Maurice Clint (1973). Program proving: Coroutines. Acta Informatica 2(1), pp. 50–63. DOI: 10.1007/BF00571463. The earliest use of auxiliary variables Owicki and Gries had found.

Bob Coecke, Tobias Fritz and Robert W. Spekkens (2016). A mathematical theory of resources. Information and Computation 250, pp. 59–86. DOI: 10.1016/j.ic.2016.02.008.

Thomas Colcombet and Daniela Petrişan (2020). Automata minimization: a functorial approach. Logical Methods in Computer Science 16(1), paper 32. DOI: 10.23638/LMCS-16(1:32)2020.

Norman Danner, Daniel R. Licata and Ramyaa Ramyaa (2015). Denotational cost semantics for functional languages with inductive types. Proceedings of ICFP 2015, pp. 140–151. DOI: 10.1145/2784731.2784749.

Norman Danner and Daniel R. Licata (2022). Denotational semantics as a foundation for cost recurrence extraction for functional languages. Journal of Functional Programming 32, e8. DOI: 10.1017/S095679682200003X.

Jean-Christophe Filliâtre, Léon Gondelman and Andrei Paskevich (2014). The spirit of ghost code. Computer Aided Verification (CAV 2014), Lecture Notes in Computer Science 8559, Springer, pp. 1–16. DOI: 10.1007/978-3-319-08867-9_1. When ghost code can be erased with no difference in what the program computes.

Tobias Fritz (2017). Resource convertibility and ordered commutative monoids. Mathematical Structures in Computer Science 27(6), pp. 850–938. DOI: 10.1017/S0960129515000444; arXiv: 1504.03661.

Joseph A. Goguen (1972). Realization is universal. Mathematical Systems Theory 6(4), pp. 359–374. DOI: 10.1007/BF01843493.

Tomáš Gonda and Robert W. Spekkens (2023). Monotones in general resource theories. Compositionality 5, article 7. DOI: 10.32408/compositionality-5-7.

Geoffrey M. Hodgson (n.d.). The secret history of the tally stick. Web page at geoffreymhodgson.uk, secret history of the tally stick. Tallies under Henry I from about 1100 until 1826, the stock and the foil, and the stockholder.

Jan Hoffmann, Klaus Aehlig and Martin Hofmann (2011). Multivariate amortized resource analysis. Proceedings of POPL 2011, pp. 357–370. DOI: 10.1145/1926385.1926427. Journal version: ACM Transactions on Programming Languages and Systems 34(3), 2012, article 14, DOI: 10.1145/2362389.2362393.

Martin Hofmann and Steffen Jost (2003). Static prediction of heap space usage for first-order functional programs. Proceedings of POPL 2003, pp. 185–197. DOI: 10.1145/604131.604148. The potential method put into the checking of programs.

Paul R. Johnson and Robert H. Thomas (1975). The Maintenance of Duplicate Databases. RFC 677, 27 January 1975. RFC 677. The timestamps Lamport started from.

Shin-ya Katsumata (2014). Parametric effect monads and semantics of effect systems. Proceedings of POPL 2014, pp. 633–645. DOI: 10.1145/2535838.2535846.

Narayana R. Kocherlakota (1998). Money is memory. Journal of Economic Theory 81(2), pp. 232–251. DOI: 10.1006/jeth.1997.2357. Where nobody can commit, anything money makes possible, memory of past dealings makes possible too.

Leslie Lamport (1978). Time, clocks, and the ordering of events in a distributed system. Communications of the ACM 21(7), pp. 558–565. DOI: 10.1145/359545.359563. “Counters with no actual timing mechanism” is on p. 559.

Leslie Lamport (n.d.). The Writings of Leslie Lamport, note on “Time, Clocks and the Ordering of Events in a Distributed System.” Web page at lamport.azurewebsites.net, publications. The Johnson and Thomas note, special relativity, and Jim Gray’s two opinions of the paper.

Elliott H. Lieb and Jakob Yngvason (1999). The physics and mathematics of the second law of thermodynamics. Physics Reports 310(1), pp. 1–96. DOI: 10.1016/S0370-1573(98)00082-9. From an order of which states can reach which, an entropy that is unique once zero and unit are chosen.

Aleksandr M. Lyapunov (1892). The general problem of the stability of motion. Doctoral thesis, Kharkov (in Russian). French translation by Édouard Davaux, naval engineer at Toulon, revised and corrected by the author: Problème général de

la stabilité du mouvement, Annales de la Faculté des Sciences de Toulouse, 2e série, 9 (1907), pp. 203–474, Numdam at numdam.org, AFST_1907_2_9__203_0. English translation by A. T. Fuller, International Journal of Control 55(3) (1992), pp. 531–772, DOI: 10.1080/00207179208934253 (opening part).

Alfred Mitchell-Innes (1913). What is money? The Banking Law Journal, May 1913, pp. 377–408. Levy Institute working paper discussing it. The crown paying its creditors by "raising a tally" on some branch of revenue, p. 398.

Richard fitz Nigel (c. 1180). The Dialogue Concerning the Exchequer (Dialogus de Scaccario). Translated in Ernest F. Henderson, Select Historical Documents of the Middle Ages, George Bell and Sons, London, 1896. Avalon Project text at avalon.law.yale.edu, excheq. Modern edition and translation by Emilie Amt, Dialogus de Scaccario, and Constitutio Domus Regis, Oxford Medieval Texts, Oxford University Press, 2007. The widths of the notches on a tally, from a palm for a thousand pounds down to a bare cut for a penny.

Yue Niu, Jonathan Sterling, Harrison Grodin and Robert Harper (2022). A cost-aware logical framework. Proceedings of the ACM on Programming Languages 6(POPL), article 9. DOI: 10.1145/3498670. The framework calf.

Dominic Orchard, Vilem-Benjamin Liepelt and Harley Eades III (2019). Quantitative program reasoning with graded modal types. Proceedings of the ACM on Programming Languages 3(ICFP), article 110. DOI: 10.1145/3341714.

Susan Owicki and David Gries (1976). An axiomatic proof technique for parallel programs I. Acta Informatica 6(4), pp. 319–340. DOI: 10.1007/BF00268134. Auxiliary variables, needed for the proof and not for the program, which only ever get assigned.

Marcel-Paul Schützenberger (1961). On the definition of a family of automata. Information and Control 4(2–3), pp. 245–270. DOI: 10.1016/S0019-9958(61)80020-X. Automata with weights on their moves.

Caroline Shenton (2012). The Day Parliament Burned Down. Oxford University Press, Oxford. The tallies burned in the furnaces under the House of Lords on 16 October 1834, and the fire that started in the flues.

Robert Endre Tarjan (1985). Amortized computational complexity. SIAM Journal on Algebraic and Discrete Methods 6(2), pp. 306–318. DOI: 10.1137/0606031. The potential method, with the physicist's view credited to Daniel Sleator, and a stack whose total work is at most twice the number of pushes.

Alan M. Turing (1949). Checking a large routine. In Report of a Conference on High Speed Automatic Calculating Machines, University Mathematical Laboratory,

Cambridge, pp. 67–69. Turing Digital Archive at turingarchive.kings.cam.ac.uk, amt-b-8. Reprinted by F. L. Morris and C. B. Jones, An early program proof by Alan Turing, Annals of the History of Computing 6(2), 1984, pp. 139–143, DOI: 10.1109/MAHC.1984.10017. A quantity that decreases continually and vanishes when the machine stops.

UK Parliament. The Great Fire of 1834. Living Heritage. Web page at parliament.uk, living heritage, the great fire. Two cart-loads of tally sticks, unused since 1826, burned in the stoves under the House of Lords.

Gavin Wood (2014). Ethereum: A Secure Decentralised Generalised Transaction Ledger (the yellow paper). Current version at ethereum.github.io, paper. Why every computation costs a fee, and the gas table in its appendix.

Chapter 12. What a ticked box lets you lean on

Sergei N. Artemov (2001). Explicit provability and constructive semantics. Bulletin of Symbolic Logic 7(1), pp. 1–36. DOI: 10.2307/2687821. The Logic of Proofs, where knowing comes with the reason written out.

A. J. Ayer (1956). The Problem of Knowledge. Macmillan, London, p. 34. One of the two statements of the definition that Gettier takes apart.

Alexandru Baltag, Bryan Renne and Sonja Smets (2014). The logic of justified belief, explicit knowledge, and conclusive evidence. Annals of Pure and Applied Logic 165(1), pp. 49–81. DOI: 10.1016/j.apal.2013.07.005.

Walter J. Boyne (2013). The Checklist. Air Force Magazine, August 2013, pp. 52–56. Article at airandspaceforces.com, 0813checklist. The Model 299 crash of 30 October 1935, the written pilot routines going back to 1911, and the judgment that “it took the Model 299 crash to institutionalize its use.”

Robert B. Brandom (1994). Making It Explicit: Reasoning, Representing, and Discursive Commitment. Harvard University Press, Cambridge, Massachusetts. Commitment, entitlement, and the score speakers keep on each other; his home page at the University of Pittsburgh is sites.pitt.edu/~rbrandom.

Martin Caidin (1968). Flying Forts. Meredith Press, New York. Has “rumors that the 299 was too much airplane for any one pilot.”

Roderick M. Chisholm (1957). Perceiving: A Philosophical Study. Cornell University Press, Ithaca, p. 16. One of the two statements of the definition that Gettier takes apart.

DeWitt S. Copp (1980). A Few Great Captains: The Men and Events That Shaped the Development of U.S. Air Power. Doubleday, Garden City, New York. The earliest

place found with the exact line “too much airplane for one man to fly,” given as word from the War Department.

Charles Dickens (1843). A Christmas Carol. In Prose. Being a Ghost Story of Christmas. Chapman and Hall, London. Stave Three, the game of Yes and No at Fred’s party. Project Gutenberg 46.

Julien Dutant (2015). The Legend of the Justified True Belief Analysis. Philosophical Perspectives 29(1), pp. 95–145. DOI: 10.1111/phpe.12061. Known here only from its abstract and summaries.

Euclid. Elements, Book X. D. E. Joyce’s online edition. The book on lengths that share no common measure, thought to be largely Theaetetus’s work.

Plato. Theaetetus. Trans. Harold N. Fowler, Plato in Twelve Volumes, vol. 12, Harvard University Press, 1921. Perseus Digital Library text, Perseus:text:1999.01.0172. Theaetetus carried home from the army at Corinth and Euclides’s notes corrected by Socrates (142a–143c), Theodorus’s square roots up to seventeen (147d–148b), and the three meanings of an account (201c–210b).

Peter Gärdenfors (1986). Belief revisions and the Ramsey test for conditionals. The Philosophical Review 95(1), pp. 81–93. DOI: 10.2307/2185133. The impossibility theorem.

Atul Gawande (2007). The Checklist. The New Yorker, 10 December 2007. The New Yorker, 10 December 2007, The Checklist. Gives the line “too much airplane for one man to fly” as a newspaper’s, without naming the paper.

Gösta Grahne (1998). Updates and counterfactuals. Journal of Logic and Computation 8(1), pp. 87–117. DOI: 10.1093/logcom/8.1.87. Gärdenfors’s theorem bites revision, not update.

Ralph V. L. Hartley (1928). Transmission of Information. Bell System Technical Journal 7(3), pp. 535–563. DOI: 10.1002/j.1538-7305.1928.tb01236.x. Information measured by the count of possibilities alone.

David Hume (1748). An Enquiry Concerning Human Understanding, Section X, “Of Miracles,” paragraph 4: “A wise man, therefore, proportions his belief to the evidence.” Hume Texts Online, E 10.4.

Jonathan Jenkins Ichikawa and Matthias Steup (2001, revised 2026). The Analysis of Knowledge. Stanford Encyclopedia of Philosophy. Stanford Encyclopedia of Philosophy, entry knowledge-analysis. The “traditional” analysis and the “convenient fiction” that it was widely held, and the older cases from Dharmottara and Peter of Mantua.

Edward Jablonski (1965). Flying Fortress: The Illustrated Biography of the B-17s and the Men Who Flew Them. Doubleday, Garden City, New York. The talk after the crash that the airplane was “too much airplane for one pilot to handle.”

Lester R. Ford Jr. and Selmer M. Johnson (1959). A Tournament Problem. American Mathematical Monthly 66(5), pp. 387–389. DOI: 10.1080/00029890.1959.11989306. Merge insertion.

Hirofumi Katsuno and Alberto O. Mendelzon (1991). On the difference between updating a knowledge base and revising it. Proceedings of the Second International Conference on Principles of Knowledge Representation and Reasoning (KR’91), pp. 387–394; reprinted in Peter Gärdenfors (ed.), Belief Revision, Cambridge University Press (1992), pp. 183–203. DOI: 10.1017/CBO9780511526664.007.

Arthur M. Keller and Marianne Winslett Wilkins (1985). On the use of an extended relational model to handle changing incomplete information. IEEE Transactions on Software Engineering SE-11(7), pp. 620–633. DOI: 10.1109/TSE.1985.232506.

Donald E. Knuth (1998). The Art of Computer Programming, Volume 3: Sorting and Searching, second edition. Addison-Wesley. Section 5.3.1: the counting floor for sorting by comparisons, merge insertion, and Wells’s result for twelve items.

Isaac Levi (1980). The Enterprise of Knowledge. MIT Press, Cambridge, Massachusetts. Where routine and deliberate expansion first appear.

Isaac Levi (1991). The Fixation of Belief and Its Undoing: Changing Beliefs through Inquiry. Cambridge University Press. DOI: 10.1017/CBO9780511663819. Routine and deliberate expansion.

George C. Necula and Peter Lee (1996). Safe Kernel Extensions Without Run-Time Checking. Proceedings of the 2nd USENIX Symposium on Operating Systems Design and Implementation (OSDI ’96), pp. 229–243. USENIX page at usenix.org.

George C. Necula (1997). Proof-Carrying Code. Proceedings of the 24th ACM SIGPLAN-SIGACT Symposium on Principles of Programming Languages (POPL ’97), pp. 106–119. DOI: 10.1145/263699.263712.

New Castle News (1935). Report on the crash of the Boeing bomber at Wright Field, 12 November 1935, New Castle, Pennsylvania. One of the papers that November reporting the locked controls.

J. J. O’Connor and E. F. Robertson (1999). Theaetetus of Athens. MacTutor History of Mathematics archive, University of St Andrews. MacTutor biography, Theaetetus. His death in about 369 BC after being wounded in the fighting at Corinth, and Book X of Euclid as largely his work.

Bertrand Russell (1948). Human Knowledge: Its Scope and Limits. George Allen and Unwin, London. The stopped clock.

Krister Segerberg (1998). Irrevocable Belief Revision in Dynamic Doxastic Logic. Notre Dame Journal of Formal Logic 39(3), pp. 287–306. DOI: 10.1305/ndjfl/1039182247.

Neil Tennant (2008). Belief-revision, the Ramsey test, monotonicity, and the so-called impossibility results. The Review of Symbolic Logic 1(4), pp. 402–423. DOI: 10.1017/S1755020308090023.

Mark B. Wells (1965). Applications of a language for computing in combinatorics. Information Processing 65, Proceedings of IFIP Congress 1965, pp. 497–498. The computer search showing twelve items need thirty comparisons.

Linda Zagzebski (1994). The inescapability of Gettier problems. The Philosophical Quarterly 44(174), pp. 65–73. DOI: 10.2307/2220147.

Chapter 13. What insight is

Karl Duncker (1935; English translation 1945). Zur Psychologie des produktiven Denkens. Julius Springer, Berlin. English translation by Lynne S. Lees: On Problem-Solving, Psychological Monographs 58(5), whole no. 270, 1945. DOI: 10.1037/h0093599. The candle problem.

Galileo Galilei (1586). La Bilancetta (The Little Balance). Described at the Galileo Project, Rice University: page at galileo.library.rice.edu, balance. The words quoted are from the English translation in Laura Fermi and Gilberto Bernardini, Galileo and the Scientific Revolution, Basic Books, 1961, as reproduced at math.nyu.edu.

Sam Glucksberg (1962). The influence of strength of drive on functional fixedness and perceptual recognition. Journal of Experimental Psychology 63(1), pp. 36–41. DOI: 10.1037/h0044683. The candle problem with prize money on it.

I. J. Good (1979). Studies in the History of Probability and Statistics XXXVII: A. M. Turing's statistical work in World War II. Biometrika 66(2), pp. 393–396. DOI: 10.1093/biomet/66.2.393. Turing's expected weight of evidence, the ban and the deciban, and Peirce's 1878 name for the measure.

Wolfgang Köhler (1917). Intelligenzprüfungen an Anthropoiden. Abhandlungen der Königlich Preussischen Akademie der Wissenschaften, Physikalisch-mathematische Klasse, 1917. English translation by Ella Winter, The Mentality of Apes, Kegan Paul, Trench, Trubner, London, 1925. Sultan and the two sticks.

Dennis V. Lindley (1956). On a Measure of the Information Provided by an Experiment. Annals of Mathematical Statistics 27(4), pp. 986–1005. DOI: 10.1214/aoms/1177728069.

Janet Metcalfe (1986). Premonitions of insight predict impending error. Journal of Experimental Psychology: Learning, Memory, and Cognition 12(4), pp. 623–634. DOI: 10.1037/0278-7393.12.4.623. High warmth before an insight answer goes with the answer being wrong.

Janet Metcalfe and David Wiebe (1987). Intuition in insight and noninsight problem solving. Memory and Cognition 15(3), pp. 238–246. DOI: 10.3758/BF03197722. Warmth that climbs on algebra and stays flat on insight problems.

Charles S. Peirce (1878). The Probability of Induction. Popular Science Monthly 12, pp. 705–718. Illustrations of the Logic of Science, fourth paper. Wikisource text at en.wikisource.org. Belief measured by a logarithm, so that independent weights of evidence add.

Vitruvius. The Ten Books on Architecture, Book IX, Introduction. Trans. Morris Hicky Morgan, Harvard University Press, 1914. Passage at math.nyu.edu, Vitruvius. Archimedes, Hiero's crown and the bath.

Chapter 14. A box over any machine

Martín Abadi and Leslie Lamport (1991). The Existence of Refinement Mappings. Theoretical Computer Science 82(2), pp. 253–284. DOI: 10.1016/0304-3975(91)90224-P. History variables.

David Z. Albert (2000). Time and Chance. Harvard University Press, Cambridge, Massachusetts. The name "past hypothesis" for an assumed low-entropy beginning; its roots in Boltzmann are traced in the Stanford Encyclopedia of Philosophy entry Thermodynamic Asymmetry in Time, Stanford Encyclopedia of Philosophy, entry time-thermo.

Pietro Baroni and Massimiliano Giacomin (2007). On principle-based evaluation of extension-based argumentation semantics. Artificial Intelligence 171(10–15), pp. 675–700. DOI: 10.1016/j.artint.2007.04.004.

John S. Bell (1975). On wave packet reduction in the Coleman-Hepp model. Helvetica Physica Acta 48, p. 93. Reprinted in Speakable and Unspeakable in Quantum Mechanics, Cambridge University Press, chapter page at cambridge.org. The reduction happens only in a limit that never comes.

Thomas Blanchard (2024). Causation and the Time-Asymmetry of Knowledge. Australasian Journal of Philosophy 102(4), pp. 959–977. DOI: 10.1080/00048402.2024.2328629.

John P. Burgess (1986). The truth is never simple. Journal of Symbolic Logic 51(3), pp. 663–681. DOI: 10.2307/2274021. Quasi-inductive definitions.

Martin Caminada (2014). Strong admissibility revisited. In Simon Parsons, Nir Oren, Chris Reed and Federico Cerutti (eds.), Computational Models of Argument: Proceedings of COMMA 2014, Frontiers in Artificial Intelligence and Applications 266, IOS Press, pp. 197–208. DOI: 10.3233/978-1-61499-436-7-197.

Wen Tsing Chow and William H. Henrich (1962). Storage matrix. US patent 3,028,659, filed 27 December 1957, granted 10 April 1962. Google Patents, US3028659A. The memory of blown fuses that chapter fourteen describes.

Vincent Danos and Jean Krivine (2004). Reversible Communicating Systems. CONCUR 2004, Lecture Notes in Computer Science 3170, pp. 292–307. DOI: 10.1007/978-3-540-28644-8_19.

Vincent Danos and Jean Krivine (2005). Transactions in RCCS. CONCUR 2005, Lecture Notes in Computer Science 3653, pp. 398–412. DOI: 10.1007/11539452_31.

David Deutsch and Chiara Marletto (2025). Constructor theory of time. Preprint. arXiv: 2505.08692.

Rod Downey, Carl Jockusch, Timothy H. McNicholl and Paul Schupp (2015). Asymptotic density and the Ershov hierarchy. Mathematical Logic Quarterly 61(3), pp. 189–195. DOI: 10.1002/malq.201300081; arXiv: 1309.0137 (revised 2014). Their account of Ershov's hierarchy, where one change of guess from no to yes gives the sets a program can list.

Phan Minh Dung (1995). On the acceptability of arguments and its fundamental role in nonmonotonic reasoning, logic programming and n-person games. Artificial Intelligence 77(2), pp. 321–357. DOI: 10.1016/0004-3702(94)00041-X.

Aram Ebtekar and Marcus Hutter (2024). Modeling the Arrows of Time with Causal Multibaker Maps. Entropy 26(9), article 776. DOI: 10.3390/e26090776.

W. H. Eccles and F. W. Jordan (1919). A trigger relay utilizing three-electrode thermionic vacuum tubes. The Electrician 83, p. 298, 19 September 1919; reprinted in Radio Review 1(3), pp. 143–146.

W. H. Eccles and F. W. Jordan (1920). Improvements in ionic relays. British patent GB 148,582, filed 21 June 1918, published 5 August 1920. Google Patents, GB148582A. The two-valve circuit that became the flip-flop.

Yu. L. Ershov (1968). A hierarchy of sets. I. Algebra and Logic 7, pp. 25–43; translated from Algebra i Logika 7(1), pp. 47–74. DOI: 10.1007/BF02218750. Cited through Downey, Jockusch, McNicholl and Schupp, not read in the original.

Murray Gell-Mann (1964). A schematic model of baryons and mesons. Physics Letters 8(3), pp. 214–215. DOI: 10.1016/S0031-9163(64)92001-3. The paper that named the quark.

Murray Gell-Mann and James B. Hartle (1995, printed 1998). Strong decoherence. In D.-H. Feng and B.-L. Hu (eds.), Proceedings of the 4th Drexel Conference on Quantum Non-Integrability: The Quantum-Classical Correspondence, International Press, Hong Kong. arXiv: gr-qc/9509054. The permanence of the past.

E. Mark Gold (1965). Limiting recursion. Journal of Symbolic Logic 30(1), pp. 28–48. DOI: 10.2307/2270580.

J. J. Halliwell (1999). Arrival Times in Quantum Theory from an Irreversible Detector Model. Progress of Theoretical Physics 102(4), pp. 707–717. DOI: 10.1143/PTP.102.707. A two-state detector made “effectively irreversible” by a large environment.

C. L. Hamblin (1962). Translation to and from Polish notation. The Computer Journal 5(3), pp. 210–213. DOI: 10.1093/comjnl/5.3.210. His reverse Polish notation for computers.

C. L. Hamblin (1970). Fallacies. Methuen, London. Chapter 8, formal dialectic, where each speaker has a store of commitments that the rules update. Known here through secondary accounts, not from the chapter itself.

Joel David Hamkins and Andy Lewis (2000). Infinite time Turing machines. Journal of Symbolic Logic 65(2), pp. 567–604. DOI: 10.2307/2586556; arXiv: math/9808093.

Klaus Havelund and Grigore Roşu (2002). Synthesizing monitors for safety properties. Tools and Algorithms for the Construction and Analysis of Systems (TACAS 2002), Lecture Notes in Computer Science 2280, pp. 342–356. DOI: 10.1007/3-540-46002-0_24.

Klaus Hepp (1972). Quantum theory of measurement and macroscopic observables. Helvetica Physica Acta 45, p. 237. The Coleman-Hepp model. Cited through Bell’s 1975 paper, not read in the original.

Erik C. W. Krabbe (2001). The problem of retraction in critical discussion. Synthese 127(1–2), pp. 141–159. DOI: 10.1023/A:1010318403544.

Saul Kripke (1975). Outline of a theory of truth. Journal of Philosophy 72(19), pp. 690–716. DOI: 10.2307/2024634. Grounded and ungrounded sentences.

Orna Lichtenstein, Amir Pnueli and Lenore Zuck (1985). The glory of the past. Logics of Programs, Lecture Notes in Computer Science 193, pp. 196–218. DOI: 10.1007/3-540-15648-8_16. A logic of the past, with the operator "once."

Robin Milner (1980). A Calculus of Communicating Systems. Lecture Notes in Computer Science 92, Springer. DOI: 10.1007/3-540-10235-3.

Henry Prakken (2005). Coherence and flexibility in dialogue games for argumentation. Journal of Logic and Computation 15(6), pp. 1009–1040. DOI: 10.1093/logcom/exi046.

Hilary Putnam (1965). Trial and error predicates and the solution to a problem of Mostowski. Journal of Symbolic Logic 30(1), pp. 49–57. DOI: 10.2307/2270581.

Wayne Richter and Peter Aczel (1974). Inductive definitions and reflecting properties of admissible ordinals. In J. E. Fenstad and P. G. Hinman (eds.), Generalized Recursion Theory, North-Holland, Studies in Logic and the Foundations of Mathematics 79, pp. 301–381. DOI: 10.1016/S0049-237X(08)70592-5.

Carlo Rovelli (2022). Memory and Entropy. Entropy 24(8), article 1022. DOI: 10.3390/e24081022; arXiv: 2003.06687.

Alin Tomescu, Vivek Bhupatiraju, Dimitrios Papadopoulos, Charalampos Papamanthou, Nikos Triandopoulos and Srinivas Devadas (2019). Transparency logs via append-only authenticated dictionaries. Proceedings of the 2019 ACM SIGSAC Conference on Computer and Communications Security (CCS '19), pp. 1299–1316. DOI: 10.1145/3319535.3345652.

Hao Wang (1957). A variant to Turing's theory of computing machines. Journal of the ACM 4(1), pp. 63–92. DOI: 10.1145/320856.320867. Machines that mark a square and never erase one.

Wikipedia. Programmable ROM, History section. Wikipedia, Programmable ROM. Chow's invention at Arma in 1956, at the Air Force's request, to hold the targeting constants of the Atlas computer, and "burn" as the patent's word.

David H. Wolpert (1992). Memory systems, computation, and the second law of thermodynamics. International Journal of Theoretical Physics 31(4), pp. 743–785. DOI: 10.1007/BF00670829.

David H. Wolpert and Jens Kipper (2024). Memory Systems, the Epistemic Arrow of Time, and the Second Law. Entropy 26(2), article 170. DOI: 10.3390/e26020170.

Chapter 16. Reading the box

A. P. Ershov (1958). On programming of arithmetic operations. Communications of the ACM 1(8), pp. 3–6. DOI: 10.1145/368892.368907. Translated by Morris D.

Friedman from Doklady Akademii Nauk SSSR 118(3), 1958, pp. 427–430. Never working out the same expression twice, in time that grows with the number of instructions instead of its square.

Kurt Gödel (1931). Über formal unentscheidbare Sätze der Principia Mathematica und verwandter Systeme I. Monatshefte für Mathematik und Physik 38, pp. 173–198. DOI: 10.1007/BF01700692.

Eiichi Goto (1974). Monocopy and associative algorithms in an extended Lisp. Technical report, Information Science Laboratory, Faculty of Science, University of Tokyo, May 1974. Scanned copy at cs.utexas.edu, monocopy-goto. No two copies of the same structure, building on a scheme proposed at the 1973 Programming Symposium.

IBM (1974, revised 1975). IBM System/370 Principles of Operation. GA22-7000-4, fourth edition, September 1974, as revised 1 September 1975 by TNL GN22-0498. Programming note to Compare Double and Swap, p. 125. Scanned manual at ftp.mirrorservice.org. The list changed behind a program's back, and the change number that brings the risk "to a negligible level."

Maged M. Michael and Michael L. Scott (1995). Correction of a Memory Management Method for Lock-Free Data Structures. Technical Report 599, Computer Science Department, University of Rochester, December 1995. PDF at cs.rochester.edu, technical report TR599, 1995. Uses the name "ABA problem."

Maged M. Michael and Michael L. Scott (1995). Simple, Fast, and Practical Non-Blocking and Blocking Concurrent Queue Algorithms. Technical Report 600, Computer Science Department, University of Rochester, December 1995. DOI: 10.21236/ADA309412.

Maged M. Michael and Michael L. Scott (1996). Simple, fast, and practical non-blocking and blocking concurrent queue algorithms. Proceedings of the Fifteenth Annual ACM Symposium on Principles of Distributed Computing (PODC '96), pp. 267–275. DOI: 10.1145/248052.248106. A value that goes from A to B and back to A, and the counter that catches it.

Donald Michie (1968). "Memo" Functions and Machine Learning. Nature 218, 6 April 1968, pp. 19–22. DOI: 10.1038/218019a0. The rule part and the rote part, and each evaluation by rule adding a fresh entry to the rote.

R. Kent Treiber (1986). Systems Programming: Coping with Parallelism. Research Report RJ 5118, IBM Almaden Research Center, April 1986. The lock-free stack and its counter beside the head of the list.

Chapter 17. What a machine can't decide about itself

Manuel Blum (1967). A machine-independent theory of the complexity of recursive functions. Journal of the ACM 14(2), pp. 322–336. DOI: 10.1145/321386.321395. The speedup theorem: functions with no fastest program.

Georg Cantor (1874). Über eine Eigenschaft des Inbegriffes aller reellen algebraischen Zahlen. Journal für die reine und angewandte Mathematik 77, pp. 258–262. DOI: 10.1515/crll.1874.77.258. The first, harder proof.

Georg Cantor (1891). Über eine elementare Frage der Mannigfaltigkeitslehre. Jahresbericht der Deutschen Mathematiker-Vereinigung 1, pp. 75–78. English: On an elementary question in the theory of manifolds, translated by Anthony Phillips, translation at math.stonybrook.edu, cantor. The diagonal argument, with the symbols m and w and the comparison with the 1874 proof on p. 75.

Solomon Feferman (2006). Are There Absolutely Unsolvable Problems? Gödel's Dichotomy. Philosophia Mathematica (III) 14(2), pp. 134–152. DOI: 10.1093/philmat/nkj003. The Gibbs lecture, Gödel's private conviction, and why he stated only the either-or.

Solomon Feferman (2006). The nature and significance of Gödel's incompleteness theorems. Lecture, Institute for Advanced Study, Gödel Centenary Program, 17 November 2006. Author's text at math.stanford.edu, Godel-IAS. The announcement at Königsberg met with general incomprehension except from von Neumann, and von Neumann's letter of 20 November 1930.

Kurt Gödel (1951). Some basic theorems on the foundations of mathematics and their implications. Gibbs Lecture, American Mathematical Society, Brown University, 26 December 1951. In Kurt Gödel, Collected Works, Vol. III: Unpublished Essays and Lectures, S. Feferman et al. (eds.), Oxford University Press, 1995, pp. 304–323.

Douglas R. Hofstadter (1979). Gödel, Escher, Bach: An Eternal Golden Braid. Basic Books, New York. Where the word "quine" comes from.

Paul A. Karger and Roger R. Schell (1974). Multics Security Evaluation: Vulnerability Analysis. Technical Report ESD-TR-74-193, Vol. II, Electronic Systems Division, Air Force Systems Command, June 1974. DTIC copy at apps.dtic.mil, ADA001120. A trap door planted in a compiler that keeps itself alive.

Stephen Cole Kleene (1938). On notation for ordinal numbers. Journal of Symbolic Logic 3(4), pp. 150–155. DOI: 10.2307/2267778. The second recursion theorem, stated and proved at the end of section 2.

F. William Lawvere (1969). Diagonal arguments and cartesian closed categories. In Category Theory, Homology Theory and their Applications II, Lecture Notes in

Mathematics 92, Springer, pp. 134–145. DOI: 10.1007/BFb0080769. Reprinted with author commentary in Reprints in Theory and Applications of Categories 15 (2006), pp. 1–13, reprint at tac.mta.ca, tr15abs.

Martin H. Löb (1955). Solution of a problem of Leon Henkin. Journal of Symbolic Logic 20(2), pp. 115–118. DOI: 10.2307/2266895.

J. R. Lucas (1961). Minds, Machines and Gödel. Philosophy 36(137), pp. 112–127. DOI: 10.1017/S0031819100057983.

Yiannis N. Moschovakis (2010). Kleene's Amazing Second Recursion Theorem. Bulletin of Symbolic Logic 16(2), pp. 189–239. DOI: 10.2178/bsl/1286889124. Extended abstract in Computer Science Logic 2009, Lecture Notes in Computer Science 5771, pp. 24–39. "This little gem is stated unbilled and proved (completely) in the last two lines of §2 of the short note Kleene [1938]."

Caspar Oesterheld (2019). Robust program equilibrium. Theory and Decision 86(1), pp. 143–159. DOI: 10.1007/s11238-018-9679-3. Programs that cooperate by simulating each other, with a grounding step to stop the regress.

Roger Penrose (1989). The Emperor's New Mind: Concerning Computers, Minds, and the Laws of Physics. Oxford University Press.

Roger Penrose (1994). Shadows of the Mind: A Search for the Missing Science of Consciousness. Oxford University Press.

W. V. Quine (1962). Paradox. Scientific American 206(4), pp. 84–96. DOI: 10.1038/scientificamerican0462-84. Reprinted as "The Ways of Paradox" in The Ways of Paradox and Other Essays (1966). The sentence "yields falsehood when preceded by its quotation."

Panu Raatikainen (2013, revised 2025). Gödel's Incompleteness Theorems. Stanford Encyclopedia of Philosophy. Stanford Encyclopedia of Philosophy, entry goedel-incompleteness. Why the Gödel sentence doesn't literally say anything about itself, and why the theorem isn't about provability in any absolute sense.

Jules Richard (1905). Les principes des mathématiques et le problème des ensembles. Revue générale des sciences pures et appliquées 16(12), pp. 541–543. English translation in Jean van Heijenoort (ed.), From Frege to Gödel, Harvard University Press, 1967, pp. 142–144. His school posts are from J. J. O'Connor and E. F. Robertson, "Jules Antoine Richard," MacTutor History of Mathematics archive, mathshistory.st-andrews.ac.uk.

J. Barkley Rosser (1936). Extensions of some theorems of Gödel and Church. Journal of Symbolic Logic 1(3), pp. 87–91. DOI: 10.2307/2269028. The improvement that

needs only that the system doesn't contradict itself. Rosser as a student of Alonzo Church: Wikipedia, Wikipedia, J. Barkley Rosser.

Bertrand Russell (1903). The Principles of Mathematics. Cambridge University Press. Chapter X, The Contradiction, §100: "I was led to it in the endeavour to reconcile Cantor's proof that there can be no greatest cardinal number with the very plausible supposition that the class of all terms ... has necessarily the greatest possible number of members." Text of §100 at fair-use.org, s100.

Bertrand Russell (1906). Les paradoxes de la logique. Revue de métaphysique et de morale 14, pp. 627–650. The first printing of Berry's paradox, credited to G. G. Berry; Berry as a junior librarian at the Bodleian: Wikipedia, Wikipedia, Berry paradox.

Ken Thompson (1984). Reflections on Trusting Trust. Communications of the ACM 27(8), pp. 761–763. DOI: 10.1145/358198.358210. Delivered as the 1983 Turing Award lecture.

David A. Wheeler (2005). Countering Trusting Trust through Diverse Double-Compiling. Proceedings of the 21st Annual Computer Security Applications Conference (ACSAC 2005), Tucson, pp. 33–48. DOI: 10.1109/CSAC.2005.17.

David A. Wheeler (2009). Fully Countering Trusting Trust through Diverse Double-Compiling. PhD dissertation, George Mason University. Author's page at dwheeler.com, trusting-trust.

Alfred North Whitehead and Bertrand Russell (1910). Principia Mathematica, Volume I. Cambridge University Press. Introduction, Chapter II, section VIII, pp. 63–64: "the least integer not nameable in fewer than nineteen syllables." Project Gutenberg 78050.

Noson S. Yanofsky (2003). A Universal Approach to Self-Referential Paradoxes, Incompleteness and Fixed Points. Bulletin of Symbolic Logic 9(3), pp. 362–386. DOI: 10.2178/bsl/1058448677; arXiv: math/0305282.

Eliezer Yudkowsky and Marcello Herreshoff (2013). Tiling agents for self-modifying AI, and the Löbian obstacle. Draft report, Machine Intelligence Research Institute. PDF at intelligence.org, TilingAgentsDraft.

Chapter 18. Five real systems

Will Arthur and David Challener, with Kenneth Goldman (2015). A Practical Guide to TPM 2.0: Using the Trusted Platform Module in the New Age of Security. Apress. Chapter 11, NV Indexes, pp. 137–150 (bit-field indexes and counters); Chapter

12, Platform Configuration Registers, pp. 151–161 (read, extend and reset). DOI: 10.1007/978-1-4302-6584-9.

Vitalik Buterin (2016). EIP-150: Gas cost changes for IO-heavy operations. Ethereum Improvement Proposals, created 24 September 2016. eips.ethereum.org/EIPS/eip-150. The repair switched on in the Tangerine Whistle fork, raising the prices of the opcodes that read the state from disk.

Vitalik Buterin (2016). Transaction spam attack: next steps. Ethereum Foundation Blog, 22 September 2016. blog.ethereum.org, 22 September 2016. The 2016 attack, which made every node read state from disk with a cheap opcode.

Vitalik Buterin and Virgil Griffith (2017). Casper the Friendly Finality Gadget. arXiv: 1710.09437v1. Justified and final checkpoints, the two slashing rules, and accountable safety, for a finality system laid over a proof-of-work chain.

Vitalik Buterin and Martin Swende (2020). EIP-2929: Gas cost increases for state access opcodes. Ethereum Improvement Proposals. eips.ethereum.org/EIPS/eip-2929. The cold-slot charge of 2,100, justified by the 2016 attacks and switched on in the Berlin upgrade of April 2021.

Vitalik Buterin, Diego Hernandez, Thor Kamphefner, Khiem Pham, Zhi Qiao, Danny Ryan, Juhyeok Sin, Ying Wang and Yan X. Zhang (2020). Combining GHOST and Casper. arXiv: 2003.03052. Gasper, which joins Casper's finality to the LMD GHOST rule for picking branches.

Byung-Gon Chun, Petros Maniatis, Scott Shenker and John Kubiatowicz (2007). Attested append-only memory: making adversaries stick to their word. Proceedings of the 21st ACM Symposium on Operating Systems Principles (SOSP '07), pp. 189–204. DOI: 10.1145/1294261.1294280.

Allen Clement, Flavio Junqueira, Aniket Kate and Rodrigo Rodrigues (2012). On the (limited) power of non-equivocation. Proceedings of the 2012 ACM Symposium on Principles of Distributed Computing (PODC '12), pp. 301–308. DOI: 10.1145/2332432.2332490. Stopping a copy from saying two different things doesn't lower the 3f + 1 bound on its own, and with signatures a third party can check it turns a crash-tolerant protocol into one that tolerates liars with no extra copies.

Victor Costan and Srinivas Devadas (2016). Intel SGX Explained. Cryptology ePrint Archive, Paper 2016/086. IACR ePrint 2016/086. How SGX enclaves are measured and how they produce reports of their own, apart from the chip's registers.

Scott A. Crosby and Dan S. Wallach (2009). Efficient data structures for tamper-evident logging. Proceedings of the 18th USENIX Security Symposium, Montreal.

Conference page at usenix.org. Logs built on hash trees at Rice University, with short proofs that an entry is in and that a later version extends an earlier one.

Joe DeBlasio, for the Chrome CT team (2025). Discussion: removal of one-6962-log requirement. Certificate Transparency Policy group, 17 November 2025. Google Groups post, chromium.org ct-policy, thread amrnTiDFdwM. The proposal to stop requiring every certificate to carry a timestamp from at least one RFC 6962 log.

Jérémie Decouchant, David Kozhaya, Vincent Rahli and Jiangshan Yu (2022). Damysus: streamlined BFT consensus leveraging trusted components. Proceedings of the Seventeenth European Conference on Computer Systems (EuroSys '22), pp. 1–16. DOI: 10.1145/3492321.3519568. The small trusted service called the Checker.

Devcon (n.d.). Past events. devcon.org, past events. Devcon 0 in Berlin, Devcon 1 in London and Devcon 2 in Shanghai, where denial-of-service attacks hit the network hours before the conference began.

Ethos (2016). Ethereum's Shanghai attacks. ethos.dev, 30 November 2016. ethos.dev/shanghai-attacks. The attacks that began during the second developer conference in Shanghai, and the October 2016 fork that repriced the cheap opcodes.

Ethereum Foundation (n.d.). Ethereum execution layer specifications (execution-specs). GitHub repository. Repository at github.com, execution-specs. The specification the copied gas and refund arithmetic for one storage slot follows.

Ethereum Foundation (n.d.). Timeline of all Ethereum forks. ethereum.org. ethereum.org, Ethereum forks. Tangerine Whistle on 18 October 2016, Berlin on 15 April 2021, London on 5 August 2021, and the Merge to proof of stake on 15 September 2022.

Yoichi Hirai (2017). Isabelle/HOL proofs of accountable safety for one-message Casper, in the repository pos. Repository at github.com, pos.

Ben Laurie, Adam Langley and Emilia Kasper (2013). Certificate Transparency. RFC 6962, Internet Engineering Task Force, June 2013. DOI: 10.17487/RFC6962. The version the running logs and browsers still follow.

Ben Laurie, Eran Messeri and Rob Stradling (2021). Certificate Transparency Version 2.0. RFC 9162, Internet Engineering Task Force, December 2021. DOI: 10.17487/RFC9162. The version copied here, which says logs let misissued certificates be detected afterwards without preventing them, and lists a log showing different trees to different parties among the ways a log can misbehave.

Dave Levin, John R. Douceur, Jacob R. Lorch and Thomas Moscibroda (2009). TrInc: small trusted hardware for large distributed systems. Proceedings of the 6th USENIX Symposium on Networked Systems Design and Implementation (NSDI '09), pp.

1–14. PDF at usenix.org, NSDI 2009, Levin. A trusted component that is only a counter that never goes down and a key.

Sinisa Matetic, Mansoor Ahmed, Kari Kostiainen, Aritra Dhar, David Sommer, Arthur Gervais, Ari Juels and Srdjan Capkun (2017). ROTE: rollback protection for trusted execution. Proceedings of the 26th USENIX Security Symposium, pp. 1289–1306. USENIX page at usenix.org, matetic. Several machines vouch for each other's counters, so the only way left to break integrity is to reset all of them to their initial state.

Mozilla (n.d.). Certificate Transparency. MDN Web Docs. developer.mozilla.org, Certificate Transparency. Browser implementations are based on RFC 6962, though RFC 9162 is the current specification.

Devon O'Brien (2018). Certificate Transparency Enforcement in Google Chrome. Certificate Transparency Policy group, 6 February 2018. Google Groups post, chromium.org ct-policy, thread wHILiYf31DE. Certificates issued after 30 April 2018 must comply with Chrome's CT policy or are not trusted, with a full-page warning for the visitor.

Karl Palmskog, Milos Gligoric, Lucas Peña, Brandon Moore and Grigore Roşu (2018). Verification of Casper in the Coq Proof Assistant. Technical report, Runtime Verification, Inc., The University of Texas at Austin and University of Illinois at Urbana-Champaign. PDF at github.com, runtimeverification/casper-proofs, report.

Bryan Parno, Jacob R. Lorch, John R. Douceur, James Mickens and Jonathan M. McCune (2011). Memoir: practical state continuity for protected modules. Proceedings of the 2011 IEEE Symposium on Security and Privacy, pp. 379–394. DOI: 10.1109/SP.2011.38. The counter that need only tick once every five seconds, the storage rated for about 100,000 writes, and the password checker rolled back to undo a password change.

Rob Smart (2021). Implementing the 10 Security Goals: Anti-rollback Explained. PSA Certified, 27 April 2021. psacertified.org blog, anti-rollback explained. A counter kept in one-time-programmable fuse bits, with enough bits for the updates expected over the root of trust's supported lifetime.

Raoul Strackx and Frank Piessens (2016). Ariadne: a minimal approach to state continuity. Proceedings of the 25th USENIX Security Symposium, pp. 875–892. USENIX page at usenix.org, strackx. A single bit flip of lasting memory per state update, which the authors call the theoretical lower limit.

Runtime Verification, Inc. and The University of Texas at Austin (2018). casper-proofs: models and proofs of Casper FFG accountable safety and plausible liveness.

Repository at github.com, casper-proofs. The accountable-safety argument copied here, built on Yoichi Hirai's proof with the two-thirds overlap taken as a condition.

Giuliana Santos Veronese, Miguel Correia, Alysson Neves Bessani, Lau Cheuk Lung and Paulo Verissimo (2013). Efficient Byzantine fault-tolerance. IEEE Transactions on Computers 62(1), pp. 16–30. DOI: 10.1109/TC.2011.221. MinBFT, which uses a trusted counter to agree with 2f + 1 copies.

Chapter 19. Which box is worth keeping

János Aczél, Bruno Forte and Che Tat Ng (1974). Why the Shannon and Hartley entropies are "natural." Advances in Applied Probability 6(1), pp. 131–146. DOI: 10.2307/1426210. The entropies with a short list of natural properties are exactly the mixtures of Shannon's and Hartley's.

John C. Baez, Tobias Fritz and Tom Leinster (2011). A characterization of entropy in terms of information loss. Entropy 13(11), pp. 1945–1957. DOI: 10.3390/e13111945; arXiv: 1106.1791. Shannon's entropy, up to a constant, as the only measure of information loss that adds along a chain of maps, mixes the way probabilities mix, and changes smoothly.

Philip Ball (2019). Quantum Darwinism, an Idea to Explain Objective Reality, Passes First Tests. Quanta Magazine, 22 July 2019. Quanta Magazine, 22 July 2019. The three experiments in Italy, China and Germany, with Zurek's and others' cautions about them.

Fernando G. S. L. Brandão, Marco Piani and Paweł Horodecki (2015). Generic emergence of classical features in quantum Darwinism. Nature Communications 6, 7908. DOI: 10.1038/ncomms8908.

Vitalik Buterin, Eric Conner, Rick Dudley, Matthew Slipper, Ian Norden and Abdelhamid Bakhta (2019). EIP-1559: Fee market change for ETH 1.0 chain. Ethereum Improvement Proposals, created 13 April 2019. eips.ethereum.org/EIPS/eip-1559. The base fee aimed at a half-full block, moving by at most an eighth per block.

Ming-Cheng Chen, Han-Sen Zhong, Yuan Li, Dian Wu, Xi-Lin Wang, Li Li, Nai-Le Liu, Chao-Yang Lu and Jian-Wei Pan (2019). Emergence of classical objectivity of quantum Darwinism in a photonic quantum simulator. Science Bulletin 64(9), pp. 580–585. DOI: 10.1016/j.scib.2019.03.032. The experiment in China, done with light.

Mario A. Ciampini, Giorgia Pinna, Paolo Mataloni and Mauro Paternostro (2018). Experimental signature of quantum Darwinism in photonic cluster states. Physical

Review A 98, 020101. DOI: 10.1103/PhysRevA.98.020101. The experiment in Italy, done with light.

Philippe Allard Guérin, Veronika Baumann, Flavio Del Santo and Časlav Brukner (2021). A no-go theorem for the persistent reality of Wigner's friend's perception. Communications Physics 4, 93. DOI: 10.1038/s42005-021-00589-1.

J. J. Halliwell (1999). Somewhere in the universe: Where is the information stored when histories decohere? Physical Review D 60, 105031. DOI: 10.1103/PhysRevD.60.105031. The link between records and decoherent histories carried from pure starting states to mixed ones, where records can still exist but may be imperfect.

Omer Tamuz, Caltech, faculty page at hss.caltech.edu, omer-tamuz; Philipp Strack, Yale, faculty page at economics.yale.edu, philipp-strack; Luciano Pomatto, Caltech, HSS new faculty 2016–17. The economists behind the 2023 cost-of-information paper.

Luciano Pomatto, Philipp Strack and Omer Tamuz (2023). The cost of information: the case of constant marginal costs. American Economic Review 113(5), pp. 1360–1393. DOI: 10.1257/aer.20190185. A uniqueness result for the cost of getting information when running an experiment twice costs twice as much.

Xiao-Liang Qi and Daniel Ranard (2021). Emergent classicality in general multipartite states and channels. Quantum 5, article 555. DOI: 10.22331/q-2021-09-28-555.

C. Jess Riedel, Wojciech H. Zurek and Michael Zwolak (2016). Objective past of a quantum universe: Redundant records of consistent histories. Physical Review A 93, 032126. DOI: 10.1103/PhysRevA.93.032126; arXiv: 1312.0331.

Tom Leinster. Wikipedia. Wikipedia, Tom Leinster. A British mathematician known for his work on category theory, at Glasgow and then Edinburgh. John C. Baez, Wikipedia, Wikipedia, John C. Baez: mathematical physicist at the University of California, Riverside, working on higher categories and applied category theory. Tobias Fritz's category-theoretic approach to probability: arXiv listing.

T. K. Unden, D. Louzon, M. Zwolak, W. H. Zurek and F. Jelezko (2019). Revealing the Emergence of Classicality Using Nitrogen-Vacancy Centers. Physical Review Letters 123, 140402. DOI: 10.1103/PhysRevLett.123.140402. The experiment at Ulm on a single defect in diamond and the carbon atoms around it.

University of Waterloo, Department of Pure Mathematics (2020). We remember Distinguished Professor Emeritus János Aczél. Notice at uwaterloo.ca. Aczél at Waterloo from 1965, founder of Aequationes Mathematicae. Bruno Forte at Waterloo: Mathematics Genealogy Project, id 30611. Che Tat Ng's doctorate, Waterloo 1971, under Aczél: Mathematics Genealogy Project, id 102040.

Wojciech H. Zurek (1981). Pointer basis of quantum apparatus: Into what mixture does the wave packet collapse? Physical Review D 24(6), pp. 1516–1525. DOI: 10.1103/PhysRevD.24.1516.

Wojciech H. Zurek (2009). Quantum Darwinism. Nature Physics 5(3), pp. 181–188. DOI: 10.1038/nphys1202; arXiv: 0903.5082v1.

Chapter 20. Heat

Erik Aurell, Carlos Mejía-Monasterio and Paolo Muratore-Ginanneschi (2011). Optimal protocols and optimal transport in stochastic thermodynamics. Physical Review Letters 106, 250601. DOI: 10.1103/PhysRevLett.106.250601; arXiv: 1012.2037.

Antoine Bérut, Artyom Petrosyan and Sergio Ciliberto (2013). Detailed Jarzynski equality applied to a logically irreversible procedure. EPL (Europhysics Letters) 103(6), 60002. DOI: 10.1209/0295-5075/103/60002; arXiv: 1302.4417. The bound of kT times ln 2 plus ln of the success rate on the work averaged over the successful erasures.

Antoine Bérut, Artyom Petrosyan and Sergio Ciliberto (2015). Information and thermodynamics: experimental verification of Landauer's erasure principle. Journal of Statistical Mechanics: Theory and Experiment, P06015. DOI: 10.1088/1742-5468/2015/06/P06015; arXiv: 1503.06537. The long version, with the protocol in full, the fit $\langle Q \rangle = kT \ln 2 + B/\tau$ with $B = 8.15\, kT \cdot \mathrm{s}$, and the hopping times of about 3,000 seconds above an $8\,kT$ barrier and about ten seconds at $2.2\,kT$.

John Earman and John D. Norton (1998, 1999). Exorcist XIV: The Wrath of Maxwell's Demon. Part I: From Maxwell to Szilard, Studies in History and Philosophy of Modern Physics 29(4), pp. 435–471, DOI: 10.1016/S1355-2198(98)00023-9; Part II: From Szilard to Landauer and Beyond, 30(1), pp. 1–40, DOI: 10.1016/S1355-2198(98)00026-4. Author copies: Part I at sites.pitt.edu, Exorcist1; Part II at sites.pitt.edu, Exorcist2. The dilemma with its sound and profound horns is set out at the start of Part II.

Rocco Gaudenzi, Enrique Burzurí, Satoru Maegawa, Herre S. J. van der Zant and Fernando Luis (2018). Quantum Landauer erasure with a molecular nanomagnet. Nature Physics 14, pp. 565–568. DOI: 10.1038/s41567-018-0070-7. Landauer's floor on a register of molecular spins at one kelvin.

Momčilo Gavrilov and John Bechhoefer (2016). Erasure without Work in an Asymmetric Double-Well Potential. Physical Review Letters 117, 200601. DOI: 10.1103/PhysRevLett.117.200601. Average erasure work below $kT \ln 2$ when the two wells are not alike.

Anson Ho, Ege Erdil and Tamay Besiroglu (2023). Limits to the Energy Efficiency of CMOS Microprocessors. arXiv: 2312.08595. Table I: switching a CMOS gate, 100 aJ to 100 fJ.

Jeongmin Hong, Brian Lambson, Scott Dhuey and Jeffrey Bokor (2016). Experimental test of Landauer's principle in single-bit operations on nanomagnetic memory bits. Science Advances 2(3), e1501492. DOI: 10.1126/sciadv.1501492. The floor on a real memory technology, $1.45 \pm 0.35\,kT$, within two standard deviations.

Christopher Jarzynski (1997). Nonequilibrium Equality for Free Energy Differences. Physical Review Letters 78(14), pp. 2690–2693. DOI: 10.1103/PhysRevLett.78.2690.

Ryoichi Kawai, Juan M. R. Parrondo and Christian Van den Broeck (2007). Dissipation: the phase-space perspective. Physical Review Letters 98, 080602. DOI: 10.1103/PhysRevLett.98.080602. The detailed version of Jarzynski's equality for a partition of the states.

Artemy Kolchinsky and David H. Wolpert (2017). Dependence of dissipation on the initial distribution over states. Journal of Statistical Mechanics: Theory and Experiment 2017, 083202. DOI: 10.1088/1742-5468/aa7ee1. The extra cost of running a fixed process on a spread it wasn't built for, which they call dissipation due to incorrect priors.

James Ladyman, Stuart Presnell, Anthony J. Short and Berry Groisman (2007). The connection between logical and thermodynamic irreversibility. Studies in History and Philosophy of Modern Physics 38(1), pp. 58–79. DOI: 10.1016/j.shpsb.2006.03.007. From Kelvin's statement of the second law, no L-machine carries out a logically irreversible operation in a thermodynamically reversible way.

Gonzalo Manzano, Gülce Kardeş, Édgar Roldán and David H. Wolpert (2024). Thermodynamics of Computations with Absolute Irreversibility, Unidirectional Transitions, and Stochastic Computation Times. Physical Review X 14, 021026. DOI: 10.1103/PhysRevX.14.021026.

O. J. E. Maroney (2009). Generalizing Landauer's Principle. Physical Review E 79, 031105. DOI: 10.1103/PhysRevE.79.031105; arXiv: quant-ph/0702094v3. Written at the Perimeter Institute; the general form of the principle, and the floor for a memory whose two states differ.

Owen Maroney (2009). Information Processing and Thermodynamic Entropy. Stanford Encyclopedia of Philosophy, first published 15 September 2009. Stanford Encyclopedia of Philosophy, entry information-entropy. It ends without a verdict on Landauer's principle.

Yûto Murashita, Ken Funo and Masahito Ueda (2014). Nonequilibrium equalities in absolutely irreversible processes. Physical Review E 90, 042110. DOI: 10.1103/PhysRevE.90.042110; arXiv: 1401.4494. Absolute irreversibility, and the free expansion of a gas from one part of a box split in the ratio l to 1 minus l.

John D. Norton (2005). Eaters of the Lotus: Landauer's Principle and the Return of Maxwell's Demon. Studies in History and Philosophy of Modern Physics 36(2), pp. 375–411. DOI: 10.1016/j.shpsb.2004.12.002.

John D. Norton (2018). Maxwell's demon does not compute. In M. E. Cuffaro and S. C. Fletcher (eds.), Physical Perspectives on Computation, Computational Perspectives on Physics, Cambridge University Press, pp. 240–256. DOI: 10.1017/9781316759745.012; PhilSci-Archive 14808. The demon exorcised with Liouville's theorem alone.

John D. Norton (2026). Too good to be true: entropy, information, computation, and the uninformed thermodynamics of erasure. Philosophy of Science, published online 22 May 2026; manuscript dated 2025. DOI: 10.1017/psa.2026.10221. The careful literature "almost invariably accepts the sound horn," and what works there is relabelled ordinary thermodynamics.

Juan M. R. Parrondo, Jordan M. Horowitz and Takahiro Sagawa (2015). Thermodynamics of information. Nature Physics 11(2), pp. 131–139. DOI: 10.1038/nphys3230.

Paul M. Riechers, Alexander B. Boyd, Gregory W. Wimsatt and James P. Crutchfield (2020). Balancing error and dissipation in computing. Physical Review Research 2, 033524. DOI: 10.1103/PhysRevResearch.2.033524; arXiv: 1909.06650. Time-symmetric control and nonreciprocal transitions.

Takahiro Sagawa (2014). Thermodynamic and logical reversibilities revisited. Journal of Statistical Mechanics: Theory and Experiment, P03025. DOI: 10.1088/1742-5468/2014/03/P03025; arXiv: 1311.1886v2. Erasure that can in principle approach producing no entropy overall.

Tim Schmiedl and Udo Seifert (2007). Optimal finite-time processes in stochastic thermodynamics. Physical Review Letters 98, 108301. DOI: 10.1103/PhysRevLett.98.108301; arXiv: cond-mat/0701554. The least work to drag a trapped bead a set distance in a set time.

J. Schnakenberg (1976). Network theory of microscopic and macroscopic behavior of master equation systems. Reviews of Modern Physics 48(4), pp. 571–585. DOI: 10.1103/RevModPhys.48.571. The entropy produced by a system that jumps between states.

Udo Seifert (2012). Stochastic thermodynamics, fluctuation theorems and molecular machines. Reports on Progress in Physics 75(12), 126001. DOI: 10.1088/0034-4885/75/12/126001; arXiv: 1205.4176. The local-detailed-balance premise linking jump ratios to heat.

Claude E. Shannon, interviewed by Robert Price (1982). Oral history, 28 July 1982, IEEE History Center. Transcript at ethw.org, Oral-History:Claude_E._Shannon. Excerpts printed as A conversation with Claude Shannon, IEEE Communications Magazine 22(5), pp. 123–126, May 1984, DOI: 10.1109/MCOM.1984.1091957. Asked whether von Neumann gave him the word "entropy," Shannon answered, "No, I don't think he did."

Naoto Shiraishi, Ken Funo and Keiji Saito (2018). Speed limit for classical stochastic processes. Physical Review Letters 121, 070601. DOI: 10.1103/PhysRevLett.121.070601; arXiv: 1802.06554.

Myron Tribus and Edward C. McIrvine (1971). Energy and Information. Scientific American 225(3), pp. 179–188. DOI: 10.1038/scientificamerican0971-179. The story of von Neumann telling Shannon to call his measure entropy.

Suriyanarayanan Vaikuntanathan and Christopher Jarzynski (2009). Dissipation and lag in irreversible processes. EPL (Europhysics Letters) 87(6), 60005. DOI: 10.1209/0295-5075/87/60005.

David H. Wolpert and Artemy Kolchinsky (2020). Thermodynamics of computing with circuits. New Journal of Physics 22, 063047. DOI: 10.1088/1367-2630/ab82b8. First posted in 2018 as Exact, complete expressions for the thermodynamic costs of circuits, arXiv: 1806.04103v1, where the extra cost of a mismatched spread is called mismatch cost.

Chapter 21. Keeping without the heat

Jacob D. Bekenstein (1981). Universal upper bound on the entropy-to-energy ratio for bounded systems. Physical Review D 23(2), pp. 287–298. DOI: 10.1103/PhysRevD.23.287. The bound ties entropy to energy and size under its own physical assumptions; it isn't a measured count of a device's states.

John S. Bell (1990). Against "measurement". Physics World 3(8), pp. 33–41. DOI: 10.1088/2058-7058/3/8/26. The list of words with no place in a physics that wants to be precise, with measurement the worst of all.

Charles H. Bennett (1989). Time/Space Trade-Offs for Reversible Computation. SIAM Journal on Computing 18(4), pp. 766–776. DOI: 10.1137/0218053. A

reversible machine saves room by undoing and redoing its own intermediate work, paying for it in time.

Niels Bohr (1958). Atomic Physics and Human Knowledge. John Wiley and Sons, New York. The essay Unity of Knowledge, an address given in October 1954 for the bicentennial of Columbia University, first printed in The Unity of Knowledge (Doubleday, 1955). The line about permanent marks on a photographic plate is on p. 73, where the sentence ends with electrons entering the emulsion, not with water drops in a cloud chamber.

Sandro Donadi, Kristian Piscicchia, Catalina Curceanu, Lajos Diósi, Matthias Laubenstein and Angelo Bassi (2021). Underground test of gravity-related wave function collapse. Nature Physics 17, pp. 74–78. DOI: 10.1038/s41567-020-1008-4. The radiation a gravity-driven collapse would give off, listened for at Gran Sasso and not found at the level the version with no adjustable settings predicts.

Mauro Dorato and Jan Faye (2026). Niels Bohr as a Physicist of Principle. Preprint. arXiv: 2609.04297. Where I first found Bohr's line; their quotation runs on into a clause about a water drop forming around an ion in a cloud chamber, which isn't on Bohr's p. 73.

Bernhard Rauer, Sebastian Erne, Thomas Schweigler, Federica Cataldini, Mohammadamin Tajik and Jörg Schmiedmayer (2018). Recurrences in an isolated quantum many-body system. Science 360(6386), pp. 307–310. DOI: 10.1126/science.aan7938; arXiv: 1705.08231. The coherence of an isolated gas of thousands of atoms came partly back around.

David Reeb and Michael M. Wolf (2014). An improved Landauer principle with finite-size corrections. New Journal of Physics 16, 103011. DOI: 10.1088/1367-2630/16/10/103011; arXiv: 1306.4352. Their heat bound assumes a reservoir that starts thermal and uncorrelated with the memory, and a combined motion that is unitary.

Andrea Vinante, Rima Mezzena, Paolo Falferi, Matteo Carlesso and Angelo Bassi (2017). Improved noninterferometric test of collapse models using ultracold cantilevers. Physical Review Letters 119, 110401. DOI: 10.1103/PhysRevLett.119.110401. A cantilever at millikelvin with a force noise of unknown origin, compatible with the heating one collapse model predicts.

John Archibald Wheeler (1989). Information, Physics, Quantum: The Search for Links. Proceedings of the 3rd International Symposium on Foundations of Quantum Mechanics, Tokyo, 1989, pp. 354–368. PDF at jawarchive.files.wordpress.com, informationquantumphysics. Also printed in Wojciech H. Zurek (ed.), Complexity, Entropy and the Physics of Information, Addison-Wesley, 1990. It from bit.

Chapter 22. What's shown, what's chosen, what's argued

Alok Aggarwal and Jeffrey Scott Vitter (1988). The input/output complexity of sorting and related problems. Communications of the ACM 31(9), pp. 1116–1127. DOI: 10.1145/48529.48535. A model that counts only the blocks moved between a slow memory and a fast one.

Pablo Arrighi and Gilles Dowek (2012). The physical Church-Turing thesis and the principles of quantum theory. International Journal of Foundations of Computer Science 23(5), pp. 1131–1145. DOI: 10.1142/S0129054112500153; arXiv: 1102.1612. Gandy's argument laid out from homogeneity of space and time, bounded density and velocity of information, and a quiet background outside a finite region.

Alexander Bastounis, Fabian Circelli and Anders C. Hansen (2026). Navier-Stokes lost in translation: Why Lean verification of AI autoformalisation does not guarantee correct natural language proofs. arXiv:2610.08144. arXiv. Argues that the version of OpenAI's proof written for the proof checker doesn't match the proof written in words.

Tristan Buckmaster (2026). Statement on the forced blowup results with Levent Alpöge and on events with OpenAI. Author's page, Courant Institute, New York University. PDF at cims.nyu.edu, tristanb/statement. Forced blowup for incompressible porous media, Boussinesq and three-dimensional Euler, and his account of the dispute over credit.

Stephen A. Cook and Robert A. Reckhow (1973). Time bounded random access machines. Journal of Computer and System Sciences 7(4), pp. 354–375. DOI: 10.1016/S0022-0000(73)80029-7.

Erik D. Demaine, Jayson Lynch, Geronimo J. Mirano and Nirvan Tyagi (2016). Energy-Efficient Algorithms. Proceedings of the 2016 ACM Conference on Innovations in Theoretical Computer Science (ITCS), pp. 321–332. DOI: 10.1145/2840728.2840756. Energy counted beside time and space, charging for the bits an instruction destroys.

Charles L. Fefferman (n.d.). Existence and Smoothness of the Navier–Stokes Equation. Official problem description, Clay Mathematics Institute. PDF at claymath.org, the official Navier–Stokes problem statement. The four statements a solution may prove; (C) and (D) are breakdown with a smooth force, on all of space and on the periodic box.

Robin Gandy (1980). Church's Thesis and Principles for Mechanisms. In The Kleene Symposium, Studies in Logic and the Foundations of Mathematics, North-Holland, Amsterdam, pp. 123–148. DOI: 10.1016/S0049-237X(08)71257-6. Principles for mechanisms from which computability follows.

Douglas Harper (n.d.). Online Etymology Dictionary, entry “ledger”. etymonline.com, ledger. A large book that lay in one place, around 1400; the account-book sense from the 1580s.

Clay Mathematics Institute (n.d.). The Millennium Prize Problems. claymath.org, Millennium Problems. Announced at a meeting in Paris on 24 May 2000, a million dollars each; P versus NP is listed as unsolved and Navier–Stokes as active.

Clay Mathematics Institute (n.d.). Navier–Stokes Equation. claymath.org, Navier–Stokes Equation. The problem’s status, listed as active.

Clay Mathematics Institute (n.d.). The Poincaré Conjecture. claymath.org, Poincaré Conjecture. Perelman’s preprints of 2002 and 2003, and the prize of 2010.

Konstantin Kakaes (2026). AI Has Solved One of Math’s $1 Million Millennium Prize Problems. Quanta Magazine, 8 September 2026. Quanta Magazine, 8 September 2026. About ten thousand agents, eighty-eight hours, and a Lean formalization that took about seventeen hours more.

Ming Li and Paul Vitányi (1996). Reversibility and adiabatic computation: trading time and space for energy. Proceedings of the Royal Society of London A 452, pp. 769–789. DOI: 10.1098/rspa.1996.0039; arXiv: quant-ph/9703022. One unit of price for each bit erased or taken in irreversibly, and nothing for reversible steps.

OpenAI (2026). Finite Time Blowup for Navier–Stokes. Preprint, 166 pages. PDF at cdn.openai.com, navier-stokes. A smooth force, compactly supported, that drives a three-dimensional flow starting from rest to unbounded velocity in finite time; the paper says this establishes Fefferman’s alternatives (C) and (D).

OpenAI (2026). On the Navier–Stokes Millennium Prize Problem. Announcement, 8 September 2026. openai.com, navier-stokes-solution.

Grisha Perelman (2002). The entropy formula for the Ricci flow and its geometric applications. Preprint. arXiv: math/0211159.

Grisha Perelman (2003). Ricci flow with surgery on three-manifolds. Preprint. arXiv: math/0303109.

Grisha Perelman (2003). Finite extinction time for the solutions to the Ricci flow on certain three-manifolds. Preprint. arXiv: math/0307245.

Andrew Chi-Chih Yao (1981). Should tables be sorted? Journal of the ACM 28(3), pp. 615–628. DOI: 10.1145/322261.322274. The cell-probe model, which counts only the looks into memory; an earlier version appeared at FOCS 1978.

Afterword

Meta (2021). Introducing Meta: A Social Technology Company. Meta Newsroom, 28 October 2021. Announcement at about.fb.com, facebook-company-is-now-meta. Facebook's parent company takes the name Meta.

What's mine, and what isn't

Robert K. Merton (1957). Priorities in Scientific Discovery: A Chapter in the Sociology of Science. American Sociological Review 22(6), pp. 635–659. DOI: 10.2307/2089193. Fights over priority in science.

Stephen M. Stigler (1980). Stigler's Law of Eponymy. Transactions of the New York Academy of Sciences 39(1 Series II), pp. 147–157. DOI: 10.1111/j.2164-0947.1980.tb02775.x. No discovery is named after the person who first found it, credited to Merton.